US Army Corps of Engineers
Tulsa District

Mountain Fork River
Broken Bow Powerhouse, Oklahoma

Broken Bow Powerhouse Roof Replacement

Design-Build Specifications

Ready to Advertise (RTA)

Solicitation No. W912BV-26-R-A015

16 APRIL 2026

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SOLICITATION, OFFER, AND AWARD *(Construction, Alteration, or Repair)*	1. SOLICITATION NUMBER W912BV26RA015	2. TYPE OF SOLICITATION ☐ SEALED BID (IFB) INVITATION FOR BID ☒ NEGOTIATED (RFP) REQUEST FOR PROPOSAL	3. DATE ISSUED 16 April 2026	PAGE 1 OF PAGES 121

IMPORTANT - The "offer" section on the reverse must be fully completed by offeror.

4. CONTRACT NUMBER	5. REQUISITION/PURCHASE REQUEST NUMBER	6. PROJECT NUMBER PANSWD-26-P-0000 027666

7. ISSUED BY CODE W912BV	8. ADDRESS OFFER TO
W076 ENDIST TULSA KO CONTRACTING DIVISION, 2488 E 81ST STREET TULSA, OK 74137-4290 UNITED STATES CLINTON YANDELL, EMAIL: CLINTON.J.YANDELL@USACE.ARMY.MIL TELEPHONE: (918)669-4920	W076 ENDIST TULSA KO CONTRACTING DIVISION, 2488 E 81ST STREET TULSA, OK 74137-4290 UNITED STATES

9. FOR INFORMATION CALL:	a. NAME CLINTON YANDELL	b. TELEPHONE NUMBER *(Include area code) (NO COLLECT CALLS)* (918)669-4920

SOLICITATION

NOTE: In sealed bid solicitations "offer" and "offeror" mean "bid and "bidder".

10. THE GOVERNMENT REQUIRES PERFORMANCE OF THE WORK DESCRIBED IN THESE DOCUMENTS *(Title, identifying number, date)*

PROJECT TITLE: BROKEN BOW POWERHOUSE ROOF REPLACEMENT
PROJECT LOCATION: MOUNTAIN FORK RIVER, 9 MILES NORTH OF BROKEN BOW, OKLAHOMA

THIS ACQUISITION WILL BE CONDUCTED AS TOTAL SMALL BUSINESS SET-ASIDE, LOWEST PRICE TECHNICALLY ACCEPTABLE (LPTA), IN ACCORDANCE WITH FAR PART 6

THE NAICS CODE IS 238160 - ROOFING CONTRACTORS, AND THE SIZE STANDARD IS $19M

THIS SOLICITATION REQUIRES OFFERORS TO ENTER A NATIONAL INSTITUE OF STANDARDS AND TECHNOLOGY (NIST) SCORE IN THE SUPPLIER PERFORMANCE RISK SYSTEM (SPRS). AN AWARD CANNOT BE MADE WITHOUT A NIST SCORE

THE MAGNITUDE OF THIS PROJECT IS BETWEEN $1,000,000.00 AND $5,000,000.00

THE GOVERNMENT POINT OF CONTACT FOR THIS ACTION IS CLINTON J. YANDELL, CLINTON.J.YANDELL@USACE.ARMY.MIL

11. The contractor shall begin performance within 10 calendar days and complete it within 730 calendar days after receiving

☐ award, ☒ notice to proceed. This performance period is ☒ mandatory ☐ negotiable. ***(See*** FAR 52.211-10 ***).***

12a. THE CONTRACTOR MUST FURNISH ANY REQUIRED PERFORMANCE AND PAYMENT BONDS? *(If "YES", indicate within how many calendar days after award in Item 12b.)* ☒ YES ☐ NO	12b. CALENDAR DAYS 10

13. ADDITIONAL SOLICITATION REQUIREMENTS:

a. Sealed offers in original and 1 copies to perform the work required are due at the place specified in Item 8 by 02:00 PM ***(hour)*** local time 18 May 2026 (date). If this is a sealed bid solicitation, offers will be publicly opened at that time. Sealed envelopes containing offers shall be marked to show the offeror's name and address, the solicitation number, and the date and time offers are due.

b. An offer guarantee ☒ is, ☐ is not required.

c. All offers are subject to the (1) work requirements, and (2) other provisions and clauses incorporated in the solicitation in full text or by reference.

d. Offers providing less than 60 calendar days for Government acceptance after the date offers are due will not be considered and will be rejected.

OFFER *(Must be fully completed by offeror)*

14. NAME AND ADDRESS OF OFFEROR *(Include ZIP Code)*

15. TELEPHONE NUMBER *(Include area code)*

16. REMITTANCE ADDRESS *(Include only if different than Item 14.)*

CODE FACILITY CODE

17. The offeror agrees to perform the work required at the prices specified below in strict accordance with the terms of this solicitation, if this offer is accepted by the Government in writing within ________ calendar days after the date offers are due. *(Insert any number equal to or greater than the minimum requirement stated in Item 13d. Failure to insert any number means the offeror accepts the minimum in Item 13d.)*

AMOUNTS ▶

18. The offeror agrees to furnish any required performance and payment bonds.

19. ACKNOWLEDGMENT OF AMENDMENTS

(The offeror acknowledges receipt of amendments to the solicitation -- give number and date of each)

AMENDMENT NUMBER										
DATE										

20a. NAME AND TITLE OF PERSON AUTHORIZED TO SIGN OFFER *(Type or print)*

20b. SIGNATURE

20c. OFFER DATE

AWARD *(To be completed by Government)*

21. ITEMS ACCEPTED:

22. AMOUNT

23. ACCOUNTING AND APPROPRIATION DATA

24. SUBMIT INVOICES TO ADDRESS SHOWN IN *(4 copies unless otherwise specified)* ▶ ITEM

25. OTHER THAN FULL AND OPEN COMPETITION PURSUANT TO THE UNITED STATES CODE AT

☐ 10 U.S.C. 3204(a) ( ) ☐ 41 U.S.C. 3304(a) ( )

26. ADMINISTERED BY

27. PAYMENT WILL BE MADE BY

CONTRACTING OFFICER WILL COMPLETE ITEM 28 OR 29 AS APPLICABLE

☐ 28. NEGOTIATED AGREEMENT *(Contractor is required to sign this document and return ________ copies to issuing office.)* Contractor agrees to furnish and deliver all items or perform all work requirements identified on this form and any continuation sheets for the consideration stated in this contract. The rights and obligations of the parties to this contract shall be governed by (a) this contract award, (b) the solicitation, and (c) the clauses, representations, certifications, and specifications incorporated by reference in or attached to this contract.

☐ 29. AWARD *(Contractor is not required to sign this document.)* Your offer on this solicitation is hereby accepted as to the items listed. This award consummates the contract, which consists of (a) the Government solicitation and your offer, and (b) this contract award. No further contractual document is necessary.

30a. NAME AND TITLE OF CONTRACTOR OR PERSON AUTHORIZED TO SIGN *(Type or print)*

31a. NAME OF CONTRACTING OFFICER *(Type or print)*

30b. SIGNATURE

30c. DATE

31b. UNITED STATES OF AMERICA

BY

31c. DATE

Section 00 00 00 - Procurement and Contracting Requirements

Instrument Name: Broken Bow Powerhouse Roof Replacement

THIS ACQUISITION WILL BE CONDUCTED AS TOTAL SMALL BUSINESS SET-ASIDE, LOWEST PRICE TECHNICALLY ACCEPTABLE (LPTA), IN ACCORDANCE WITH FAR PART 6 THE NAICS CODE IS 238160 - ROOFING CONTRACTORS, AND THE SIZE STANDARD IS $19M. THIS SOLICITATION REQUIRES OFFERORS TO ENTER A NATIONAL INSTITUE OF STANDARDS AND TECHNOLOGY (NIST) SCORE IN THE SUPPLIER PERFORMANCE RISK SYSTEM (SPRS). AN AWARD CANNOT BE MADE WITHOUT A NIST SCORE. THE MAGNITUDE OF THIS PROJECT IS BETWEEN $1,000,000.00 AND $5,000,000.00. THE GOVERNMENT POINT OF CONTACT FOR THIS ACTION IS CLINTON J. YANDELL, CLINTON.J.YANDELL@USACE.ARMY.MIL

SECTION 00 11 00

PRICING (CLIN) SCHEDULE

SWT – 04/26

Base Bid

CLIN No.	DESCRIPTION	UNIT	AMOUNT
0001	Upper Roof Area Design	JOB	$________________________
0002	Upper Roof Area Construction	JOB	$________________________
0003	Lower Roof Area Design	JOB	$________________________
0004	Lower Roof Area Construction	JOB	$________________________
Total – Base Bid (CLIN No. 0001 to 0004)			$________________________

Option 1

CLIN No.	DESCRIPTION	UNIT	AMOUNT
1001	Guardrail on Parapets Design	JOB	$________________________
1002	Guardrail on Parapets Construction	JOB	$________________________
Total – Option 1 (CLIN No. 1001 to 1002)			$________________________

Option 2

CLIN No.	DESCRIPTION	UNIT	AMOUNT
2001	Replace Roof Access Hatch and Ladder to Upper Roof Design	JOB	$________________________
2002	Replace Roof Access Hatch and Ladder to Upper Roof Construction	JOB	$________________________
Total – Option 2 (CLIN No. 2001 to 2002)			$________________________

Option 3

CLIN No.	DESCRIPTION	UNIT	AMOUNT
3001	Replace Exhaust Fans	JOB	$________________________
	Total – Option 3 (CLIN No. 3001)		$________________________

Contract Duration (Period of Performance) in Calendar Days (See Note 6): ___________

PRICING SCHEDULE NOTES

1. If this offer is accepted in writing by the Government within 60 calendar days after the offers are due, the offeror agrees to perform the work required, at the prices specified above, in strict accordance with the terms of this Scope of Work.

2. The Offeror shall submit pricing data on the latest Pricing Schedule as published in the solicitation or the latest amendment there to. If the Offeror has additions/deductions to line items, the revised prices for each item must be clearly stated in the Pricing Schedule.

3. Offerors must insert a price on all numbered items of the Pricing Schedule. Failure to do so will disqualify the Offer.

4. All quantities are estimated except where the unit is given as JOB. If the Offeror proposes an alternative to a lump-sum bid item, the price difference between the alternatives must be clearly stated or the Offeror agrees that the lump-sum adjustment and price offset will be applied on a prorated basis to every other item in the Bidding Schedule.

5. All the extensions of the unit prices shown (if applicable) will be subject to verification by the Government. In case of variation between the unit price and the extension, the unit price will be considered as the offer.

6. The Offeror shall propose the contract duration (period of performance) in calendar days in the location requesting it above.

   The Government estimates the contract duration at 730 calendar days unless the options are executed.

   The Government may issue the Notice to Proceed (NTP) via e-mail or Facsimile (FAX) or by other means.

   Day number 1 is the next business day following the receipt of the NTP.

6. Only one contract for the entire Scope of Work and corresponding schedule will be awarded under this Solicitation.

-- End of Section --

SECTION 00 21 00
INSTRUCTIONS, CONDITIONS AND NOTICES TO OFFERORS

1.1. GENERAL DESCRIPTION OF WORK

The objective of this project is to replace the Broken Bow Powerhouse roof in Broken Bow, Oklahoma. The Contractor shall provide all plant, materials, equipment, labor, instruments, tools, subcontractors, supervision, management and travel necessary to complete all the work in accordance with this summary of work, the design drawings, the specifications and all other contract documents and requirements.

This is a solicitation for a construction contract for the design and construction of a new approximately 15,000 square foot Broken Bow Powerhouse roof, installed in 1981. The existing roof is deteriorating due to membrane movement, joint separation, environmental components, and water infiltration due to poor drainage. Significant leaking is infiltrating architectural, structural, and electrical elements in the powerhouse. The design work to be performed includes replacement of the powerhouse roof and rehabilitation of the roof drainage system. In addition, include structural analysis to show the existing structure can accommodate newly applied loads, and a test cut and inspection of the existing roof system. The roof system shall be designed and installed so as to require no structural modifications to the existing structure.

This project will be a total turn-key, Design-Build (D-B) contract for roofing replacement with all design and construction work being performed by the Contractor. The Contractor shall provide all investigation, verification of site conditions and dimensions, design, research, travel, plant, materials, equipment, labor, instruments, tools, subcontractors, supervision, and management to complete all work required by these contract documents.

The Government will evaluate proposals in accordance with the criteria described herein and award a firm fixed price contract to the responsible firm whose proposal conforms to all terms and conditions of the solicitation.

1.2. GOVERNMENT SECURITY REQUIREMENTS

The Contractor shall submit all volumes of its proposal electronically via email to **clinton.j.yandell@usace.army.mil**. **There will be no exceptions.**

USACE Tulsa District
C/O: Clinton J. Yandell
Phone: 918-669-4920
Email: clinton.j.yandell@usace.army.mil

1.3. COPIES OF SOLICITATION DOCUMENTS AND AMENDMENTS

Copies of the solicitation and any amendments will be posted by electronic means via the System for Award Management (SAM) at https://sam.gov. The solicitation number is W912BV26RA015. It shall be the offeror's responsibility to check SAM.gov for any amendments. The offeror shall submit all requested information specified in the solicitation. Any information given to an offeror which impacts the solicitation and/or offer will be given in the form of a written amendment to the solicitation.

1.4. OFFEROR'S QUESTIONS AND COMMENTS

All contractual matters, questions and or comments relative to these documents should be submitted via Bidder Inquiry in ProjNet at http://www.projnet.org/projnet/binKornHome/index.cfm?strKornCob=HomePagePublic no later than 10 calendar days prior to the proposal due date, in order that they may be given consideration or actions taken prior to receipt of offers.

The bidder inquiry key for this project is: **CBKFDQ-WTKBYD**

Instructions for registering in the ProjNet Bidder Inquiry system are available on the above website.

1.5. SMALL BUSINESS SIZE STANDARD/NAICS CODE

The NAICS Code selected for this project is 238160 – Roofing Contractors, and the size standard is $19,000,000. Also see Revolutionary Federal Acquisition Regulation Overhaul (RFO) 52.204-8, Annual Representations and Certifications.

1.6. PROPOSAL EXPENSES AND PRE-CONTRACT COSTS

This request for proposal (RFP) does not commit the Government to pay, as a direct charge, any costs incurred in the preparation and submission of a proposal.

1.7. SITE VISIT

The vendor is highly encouraged to attend a site visit. See RFO 52.236-27 ALT I for site visit information.

1.8. ACCURACY IN PROPOSALS

Proposals must set forth full, accurate, and complete information as required by this RFP, including attachments. The penalty for making false statements is prescribed in 18 U.S.C. 1001.

1.9. PROPOSAL SUBMITTAL

1.9.1. The submittal method for proposals is electronically via email to clinton.j.yandell@usace.army.mil using the applicable solicitation number. Proposals submitted by mail or hand-carried will not be evaluated. Proposals sent through proprietary or third-party File Transfer Protocol (FTP) sites or DoD SAFE will not be retrieved. It is the responsibility of the Offeror to confirm receipt of proposals.

The following email addresses must be entered for notification of proposal receipt and download availability:
clinton.j.yandell@usace.army.mil
shawn.g.brady@usace.army.mil

1.9.2. Proposals will be received until the date and time specified in this request for proposal. Mail all bonds to:

U.S. Army Corps of Engineers, Tulsa District
Attn: CESWT-CT-E (Clinton J. Yandell)
2488 E 81st Street
Tulsa, OK 7413-4290

1.9.3. The packaging shall be marked Solicitation Number: W912BV26RA015

1.10. PROPOSALS

Offer Closing Date: 18 May 2026
Offer Closing Time: 2:00 PM CDT

1.11. DEVIATIONS AND EXCEPTIONS

Deviations and exceptions to the terms and conditions of the solicitation are neither encouraged nor desired. Should the offer include any standard company terms and conditions that conflict with the terms and conditions of the solicitation, the offer may be determined "unacceptable" and thus ineligible for the award. Should the offeror have any questions related to specific terms and conditions, these must be resolved prior to submission of the offer.

Notwithstanding the above, if deviations and exceptions are included with the offer, the offeror shall list and describe in detail the deviations and/or exceptions. All deviations and exceptions shall be fully supported with the Offeror's rationale and shall fully explain the impact, if any, on the performance and/or specific requirement of the RFP.

PROPOSAL PREPARATION CHECKLIST

DO NOT RETURN THIS CHECKLIST WITH YOUR PROPOSAL. Important items for you to check are included, but not limited to, the items listed below. This checklist is furnished only to assist you in submitting a proper quote.

_____Have you registered in the Procurement Integrated Enterprise Environment (PIEE) website located at https://piee.eb.mil/?

_____Have you obtained a National Institute of Standards and Technology (NIST) score? More information about creating an account for the first time is in PIEE. Refer to the website's "Vendors – Getting Started Help" page at the following URL: https://piee.eb.mil/xhtml/unauth/web/homepage/vendorGettingStartedHelp.xhtml. Note, a Cyber Vendor Role in PIEE is required to enter/edit basic self-assessment information.

_____Have you obtained a UIE number? Provide your UIE number on the first page or cover page.

_____Have you registered in System for Award Management (SAM)? The name you use to submit a quote must be the exact name you register in SAM. Do not delay returning your proposal while processing your SAM entry. It does not cost to register to SAM at https://sam.gov.

_____Is the NAICS Code in your current SAM profile? This solicitation is being advertised under NAICS code 238160 (listed in box 10 on page 1 of the SF 1442). You must ensure this NAICS Code is incorporated into your current SAM profile if your company can provide the type of product or service applicable to this NAICS code. Failure to have this NAICS Code in your SAM profile may result in not being considered for award. NAICS Codes may be viewed at the U.S. Census Bureau's website at https://www.census.gov/eos/www/naics.

_____Have you entered a unit price for each line item? The quantity multiplied by the unit price equals the total amount. Ensure all prices on line items are filled in. All line items must be priced to be considered for award. Unit prices must be rounded to the nearest hundredth of a cent. For example, see below:

$21.33 IS an acceptable price
$21.338 IS NOT an acceptable price.

Do not round total amount numbers. If your price exceeds $2,000 for construction work or $2,500 for services, you will be required to pay employees at least the wage rates specified in the applicable wage decision/determination.

_____Cross outs or strikethroughs WILL NOT be accepted. Please submit a legible proposal with no correction marks.

_____Did you sign the original solicitation? Ensure to include the first page of the solicitation and fill out block 12, 17a, 30a, 30b and 30c. A signature of an authorized individual is required for your proposal.

_____Have you acknowledged all amendments, if any have been issued? Acknowledge amendments by signing and dating the first page of the amendment. Ensure to read the amendment to determine if further documents are required to be submitted and to see what changes have been made to the solicitation. Please reference the solicitation number W912BV26RA015 in the subject line when submitting.

_____Have you submitted all information needed to evaluate your offer? Please refer to the Instructions, Conditions and Notices to Offerors and Statement of Work sections.

SECTION 00 22 00
PROPOSAL PREPARATION AND EVALUATION CRITERIA

1.0 OVERVIEW

A. GENERAL

(1) This solicitation is a full and open competitive acquisition using the Lowest Price Technically Acceptable (LPTA) source selection process for the award of a firm-fixed-price contract titled "Broken Bow Powerhouse Roof Replacement" at Broken Bow, Oklahoma. Award will be made based on the lowest evaluated price of proposals meeting or exceeding the acceptability standards for non-cost factors. This acquisition will utilize Revolutionary Federal Acquisition Regulation Overhaul (RFO) part 15 LPTA source selection procedures, leading to selection of the proposal representing the best value to the Government.

(2) In the context of this proposal, the "Offeror" refers to the proposed prime contractor.

B. WHO CAN SUBMIT

(1) Firms formally organized as construction contractors that have associated specifically for this project, consortia of firms or any other interested parties may submit proposals. Associations may be as joint ventures or as key team subcontractors.

(2) Joint Ventures (JV). A joint venture is defined as a legal business entity formed between two or more companies (parties) to undertake the performance activities of a contract together. This does not include other arrangements such as "teaming agreements" or "strategic alliances," which are not recognized as bona fide joint ventures for the purposes of this solicitation. **An Offeror that is a joint venture must submit with its proposal a copy of its legally binding joint venture agreement signed by an authorized officer from each of the firms comprising the joint venture with the chief executive of each entity identified.** If submitting a proposal as a joint venture, the experience and past performance of each of the joint venture partners can be submitted for the joint venture entity. The experience for each joint venture partner will be considered the experience of the joint venture entity. Page and project form limits apply to the joint venture as a whole.

2.0 PROPOSAL SUBMISSION REQUIREMENTS AND INSTRUCTIONS

A. REQUIREMENT FOR SEPARATE PRICE AND TECHNICAL PROPOSALS

(1) Each Offeror must submit two distinct volumes: a Price Proposal and a Technical Proposal. These volumes must be submitted as separate PDF files, each clearly labeled to denote its contents and the identity of the Offeror. Additionally, clearly identify the "original" price proposal and the "original" technical proposal on the outside cover.

(2) The Price Proposal and Technical Proposal must be received by the closing date and time set for receipt of proposals.

(3) No dollar amounts from the Price Proposal are to be included in the Technical Proposal.

(4) All information intended to be evaluated as part of the Technical Proposal must be submitted as part of the Technical Proposal (Volume I). Do not merely cross-reference similar material in the Price Proposal, or vice versa. Also, do not include links to websites in lieu of incorporating information into your proposal.

(5) Do not include exceptions or deviations to the terms and conditions of the solicitation in either the technical or price proposal. Should the offer include any standard company terms and conditions that conflict with the terms and conditions of the solicitation, the offer may be determined "unacceptable" and thus ineligible for award. Should the Offeror have any questions related to specific terms and conditions, these should be resolved prior to submission of the offer. If the Offeror must make any assumptions prior to submitting the proposal, the Offeror shall list and describe in detail the assumptions on the first page of each Tab. All

assumptions shall be fully supported with the Offeror's rationale and shall fully explain the impact, if any, on the performance and/or specific requirement of the RFP. The Offeror shall clearly describe on the Proposal Cover Sheet submitted with the Price Proposal any assumption to the contractual and/or technical terms and include the monetary impact on the solicitation with the price proposal. Proposal exceptions/deviations will not be considered a part of the contract after award unless specifically approved by the Contracting Officer (KO) in writing at the time of contract award. This information will not be evaluated separately but may impact the evaluation of other factors.

B. DISCUSSIONS

(1) The Government **does not** intend to enter into discussions with Offerors prior to making an award. Offerors are encouraged to present their best technical proposal and prices in their initial proposal submission. However, in accordance with RFO 15.204, should discussions become necessary, the Government reserves the right to hold them. If this occurs, a competitive range will be determined, and Offerors notified. The competitive range may be limited for purposes of efficiency IAW RFO 15-204-1(b).

C. COST OR PRICING DATA

(1) Offerors are not required to submit Cost or Pricing Data with their offers. However, if deemed necessary, supplemental price breakdown information will be used to assist the Government in performing the price evaluation. This will not be requested until after the initial proposal submission.

D. GENERAL INSTRUCTIONS

(1) Submit only the electronic files specifically authorized and/or required elsewhere in this section. Do not submit excess information, to include audio-visual materials, electronic media, etc. All pages should be numbered and include working bookmarks.

(2) "Confidential" projects cannot be submitted to demonstrate capability unless all of the information required for evaluation as specified herein can be provided to the Government as part of the Offeror's technical proposal. Offerors that include in their proposals information that they do not want disclosed to the public for any purpose, or used by the Government except for evaluation purposes, must be clearly marked in accordance with the instructions at RFO 52.215-1, "Instructions to Offerors—Competitive Acquisition", paragraph (e), "Restriction on disclosure and use of data".

(3) Proposal revisions shall be submitted as page replacements with revised text readily identifiable, e.g., bold face print or underlining. The source of the revision, e.g., Error, Omission, or Clarification, or amendment shall be included and be annotated for each revision. Proposal replacement pages shall be numbered, shall be clearly marked "REVISED", shall show the date of revision, shall be submitted in appropriate number of copies.

E. SPECIFIC INSTRUCTIONS FOR THE TECHNICAL PROPOSAL

(1) Number of Sets of the Technical Proposal. Submit one (1) electronic copy of the Technical Proposal. Submission shall contain the entire proposal in .pdf format using Adobe Acrobat software to print to a .pdf file; do NOT scan the document(s) into a .pdf file. The text portion of the complete proposal shall be contained as a single .pdf file. Sections of the proposal shall be bookmarked (linked to the index) in logical order.

(2) Format and Contents of the Technical Proposal and List of Tabs. The Technical Proposal shall be organized using PDF bookmarks specified in the chart below. Bookmarks are to be visible immediately after opening.

VOLUME 1 – TECHNICAL PROPOSAL		
LOCATION	**FACTOR**	**DESCRIPTION**
Tab A	Factor 1	Key Subcontractors & Key Personnel
Tab B	Subfactor 1	Technical Approach
Tab C	Subfactor 2	Relevant Experience
Tab D	Factor 2	Past Performance

(3) Page Limitations. The following page limitations are established for each factor described above:

(a) Factor #1, Key Subcontractors & Key Personnel – Limited to two (2) pages for each resume provided
(b) Subfactor #1, Technical Approach – Limited to 10 pages.
(c) Subfactor #2, Relevant Experience – Limited to 20 pages (a maximum of 5 forms per contractor/subcontractor, see Paragraph 2.E.(4)(a)(i) below)
(d) Factor #2, Past Performance – No page limitation

Tables of content, proposal cover letters, and tabs between proposal information do not count toward any page limitations in the proposal.

(4) Detailed Submission Requirements for the Technical Proposal. The following is a detailed description of the information to be submitted under each TAB.

(a) **Tab A, Factor 1, Key Subcontractors & Key Personnel**

(i) By naming these firms and personnel, the Offeror is making a commitment that barring unforeseen circumstances, they are the firms and personnel that will be working on this project. Any substitutions shall be approved by the Contracting Officer. All key personnel and subcontractor personnel working on this project must be U.S. citizens or documented U.S. workers and must be able to provide proof of U.S. citizenship or other authorized documentation.

(ii) **Key Subcontractors:** Identify the potential Key Subcontractors chosen for demolition, design, fabrication, and fitting, describing the extent of their involvement in the project. Submit no more than five Relevant Experience forms (Attachment 1) for each Key Subcontractor. Subcontractors shall be selected from the list of potential subcontractors.

(iii) **Key Personnel**: The Offeror must provide resume data for the following key construction personnel: Project Manager (if applicable), Project Scheduler, Quality Control (QC) Manager, Site Superintendent, Site Safety and Health Officer (SSHO), and Design Quality Control (DQC) Manager. Minimum qualifications for the Construction Site Superintendent and Quality Control (QC) Manager shall be as required by Section 01 30 00 ADMINISTRATIVE REQUIREMENTS and Section 01 45 00 QUALITY CONTROL. SSHO minimum requirements shall be as shown in Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

The Offeror must provide resume data for the following key design personnel: Lead Architect, Lead Structural Engineer, Lead Mechanical Engineer, Lead Electrical Engineer (as applicable), Project Manager (if applicable), and other team members that demonstrate the qualifications necessary for this project, which includes proposed role, relevant education and training, professional licensure, and a minimum of 5 years relevant experience in similar type work, to manage, control, and perform design.

Resume information to be provided shall be limited to no more than two (2) pages per person and shall include the following information as a minimum (Attachment 3):

(a) Name and title
(b) Project assignment
(c) Name of firm with which associated
(d) Years of experience with this firm and with other firms
(e) Education degree(s), year, specialization, if applicable
(f) Active professional registration, year first registered, if applicable
(g) Other experience and qualifications relevant to same/similar work required under this contract

(iv) **Organization/Management:** Provide information that describes the offeror's organization. Limit the information to 5 pages or less to include the required organization charts; clearly but concisely describe the organization the 5-page limit does not include unequivocal letters of commitment submitted for key personnel, unequivocal letters of commitment for teaming subcontractors, or the joint venture agreement.

(a) Organization: Describe what firms, their resources and how their resources will be utilized, their roles and responsibilities and any contractual arrangements that have been established. Clearly describe any teaming or joint venture arrangements, including a clear description of each firm's roles and responsibilities on the project. A copy of the teaming or joint venture agreement(s) shall be appended to the plan (not included in the page limitation). Include a simple organizational chart, illustrating the organization, including the proposed quality control group(s). Present a matrix of responsibilities for each firm in executing the key work breakdown structure activities of the project, including construction activities for each major feature. Identify the firms chosen for the project, if not to be self-performed. Offerors will be required to identify the specific firm(s) chosen for Architectural, Structural, Electrical, and Mechanical work as applicable. The Offeror shall be required to obtain the Contracting Officer's approval for any substitution of firms listed. Describe the proposed management structure for the team, describing how the construction process will be managed and the authorities and the delegations of authority within the team. Include a key personnel organization chart that clearly depicts the key positions and the names of the personnel, their firm affiliations and their job locations, their job/position title within the organization. The key personnel organization chart shall be consistent with the corporate organization chart, with the matrix of responsibilities assigned to the firms, and with the list of key personnel.

(b) Subcontractor Approach and Management: Describe how subcontractors are managed by the contractor. Discuss how subcontract relationships are structured to assure proper balance and flow-down of contract requirements, duties, responsibilities, and properly balanced and proportioned risk management. Discuss how non-performance by subcontractors is handled by the general contractor. The Government wants to understand that the general contractor places emphasis on strong business relationships with subcontractors to maximize the number of quotes the general contractor obtains that demonstrates market competition and the best valued prices are being provided to the government. General Contractors that do not maintain solid business practices with subcontractors increase business costs to the Government. Limit one (1) page.

(v) **Evaluation Criteria:** The Government will evaluate the resumes to determine how well the Offeror identifies and demonstrates key personnel meet or exceed minimum qualifications necessary to perform their assigned roles. Offerors providing resumes for key personnel for this project that have experience in similar key positions on the projects shown under Factor 1 will be viewed more favorably by the Government. Performance of key personnel proposed for this project may be considered when it comes to the attention of the Government. The Government will evaluate the clarity and strength of the overall organization and how well it is organized, structured, and staffed to execute the entire scope of work. If applicable, joint venture participants'/teaming subcontractor's contribution to the organization should be commensurate with their skills and background.

(b) **Tab B, Subfactor 1, Technical Approach**

(i) **Technical Approach for Design and Construction:** Describe the technical approach to design and construction for the new powerhouse roof. Discuss how design and construction is completed. Describe your proposed phasing plan for design and construction for this work. Limit the information to 10 pages or less concisely describe the technical approach to project management and execution.

(ii) **Offeror's Understanding:** Describe challenges, risk, and constraints associated with the project and how the offeror intends to address each with means and methods, resources, or internal organizational processes. Describe the team's safety approach, corporate systems and capabilities to maintain safety requirements during construction. Describe the proposed safety organization, including the proposed staffing plan. The intent is to provide enough detail to demonstrate the Offeror's understanding; however, it is not the intent to provide the amount of detail that will be required in the pre-work submittal as outlined in the Division 01 specifications.

(iii) **Collaborative Approach for Design and Construction:** Describe interactions within the team and with the Corps of Engineers during the design and construction. Discuss how the configuration management system will track and control design evolution and changes during design for quality control and to facilitate quicker Government reviews. Describe the role of the construction team members during design. Describe the role and interaction of the design team with the construction team during construction, addressing, as a minimum, maintaining configuration management of the design during construction, including control and approval of revisions to the accepted design; requests for information; shop drawing and submittal reviews and approvals; progress meetings; site visits, if any; contract completion, closeout, as-built and completion documentation. Limit three (3) pages.

(iv) **Self-Performed Work:** Generally, describe the items the Offeror will self-perform to comply with the requirements in Revolutionary Federal Acquisition Regulation Overhaul (RFO) 52.236-1 for self-performed work. All other items not self-performed will be assumed by the Government to be subcontracted. It is the position of the government that self-performance of items allows the general contractor greater control of the work and yields in cost savings to the government as the amount of subcontracted markups are reduced and provide a cost savings to the government and government customers.

(v) **Planning and Scheduling:** Describe the time control capabilities and systems to be used to plan, design, and construct the facility and how the schedule will be used to manage the design and construction. Provide enough detail to demonstrate an understanding of the requirement including any constraints that can be anticipated (e.g., labor or material availability, permits, weather, site restrictions, security, etc.). Show planned sequencing and milestones of major activities and phases. Discuss internal procedures for handling delays to minimize time growth or "schedule creep". Discuss experience with interfacing scheduling software with Resident Management System (RMS) if applicable. Limit one (1) page.

(vi) **Quality Control:** Describe the team's quality control approach, corporate systems and capabilities to maintain quality control of the design and construction. Describe the proposed quality control organization, including the proposed staffing plan. Provide an organizational chart showing lines of communications and authority. Provide specific information on how you will manage design quality control, track design evolution and changes during design to meet the schedule and to facilitate quicker Government reviews. Provide information on how you will handle internal and external requests for information, shop drawings, submittal reviews, progress meetings, site visits, contract completion, closeout, as-built, and completion documentation. In addition to the required designer-of-record roles specified within the RFP for maintaining integrity of the design, describe any other DOR involvement in the quality control process, if any. There is no need to submit a quality control plan as the successful offeror will provide that after award. In phase 1, the Government is interested in demonstrable capabilities to assure and control quality and how the offeror can achieve or exceed the contract's minimum quality control system requirements. Limit one (1) page.

(vii) **Evaluation Criteria:** The Government will evaluate the strengths, weaknesses and any deficiencies in the organization and technical approach. The Government will evaluate the firm's understanding of Design-Build (D-B),the requirements in described in the Division 01 requirements of the Solicitation, and the capability to execute the project. Some additional specific evaluation considerations are listed below. This list is not all-inclusive.

The Government will evaluate the offeror's understanding of the challenges and risks associated with this project and the specific methods, resources, or internal processes that will be employed to mitigate those challenges and risks.

The Government will evaluate clarity and strength of the overall organization, the structure and staffing to execute the entire scope of work. The government will evaluate the general contractor's ability to develop and maintain strong subcontractor and small business relationships. The government will evaluate the general contractor's subcontractor management model and processes. The Offeror is required to select and commit to design firms to achieve an "acceptable" rating.

The Government will evaluate the Offeror's capabilities and understanding of the contractually required quality control processes for construction. The Government will evaluate the adequacy of the staffing plan to cover all required tasks and responsibilities.

(c) **Tab C, Subfactor 2, Relevant Experience**

(i) The Offeror and subcontracted design firm(s) (or prime contractor if design is to be self-performed) shall each demonstrate the experience on projects same/similar in size and complexity to that described in the solicitation for same/similar design and construction services. If using subcontractors as part of the proposed team to demonstrate experience, describe the role they will play in this project and the percentage of work they will perform.

(ii) The Offeror and subcontracted design firm(s) shall complete a minimum of three (3), but not more than ten (10) "Relevant Experience Information" forms, attached at the end of this section (Attachment 1), in response to this factor. All blocks must be filled in and all data should be accurate, current, and complete. All projects submitted must be at least 50% complete and completed within the last 5 years. If the Offeror's firm has multiple locations, projects should have been completed by the regional office that would be performing the work for this solicitation. The Government preference is that project examples submitted be of completed projects (i.e. construction complete). Design firms may list prime contractors they have worked for or Government, private or commercial customers. The Offeror shall select the design firm(s). The offeror should describe any previous teaming experience between current team members, if not described in the project list. If any, identify experience in design-build projects.

The Offeror and design firm(s) must each submit for projects of similar size and scope as outlined in the RFP. At least two (2) of the projects provided must be valued at over $1,000,000. For evaluation purposes, "similar" is defined as:

(a) Projects demonstrating experience with Design-Build delivery.
(b) Projects demonstrating experience with the demolition of existing roof assemblies.
(c) Projects demonstrating experience with the design and construction of new TPO roof assemblies on existing structures, including the structural analysis of the new roof assembly on the existing structure.
(d) Projects demonstrating experience with the design and construction of roof drainage systems.
(e) Projects demonstrating experience with the design and construction of misc. roof appurtenances such as roof ladders, roof ventilation, and parapet coping.

(f) Demonstrated experience with local conditions.
(g) Demonstrated experience with applicable codes and Government regulations and design criteria.

An Offeror may demonstrate its experience in all four of the above areas through either one project or a combination of projects submitted. Each project summary must include the following for each project: a description of the project; scope; location; firms on the proposed team that constructed the project; construction contract award amount and final construction cost; the original contract finish date and actual finish date (if finished).

(iii) Narratives of each project should include a brief overview of each project and its relevance to this project. All summaries should also contain the name, address, telephone, and e-mail, of a representative of the customer familiar with the proposer's experience on the project and can verify the experience cited.

(iv) If any of the information required is not included in the form, then the Offeror will be considered non-responsive and evaluated as unacceptable.

(v) **Evaluation Criteria:** The Government will evaluate the extent of recent, related experience of the Offeror, design firms, and subcontractors in design, construction, and design-build construction, as relevant to their role on this project. If the design will be accomplished in-house, rather than by subcontract, then the design element of relevant experience will still be evaluated, realizing that the work is being done in-house. Experience on the similar projects identified in the project lists will receive more consideration than experience provided in the supplemental narrative. The Government will place greater importance on projects performed as a prime contractor and design firm, than as a subcontractor.

The Government will consider extent of recent experience, degree of relationship of such experience to this project, demonstrated familiarity with applicable codes and local conditions. Some examples of relevancy to this project may include, but not be limited to:
(a) Number, size, type work, complexity, location.
(b) Projects completed after 2020.
(c) Firm's role and extent of work performed.
(d) Project delivery type (Design-Build).
(e) Projects constructed at Powerhouses or on Government property.
(f) Projects involving pre-cast or modular elements.
(g) Projects with a value over $1M.

Previous design-build experience is not necessary for an acceptable rating. The Government may consider previous D-B experience a strength, even if the experience is on different type projects than this project. Similarly, the Government may consider previous recent teaming experience among the team members as value added, even if on different type design and/or construction projects than this project.

(vi) The Government reserves the right to verify the experience record of cited projects or other recent projects by reviewing the Corps of Engineers Construction Contractor (or Architect-Engineer) Appraisal Support System, other DOD or Government appraisal systems or to interview owners or references. The Government may check any or all cited references to verify supplied information.

(d) **Tab D, Factor 2, Past Performance**

(i) For each project submitted, Offerors shall submit available performance documentation. If no performance documentation is available, state so on the project information worksheet. The Government will review the experience forms submitted as well as query the Contractor Performance Assessment Reporting System to validate past performance ratings on Department of Defense contracts and any other past performance information the Government deems necessary to evaluate a contractor's past performance. Firms without a history of past performance will be given a neutral rating. For non-Corps of Engineers contracts, provide a copy of the performance rating issued by the contracting agent.

(ii) For each project that does not have a final CPARS evaluation, Offerors shall provide a questionnaire (See Attachment 2) to the point of contact, clearly identifying the project to be evaluated. A sample Past Performance Evaluation Questionnaire is included at the end of this section. When completed, these forms shall be mailed, faxed or e-mailed to the Tulsa District Contract Specialist identified in the RFP letter prior to proposal closing date. It is the contractor's responsibility to ensure that the reference documentation is provided, as the Government may not make additional requests for past performance information from the references. The evaluation form shall be provided to the Contract Specialist directly from the reference.

(iii) The information contained in the questionnaires will be used to evaluate the Offeror's past performance. If any negative past performance information is received, the contractor will be given an opportunity to provide rebuttal. Each contractor shall provide at least three (3) past performance ratings. Any rating an Offeror has IN CPARS for the projects listed under Factor 1, Subfactor 2 will count toward this requirement.

The Offeror is responsible for ensuring that the KO has received a minimum of three (3) questionnaires prior to that date/time or has provided written notification to the KO that they have current ratings in CPARS. Entities that have no past customers shall notify the KO in writing, prior to the date and time established for the RFP closing, that they have had no previous clients and that the minimum number of questionnaires cannot be provided. It is the contractor's responsibility to ensure that the reference documentation is provided, as the Government may not make additional requests for past performance information from the references.

Do not request past performance questionnaires on projects that have final CPARS evaluations. If a past performance questionnaire is received on a project where a final CPARS evaluation exists, the CPARS evaluation will serve as the official rating of record for evaluation purposes.

(iv) The Government may contact references provided as part of Factor 1, Subfactor 2 – Relevant Experience, for information regarding the Offeror's past performance on the project and for the purposes of assessing and verifying the scope of the work performed. Offerors should provide accurate, current, and complete contact information for references provided in the project descriptions.

The Government may review any other sources of information for evaluating past performance. Other sources may include, but are not limited to, past performance information retrieved through the Past Performance Information Retrieval System (PPIRS), using all UEI numbers of team members (partnership, joint venture, teaming arrangement, or parent company/subsidiary/affiliate) identified in the Offeror's proposal, inquiries of owner representative(s), Federal Awardee Performance and Integrity Information System (FAPIIS), Electronic Subcontract Reporting System (ESRS), and any other sources not provided by the offeror.

(v) Each entity (firm) will be rated on its own performance or that of its predecessor, as defined in RFO 52.204-20, if relevant. An entity may establish past performance based on the past performance of its proposed key personnel, apart from that of the entity. If the Government does not obtain past performance information for the projects identified by the Offeror and cannot establish a past performance record for the Offeror through other sources, past performance will be rated neither favorably nor unfavorably. The performance confidence assessment will be considered "Unknown Confidence."

F. SPECIFIC INSTRUCTIONS FOR THE PRICE PROPOSAL

(1) Number of Sets of the Price Proposal. Submit one (1) electronic copy of the Price Proposal. Submission shall contain the entire proposal in .pdf format using Adobe Acrobat software to print to a .pdf file; do NOT scan the document(s) into a .pdf file. The text portion of the complete proposal shall be contained as a single .pdf file. Sections of the proposal shall be bookmarked (linked to the index) in logical order.

(2) Size Restrictions and Page Limits. Use only 8 ½" x 11" pages. There are no page limits set for the price proposal. However, limit your response to information required by this solicitation. Excess information will not be considered in the Government's evaluation. Do not use a font size smaller than 10, an unusual font style such as script, or condensed print for any submission.

(3) Format and Contents of the Price Proposal and List of Tabs. The Price Proposal shall be appropriately labeled as such and shall be organized as indicated in the following chart. Note: If the Offeror is not required to submit any information under a listed Tab in accordance with the instructions below, that tab can be omitted. However, do not renumber the subsequent tabs.

VOLUME 2 – PRICE PROPOSAL		
LOCATION	**FACTOR**	**DESCRIPTION**
Tab 1	N/A	Proposal Cover Sheet
Tab 2	N/A	SF1442 and Acknowledgement of Amendments
Tab 3	Factor 3	SECTION 00 11 00, PRICING SCHEDULE
Tab 4	N/A	Bid Guarantee (Bid Bond)
Tab 5	N/A	Representations & Certifications
Tab 6	N/A	JV Agreement (if applicable)

(4) Detailed Submission Instructions for the Price Proposal

(a) **Tab 1**: The proposal cover sheet is required by RFO 52.215-1(c)(2)(i)-(v) and must be submitted by all Offerors. This provision, titled "Instructions to Offerors—Competitive Acquisition," and the format for the proposal cover sheet as shown here:

PROPOSAL COVER SHEET

1. Solicitation Number:

2. The name, address, and telephone and facsimile numbers of the Offeror (and electronic address if available):

3. Names, titles, and telephone and facsimile numbers (and electronic addresses if available) of persons authorized to negotiate on the Offeror's behalf with the Government in connection with this solicitation:

4. Name, title, and signature of person authorized to sign the proposal. Proposals signed by an agent shall be accompanied by evidence of that agent's authority unless that evidence has been previously furnished to the issuing office.

5. Tax Identification number (TIN or EIN) of Offeror.

(b) **Tab 2**: The SF1442 is to be completed by all Offerors and duly executed with an original signature by an official authorized to bind the company in accordance with RFO 4.207. Provide signed

SF1442 with all amendments acknowledged by all Offerors in accordance with the instructions on the Standard Form 30, Amendment of Solicitation.

(c) **Tab 3:** Section 00 11 00, CLIN Schedule (Pricing Schedule) is to be completed in its entirety by all Offerors. See Section 00 11 00 with attached notes, for further instructions. Include price breakdown information behind the Pricing Schedule.

(d) **Tab 4:** Provide a fully executed Bid Bond as required by RFO 52.228-1, Bid Guarantee. The completed bid bond shall be provided electronically. A paper copy may be requested by the contracting officer.

(e) **Tab 5:** Representations and Certifications: All Offerors must have electronically completed the annual representations and certifications in the SAM website (www.sam.gov). The Offerors are responsible for ensuring that these on-line Representations and Certifications are updated as necessary to reflect changes, but at least annually to ensure that they are kept current, accurate and complete. If the Offeror is a Joint Venture, all participants must separately complete the on-line Representations and Certifications.

(f) **Tab 6:** Joint Venture Agreement (if applicable): If the Offeror is a Joint Venture (JV), include a copy of the JV Agreement. If a JV Agreement has not yet been finalized/approved, indicate its status. JV Agreements must clearly indicate the percentage of the JV participants, in particular the percent of the controlling party, and a clear delineation of responsibilities and authorities between the JV parties.

3.0 EVALUATION OF OFFEROR PROPOSALS

A. <u>SOURCE SELCTION USING LOW-PRICED, TECHNICALLY ACCEPTABLE PROCESS (LPTA)</u>

(1) An evaluation for acceptability will be performed on the low-price proposal in accordance with RFO 15.103-2. The proposal that provides the lowest overall price and is otherwise technically acceptable in all factors will be selected for award. To be considered technically acceptable, no technical factor in the proposal may be determined to be unacceptable. The failure of a proposal to meet any of the factors will result in a technically unacceptable rating and preclude award.

B. <u>BASIS OF AWARD</u>

(1) Award will be made to the acceptable proposal meeting or exceeding the technical acceptability standards for non- cost factors and is the lowest price of all acceptable proposals and is found fair and reasonable by the Contracting Officer. Tradeoffs are not permitted. Proposals are evaluated for acceptability but not ranked using non-cost/price factors. The Government will not award a contract to a Firm whose proposal contains a deficiency, as defined in RFO 15.001. The Government reserves the right to reject any offer if the offer is not in the best interest of the Government.

C. <u>GENERAL TECHNICAL CRITERIA</u>

(1) Material omission(s) may cause the technical proposal to be rejected as unacceptable.

(2) Proposals which are generic, vague, or lacking in detail may be considered unacceptable. The offeror's failure to include information that the Government has indicated should be included may result in the proposal being found deficient if inadequate detail is provided.

(3) The Government cannot make award based on a deficient offer. Therefore, receipt of a "Non-Acceptable" determination of acceptability for any factor will make the offer ineligible for award, <u>unless</u> the Government elects to enter into discussions with that Offeror and all deficiencies are remedied in a revised proposal. Note: It is the Government's intent to award without discussions.

(4) If awarded the contract, the Contractor shall construct the project in accordance with the plans and specifications issued with this RFP.

D. EVALUATION OF THE PRICE PROPOSALS

(1) Price will be evaluated and considered but will not be scored or combined with other aspects of the proposal evaluation. The proposed prices will be analyzed for completeness, fairness, and reasonableness. They may also be analyzed to determine whether they are realistic for the work to be performed, reflect a clear understanding of the requirements, and are consistent with the information provided by the Offeror. Firms are cautioned to distribute direct costs, such as material, labor, equipment, subcontracts, etc. and to evenly distribute indirect costs, such as job overhead, home office overhead, bond, etc., to the appropriate contract line items. Additionally, all offers will be analyzed for unbalanced pricing in accordance with RFO 15.404-6(b) and may be rejected if it is determined that the lack of balance poses an unacceptable risk to the Government.

(2) The otherwise technically acceptable, lowest-priced Offeror may be required to confirm its price on either a CLIN, element, or total price basis, and/or provide additional information in support of their price, prior to contract award at the Government's request and discretion.

ATTACHMENT 1

<table>
<tr><td colspan="2">RELEVANT EXPERIENCE INFORMATION
(To be completed by Contractor)</td></tr>
<tr><td>1. Contract/Task Order Number:</td><td>2. Contractor Name:

Address:</td></tr>
<tr><td rowspan="2">Include Brief Explanation for Block 4 discussing reasons for time growth as necessary:</td><td>3. Contract Award Value:</td></tr>
<tr><td>4. Contract/TO /PO Status: ☐ Active ☐ Complete

Original Contract Completion Date:

Actual Completion Date:

Liquidated Damages Assessed(Y/N):
(if yes, state $ amount)</td></tr>
<tr><td colspan="2">5. Project Title:

Location:

6. Project Description-- to include the role of the contractor on the project and specific responsibilities of the contractor in performance of the effort and a brief overview of relevance to this project:</td></tr>
<tr><td colspan="2">7. Project Owner or Project Manager for the Client – provide:
Name:
Address:
Telephone Number and E-mail:</td></tr>
</table>

ATTACHMENT 2

NAVFAC/USACE PAST PERFORMANCE QUESTIONNAIRE (Form PPQ-0)

CONTRACT INFORMATION (Contractor to complete Blocks 1-4)

1. Contractor Information
Firm Name: CAGE Code:
Address: DUNs Number:
Phone Number:
Email Address:
Point of Contact: Contact Phone Number:

2. Work Performed as: ☐ Prime Contractor ☐ Sub Contractor ☐ Joint Venture ☐ Other (Explain)
Percent of project work performed:
If subcontractor, who was the prime (Name/Phone #):

3. Contract Information
Contract Number:
Delivery/Task Order Number (if applicable):
Contract Type: ☐ Firm Fixed Price ☐ Cost Reimbursement ☐ Other (Please specify):
Contract Title:
Contract Location:

Award Date (mm/dd/yy):
Contract Completion Date (mm/dd/yy):
Actual Completion Date (mm/dd/yy):
Explain Differences:

Original Contract Price (Award Amount):
Final Contract Price (*to include all modifications, if applicable*):
Explain Differences:

4. Project Description:
Complexity of Work ☐ High ☐ Med ☐ Routine
How is this project relevant to project of submission? (*Please provide details such as similar equipment, requirements, conditions, etc.*)

CLIENT INFORMATION (Client to complete Blocks 5-8)

5. Client Information
Name:
Title:
Phone Number:
Email Address:

6. Describe the client's role in the project:

7. Date Questionnaire was completed (mm/dd/yy)**:**

8. Client's Signature:

NOTE: NAVFAC/USACE REQUESTS THAT THE CLIENT COMPLETES THIS QUESTIONNAIRE AND SUBMITS DIRECTLY BACK TO THE GOVERNMENT'S CONTRACT SPECIALIST. THE OFFEROR WILL SUBMIT THE COMPLETED QUESTIONNAIRE TO USACE WITH THEIR PROPOSAL, AND MAY DUPLICATE THIS QUESTIONNAIRE FOR FUTURE SUBMISSION ON USACE SOLICITATIONS. CLIENTS ARE HIGHLY ENCOURAGED TO SUBMIT QUESTIONNAIRES DIRECTLY TO THE OFFEROR. HOWEVER, QUESTIONNAIRES MAY BE SUBMITTED DIRECTLY TO USACE. PLEASE CONTACT THE OFFEROR FOR USACE POC INFORMATION. THE GOVERNMENT RESERVES THE RIGHT TO VERIFY ANY AND ALL INFORMATION ON THIS FORM.

ADJECTIVE RATINGS AND DEFINITIONS TO BE USED TO BEST REFLECT YOUR EVALUATION OF THE CONTRACTOR'S PERFORMANCE		
RATING	**DEFINITION**	**NOTE**
(E) Exceptional	Performance meets contractual requirements and exceeds many to the Government/Owner's benefit. The contractual performance of the element or sub-element being assessed was accomplished with few minor problems for which corrective actions taken by the contractor was highly effective.	An Exceptional rating is appropriate when the Contractor successfully performed multiple significant events that were of benefit to the Government/Owner. A singular benefit, however, could be of such magnitude that it alone constitutes an Exceptional rating. Also, there should have been NO significant weaknesses identified.
(VG) Very Good	Performance meets contractual requirements and exceeds some to the Government's/Owner's benefit. The contractual performance of the element or sub- element being assessed was accomplished with some minor problems for which corrective actions taken by the contractor were effective.	A Very Good rating is appropriate when the Contractor successfully performed a significant event that was a benefit to the Government/Owner. There should have been no significant weaknesses identified.
(S) Satisfactory	Performance meets minimum contractual requirements. The contractual performance of the element or sub-element contains some minor problems for which corrective actions taken by the contractor appear or were satisfactory.	A Satisfactory rating is appropriate when there were only minor problems, or major problems that the contractor recovered from without impact to the contract. There should have been NO significant weaknesses identified. Per DOD policy, a fundamental principle of assigning ratings is that contractors will not be assessed a rating lower than Satisfactory solely for not performing beyond the requirements of the contract.
(M) Marginal	Performance does not meet some contractual requirements. The contractual performance of the element or sub-element being assessed reflects a serious problem for which the contractor has not yet identified corrective actions. The contractor's proposed actions appear only marginally effective or were not fully implemented.	A Marginal is appropriate when a significant event occurred that the contractor had trouble overcoming which impacted the Government/Owner.
(U) Unsatisfactory	Performance does not meet most contractual requirements and recovery is not likely in a timely manner. The contractual performance of the element or sub-element contains serious problem(s) for which the contractor's corrective actions appear or were ineffective.	An Unsatisfactory rating is appropriate when multiple significant events occurred that the contractor had trouble overcoming and which impacted the Government/Owner. A singular problem, however, could be of such serious magnitude that it alone constitutes an unsatisfactory rating.
(N) Not Applicable	No information or did not apply to your contract	Rating will be neither positive nor negative.

TO BE COMPLETED BY CLIENT

PLEASE CIRLCE THE ADJECTIVE RATING WHICH BEST REFLECTS YOUR EVALUATION OF THE CONTRACTOR'S PERFORMANCE.						
1. QUALITY:						
a) Quality of technical data/report preparation efforts	E	VG	S	M	U	N
b) Ability to meet quality standards specified for technical performance	E	VG	S	M	U	N
c) Timeliness/effectiveness of contract problem resolution without extensive customer guidance	E	VG	S	M	U	N
d) Adequacy/effectiveness of quality control program and adherence to contract quality assurance requirements (without adverse effect on performance)	E	VG	S	M	U	N
2. SCHEDULE/TIMELINESS OF PERFORMANCE:						
a) Compliance with contract delivery/completion schedules including any significant intermediate milestones. *(If liquidated damages were assessed or the schedule was not met, please address below)*	E	VG	S	M	U	N
b) Rate the contractor's use of available resources to accomplish tasks identified in the contract	E	VG	S	M	U	N
3. CUSTOMER SATISFACTION:						
a) To what extent were the end users satisfied with the project?	E	VG	S	M	U	N
b) Contractor was reasonable and cooperative in dealing with your staff (including the ability to successfully resolve disagreements/disputes; responsiveness to administrative reports, businesslike and communication)	E	VG	S	M	U	N
c) To what extent was the contractor cooperative, businesslike, and concerned with the interests of the customer?	E	VG	S	M	U	N
d) Overall customer satisfaction	E	VG	S	M	U	N
4. MANAGEMENT/ PERSONNEL/LABOR						
a) Effectiveness of on-site management, including management of subcontractors, suppliers, materials, and/or labor force?	E	VG	S	M	U	N
b) Ability to hire, apply, and retain a qualified workforce to this effort	E	VG	S	M	U	N
c) Government Property Control	E	VG	S	M	U	N
d) Knowledge/expertise demonstrated by contractor personnel	E	VG	S	M	U	N
e) Utilization of Small Business concerns	E	VG	S	M	U	N
f) Ability to simultaneously manage multiple projects with multiple disciplines	E	VG	S	M	U	N
g) Ability to assimilate and incorporate changes in requirements and/or priority, including planning, execution and response to Government changes	E	VG	S	M	U	N
h) Effectiveness of overall management (including ability to effectively lead, manage and control the program)	E	VG	S	M	U	N
5. COST/FINANCIAL MANAGEMENT						
a) Ability to meet the terms and conditions within the contractually agreed price(s)?	E	VG	S	M	U	N
b) Contractor proposed innovative alternative methods/processes that reduced cost, improved maintainability or other factors that benefited the client	E	VG	S	M	U	N
c) If this is/was a Government cost type contract, please rate the Contractor's timeliness and accuracy in submitting monthly invoices with appropriate back-up documentation, monthly status reports/budget variance reports, compliance with established budgets and avoidance of significant and/or unexplained variances (under runs or overruns)	E	VG	S	M	U	N
d) Is the Contractor's accounting system adequate for management and tracking of costs? *If no, please explain in Remarks section.*		Yes		No		

e) If this is/was a Government contract, has/was this contract been partially or completely terminated for default or convenience or are there any pending terminations? *Indicate if show cause or cure notices were issued, or any default action in comment section below.*	Yes No
f) Have there been any indications that the contractor has had any financial problems? *If yes, please explain below.*	Yes No
6. SAFETY/SECURITY	
a) To what extent was the contractor able to maintain an environment of safety, adhere to its approved safety plan, and respond to safety issues? (Includes: following the user's rules, regulations, and requirements regarding housekeeping, safety, correction of noted deficiencies, etc.)	E VG S M U N
b) Contractor complied with all security requirements for the project and personnel security requirements.	E VG S M U N
7. GENERAL	
a) Ability to successfully respond to emergency and/or surge situations (including notifying COR, PM or Contracting Officer in a timely manner regarding urgent contractual issues).	E VG S M U N
b) Compliance with contractual terms/provisions *(explain if specific issues)*	E VG S M U N
c) Would you hire or work with this firm again? *(If no, please explain below)*	Yes No
d) In summary, provide an overall rating for the work performed by this contractor.	E VG S M U N

Please provide responses to the questions above (*if applicable*) and/or additional remarks. Furthermore, please provide a brief narrative addressing specific strengths, weaknesses, deficiencies, or other comments which may assist our office in evaluating performance risk *(please attach additional pages if necessary)***:**

ATTACHMENT 3
KEY PERSONNEL RESUME

Provide information, listed below, on separate sheets showing qualifications of: On Site Construction Project Manager, Quality Control Manager (CQC System Manager), Site Superintendent, and Site Safety and Health Officer. NOTE: Match the positions on this page to the list of key personnel in the narrative submission requirements and evaluation criteria.

Name and Title ____

Project Assignment ____

Name of Your Firm ____

Years experience in the role identified ________________________________

No. of Years: With this Firm__________With other Firms___________

Education: Degree(s)/Year/Specialization ______________________________
____

Active Registration, if any: No._______, State(s) ______,
First Year/ Current Year ______/______

Other experience and qualifications relevant to same/similar work required under this contract:
____
____
____
____
____
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____
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____
____

ATTACHMENT 4
LETTER OF COMMITMENT FOR KEY PERSONNEL

TO: Contracting Officer

SUBJECT: Letter of Commitment for Proposed Contract for ______________________________________

Dear Sir or Madam:

I hereby make the unequivocal commitment that, in the event of an award of a contract to (Fill in name of Proposer), that I will fulfill the duty of (Job Title).

Sincerely,
(prospective employee signs)

Date: _____________

ATTACHMENT 5

LETTER OF COMMITMENT FOR KEY SUBCONTRACTORS

(USE COMPANY LETTERHEAD)

TO: Contracting Officer

SUBJECT: Letter of Commitment for Proposed Contract for

____

Dear Sir or Madam:

I hereby make the unequivocal commitment that, in the event of an award of a contract to (Fill in name of Proposer), that (insert name of key Subcontractors firm) will fulfill the duties of (state role on a project).

Sincerely, (Authorized official)

Date:________________

Section 00 21 00 - Instructions

RFO Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.204-7	System for Award Management-Registration. (Deviation)	2026-02		
52.214-34	Submission of Offers in the English Language.	1991-04		
52.214-35	Submission of Offers in U.S. Currency.	1991-04		
52.215-1	Instructions to Offerors-Competitive Acquisition. (Deviation)	2026-02		
52.222-5	Construction Wage Rate Requirements-Secondary Site of the Work. (Deviation)	2026-02		
52.225-10	Notice of Buy American Requirement-Construction Materials.	2014-05		
52.232-28	Invitation to Propose Performance-Based Payments.	2000-03		
52.233-2	Service of Protest. (Deviation)	2026-02		

DFARS Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.204-7019	Notice of NIST SP 800-171 DoD Assessment Requirements.	2023-11		
252.204-7024	Notice on the Use of the Supplier Performance Risk System.	2023-03		

252.215-7013	Supplies and Services Provided by Nontraditional Defense Contractors.	2023-01

RFO Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.216-1	Type of Contract. (Deviation)	2026-02		

Type of Contract (Feb 2026) (Deviation)

The Government contemplates award of a firm fixed price contract resulting from this solicitation.

(End of provision)

52.228-1	Bid Guarantee.	1996-09

Bid Guarantee (Sept 1996)

(a) Failure to furnish a bid guarantee in the proper form and amount, by the time set for opening of bids, may be cause for rejection of the bid.

(b) The bidder shall furnish a bid guarantee in the form of a firm commitment, e.g., bid bond supported by good and sufficient surety or sureties acceptable to the Government, postal money order, certified check, cashier's check, irrevocable letter of credit, or, under Treasury Department regulations, certain bonds or notes of the United States. The Contracting Officer will return bid guarantees, other than bid bonds-

(1) To unsuccessful bidders as soon as practicable after the opening of bids; and

(2) To the successful bidder upon execution of contractual documents and bonds (including

any necessary coinsurance or reinsurance agreements), as required by the bid as accepted.

(c) The amount of the bid guarantee shall be 20 percent of the bid price or $3,000,000.00, whichever is less.

(d) If the successful bidder, upon acceptance of its bid by the Government within the period specified for acceptance, fails to execute all contractual documents or furnish executed bond (s) within 10 days after receipt of the forms by the bidder, the Contracting Officer may terminate the contract for default.

(e) In the event the contract is terminated for default, the bidder is liable for any cost of acquiring the work that exceeds the amount of its bid, and the bid guarantee is available to offset the difference.

(End of clause)

52.252-1	Solicitation Provisions Incorporated by Reference.	1998-02

Solicitation Provisions Incorporated by Reference (Feb 1998)

This solicitation incorporates one or more solicitation provisions by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. The offerer is cautioned that the listed provisions may include blocks that must be completed by the offerer and submitted with its quotation or offer. In lieu of submitting the full text of those provisions, the offerer may identify the provision by paragraph identifier and provide the appropriate information with its quotation or offer. Also, the full text of a solicitation provision may be accessed electronically at this/these address (es):

www.acquisition.gov

(End of provision)

52.252-3	Alterations in Solicitation.	1984-04

Alterations in Solicitation (Apr 1984)

Portions of this solicitation are altered as follows:

None

(End of clause)

52.252-5	Authorized Deviations in Provisions.	2020-11

Authorized Deviations in Provisions (Nov 2020)

(a) The use in this solicitation of any Federal Acquisition Regulation (48 CFR Chapter 1) provision with an authorized deviation is indicated by the addition of "(DEVIATION)" after the date of the provision.

(b) The use in this solicitation of any Defense Federal Acquisition Regulation Supplement (DFARS)(48 CFR Chapter2) provision with an authorized deviation is indicated by the addition of "(DEVIATION)" after the name of the regulation.

(End of provision)

DFARS Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.215-7008	Only One	2022-12		

Offer.

ONLY ONE OFFER (DEC 2022)

(a) Cost or pricing data requirements. After initial submission of offers, if the Contracting Officer notifies the Offerer that only one offer was received, the Offerer agrees to-

(1) Submit any additional cost or pricing data that is required in order to determine whether the price is fair and reasonable (10 U.S.C. 3705) or to comply with the statutory requirement for certified cost or pricing data (10 U.S.C. 3702 and FAR 15.403-3); and

(2) Except as provided in paragraph (b) of this provision, if the acquisition exceeds the certified cost or pricing data threshold and an exception to the requirement for certified cost or pricing data at FAR 15.403-1(b)(2) through (5) does not apply, certify all cost or pricing data in accordance with paragraph (c) of DFARS provision 252.215-7010, Requirements for Certified Cost or Pricing Data and Data Other Than Certified Cost or Pricing Data, of this solicitation.

(b) Canadian Commercial Corporation. If the Offerer is the Canadian Commercial Corporation, certified cost or pricing data are not required. If the Contracting Officer notifies the Canadian Commercial Corporation that additional data other than certified cost or pricing data are required in accordance with DFARS 225.870-4(c), the Canadian Commercial Corporation shall obtain and provide the following:

(1) Profit rate or fee (as applicable).

(2) Analysis provided by Public Works and Government Services Canada to the Canadian Commercial Corporation to determine a fair and reasonable price (comparable to the analysis required at FAR 15.404-1).

(3) Data other than certified cost or pricing data necessary to permit a determination by the U.S. Contracting Officer that the proposed price is fair and reasonable Documents supporting cost or price[U.S. Contracting Officer to provide description of the data required in accordance with FAR 15.403-3(a)(1) with the notification].

(4) As specified in FAR 15.403-3(a)(4), an offerer who does not comply with a requirement to submit data that the U.S. Contracting Officer has deemed necessary to determine price reasonableness or cost realism is ineligible for award unless the head of the contracting activity determines that it is in the best interest of the Government to make the award to that

offerer.

(c) Subcontracts. Unless the Offerer is the Canadian Commercial Corporation, the Offerer shall insert the substance of this provision, including this paragraph (c), in all subcontracts exceeding the simplified acquisition threshold defined in FAR part 2.

(End of provision)

Section 00 45 00 - Representations and Certifications

RFO Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.229-12	Tax on Certain Foreign Procurements.	2021-02		

DFARS Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.203-7005	Representation Relating to Compensation of Former DoD Officials.	2022-09		
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls.	2016-10		
252.225-7055	Representation Regarding Business Operations with the Maduro Regime.	2022-05		
252.225-7059	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region-Representation.	2023-06		
252.232-7016	Notice of Progress Payments or Performance-Based Payments.	2020-04		

RFO Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.209-7	Information Regarding Responsibility Matters. (Deviation)	2026-02		

Information Regarding Responsibility Matters (Feb 2026) (Deviation)

(a) Definitions. As used in this provision-

Administrative proceeding means a non-judicial process that is adjudicatory in nature in order to make a determination of fault or liability (e.g., Securities and Exchange Commission Administrative Proceedings, Civilian Board of Contract Appeals Proceedings, and Armed Services Board of Contract Appeals Proceedings). This includes administrative proceedings at the Federal and State level but only in connection with performance of a Federal contract or grant. It does not include agency actions such as contract audits, site visits, corrective plans, or inspection of deliverables.

Federal contracts and grants with total value greater than $10,000,000 means-

(1) The total value of all current, active contracts and grants, including all priced options; and

(2) The total value of all current, active orders including all priced options under indefinite-delivery, indefinite-quantity, 8(a), or requirements contracts (including task and delivery and multiple-award Schedules).

Principal means an officer, director, owner, partner, or a person having primary management or supervisory responsibilities within a business entity (e.g., general manager; plant manager; head of a division or business segment; and similar positions).

(b) The offerer [] has [] does not have current active Federal contracts and grants with total value greater than $10,000,000.

(c) If the offerer checked "has" in paragraph (b) of this provision, the offerer represents, by submission of this offer, that the information it has entered in the Federal Awardee Performance and Integrity Information System (FAPIIS) is current, accurate, and complete as of the date of submission of this offer with regard to the following information:

(1) Whether the offerer, and/or any of its principals, has or has not, within the last five years, in connection with the award to or performance by the offerer of a Federal contract or grant, been the subject of a proceeding, at the Federal or State level that resulted in any of the following dispositions:

(i) In a criminal proceeding, a conviction.

(ii) In a civil proceeding, a finding of fault and liability that results in the payment of a monetary fine, penalty, reimbursement, restitution, or damages of $5,000 or more.

(iii) In an administrative proceeding, a finding of fault and liability that results in-

(A) The payment of a monetary fine or penalty of $5,000 or more; or

(B) The payment of a reimbursement, restitution, or damages in excess of $100,000.

(iv) In a criminal, civil, or administrative proceeding, a disposition of the matter by consent or compromise with an acknowledgment of fault by the Contractor if the proceeding could have led to any of the outcomes specified in paragraphs (c)(1)(i), (c)(1)(ii), or (c)(1)(iii) of this provision.

(2) If the offerer has been involved in the last five years in any of the occurrences listed in (c)(1) of this provision, whether the offerer has provided the requested information with regard to each occurrence.

(d) The offerer shall post the information in paragraphs (c)(1)(i) through (c)(1)(iv) of this provision in FAPIIS as required through maintaining an active registration in the System for Award Management, which can be accessed via https://www.sam.gov (see 52.204-7).

(End of provision)

52.209-13	Violation of Arms Control Treaties or Agreements-Certification. (Deviation)	2026-02

Violation of Arms Control Treaties or Agreements-Certification (Feb 2026) (Deviation)

(a) This provision does not apply to acquisitions at or below the simplified acquisition threshold or to acquisitions of commercial products and commercial services as defined in Federal Acquisition Regulation 2.101.

(b) Certification. [Offerer shall check either (1) or (2).]

[ ] (1) The Offerer certifies that-

(i) It does not engage and has not engaged in any activity that contributed to or was a significant factor in the President's or Secretary of State's determination that a foreign country is in violation of its obligations undertaken in any arms control, nonproliferation, or disarmament agreement to which the United States is a party, or is not adhering to its arms control, nonproliferation, or disarmament commitments in which the United States is a participating state. The determinations are described in the most recent unclassified annual report provided to Congress pursuant to section 403 of the Arms Control and Disarmament Act (22 U.S.C. 2593a). The report is available at https://www.state.gov/bureaus-offices /under-secretary-for-arms-control-and-international-security-affairs/bureau-of-arms-control-verification-and-compliance/; and

(ii) No entity owned or controlled by the Offerer has engaged in any activity that contributed to or was a significant factor in the President's or Secretary of State's determination that a foreign country is in violation of its obligations undertaken in any arms control, nonproliferation, or disarmament agreement to which the United States is a party, or is not adhering to its arms control, nonproliferation, or disarmament commitments in which the United States is a participating state. The determinations are described in the most recent unclassified annual report provided to Congress pursuant to section 403 of the Arms Control and Disarmament Act (22 U.S.C. 2593a). The report is available at https://www.state.gov /bureaus-offices/under-secretary-for-arms-control-and-international-security-affairs/bureau-of-arms-control-verification-and-compliance/; or

[ ] (2) The Offerer is providing separate information with its offer in accordance with paragraph (d)(2) of this provision.

(c) Procedures for reviewing the annual unclassified report (see paragraph (b)(1) of this provision). For clarity, references to the report in this section refer to the entirety of the annual unclassified report, including any separate reports that are incorporated by reference into the annual unclassified report.

(1) Check the table of contents of the annual unclassified report and the country section headings of the reports incorporated by reference to identify the foreign countries listed there. Determine whether the Offerer or any person owned or controlled by the Offerer may have engaged in any activity related to one or more of such foreign countries.

(2) If such activity might have occurred, review all findings in the report associated with those foreign countries to determine whether or not each such foreign country was determined to be in violation of its obligations undertaken in an arms control, nonproliferation, or disarmament agreement to which the United States is a party, or to be not adhering to its arms control, nonproliferation, or disarmament commitments in which the United States is a participating state. For clarity, in the annual report an explicit certification of non-compliance is equivalent to a determination of violation. However, the following statements in the annual report are not equivalent to a determination of violation:

(i) An inability to certify compliance.

(ii) An inability to conclude compliance.

(iii) A statement about compliance concerns.

(3) If so, determine whether the Offerer or any person owned or controlled by the Offerer has engaged in any activity that contributed to or is a significant factor in the determination in the report that one or more of these foreign countries is in violation of its obligations undertaken in an arms control, nonproliferation, or disarmament agreement to which the United States is a party, or is not adhering to its arms control, nonproliferation, or disarmament commitments in which the United States is a participating state. Review the narrative for any such findings reflecting a determination of violation or non-adherence related to those foreign countries in the report, including the finding itself, and to the extent necessary, the conduct giving rise to the compliance or adherence concerns, the analysis of compliance or adherence concerns, and efforts to resolve compliance or adherence concerns.

(4) The Offerer may submit any questions with regard to this report by email to NOAA1290Cert@state.gov. To the extent feasible, the Department of State will respond to such email inquiries within 3 business days.

(d) Do not submit an offer unless-

(1) A certification is provided in paragraph (b)(1) of this provision and submitted with the offer; or

(2) In accordance with paragraph (b)(2) of this provision, the Offerer provides with its offer information that the President of the United States has

(i) Waived application under 22 U.S.C. 2593e(d) or (e); or

(ii) Determined under 22 U.S.C. 2593e(g)(2) that the entity has ceased all activities for which measures were imposed under 22 U.S.C. 2593e(b).

(e) Remedies. The certification in paragraph (b)(1) of this provision is a material representation of fact upon which reliance was placed when making award. If the Government later determines that the Offerer knowingly submitted a false certification, in addition to other remedies available to the Government, such as suspension or debarment, the Contracting Officer may terminate any contract resulting from the false certification.

(End of provision)

52.230-1 Cost Accounting Standards Notices and Certification. (Deviation) 2026-02

Cost Accounting Standards Notices and Certification (Feb 2026) (Deviation)

Note: This notice does not apply to small businesses or foreign governments. This notice is in three parts, identified by Roman numerals I through III.

Offerers shall examine each part and provide the requested information in order to determine Cost Accounting Standards (CAS) requirements applicable to any resultant contract.

If the offerer is an educational institution, part II does not apply unless the contemplated contract will be subject to full or modified CAS coverage pursuant to 48 CFR 9903.201-2(c)(5) or 9903.201-2(c)(6), respectively.

I. Disclosure Statement-Cost Accounting Practices and Certification

(a) Any contract in excess of the lower CAS threshold specified in Federal Acquisition Regulation (FAR) 30.205(b) resulting from this solicitation will be subject to the requirements of the Cost Accounting Standards Board (48 CFR chapter 99), except for those contracts which are exempt as specified in 48 CFR 9903.201-1.

(b) Any offerer submitting a proposal which, if accepted, will result in a contract subject to the requirements of 48 CFR chapter 99 must, as a condition of contracting, submit a Disclosure Statement as required by 48 CFR 9903.202. When required, the Disclosure Statement must be submitted as a part of the offerer's proposal under this solicitation unless the offerer has already submitted a Disclosure Statement disclosing the practices used in connection with the pricing of this proposal. If an applicable Disclosure Statement has already been submitted, the offerer may satisfy the requirement for submission by providing the information requested in paragraph (c) of part I of this provision.

Caution: In the absence of specific regulations or agreement, a practice disclosed in a Disclosure Statement shall not, by virtue of such disclosure, be deemed to be a proper, approved, or agreed-to practice for pricing proposals or accumulating and reporting contract performance cost data.

(c) Check the appropriate box below:

(1) [] Certificate of Concurrent Submission of Disclosure Statement. The offerer hereby certifies that, as a part of the offer, copies of the Disclosure Statement have been submitted as follows:

(i) Original and one copy to the cognizant Administrative Contracting Officer (ACO) or cognizant Federal agency official authorized to act in that capacity (Federal official), as applicable; and

(ii) One copy to the cognizant Federal auditor.

(Disclosure must be on Form No. CASB DS-1 or CASB DS-2, as applicable. Forms may be obtained from the cognizant ACO or Federal official.)

Date of Disclosure Statement: ___ Name and Address of Cognizant ACO or Federal Official Where Filed: ______________

The offerer further certifies that the practices used in estimating costs in pricing this proposal are consistent with the cost accounting practices disclosed in the Disclosure Statement.

(2) [ ] Certificate of Previously Submitted Disclosure Statement. The offerer hereby certifies that the required Disclosure Statement was filed as follows:

Date of Disclosure Statement: ___ Name and Address of Cognizant ACO or Federal Official Where Filed: ______________

The offerer further certifies that the practices used in estimating costs in pricing this proposal are consistent with the cost accounting practices disclosed in the applicable Disclosure Statement.

(3) [ ] Certificate of Monetary Exemption. The offerer hereby certifies that the offerer, together with all divisions, subsidiaries, and affiliates under common control, did not receive net awards of negotiated prime contracts and subcontracts subject to CAS totaling $50 million or more in the cost accounting period immediately preceding the period in which this proposal was submitted. The offerer further certifies that if such status changes before an award resulting from this proposal, the offerer will advise the Contracting Officer immediately.

(4) [] Certificate of Interim Exemption. The offerer hereby certifies that (i) the offerer first exceeded the monetary exemption for disclosure, as defined in (3) of this subsection, in the cost accounting period immediately preceding the period in which this offer was submitted

and (ii) in accordance with 48 CFR 9903.202-1, the offerer is not yet required to submit a Disclosure Statement. The offerer further certifies that if an award resulting from this proposal has not been made within 90 days after the end of that period, the offerer will immediately submit a revised certificate to the Contracting Officer, in the form specified under paragraph (c)(1) or (c)(2) of part I of this provision, as appropriate, to verify submission of a completed Disclosure Statement.

Caution: Offerers currently required to disclose because they were awarded a CAS-covered prime contract or subcontract of $50 million or more in the current cost accounting period may not claim this exemption (4). Further, the exemption applies only in connection with proposals submitted before expiration of the 90-day period following the cost accounting period in which the monetary exemption was exceeded.

II. Cost Accounting Standards-Eligibility for Modified Contract Coverage

If the offerer is eligible to use the modified provisions of 48 CFR 9903.201-2(b) and elects to do so, the offerer shall indicate by checking the box below. Checking the box below shall mean that the resultant contract is subject to the Disclosure and Consistency of Cost Accounting Practices clause in lieu of the Cost Accounting Standards clause.

[ ] The offerer hereby claims an exemption from the Cost Accounting Standards clause under the provisions of 48 CFR 9903.201-2(b) and certifies that the offerer is eligible for use of the Disclosure and Consistency of Cost Accounting Practices clause because during the cost accounting period immediately preceding the period in which this proposal was submitted, the offerer received less than $50 million in awards of CAS-covered prime contracts and subcontracts. The offerer further certifies that if such status changes before an award resulting from this proposal, the offerer will advise the Contracting Officer immediately.

Caution: An offerer may not claim the above eligibility for modified contract coverage if this proposal is expected to result in the award of a CAS-covered contract of $50 million or more or if, during its current cost accounting period, the offerer has been awarded a single CAS-covered prime contract or subcontract of $50 million or more.

III. Additional Cost Accounting Standards Applicable to Existing Contracts

The offerer shall indicate below whether award of the contemplated contract would, in accordance with paragraph (a)(3) of the Cost Accounting Standards clause, require a change in established cost accounting practices affecting existing contracts and subcontracts.

[] Yes [] No

(End of provision)

52.230-7 Proposal Disclosure-Cost Accounting Practice Changes. 2005-04

Proposal Disclosure-Cost Accounting Practice Changes (Apr 2005)

The offerer shall check "yes" below if the contract award will result in a required or unilateral change in cost accounting practice, including unilateral changes requested to be desirable changes.

[]Yes[]No

If the offerer checked "Yes" above, the offerer shall-

(1) Prepare the price proposal in response to the solicitation using the changed practice for the period of performance for which the practice will be used; and

(2) Submit a description of the changed cost accounting practice to the Contracting Officer and the Cognizant Federal Agency Official as pricing support for the proposal.

(End of provision)

Section 00 70 00 - Conditions of the Contract

RFO Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.222-3	Convict Labor. (Deviation)	2026-02		
52.222-7	Withholding of Funds. (Deviation)	2026-02		
52.240-90	Security Prohibitions and Exclusions Representations and Certifications. (Deviation)	2026-02		
52.240-91	Security Prohibitions and Exclusions. (Deviation)	2026-02		

DFARS Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting. (DEVIATION 2024-00013 REVISION 1)	2024-05	Deviation 2024-00013	2024-05
252.226-7003	Drug-Free Work Force.	2024-08		

RFO Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.215-20	Requirements for Certified Cost or Pricing Data and Data Other Than Certified Cost or Pricing Data. (Deviation) (Alternate IV)	2026-02	Alternate IV	2026-02

Alternate IV-(Feb 2026) (Deviation). As prescribed in 15.110(t)(4), replace the text of the provision with the following:

(a) Submission of certified cost or pricing data is not required.

(b) Provide data described below: Documents supporting cost or price

52.219-28	Postaward Small Business Program Rerepresentation. (Deviation) (Alternate I)	2026-02	Alternate I	2026-02

Alternate I (Feb 2026) (Deviation). As prescribed in 19.101(a)(2)(iii)(B), substitute the following paragraph (g)(1) for paragraph (g)(1) of the basic clause:

(g)(1) The Contractor represents its small business size status for each one of the NAICS codes assigned to this contract.

NAICS Code	Small business concern (yes/no)
238160	

DFARS Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date

252.204-7998	Alternate A, Annual Representations and Certifications. (DEVIATION 2026-00043)	2026-02	Alternate A Deviation 2026-00043	2026-02 2026-02

Alternate A, Annual Representations and Certifications (DEVIATION 2026-O0043)(FEB 2026)

Include the following paragraphs (e), (f), and (g) in the provision at FAR 52.204-7:

(e)(1) If the provision at FAR 52.204-7, System for Award Management - Registration, is included in this solicitation, paragraph (g) of this provision applies.

(2) If the provision at FAR 52.204-7, System for Award Management - Registration, is not included in this solicitation, and the Offerer has an active registration in the System for Award Management (SAM), the Offerer may choose to use paragraph (g) of this provision instead of completing the corresponding individual representations and certifications in the solicitation. The Offerer shall indicate which option applies by checking one of the following boxes:

[ ] (i) Paragraph (g) applies.

[ ] (ii) Paragraph (g) does not apply and the Offerer has completed the individual representations and certifications in the solicitation.

(f)(1) The following representations or certifications in the SAM database are applicable to this solicitation as indicated:

(i) 252.204-7016, Covered Defense Telecommunications Equipment or Services--Representation. Applies to all solicitations.

(ii) 252.216-7008, Economic Price Adjustment--Wage Rates or Material Prices Controlled by a Foreign Government. Applies to solicitations for fixed-price supply and service contracts when the contract is to be performed wholly or in part in a foreign country, and a foreign government controls wage rates or material prices and may during contract performance impose a mandatory change in wages or prices of materials.

(iii) 252.225-7042, Authorization to Perform. Applies to all solicitations when performance will be wholly or in part in a foreign country.

(iv) 252.225-7049, Prohibition on Acquisition of Certain Foreign Commercial Satellite Services--Representations. Applies to solicitations for the acquisition of commercial satellite services.

(v) 252.225-7050, Disclosure of Ownership or Control by the Government of a Country that is a State Sponsor of Terrorism. Applies to all solicitations expected to result in contracts of $150,000 or more.

(vi) 252.229-7012, Tax Exemptions (Italy)--Representation. Applies to solicitations when contract performance will be in Italy.

(vii) 252.229-7013, Tax Exemptions (Spain)--Representation. Applies to solicitations when contract performance will be in Spain.

(2) The following representations or certifications in SAM are applicable to this solicitation as indicated by the Contracting Officer: [Contracting Officer check as appropriate.]

[ ](i) 252.209-7002, Disclosure of Ownership or Control by a Foreign Government.

[ ](ii) 252.225-7000, Buy American--Balance of Payments Program Certificate.

[ ](iii) 252.225-7020, Trade Agreements Certificate.

[ ]Use with Alternate I.

[ ](iv) 252.225-7031, Secondary Arab Boycott of Israel.

[ ](v) 252.225-7035, Buy American--Free Trade Agreements--Balance of Payments Program Certificate.

[ ]Use with Alternate I.

[ ]Use with Alternate II.

[ ]Use with Alternate III.

[ ]Use with Alternate IV.

[ ]Use with Alternate V.

[] (vi) 252.226-7002, Representation for Demonstration Project for Contractors Employing Persons with Disabilities.

[] (vii) 252.232-7015, Performance-Based Payments--Representation.

(g) The Offerer has completed the annual representations and certifications electronically via the SAM website at https://www.sam.gov After reviewing the SAM database information, the Offerer verifies by submission of the offer that the representations and certifications currently posted electronically that apply to this solicitation as indicated in FAR 52.204-7 and paragraph (f) of this provision have been entered or updated within the last 12 months, are current, accurate, complete, and applicable to this solicitation (including the business size standard applicable to the NAICS code referenced for this solicitation), as of the date of this offer, and are incorporated in this offer by reference (see FAR 4.203-1); except for the changes identified below [Offerer to insert changes, identifying change by provision number, title, date]. These amended representation(s) and/or certification(s) are also incorporated in this offer and are current, accurate, and complete as of the date of this offer.

FAR/DFARS provision No.	Title	Date	Change

Any changes provided by the Offerer are applicable to this solicitation only, and do not result in an update to the representations and certifications located in the SAM database.

(End of provision)

Section 00 72 00 - General Conditions

RFO Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.202-1	Definitions.	2020-06		
52.203-3	Gratuities.	1984-04		
52.203-5	Covenant Against Contingent Fees.	2014-05		
52.203-6	Restrictions on Subcontractor Sales to the Government.	2020-06		
52.203-7	Anti-Kickback Procedures.	2020-06		
52.203-8	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity.	2014-05		
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity.	2014-05		
52.203-12	Limitation on Payments to Influence Certain Federal Transactions.	2020-06		
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements.	2017-01		
52.204-10	Reporting Executive Compensation and First-Tier Subcontract Awards. (Deviation)	2026-02		
52.204-13	System for Award Management-Maintenance. (Deviation)	2026-02		
52.204-19	Incorporation by Reference of Representations and Certifications.	2014-12		
52.204-27	Prohibition on a ByteDance Covered Application.	2023-06		

52.209-6	Protecting the Government's Interest When Subcontracting With Contractors Debarred, Suspended, Proposed for Debarment, or Voluntarily Excluded. (Deviation)	2026-02
52.209-9	Updates of Publicly Available Information Regarding Responsibility Matters. (Deviation)	2026-02
52.209-10	Prohibition on Contracting With Inverted Domestic Corporations. (Deviation)	2026-02
52.215-2	Audit and Records-Negotiation. (Deviation)	2026-02
52.215-8	Order of Precedence-Uniform Contract Format. (Deviation)	2026-02
52.215-15	Pension Adjustments and Asset Reversions. (Deviation)	2026-02
52.215-18	Reversion or Adjustment of Plans for Postretirement Benefits (PRB) Other Than Pensions. (Deviation)	2026-02
52.217-2	Cancellation Under Multi-year Contracts.	1997-10
52.217-5	Evaluation of Options (Deviation)	2026-02
52.219-3	Notice of HUBZone Set-Aside or Sole-Source Award. (Deviation)	2026-02
52.219-6	Notice of Total Small Business Set-Aside. (Deviation)	2026-02
52.219-8	Utilization of Small Business Concerns. (Deviation)	2026-02
52.219-27	Notice of Set-Aside for, or Sole-Source Award to, Service-Disabled Veteran-Owned Small Business (SDVOSB) Concerns Eligible Under the SDVOSB Program. (Deviation)	2026-02
52.219-29	Notice of Set-Aside for, or Sole-Source	2026-02

	Award to, Economically Disadvantaged Women-Owned Small Business Concerns. (Deviation)	
52.219-30	Notice of Set-Aside for, or Sole-Source Award to, Women-Owned Small Business Concerns Eligible Under the Women-Owned Small Business Program. (Deviation)	2026-02
52.219-33	Nonmanufacturer Rule. (Deviation)	2026-02
52.222-4	Contract Work Hours and Safety Standards-Overtime Compensation. (Deviation)	2026-02
52.222-6	Construction Wage Rate Requirements. (Deviation)	2026-02
52.222-8	Payrolls and Basic Records. (Deviation)	2026-02
52.222-9	Apprentices and Trainees. (Deviation)	2026-02
52.222-10	Compliance with Copeland Act Requirements. (Deviation)	2026-02
52.222-11	Subcontracts (Labor Standards). (Deviation)	2026-02
52.222-12	Contract Termination-Debarment.	2014-05
52.222-13	Compliance with Construction Wage Rate Requirements and Related Regulations.	2014-05
52.222-14	Disputes Concerning Labor Standards. (Deviation)	2026-02
52.222-15	Certification of Eligibility.	2014-05
52.222-30	Construction Wage Rate Requirements-Price Adjustment (None or Separately Specified Method).	2018-08

52.222-35	Equal Opportunity for Veterans. (Deviation)	2026-02
52.222-36	Equal Opportunity for Workers with Disabilities. (Deviation)	2026-02
52.222-37	Employment Reports on Veterans. (Deviation)	2026-02
52.222-40	Notification of Employee Rights Under the National Labor Relations Act. (Deviation)	2026-02
52.222-50	Combating Trafficking in Persons. (Deviation)	2026-02
52.222-54	Employment Eligibility Verification. (Deviation)	2026-02
52.222-55	Minimum Wages for Contractor Workers Under Executive Order 14026. (Deviation)	2026-02
52.222-62	Paid Sick Leave Under Executive Order 13706. (Deviation)	2026-02
52.223-5	Pollution Prevention and Right-to-Know Information.	2024-05
52.223-23	Sustainable Products. (Deviation)	2026-02
52.226-7	Drug-Free Workplace.	2024-05
52.226-8	Encouraging Contractor Policies to Ban Text Messaging While Driving.	2024-05
52.228-2	Additional Bond Security.	1997-10
52.228-5	Insurance-Work on a Government Installation.	1997-01
52.228-11	Individual Surety-Pledge of Assets.	2021-02
52.228-12	Prospective Subcontractor Requests for Bonds.	2022-12

52.228-14	Irrevocable Letter of Credit.	2014-11
52.228-15	Performance and Payment Bonds-Construction.	2020-06
52.228-17	Individual Surety-Pledge of Assets (Bid Guarantee).	2021-02
52.229-3	Federal, State, and Local Taxes.	2013-02
52.230-2	Cost Accounting Standards. (Deviation)	2026-02
52.230-6	Administration of Cost Accounting Standards. (Deviation)	2026-02
52.232-8	Discounts for Prompt Payment.	2002-02
52.232-17	Interest.	2014-05
52.232-23	Assignment of Claims.	2014-05
52.232-27	Prompt Payment for Construction Contracts.	2017-01
52.232-33	Payment by Electronic Funds Transfer-System for Award Management.	2018-10
52.232-39	Unenforceability of Unauthorized Obligations.	2013-06
52.232-40	Providing Accelerated Payments to Small Business Subcontractors.	2023-03
52.233-1	Disputes (Deviation)	2026-02
52.233-3	Protest after Award. (Deviation)	2026-02
52.233-4	Applicable Law for Breach of Contract Claim. (Deviation)	2026-02
52.236-2	Differing Site Conditions. (Deviation)	2026-02
52.236-3	Site Investigation and Conditions Affecting the Work. (Deviation)	2026-02
52.236-5	Material and Workmanship. (Deviation)	2026-02

52.236-6	Superintendence by the Contractor. (Deviation)	2026-02
52.236-7	Permits and Responsibilities. (Deviation)	2026-02
52.236-8	Other Contracts. (Deviation)	2026-02
52.236-9	Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements. (Deviation)	2026-02
52.236-10	Operations and Storage Areas. (Deviation)	2026-02
52.236-11	Use and Possession Prior to Completion. (Deviation)	2026-02
52.236-12	Cleaning Up. (Deviation)	2026-02
52.236-13	Accident Prevention. (Deviation)	2026-02
52.236-15	Schedules for Construction Contracts. (Deviation)	2026-02
52.236-21	Specifications and Drawings for Construction. (Deviation)	2026-02
52.242-13	Bankruptcy.	1995-07
52.242-14	Suspension of Work.	1984-04
52.243-4	Changes. (Deviation)	2026-02
52.243-5	Changes and Changed Conditions.	1984-04
52.244-6	Subcontracts for Commercial Products and Commercial Services. (Deviation)	2026-02
52.246-12	Inspection of Construction.	1996-08
52.246-21	Warranty of Construction.	1994-03
52.247-5	Familiarization with Conditions.	1984-04

52.247-15	Contractor Responsibility for Loading and Unloading.	1984-04
52.249-1	Termination for Convenience of the Government (Fixed-Price) (Short Form).	1984-04
52.249-2	Alternate 1 - Termination for Convenience of the Government (Fixed-Price).	1996-09
52.249-10	Default (Fixed-Price Construction).	1984-04
52.253-1	Computer Generated Forms. (Deviation)	2026-02

DFARS Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.201-7000	Contracting Officer's Representative.	1991-12		
252.203-7000	Requirements Relating to Compensation of Former DoD Officials.	2011-09		
252.203-7001	Prohibition on Persons Convicted of Fraud or Other Defense-Contract-Related Felonies.	2023-01		
252.203-7002	Requirement to Inform Employees of Whistleblower Rights.	2022-12		
252.204-7003	Control of Government Personnel Work Product.	1992-04		
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting.	2024-05		
252.204-7018	Prohibition on the Acquisition of	2023-01		

	Covered Defense Telecommunications Equipment or Services.	
252.204-7020	NIST SP 800-171 DoD Assessment Requirements.	2023-11
252.209-7004	Subcontracting with Firms that are Owned or Controlled by the Government of a Country that is a State Sponsor of Terrorism.	2019-05
252.222-7006	Restrictions on the Use of Mandatory Arbitration Agreements.	2023-01
252.223-7006	Prohibition on Storage, Treatment, and Disposal of Toxic or Hazardous Materials.	2014-09
252.223-7008	Prohibition of Hexavalent Chromium.	2023-01
252.225-7012	Preference for Certain Domestic Commodities.	2022-04
252.225-7048	Export-Controlled Items.	2013-06
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime.	2023-01
252.225-7060	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region.	2023-06
252.226-7001	Utilization of Indian Organizations, Indian-Owned Economic Enterprises, and Native Hawaiian Small Business Concerns.	2023-01
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports.	2018-12
252.232-7004	DoD Progress Payment Rates.	2014-10
252.232-7010	Levies on Contract Payments.	2006-12

252.236-7000	Modification Proposals--Price Breakdown.	1991-12
252.236-7002	Obstruction of Navigable Waterways.	1991-12
252.242-7005	Contractor Business Systems.	2025-01
252.242-7006	Accounting System Administration.	2025-01
252.243-7001	Pricing of Contract Modifications.	1991-12
252.243-7002	Requests for Equitable Adjustment.	2022-12
252.247-7023	Transportation of Supplies by Sea.	2024-10

RFO Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Da
52.204-1	Approval of Contract.	1989-12		

Approval of Contract (Dec 1989)

This contract is subject to the written approval of Shawn G. Brady and shall not be binding until so approved.

(End of clause)

52.211-12 Liquidated Damages-Construction. 2000-09

Liquidated Damages-Construction (Sept 2000)

(a) If the Contractor fails to complete the work within the time specified in the contract, the Contractor shall pay liquidated damages to the Government in the amount of $1,947.00 for each calendar day of delay until the work is completed or accepted.

(b) If the Government terminates the Contractor's right to proceed, liquidated damages will continue to accrue until the work is completed. These liquidated damages are in addition to excess costs of repurchase under the Termination clause.

(End of clause)

52.219-14 Limitations on Subcontracting. (Deviation) 2026-02

Limitations on Subcontracting (Feb 2026) (Deviation)

(a) This clause does not apply to the unrestricted portion of a partial set-aside.

(b) Definition. Similarly situated entity, as used in this clause, means a first-tier subcontractor, including an independent contractor, that-

(1) Has the same small business program status as that which qualified the prime contractor for the award (e.g., for a small business set-aside contract, any small business concern, without regard to its socioeconomic status); and

(2) Is considered small for the size standard under the North American Industry Classification System (NAICS) code the prime contractor assigned to the subcontract.

(c) Applicability. This clause applies only to-

(1) Contracts that have been set aside for any of the small business concerns identified in 19.000(a)(3);

(2) Part or parts of a multiple-award contract that have been set aside for any of the small business concerns identified in 19.000(a)(3);

(3) Contracts that have been awarded on a sole-source basis in accordance with sections 19.105, 19.106, 19.107, and 19.108;

(4) Orders expected to exceed the simplified acquisition threshold and that are set aside for small business concerns under multiple-award contracts, as described in 8.4 and 16.5;

(5) Orders, regardless of dollar value, that are set aside in accordance with sections 19.105, 19.106, 19.107, and 19.108 under multiple-award contracts, as described in 8.4 and 16.5; and

(6) Contracts using the HUBZone price evaluation preference to award to a HUBZone small business concern unless the concern waived the evaluation preference.

(d) Independent contractors. An independent contractor shall be considered a subcontractor.

(e) By submission of an offer and execution of a contract, the Contractor agrees that in performance of a contract assigned a North American Industry Classification System **(NAICS)** code for-

(1) Services (except construction), it will not pay more than 50 percent of the amount paid by the Government for contract performance to subcontractors that are not similarly situated

entities. Any work that a similarly situated entity further subcontracts will count towards the prime contractor's 50 percent subcontract amount that cannot be exceeded. When a contract includes both services and supplies, the 50 percent limitation shall apply only to the service portion of the contract;

(2) Supplies (other than procurement from a nonmanufacturer of such supplies), it will not pay more than 50 percent of the amount paid by the Government for contract performance, excluding the cost of materials, to subcontractors that are not similarly situated entities. Any work that a similarly situated entity further subcontracts will count towards the prime contractor's 50 percent subcontract amount that cannot be exceeded. When a contract includes both supplies and services, the 50 percent limitation shall apply only to the supply portion of the contract;

(3) General construction, it will not pay more than 85 percent of the amount paid by the Government for contract performance, excluding the cost of materials, to subcontractors that are not similarly situated entities. Any work that a similarly situated entity further subcontracts will count towards the prime contractor's 85 percent subcontract amount that cannot be exceeded; or

(4) Construction by special trade contractors, it will not pay more than 75 percent of the amount paid by the Government for contract performance, excluding the cost of materials, to subcontractors that are not similarly situated entities. Any work that a similarly situated entity further subcontracts will count towards the prime contractor's 75 percent subcontract amount that cannot be exceeded.

(f) The Contractor shall comply with the limitations on subcontracting as follows:

(1) For contracts, in accordance with paragraphs (c)(1), (2), (3) and (6) of this clause-

[X] By the end of the base term of the contract and then by the end of each subsequent option period; or

[ ] By the end of the performance period for each order issued under the contract.

(2) For orders, in accordance with paragraphs (c)(4) and (5) of this clause, by the end of the performance period for the order.

(g) A joint venture agrees that, in the performance of the contract, the applicable percentage specified in paragraph (e) of this clause will be performed by the aggregate of the joint venture participants.

(1) In a joint venture comprised of a small business protege and its mentor approved by the Small Business Administration, the small business protege shall perform at least 40 percent of the work performed by the joint venture. Work performed by the small business protege in the joint venture must be more than administrative functions.

(2) In an 8(a) joint venture, the 8(a) participant(s) shall perform at least 40 percent of the work performed by the joint venture. Work performed by the 8(a) participants in the joint venture must be more than administrative functions.

(End of clause)

52.219-28	Postaward Small Business Program Rerepresentation. (Deviation)	2026-02

Postaward Small Business Program Rerepresentation (Feb 2026) (Deviation)

(a) Definitions. As used in this clause-

Long-term contract means a contract of more than five years in duration, including options. However, the term does not include contracts that exceed five years in duration because the period of performance has been extended for a cumulative period not to exceed six months under the clause at 52.217-8, Option to Extend Services, or other appropriate authority.

Small business concern-

(1) Means a concern, including its affiliates, that is independently owned and operated, not dominant in its field of operation, and qualified as a small business under the criteria in 13 CFR part 121 and the size standard in paragraph (c) of this clause.

(2) Affiliates, as used in this definition, means business concerns, one of whom directly or indirectly controls or has the power to control the others, or a third party or parties control or have the power to control the others. In determining whether affiliation exists, consideration is given to all appropriate factors including common ownership, common management, and contractual relationships. SBA determines affiliation based on the factors set forth at 13 CFR 121.103.

(b) If the Contractor represented that it was a small business concern, a small disadvantaged business concern, or a joint venture that was any of the small business concerns identified in 19.000(a)(3) prior to award of this contract, the Contractor shall rerepresent its size and socioeconomic status according to paragraph (e) of this clause or, if applicable, paragraph (g) of this clause, upon occurrence of any of the following:

(1) Within 30 days after execution of a novation agreement or within 30 days after modification of the contract to include this clause, if the novation agreement was executed prior to inclusion of this clause in the contract.

(2) Within 30 days after a merger or acquisition that does not require a novation or within 30 days after modification of the contract to include this clause, if the merger or acquisition occurred prior to inclusion of this clause in the contract.

(3) For long-term contracts-

(i) Within 60 to 120 days prior to the end of the fifth year of the contract; and

(ii) Within 60 to 120 days prior to the date specified in the contract for exercising any option thereafter.

(c) The Contractor shall rerepresent its size status in accordance with the size standard in *effect* at the time of this rerepresentation that corresponds to the North American Industry Classification System (NAICS) code(s) assigned to this contract. The small business size standard corresponding to this NAICS code(s) can be found at https://www.sba.gov /document/support--table-size-standards.

(d) The small business size standard for a Contractor providing an end item that it does not manufacture, process, or produce itself, for a contract other than a construction or service contract, is 500 employees, or 150 employees for information technology value-added resellers under NAICS code 541519, if the acquisition-

(1) Was set aside for small business and has a value above the simplified acquisition threshold;

(2) Used the HUBZone price evaluation preference regardless of dollar value, unless the Contractor waived the price evaluation preference; or

(3) Was an 8(a), HUBZone, service-disabled veteran-owned, economically disadvantaged women-owned, or women-owned small business set-aside or sole-source award regardless of dollar value.

(e) Except as provided in paragraph (g) of this clause, the Contractor shall make the representation(s) required by paragraph (b) of this clause by validating or updating all its representations in the Representations and Certifications section of the System for Award Management (SAM) and its other data in SAM, as necessary, to ensure that they reflect the Contractor's current status. The Contractor shall notify the contracting officer in writing within the timeframes specified in paragraph (b) of this clause, that the data have been validated or updated, and provide the date of the validation or update.

(f) If the Contractor represented that it was other than a small business concern prior to award of this contract, the Contractor may, but is not required to, take the actions required by paragraphs (e) or (g) of this clause.

(g) If the Contractor does not have representations and certifications in SAM, or does not have a representation in SAM for the NAICS code applicable to this contract, the Contractor is required to complete the following rerepresentation and submit it to the contracting office, along with the contract number and the date on which the rerepresentation was completed:

(1) The Contractor represents that it [] is, [ ] is not a small business concern under ___ NAICS Code assigned to ___ contract number.

(2) [Complete only if the Contractor represented itself as a small business concern in paragraph (g)(1) of this clause.] The Contractor represents that it [] is, [ ] is not, a small disadvantaged business concern as defined in 13 CFR 124.1001.

(3) Women-owned small business (WOSB) joint venture eligible under the WOSB Program. The Contractor represents that it [ ] is, [ ] is not a joint venture that complies with the requirements of 13 CFR 127.506(a) through (c). [___ The Contractor shall enter the name and unique entity identifier of each party to the joint venture: _.]

(4) Economically disadvantaged women-owned small business (EDWOSB) joint venture. The Contractor represents that it [ ] is, [ ] is not a joint venture that complies with the requirements of 13 CFR 127.506(a) through (c). [___ The Contractor shall enter the name and unique entity identifier of each party to the joint venture: _ .]

(5) Service-disabled veteran-owned small business (SDVOSB) joint venture eligible under the SDVOSB Program. The Contractor represents that it [ ] is, [] is not an SDVOSB joint venture eligible under the SDVOSB Program that complies with the requirements of 13 CFR 128.402. [_ The Contractor shall enter the name and unique entity identifier of each party to the joint venture: _.]

(6) HUBZone joint venture eligible under the HUBZone Program.[Complete only if the offerer is a HUBZone small business concern.] The offerer represents, as part of its offer,

that It [ ] is, [ ] is not a HUBZone joint venture that complies with the requirements of 13 CFR 126.616(a) through (c). [_The Contractor shall enter the name and unique entity identifier of each party to the joint venture: .] Each HUBZone small business concern participating in the HUBZone joint venture must be certified as a HUBZone concern. [_Contractor to sign and date and insert authorized signer's name and title.]

(End of clause)

52.225-9	Buy American-Construction Materials. (Deviation)	2026-02

Buy American-Construction Materials (Feb 2026) (Deviation)

(a) Definitions. As used in this clause-

Commercially available off-the-shelf (COTS) item-

(1) Means any item of supply (including construction material) that is-

(i) A commercial product (as defined in paragraph (1) of the definition of "commercial product" at Federal Acquisition Regulation (FAR) 2.101);

(ii) Sold in substantial quantities in the commercial marketplace; and

(iii) Offered to the Government, under a contract or subcontract at any tier, without modification, in the same form in which it is sold in the commercial marketplace; and

(2) Does not include bulk cargo, as defined in 46 U.S.C. 40102(4), such as agricultural products and petroleum products.

"Construction material" means an article, material, or supply brought to the construction site by the Contractor or a subcontractor for incorporation into the building or work. The term also includes an item brought to the site preassembled from articles, materials, or supplies. However, emergency life safety systems, such as emergency lighting, fire alarm, and audio evacuation systems, that are discrete systems incorporated into a public building or work and that are produced as complete systems, are evaluated as a single and distinct construction material regardless of when or how the individual parts or components of those systems are delivered to the construction site. Materials purchased directly by the Government are supplies, not construction material.

Cost of components means-

(1) For components purchased by the Contractor, the acquisition cost, including transportation costs to the place of incorporation into the construction material (whether or not such costs are paid to a domestic firm), and any applicable duty (whether or not a duty-free entry certificate is issued); or

(2) For components manufactured by the Contractor, all costs associated with the manufacture of the component, including transportation costs as described in paragraph (1) of this definition, plus allocable overhead costs, but excluding profit. Cost of components does not include any costs associated with the manufacture of the construction material.

Critical component means a component that is mined, produced, or manufactured in the United States and deemed critical to the U.S. supply chain. The list of critical components is at FAR 25.105.

Critical item means a domestic construction material or domestic end product that is deemed critical to U.S. supply chain resiliency. The list of critical items is at FAR 25.105.

Domestic construction material means-

(1) For construction material that does not consist wholly or predominantly of iron or steel or a combination of both-

(i) An unmanufactured construction material mined or produced in the United States; or

(ii) A construction material manufactured in the United States, if-

(A) The cost of its components mined, produced, or manufactured in the United States exceeds 60 percent of the cost of all its components, except that the percentage will be 65 percent for items delivered in calendar years 2024 through 2028 and 75 percent for items delivered starting in calendar year 2029. Components of foreign origin of the same class or kind for which nonavailability determinations have been made are treated as domestic. Components of unknown origin are treated as foreign; or

(B) The construction material is a COTS item; or

(2) For construction material that consists wholly or predominantly of iron or steel or a combination of both, a construction material manufactured in the United States if the cost of

foreign iron and steel constitutes less than 5 percent of the cost of all components used in such construction material. The cost of foreign iron and steel includes but is not limited to the cost of foreign iron or steel mill products (such as bar, billet, slab, wire, plate, or sheet), castings, or forgings utilized in the manufacture of the construction material and a good faith estimate of the cost of all foreign iron or steel components excluding COTS fasteners. Iron or steel components of unknown origin are treated as foreign. If the construction material contains multiple components, the cost of all the materials used in such construction material is calculated in accordance with the definition of "cost of components".

Fastener means a hardware device that mechanically joins or affixes two or more objects together. Examples of fasteners are nuts, bolts, pins, rivets, nails, clips, and screws.

Foreign construction material means a construction material other than a domestic construction material.

Foreign iron and steel means iron or steel products not produced in the United States. Produced in the United States means that all manufacturing processes of the iron or steel must take place in the United States, from the initial melting stage through the application of coatings, except metallurgical processes involving refinement of steel additives. The origin of the elements of the iron or steel is not relevant to the determination of whether it is domestic or foreign.

Predominantly of iron or steel or a combination of both means that the cost of the iron and steel content exceeds 50 percent of the total cost of all its components. The cost of iron and steel is the cost of the iron or steel mill products (such as bar, billet, slab, wire, plate, or sheet), castings, or forgings utilized in the manufacture of the product and a good faith estimate of the cost of iron or steel components excluding COTS fasteners.

Steel means an alloy that includes at least 50 percent iron, between 0.02 and 2 percent carbon, and may include other elements.

"United States" means the 50 States, the District of Columbia, and outlying areas.

(b) Domestic preference.

(1) This clause implements 41 U.S.C. chapter 83, Buy American, by providing a preference for domestic construction material. In accordance with 41 U.S.C. 1907, the domestic content test of the Buy American statute is waived for construction material that is a COTS item, except that for construction material that consists wholly or predominantly of iron or steel or a combination of both, the domestic content test is applied only to the iron and steel content of the construction materials, excluding COTS fasteners. The Contractor shall use only domestic construction material in performing this contract, except as provided in paragraphs

(b)(2) and (b)(3) of this clause.

(2) This requirement does not apply to information technology that is a commercial product or to the construction materials or components listed by the Government as follows:

None

(3) The Contracting Officer may add other foreign construction material to the list in paragraph (b)(2) of this clause if the Government determines that-

(i) The cost of domestic construction material would be unreasonable.

(A) For domestic construction material that is not a critical item or does not contain critical components.

(1) The cost of a particular domestic construction material subject to the requirements of the Buy American statute is unreasonable when the cost of such material exceeds the cost of foreign material by more than 20 percent;

(2) For construction material that is not a COTS item and does not consist wholly or predominantly of iron or steel or a combination of both, if the cost of a particular domestic construction material is determined to be unreasonable or there is no domestic offer received, and the low offer is for foreign construction material that is manufactured in the United States and does not exceed 55 percent domestic content, the Contracting Officer will treat the lowest offer of foreign construction material that exceeds 55 percent domestic content as a domestic offer and determine whether the cost of that offer is unreasonable by applying the evaluation factor listed in paragraph (b)(3)(i)(A)(1) of this clause.

(3) The procedures in paragraph (b)(3)(i)(A)(2) of this clause will no longer apply as of January 1, 2030.

(B) For domestic construction material that is a critical item or contains critical components.

(1) The cost of a particular domestic construction material that is a critical item or contains critical components, subject to the requirements of the Buy American statute, is unreasonable when the cost of such material exceeds the cost of foreign material by more than 20 percent plus the additional preference factor identified for the critical item or construction material containing critical components listed at FAR 25.105.

(2) For construction material that does not consist wholly or predominantly of iron or steel or a combination of both, if the cost of a particular domestic construction material is determined to be unreasonable or there is no domestic offer received, and the low offer is for foreign

construction material that does not exceed 55 percent domestic content, the Contracting Officer will treat the lowest foreign offer of construction material that is manufactured in the United States and exceeds 55 percent domestic content as a domestic offer, and determine whether the cost of that offer is unreasonable by applying the evaluation factor listed in paragraph (b)(3)(i)(B)(1) of this clause.

(3) The procedures in paragraph (b)(3)(i)(B)(2) of this clause will no longer apply as of January 1, 2030.

(ii) The application of the restriction of the Buy American statute to a particular construction material would be impracticable or inconsistent with the public interest; or

(iii) The construction material is not mined, produced, or manufactured in the United States in sufficient and reasonably available commercial quantities of a satisfactory quality.

(c) Request for determination of inapplicability of the Buy American statute.

(1)

(i) Any Contractor request to use foreign construction material in accordance with paragraph (b)(3) of this clause shall include adequate information for Government evaluation of the request, including-

(A) A description of the foreign and domestic construction materials;

(B) Unit of measure;

(C) Quantity;

(D) Price;

(E) Time of delivery or availability;

(F) Location of the construction project;

(G) Name and address of the proposed supplier; and

(H) A detailed justification of the reason for use of foreign construction materials cited in accordance with paragraph (b)(3) of this clause.

(ii) A request based on unreasonable cost shall include a reasonable survey of the market and a completed price comparison table in the format in paragraph (d) of this clause.

(iii) The price of construction material shall include all delivery costs to the construction site and any applicable duty (whether or not a duty-free certificate may be issued).

(iv) Any Contractor request for a determination submitted after contract award shall explain why the Contractor could not reasonably foresee the need for such determination and could not have requested the determination before contract award. If the Contractor does not submit a satisfactory explanation, the Contracting Officer need not make a determination.

(2) If the Government determines after contract award that an exception to the Buy American statute applies and the Contracting Officer and the Contractor negotiate adequate consideration, the Contracting Officer will modify the contract to allow use of the foreign construction material. However, when the basis for the exception is the unreasonable price of a domestic construction material, adequate consideration is not less than the differential established in paragraph (b)(3)(i) of this clause.

(3) Unless the Government determines that an exception to the Buy American statute applies, use of foreign construction material is noncompliant with the Buy American statute or Balance of Payments Program.

(d) Data. To permit evaluation of requests under paragraph (c) of this clause based on unreasonable cost, the Contractor shall include the following information and any applicable supporting data based on the survey of suppliers:

Foreign and Domestic Construction Materials Price Comparison
Construction material description Unit of measure
Quantity Price (dollars)*
Item 1:
Foreign construction material.
Domestic construction material.

Item 2:
Foreign construction material.
Domestic construction material.
[* Include all delivery costs to the construction site and any applicable duty (whether or not a duty-free entry certificate is issued)].[List name, address, telephone number, and contact for suppliers surveyed. Attach copy of response; if oral, attach summary.][Include other applicable supporting information.]
(End of clause)

52.229-11 Tax on Certain Foreign Procurements-Notice and Representation. 2020-06

Tax on Certain Foreign Procurements-Notice and Representation (Jun 2020)

(a) Definitions. As used in this provision-

Foreign person means any person other than a United States person.

Specified Federal procurement payment means any payment made pursuant to a contract with a foreign contracting party that is for goods, manufactured or produced, or services provided in a foreign country that is not a party to an international procurement agreement with the United States. For purposes of the prior sentence, a foreign country does not include an outlying area.

United States person as defined in 26 U.S.C. 7701(a)(30) means

(1) A citizen or resident of the United States;

(2) A domestic partnership;

(3) A domestic corporation;

(4) Any estate (other than a foreign estate, within the meaning of 26 U.S.C. 701(a)(31)); and

(5) Any trust if-

(i) A court within the United States is able to exercise primary supervision over the administration of the trust; and

(ii) One or more United States persons have the authority to control all substantial decisions of the trust.

(b) Unless exempted, there is a 2 percent tax of the amount of a specified Federal procurement payment on any foreign person receiving such payment. See 26 U.S.C. 5000C and its implementing regulations at 26 CFR 1.5000C-1 through 1.5000C-7.

(c) Exemptions from withholding under this provision are described at 26 CFR 1.5000C-1(d)(5) through (7). The Offerer would claim an exemption from the withholding by using the

Department of the Treasury Internal Revenue Service Form W-14, Certificate of Foreign Contracting Party Receiving Federal Procurement Payments, available via the internet at www.irs.gov/w14. Any exemption claimed and self-certified on the IRS Form W-14 is subject to audit by the IRS. Any disputes regarding the imposition and collection of the 26 U.S.C. S000C tax are adjudicated by the IRS as the 26 U.S.C. S000C tax is a tax matter, not a contract issue. The IRS Form W-14 is provided to the acquiring agency rather than to the IRS.

(d) For purposes of withholding under 26 U.S.C. S000C, the Offerer represents that

(1) It [] is [] is not a foreign person; and

(2) If the Offerer indicates "is" in paragraph (d)(1) of this provision, then the Offerer represents that-I am claiming on the IRS Form W-14 [] a full exemption, or [] partial or no exemption [Offerer shall select one] from the excise tax.

(e) If the Offerer represents it is a foreign person in paragraph (d)(1) of this provision, then-

(1) The clause at FAR 52.229-12, Tax on Certain Foreign Procurements, will be included in any resulting contract; and

(2) The Offerer shall submit with its offer the IRS Form W-14. If the IRS Form W-14 is not submitted with the offer, exemptions will not be applied to any resulting contract and the Government will withhold a full 2 percent of each payment.

(f) If the Offerer selects "is" in paragraph (d)(1) and "partial or no exemption" in paragraph (d)(2) of this provision, the Offerer will be subject to withholding in accordance with the clause at FAR 52.229-12, Tax on Certain Foreign Procurements, in any resulting contract.

(g) A taxpayer may, for a fee, seek advice from the Internal Revenue Service (IRS) as to the proper tax treatment of a transaction. This is called a private letter ruling. Also, the IRS may publish a revenue ruling, which is an official interpretation by the IRS of the Internal Revenue Code, related statutes, tax treaties, and regulations. A revenue ruling is the conclusion of the IRS on how the law is applied to a specific set of facts. For questions relating to the interpretation of the IRS regulations go to https://www.irs.gov/help/tax-law-questions.

(End of provision)

52.232-5 Payments under Fixed-Price 2014-05

Construction Contracts.

Payments under Fixed-Price Construction Contracts (May 2014)

(a) Payment of price. The Government shall pay the Contractor the contract price as provided in this contract.

(b) Progress payments. The Government shall make progress payments monthly as the work proceeds, or at more frequent intervals as determined by the Contracting Officer, on estimates of work accomplished which meets the standards of quality established under the contract, as approved by the Contracting Officer.

(1) The Contractor's request for progress payments shall include the following substantiation:

(i) An itemization of the amounts requested, related to the various elements of work required by the contract covered by the payment requested.

(ii) A listing of the amount included for work performed by each subcontractor under the contract.

(iii) A listing of the total amount of each subcontract under the contract.

(iv) A listing of the amounts previously paid to each such subcontractor under the contract.

(v) Additional supporting data in a form and detail required by the Contracting Officer.

(2) In the preparation of estimates, the Contracting Officer may authorize material delivered on the site and preparatory work done to be taken into consideration. Material delivered to the Contractor at locations other than the site also may be taken into consideration if-

(i) Consideration is specifically authorized by this contract; and

(ii) The Contractor furnishes satisfactory evidence that it has acquired title to such material and that the material will be used to perform this contract.

(c) Contractor certification. Along with each request for progress payments, the Contractor shall furnish the following certification, or payment shall not be made: (However, if the Contractor elects to delete paragraph (c)(4) from the certification, the certification is still acceptable.)

I hereby certify, to the best of my knowledge and belief, that-

(1) The amounts requested are only for performance in accordance with the specifications, terms, and conditions of the contract;

(2) All payments due to subcontractors and suppliers from previous payments received under the contract have been made, and timely payments will be made from the proceeds of the payment covered by this certification, in accordance with subcontract agreements and the requirements of Chapter 39 of Title 31, United States Code;

(3) This request for progress payments does not include any amounts which the prime contractor intends to withhold or retain from a subcontractor or supplier in accordance with the terms and conditions of the subcontract; and

(4) This certification is not to be construed as final acceptance of a subcontractor's performance.

__ (Name)

__ (Title)

__ (Date)

(d) Refund of unearned amounts. If the Contractor, after making a certified request for progress payments, discovers that a portion or all of such request constitutes a payment for performance by the Contractor that fails to conform to the specifications, terms, and conditions of this contract (hereinafter referred to as the "unearned amount"), the Contractor shall-

(1) Notify the Contracting Officer of such performance deficiency; and

(2) Be obligated to pay the Government an amount (computed by the Contracting Officer in the manner provided in paragraph U) of this clause) equal to interest on the unearned amount from the 8 thday after the date of receipt of the unearned amount until-

(i) The date the Contractor notifies the Contracting Officer that the performance deficiency has been corrected; or

(ii) The date the Contractor reduces the amount of any subsequent certified request for progress payments by an amount equal to the unearned amount.

(e) Retainage. If the Contracting Officer finds that satisfactory progress was achieved during any period for which a progress payment is to be made, the Contracting Officer shall authorize payment to be made in full. However, if satisfactory progress has not been made, the Contracting Officer may retain a maximum of 10 percent of the amount of the payment until satisfactory progress is achieved. When the work is substantially complete, the Contracting Officer may retain from previously withheld funds and future progress payments that amount the Contracting Officer considers adequate for protection of the Government and shall release to the Contractor all the remaining withheld funds. Also, on completion and acceptance of each separate building, public work, or other division of the contract, for which the price is stated separately in the contract, payment shall be made for the completed work without retention of a percentage.

(f) Title, liability, and reservation of rights. All material and work covered by progress payments made shall, at the time of payment, become the sole property of the Government, but this shall not be construed as-

(1) Relieving the Contractor from the sole responsibility for all material and work upon which payments have been made or the restoration of any damaged work; or

(2) Waiving the right of the Government to require the fulfillment of all of the terms of the contract.

(g) Reimbursement for bond premiums. In making these progress payments, the Government shall, upon request, reimburse the Contractor for the amount of premiums paid for performance and payment bonds (including coinsurance and reinsurance agreements, when applicable) after the Contractor has furnished evidence of full payment to the surety. The retainage provisions in paragraph (e) of this clause shall not apply to that portion of progress payments attributable to bond premiums.

(h) Final payment. The Government shall pay the amount due the Contractor under this contract after-

(1) Completion and acceptance of all work;

(2) Presentation of a properly executed voucher; and

(3) Presentation of release of all claims against the Government arising by virtue of this contract, other than claims, in stated amounts, that the Contractor has specifically excepted from the operation of the release. A release may also be required of the assignee if the Contractor's claim to amounts payable under this contract has been assigned under the Assignment of Claims Act of1940 (31 U.S.C.3727 and 41 U.S.C.6305).

(i) Limitation because of undefinitized work. Notwithstanding any provision of this contract, progress payments shall not exceed 80 percent on work accomplished on undefinitized contract actions. A "contract action" is any action resulting in a contract, as defined in FAR subpart 2.1, including contract modifications for additional supplies or services, but not including contract modifications that are within the scope and under the terms of the contract, such as contract modifications issued pursuant to the Changes clause, or funding and other administrative changes.

U) Interest computation on unearned amounts. In accordance with 31 U.S.C.3903(c)(1), the amount payable under paragraph (d)(2) of this clause shall be-

(1) Computed at the rate of average bond equivalent rates of 91-day Treasury bills auctioned at the most recent auction of such bills prior to the date the Contractor receives the unearned amount; and

(2) Deducted from the next available payment to the Contractor.

(End of clause)

52.232-12 Advance Payments. (Deviation) 2026-02

Advance Payments (Feb 2026) (Deviation)

(a) Requirements for payment. Advance payments will be made under this contract (1) upon submission of properly certified invoices or vouchers by the Contractor, and approval by the administering office, ___ [Insert the name of the office designated under agency procedures], or (2) under a letter of credit. The amount of the invoice or voucher submitted plus all advance payments previously approved shall not exceed $____ If a letter of credit is used, the Contractor shall withdraw cash only when needed for disbursements acceptable under this contract and report cash disbursements and balances as required by the administering office. The Contractor shall apply terms similar to this clause to any advance payments to subcontractors.

(b) Special account. Until (1) the Contractor has liquidated all advance payments made under the contract and related interest charges and (2) the administering office has approved in writing the release of any funds due and payable to the Contractor, all advance payments and other payments under this contract shall be made by check payable to the

Contractor marked for deposit only in the Contractor's special account with the ____ [insert the name of the financial institution]. None of the funds in the special account shall be mingled with other funds of the Contractor. Withdrawals from the special account may be made only by check of the Contractor countersigned by the Contracting Officer or a Government countersigning agent designated in writing by the Contracting Officer.

(c) Use of funds. The Contractor may withdraw funds from the special account only to pay for properly allocable, allowable, and reasonable costs for direct materials, direct labor, and indirect costs. Other withdrawals require approval in writing by the administering office. Determinations of whether costs are properly allocable, allowable, and reasonable shall be in accordance with generally accepted accounting principles, subject to any applicable subparts of part 31 of the Federal Acquisition Regulation.

(d) Repayment to the Government. At any time, the Contractor may repay all or any part of the funds advanced by the Government. Whenever requested in writing to do so by the administering office, the Contractor shall repay to the Government any part of unliquidated advance payments considered by the administering office to exceed the Contractor's current requirements or the amount specified in paragraph (a) above. If the Contractor fails to repay the amount requested by the administering office, all or any part of the unliquidated advance payments may be withdrawn from the special account by check signed by only the countersigning agent and applied to reduction of the unliquidated advance payments under this contract.

(e) Maximum payment. When the sum of all unliquidated advance payments, unpaid interest charges, and other payments exceed ___ percent of the contract price, the Government shall withhold further payments to the Contractor. On completion or termination of the contract, the Government shall deduct from the amount due to the Contractor all unliquidated advance payments and all interest charges payable. If previous payments to the Contractor exceed the amount due, the excess amount shall be paid to the Government on demand. For purposes of this paragraph, the contract price shall be considered to be the stated contract price of $___ , less any subsequent price reductions under the contract, plus (1) any price increases resulting from any terms of this contract for price redetermination or escalation, and (2) any other price increases that do not, in the aggregate, exceed $____ [Insert an amount not higher than 10 percent of the stated contract amount inserted in this paragraph]. Any payments withheld under this paragraph shall be applied to reduce the unliquidated advance payments. If full liquidation has been made, payments under the contract shall resume.

(f) Interest.

(1) The Contractor shall pay interest to the Government on the daily unliquidated advance payments at the daily rate specified in subparagraph (f)(3) of this clause. Interest shall be

computed at the end of each calendar month for the actual number of days involved. For the purpose of computing the interest charge-

(i) Advance payments shall be considered as increasing the unliquidated balance as of the date of the advance payment check;

(ii) Repayments by Contractor check shall be considered as decreasing the unliquidated balance as of the date on which the check is received by the Government authority designated by the Contracting Officer; and

(iii) Liquidations by deductions from Government payments to the Contractor shall be considered as decreasing the unliquidated balance as of the date of the check for the reduced payment.

(2) Interest charges resulting from the monthly computation shall be deducted from payments, other than advance payments, due the Contractor. If the accrued interest exceeds the payment due, any excess interest shall be carried forward and deducted from subsequent payments. Interest carried forward shall not be compounded. Interest on advance payments shall cease to accrue upon satisfactory completion or termination of the contract for the convenience of the Government. The Contractor shall charge interest on advance payments to subcontractors in the manner described above and credit the interest to the Government. Interest need not be charged on advance payments to nonprofit educational or research subcontractors for experimental, developmental, or research work.

(3) If interest is required under the contract, the Contracting Officer shall determine a daily interest rate based on the higher of

(i) the published prime rate of the financial institution (depository) in which the special account is established or

(ii) the rate established by the Secretary of the Treasury under Pub. L. 92-41 (41 U.S.C. 7109 (b)). The Contracting Officer shall revise the daily interest rate during the contract period in keeping with any changes in the cited interest rates.

(4) If the full amount of interest charged under this paragraph has not been paid by deduction or otherwise upon completion or termination of this contract, the Contractor shall pay the remaining interest to the Government on demand.

(g) Financial institution agreement. Before an advance payment is made under this contract, the Contractor shall transmit to the administering office, in the form prescribed by the administering office, an agreement in triplicate from the financial institution in which the special account is established, clearly setting forth the special character of the account and the responsibilities of the financial institution under the account. The Contractor shall select a financial institution that is a member bank of the Federal Reserve System, an "insured" bank within the meaning of the Federal Deposit Insurance Corporation Act (12 U.S.C. 1811), or a credit union insured by the National Credit Union Administration.

(h) Lien on Special Bank Account. The Government shall have a lien upon any balance in the special account paramount to all other liens. The Government lien shall secure the repayment of any advance payments made under this contract and any related interest charges.

(i) Lien on property under contract.

(1) All advance payments under this contract, together with interest charges, shall be secured, when made, by a lien in favor of the Government, paramount to all other liens, on the supplies or other things covered by this contract and on all material and other property acquired for or allocated to the performance of this contract, except to the extent that the Government by virtue of any other terms of this contract, or otherwise, shall have valid title to the supplies, materials, or other property as against other creditors of the Contractor.

(2) The Contractor shall identify, by marking or segregation, all property that is subject to a lien in favor of the Government by virtue of any terms of this contract in such a way as to indicate that it is subject to a lien and that it has been acquired for or allocated to performing this contract. If, for any reason, the supplies, materials, or other property are not identified by marking or segregation, the Government shall be considered to have a lien to the extent of the Government's interest under this contract on any mass of property with which the supplies, materials, or other property are commingled. The Contractor shall maintain adequate accounting control over the property on its books and records.

(3) If, at any time during the progress of the work on the contract, it becomes necessary to deliver to a third person any items or materials on which the Government has a lien, the Contractor shall notify the third person of the lien and shall obtain from the third person a receipt in duplicate acknowledging the existence of the lien. The Contractor shall provide a copy of each receipt to the Contracting Officer.

(4) If, under the termination clause, the Contracting Officer authorizes the Contractor to sell or retain termination inventory, the approval shall constitute a release of the Government's lien to the extent that-

(i) The termination inventory is sold or retained; and

(ii) The sale proceeds or retention credits are applied to reduce any outstanding advance payments.

U) Insurance.

(1) The Contractor shall maintain with responsible insurance carriers-

(i) Insurance on plant and equipment against fire and other hazards, to the extent that similar properties are usually insured by others operating plants and properties of similar character in the same general locality;

(ii) Adequate insurance against liability on account of damage to persons or property; and

(iii) Adequate insurance under all applicable workers' compensation laws.

(2) Until work under this contract has been completed and all advance payments made under the contract have been liquidated, the Contractor shall-

(i) Maintain this insurance;

(ii) Maintain adequate insurance on any materials, parts, assemblies, subassemblies, supplies, equipment, and other property acquired for or allocable to this contract and subject to the Government lien under paragraph (i) of this clause; and

(iii) Furnish any evidence with respect to its insurance that the administering office may require.

(k) Default.

(1) If any of the following events occurs, the Government may, by written notice to the Contractor, withhold further withdrawals from the special account and further payments on this contract:

(i) Termination of this contract for a fault of the Contractor.

(ii) A finding by the administering office that the Contractor has failed to-

(A) Observe any of the conditions of the advance payment terms;

(B) Comply with any material term of this contract;

(C) Make progress or maintain a financial condition adequate for performance of this contract;

(D) Limit inventory allocated to this contract to reasonable requirements; or

(E) Avoid delinquency in payment of taxes or of the costs of performing this contract in the ordinary course of business.

(iii) The appointment of a trustee, receiver, or liquidator for all or a substantial part of the Contractor's property, or the institution of proceedings by or against the Contractor for bankruptcy, reorganization, arrangement, or liquidation.

(iv) The service of any writ of attachment, levy of execution, or commencement of garnishment proceedings concerning the special account.

(v) The commission of an act of bankruptcy.

(2) If any of the events described in subparagraph (k)(1) of this clause continue for 30 days after the written notice to the Contractor, the Government may take any of the following additional actions:

(i) Withdraw by checks payable to the Treasurer of the United States, signed only by the countersigning agency, all or any part of the balance in the special account and apply the amounts to reduce outstanding advance payments and any other claims of the Government against the Contractor.

(ii) Charge interest, in the manner prescribed in paragraph (f) of this clause, on outstanding advance payments during the period of any event described in subparagraph (k)(1) of this clause.

(iii) Demand immediate repayment by the Contractor of the unliquidated balance of advance payments.

(iv) Take possession of and, with or without advertisement, sell at public or private sale all or any part of the property on which the Government has a lien under this contract and, after deducting any expenses incident to the sale, apply the net proceeds of the sale to reduce the unliquidated balance of advance payments or other Government claims against the Contractor.

(3) The Government may take any of the actions described in subparagraphs (k) (1) and (2) of this clause it considers appropriate at its discretion and without limiting any other rights of the Government.

(l) Prohibition against assignment. Notwithstanding any other terms of this contract, the Contractor shall not assign this contract, any interest therein, or any claim under the contract to any party.

(m) Information and access to records. The Contractor shall furnish to the administering office (1) monthly or at other intervals as required, signed or certified balance sheets and profit and loss statements together with a report on the operation of the special account in the form prescribed by the administering office; and (2) if requested, other information concerning the operation of the Contractor's business. The Contractor shall provide the authorized Government representatives proper facilities for inspection of the Contractor's books, records, and accounts.

(n) Other security. The terms of this contract are considered to provide adequate security to the Government for advance payments; however, if the administering office considers the security inadequate, the Contractor shall furnish additional security satisfactory to the administering office, to the extent that the security is available.

(o) Representations. The Contractor represents the following:

(1) The balance sheet, the profit and loss statement, and any other supporting financial statements furnished to the administering office fairly reflect the financial condition of the Contractor at the date shown or the period covered, and there has been no subsequent materially adverse change in the financial condition of the Contractor.

(2) No litigation or proceedings are presently pending or threatened against the Contractor, except as shown in the financial statements.

(3) The Contractor has disclosed all contingent liabilities, except for liability resulting from the renegotiation of defense production contracts, in the financial statements furnished to the administering office.

(4) None of the terms in this clause conflict with the authority under which the Contractor is doing business or with the provision of any existing indenture or agreement of the Contractor.

(5) The Contractor has the power to enter into this contract and accept advance payments, and has taken all necessary action to authorize the acceptance under the terms of this contract.

(6) The assets of the Contractor are not subject to any lien or encumbrance of any character except for current taxes not delinquent, and except as shown in the financial statements furnished by the Contractor. There is no current assignment of claims under any contract affected by these advance payment provisions.

(7) All information furnished by the Contractor to the administering office in connection with

each request for advance payments is true and correct.

(8) These representations shall be continuing and shall be considered to have been repeated by the submission of each invoice for advance payments.

(p) Covenants. To the extent the Government considers it necessary while any advance payments made under this contract remain outstanding, the Contractor, without the prior written consent of the administering office, shall not-

(1) Mortgage, pledge, or otherwise encumber or allow to be encumbered, any of the assets of the Contractor now owned or subsequently acquired, or permit any preexisting mortgages, liens, or other encumbrances to remain on or attach to any assets of the Contractor which are allocated to performing this contract and with respect to which the Government has a lien under this contract;

(2) Sell, assign, transfer, or otherwise dispose of accounts receivable, notes, or claims for money due or to become due;

(3) Declare or pay any dividends, except dividends payable in stock of the corporation, or make any other distribution on account of any shares of its capital stock, or purchase, redeem, or otherwise acquire for value any of its stock, except as required by sinking fund or redemption arrangements reported to the administering office incident to the establishment of these advance payment provisions;

(4) Sell, convey, or lease all or a substantial part of its assets;

(5) Acquire for value the stock or other securities of any corporation, municipality, or governmental authority, except direct obligations of the United States;

(6) Make any advance or loan or incur any liability as guarantor, surety, or accommodation endorser for any party;

(7) Permit a writ of attachment or any similar process to be issued against its property without getting a release or bonding the property within 30 days after the entry of the writ of attachment or other process;

(8) Pay any remuneration in any form to its directors, officers, or key employees higher than rates provided in existing agreements of which notice has been given to the administering office; accrue excess remuneration without first obtaining an agreement subordinating it to all claims of the Government; or employ any person at a rate of compensation over $____ a year;

(9) Change substantially the management, ownership, or control of the corporation;

(10) Merge or consolidate with any other firm or corporation, change the type of business, or engage in any transaction outside the ordinary course of the Contractor's business as presently conducted;

(11) Deposit any of its funds except in a bank or trust company insured by the Federal Deposit Insurance Corporation or a credit union insured by the National Credit Union Administration;

(12) Create or incur indebtedness for advances, other than advances to be made under the terms of this contract, or for borrowings;

(13) Make or covenant for capital expenditures exceeding $___ in total;

(14) Permit its net current assets, computed in accordance with generally accepted accounting principles, to become less than $____ ; or

(15) Make any payments on account of the obligations listed below, except in the manner and to the extent provided in this _____________[Contracting officer to list the pertinent obligations.] .

(End of clause)

52.232-16 Progress Payments. (Deviation) 2026-02

Progress Payments (Feb 2026) (Deviation)

The Government will make progress payments to the Contractor when requested as work progresses, but not more frequently than monthly, in amounts of $2,500 or more approved by the Contracting Officer, under the following conditions:

(a) Computation of amounts.

(1) Unless the Contractor requests a smaller amount, the Government will compute each progress payment as 80 percent of the Contractor's total costs incurred under this contract whether or not actually paid, plus financing payments to subcontractors (see paragraph U) of

this clause), less the sum of all previous progress payments made by the Government under this contract. The Contracting Officer will consider cost of money that would be allowable under Federal Acquisition Regulation (FAR) 31.205-10 as an incurred cost for progress payment purposes.

(2) The amount of financing and other payments for supplies and services purchased directly for the contract are limited to the amounts that have been paid by cash, check, or other forms of payment, or that are determined due and will be paid to subcontractors-

(i) In accordance with the terms and conditions of a subcontract or invoice; and

(ii) Ordinarily within 30 days of the submission of the Contractor's payment request to the Government.

(3) The Government will exclude accrued costs of Contractor contributions under employee pension plans until actually paid unless-

(i) The Contractor's practice is to make contributions to the retirement fund quarterly or more frequently; and

(ii) The contribution does not remain unpaid 30 days after the end of the applicable quarter or shorter payment period (any contribution remaining unpaid shall be excluded from the Contractor's total costs for progress payments until paid).

(4) The Contractor shall not include the following in total costs for progress payment purposes in paragraph (a)(1) of this clause:

(i) Costs that are not reasonable, allocable to this contract, and consistent with sound and generally accepted accounting principles and practices.

(ii) Costs incurred by subcontractors or suppliers.

(iii) Costs ordinarily capitalized and subject to depreciation or amortization except for the properly depreciated or amortized portion of such costs.

(iv) Payments made or amounts payable to subcontractors or suppliers, except for-

(A) Completed work, including partial deliveries, to which the Contractor has acquired title; and

(B) Work under cost-reimbursement or time-and-material subcontracts to which the Contractor has acquired title.

(5) The amount of unliquidated progress payments may exceed neither (i) the progress payments made against incomplete work (including allowable unliquidated progress payments to subcontractors) nor (ii) the value, for progress payment purposes, of the incomplete work. Incomplete work shall be considered to be the supplies and services required by this contract, for which delivery and invoicing by the Contractor and acceptance by the Government are incomplete.

(6) The total amount of progress payments shall not exceed 80 percent of the total contract price.

(7) If a progress payment or the unliquidated progress payments exceed the amounts permitted by subparagraphs (a)(4) or (a)(5) of this clause, the Contractor shall repay the amount of such excess to the Government on demand.

(8) Notwithstanding any other terms of the contract, the Contractor agrees not to request progress payments in dollar amounts of less than $2,500. The Contracting Officer may make exceptions.

(9) The costs applicable to items delivered, invoiced, and accepted shall not include costs in excess of the contract price of the items.

(b) Liquidation. Except as provided in the Termination for Convenience of the Government clause, all progress payments shall be liquidated by deducting from any payment under this contract, other than advance or progress payments, the unliquidated progress payments, or 80 percent of the amount invoiced, whichever is less. The Contractor shall repay to the Government any amounts required by a retroactive price reduction, after computing liquidations and payments on past invoices at the reduced prices and adjusting the unliquidated progress payments accordingly. The Government reserves the right to unilaterally change from the ordinary liquidation rate to an alternate rate when deemed appropriate for proper contract financing.

(c) Reduction or suspension. The Contracting Officer may reduce or suspend progress payments, increase the rate of liquidation, or take a combination of these actions, after finding on substantial evidence any of the following conditions:

(1) The Contractor failed to comply with any material requirement of this contract (which includes paragraphs (f) and (g) of this clause).

(2) Performance of this contract is endangered by the Contractor's (i) failure to make progress or (ii) unsatisfactory financial condition.

(3) Inventory allocated to this contract substantially exceeds reasonable requirements.

(4) The Contractor is delinquent in payment of the costs of performing this contract in the ordinary course of business.

(5) The fair value of the undelivered work is less than the amount of unliquidated progress payments for that work.

(6) The Contractor is realizing less profit than that reflected in the establishment of any alternate liquidation rate in paragraph (b) of this clause, and that rate is less than the progress payment rate stated in subparagraph (a)(1) of this clause.

(d) Title.

(1) Title to the property described in this paragraph (d) shall vest in the Government. Vestiture shall be immediately upon the date of this contract, for property acquired or produced before that date. Otherwise, vestiture shall occur when the property is or should have been allocable or properly chargeable to this contract.

(2) Property, as used in this clause, includes all of the below-described items acquired or produced by the Contractor that are or should be allocable or properly chargeable to this contract under sound and generally accepted accounting principles and practices.

(i) Parts, materials, inventories, and work in process;

(ii) Special tooling and special test equipment to which the Government is to acquire title;

(iii) Nondurable (i.e., noncapital) tools, jigs, dies, fixtures, molds, patterns, taps, gauges, test equipment, and other similar manufacturing aids, title to which would not be obtained as special tooling under paragraph (d)(2)(ii) of this clause; and

(iv) Drawings and technical data, to the extent the Contractor or subcontractors are required to deliver them to the Government by other clauses of this contract.

(3) Although title to property is in the Government under this clause, other applicable clauses of this contract, e.g., the termination clauses, shall determine the handling and disposition of the property.

(4) The Contractor may sell any scrap resulting from production under this contract without requesting the Contracting Officer's approval, but the proceeds shall be credited against the costs of performance.

(5) To acquire for its own use or dispose of property to which title is vested in the Government under this clause, the Contractor must obtain the Contracting Officer's advance approval of the action and the terms. The Contractor shall (i) exclude the allocable costs of the property from the costs of contract performance, and (ii) repay to the Government any amount of unliquidated progress payments allocable to the property. Repayment may be by cash or credit memorandum.

(6) When the Contractor completes all of the obligations under this contract, including liquidation of all progress payments, title shall vest in the Contractor for all property (or the proceeds thereof) not-

(i) Delivered to, and accepted by, the Government under this contract; or

(ii) Incorporated in supplies delivered to, and accepted by, the Government under this contract and to which title is vested in the Government under this clause.

(7) The terms of this contract concerning liability for Government-furnished property shall not apply to property to which the Government acquired title solely under this clause.

(e) Risk of loss. Before delivery to and acceptance by the Government, the Contractor shall bear the risk of loss for property, the title to which vests in the Government under this clause, except to the extent the Government expressly assumes the risk. The Contractor shall repay the Government an amount equal to the unliquidated progress payments that are based on costs allocable to property that is lost (see 45.101).

(f) Control of costs and property. The Contractor shall maintain an accounting system and controls adequate for the proper administration of this clause.

(g) Reports, forms, and access to records.

(1) The Contractor shall promptly furnish reports, certificates, financial statements, and other pertinent information (including estimates to complete) reasonably requested by the Contracting Officer for the administration of this clause. Also, the Contractor shall give the Government reasonable opportunity to examine and verify the Contractor's books, records, and accounts.

(2) The Contractor shall furnish estimates to complete that have been developed or updated within six months of the date of the progress payment request. The estimates to complete shall represent the Contractor's best estimate of total costs to complete all remaining contract work required under the contract. The estimates shall include sufficient detail to permit Government verification.

(3) Each Contractor request for progress payment shall:

(i) Be submitted on Standard Form 1443, Contractor's Request for Progress Payment, or the electronic equivalent as required by agency regulations, in accordance with the form instructions and the contract terms; and

(ii) Include any additional supporting documentation requested by the Contracting Officer.

(h) Special terms regarding default. If this contract is terminated under the Default clause, (i) the Contractor shall, on demand, repay to the Government the amount of unliquidated progress payments and (ii) title shall vest in the Contractor, on full liquidation of progress payments, for all property for which the Government elects not to require delivery under the Default clause. The Government shall be liable for no payment except as provided by the Default clause.

(i) Reservations of rights.

(1) No payment or vesting of title under this clause shall (i) excuse the Contractor from performance of obligations under this contract or (ii) constitute a waiver of any of the rights or remedies of the parties under the contract.

(2) The Government's rights and remedies under this clause (i) shall not be exclusive but rather shall be in addition to any other rights and remedies provided by law or this contract and (ii) shall not be affected by delayed, partial, or omitted exercise of any right, remedy, power, or privilege, nor shall such exercise or any single exercise preclude or impair any further exercise under this clause or the exercise of any other right, power, or privilege of the Government.

U) Financing payments to subcontractors. The financing payments to subcontractors mentioned in paragraphs (a)(1) and (a)(2) of this clause shall be all financing payments to subcontractors or divisions, if the following conditions are met:

(1) The amounts included are limited to-

(i) The unliquidated remainder of financing payments made; plus

(ii) Any unpaid subcontractor requests for financing payments.

(2) The subcontract or interdivisional order is expected to involve a minimum of approximately 6 months between the beginning of work and the first delivery; or, if the subcontractor is a small business concern, 4 months.

(3) If the financing payments are in the form of progress payments, the terms of the subcontract or interdivisional order concerning progress payments-

(i) Are substantially similar to the terms of this clause for any subcontractor that is a large business concern, or this clause with its Alternate I for any subcontractor that is a small business concern;

(ii) Are at least as favorable to the Government as the terms of this clause;

(iii) Are not more favorable to the subcontractor or division than the terms of this clause are to the Contractor;

(iv) Are in conformance with the requirements of FAR 32.504(e); and

(v) Subordinate all subcontractor rights concerning property to which the Government has title under the subcontract to the Government's right to require delivery of the property to the Government if-

(A) The Contractor defaults; or

(B) The subcontractor becomes bankrupt or insolvent.

(4) If the financing payments are in the form of performance-based payments, the terms of the subcontract or interdivisional order concerning payments-

(i) Are substantially similar to the Performance-Based Payments clause at FAR 52.232-32 and *meet* the criteria for, and definition of, performance-based payments in FAR Part 32;

(ii) Are in conformance with the requirements of FAR 32.504(f); and

(iii) Subordinate all subcontractor rights concerning property to which the Government has title under the subcontract to the Government's right to require delivery of the property to the Government if-

(A) The Contractor defaults; or

(B) The subcontractor becomes bankrupt or insolvent.

(5) If the financing payments are in the form of commercial product or commercial service financing payments, the terms of the subcontract or interdivisional order concerning payments-

(i) Are constructed in accordance with FAR 32.206(c) and included in a subcontract for a commercial product or commercial service purchase that meets the definition and standards for acquisition of commercial products and commercial services in FAR parts 2 and 12;

(ii) Are in conformance with the requirements of FAR 32.504(9); and

(iii) Subordinate all subcontractor rights concerning property to which the Government has title under the subcontract to the Government's right to require delivery of the property to the Government if-

(A) The Contractor defaults; or

(B) The subcontractor becomes bankrupt or insolvent.

(6) If financing is in the form of progress payments, the progress payment rate in the subcontract is the customary rate used by the contracting agency, depending on whether the subcontractor is or is not a small business concern.

(7) Concerning any proceeds received by the Government for property to which title has vested in the Government under the subcontract terms, the parties agree that the proceeds shall be applied to reducing any unliquidated financing payments by the Government to the Contractor under this contract.

(8) If no unliquidated financing payments to the Contractor remain, but there are unliquidated financing payments that the Contractor has made to any subcontractor, the Contractor shall be subrogated to all the rights the Government obtained through the terms required by this clause to be in any subcontract, as if all such rights had been assigned and transferred to the Contractor.

(9) To facilitate small business participation in subcontracting under this contract, the Contractor shall provide financing payments to small business concerns, in conformity with the standards for customary contract financing payments stated in FAR 32.113. The Contractor shall not consider the need for such financing payments as a handicap or adverse factor in the award of subcontracts.

(k) Limitations on undefinitized contract actions. Notwithstanding any other progress payment provisions in this contract, progress payments may not exceed 80 percent of costs incurred on work accomplished under undefinitized contract actions. A contract action is any action resulting in a contract, as defined in subpart 2.1, including contract modifications for additional supplies or services, but not including contract modifications that are within the scope and under the terms of the contract, such as contract modifications issued pursuant to the Changes clause, or funding and other administrative changes. This limitation shall apply to the costs incurred, as computed in accordance with paragraph (a) of this clause, and shall remain in effect until the contract action is definitized. Costs incurred which are subject to this limitation shall be segregated on Contractor progress payment requests and invoices from those costs eligible for higher progress payment rates. For purposes of progress payment liquidation, as described in paragraph (b) of this clause, progress payments for undefinitized contract actions shall be liquidated at 80 percent of the amount invoiced for work performed under the undefinitized contract action as long as the contract action remains undefinitized. The amount of unliquidated progress payments for undefinitized contract actions shall not exceed 80 percent of the maximum liability of the Government under the undefinitized contract action or such lower limit specified elsewhere in the contract. Separate limits may be specified for separate actions.

(l) Due date. The designated payment office will make progress payments on the 30th [Contracting Officer insert date as prescribed by agency head; if not prescribed, insert "30th"] day after the designated billing office receives a proper progress payment request. In the event that the Government requires an audit or other review of a specific progress payment request to ensure compliance with the terms and conditions of the contract, the designated payment office is not compelled to make payment by the specified due date. Progress payments are considered contract financing and are not subject to the interest penalty provisions of the Prompt Payment Act.

(m) Progress payments under indefinite-delivery contracts. The Contractor shall account for and submit progress payment requests under individual orders as if the order constituted a separate contract, unless otherwise specified in this contract.

(End of clause)

52.232-32 Performance-Based Payments. 2012-04

Performance-Based Payments (Apr 2012)

(a) Amount of payments and limitations on payments. Subject to such other limitations and conditions as are specified in this contract and this clause, the amount of payments and limitations on payments shall be specified in the contract's description of the basis for payment.

(b) Contractor request for performance-based payment. The Contractor may submit requests for payment of performance-based payments not more frequently than monthly, in a form and manner acceptable to the Contracting Officer. Unless otherwise authorized by the Contracting Officer, all performance-based payments in any period for which payment is being requested shall be included in a single request, appropriately itemized and totaled. The Contractor's request shall contain the information and certification detailed in paragraphs (l) and (m) of this clause.

(c) Approval and payment of requests.

(1) The Contractor shall not be entitled to payment of a request for performance-based payment prior to successful accomplishment of the event or performance criterion for which payment is requested. The Contracting Officer shall determine whether the event or performance criterion for which payment is requested has been successfully accomplished in accordance with the terms of the contract. The Contracting Officer may, at any time, require the Contractor to substantiate the successful performance of any event or performance criterion which has been or is represented as being payable.

(2) A payment under this performance-based payment clause is a contract financing payment under the Prompt Payment clause of this contract and not subject to the interest penalty provisions of the Prompt Payment Act. The designated payment office will pay approved requests on the 14th[Contracting Officer insert day as prescribed by agency head; if not prescribed, insert "30th"] day after receipt of the request for performance-based payment by the designated payment office. However, the designated payment office is not required to provide payment if the Contracting Officer requires substantiation as provided in paragraph (c)(1) of this clause, or inquires into the status of an event or performance criterion, or into any of the conditions listed in paragraph (e) of this clause, or into the Contractor certification. The payment period will not begin until the Contracting Officer approves the request.

(3) The approval by the Contracting Officer of a request for performance-based payment does not constitute an acceptance by the Government and does not excuse the Contractor from performance of obligations under this contract.

(d) Liquidation of performance-based payments.

(1) Performance-based finance amounts paid prior to payment for delivery of an item shall be liquidated by deducting a percentage or a designated dollar amount from the delivery payment. If the performance-based finance payments are on a delivery item basis, the

liquidation amount for each such line item shall be the percent of that delivery item price that was previously paid under performance-based finance payments or the designated dollar amount. If the performance-based finance payments are on a whole contract basis, liquidation shall be by either predesignated liquidation amounts or a liquidation percentage.

(2) If at any time the amount of payments under this contract exceeds any limitation in this contract, the Contractor shall repay to the Government the excess. Unless otherwise determined by the Contracting Officer, such excess shall be credited as a reduction in the unliquidated performance-based payment balance(s), after adjustment of invoice payments and balances for any retroactive price adjustments.

(e) Reduction or suspension of performance-based payments. The Contracting Officer may reduce or suspend performance-based payments, liquidate performance-based payments by deduction from any payment under the contract, or take a combination of these actions after finding upon substantial evidence any of the following conditions:

(1) The Contractor failed to comply with any material requirement of this contract (which includes paragraphs (h) and (i) of this clause).

(2) Performance of this contract is endangered by the Contractor's-

(i) Failure to make progress; or

(ii) Unsatisfactory financial condition.

(3) The Contractor is delinquent in payment of any subcontractor or supplier under this contract in the ordinary course of business.

(f) Title.

(1) Titleto the property described in this paragraph (f) shall vest in the Government. Vestiture shall be immediately upon the date of the first performance-based payment under this contract, for property acquired or produced before that date. Otherwise, vestiture shall occur when the property is or should have been allocable or properly chargeable to this contract.

(2) "Property," as used in this clause, includes all of the following described items acquired or produced by the Contractor that are or should be allocable or properly chargeable to this contract under sound and generally accepted accounting principles and practices:

(i) Parts, materials, inventories, and work in process;

(ii) Special tooling and special test equipment to which the Government is to acquire title;

(iii) Nondurable (i.e., noncapital) tools, jigs, dies, fixtures, molds, patterns, taps, gauges, test equipment and other similar manufacturing aids, title to which would not be obtained as special tooling under paragraph (f)(2)(ii) of this clause; and

(iv) Drawings and technical data, to the extent the Contractor or subcontractors are required to deliver them to the Government by other clauses of this contract.

(3) Although title to property is in the Government under this clause, other applicable clauses of this contract (e.g., the termination clauses) shall determine the handling and disposition of the property.

(4) The Contractor may sell any scrap resulting from production under this contract, without requesting the Contracting Officer's approval, provided that any significant reduction in the value of the property to which the Government has title under this clause is reported in writing to the Contracting Officer.

(5) In order to acquire for its own use or dispose of property to which title is vested in the Government under this clause, the Contractor shall obtain the Contracting Officer's advance approval of the action and the terms. If approved, the basis for payment (the events or performance criteria) to which the property is related shall be deemed to be not in compliance with the terms of the contract and not payable (if the property is part of or needed for performance), and the Contractor shall refund the related performance-based payments in accordance with paragraph (d) of this clause.

(6) When the Contractor completes all of the obligations under this contract, including liquidation of all performance-based payments, title shall vest in the Contractor for all property (or the proceeds thereof) not-

(i) Delivered to, and accepted by, the Government under this contract; or

(ii) Incorporated in supplies delivered to, and accepted by, the Government under this contract and to which title is vested in the Government under this clause.

(7) The terms of this contract concerning liability for Government-furnished property shall not apply to property to which the Government acquired title solely under this clause.

(g) Risk of loss. Before delivery to and acceptance by the Government, the Contractor shall bear the risk of loss for property, the title to which vests in the Government under this clause, except to the extent the Government expressly assumes the risk. If any property is lost (see 45.101), the basis of payment (the events or performance criteria) to which the property is related shall be deemed to be not in compliance with the terms of the contract and not payable (if the property is part of or needed for performance), and the Contractor shall refund the related performance-based payments in accordance with paragraph (d) of this clause.

(h) Records and controls. The Contractor shall maintain records and controls adequate for administration of this clause. The Contractor shall have no entitlement to performance-based payments during any time the Contractor's records or controls are determined by the Contracting Officer to be inadequate for administration of this clause.

(i) Reports and Government access. The Contractor shall promptly furnish reports, certificates, financial statements, and other pertinent information requested by the Contracting Officer for the administration of this clause and to determine that an event or other criterion prompting a financing payment has been successfully accomplished. The Contractor shall give the Government reasonable opportunity to examine and verify the Contractor's records and to examine and verify the Contractor's performance of this contract for administration of this clause.

U) Special terms regarding default. If this contract is terminated under the Default clause, (1) the Contractor shall, on demand, repay to the Government the amount of unliquidated performance-based payments, and (2) title shall vest in the Contractor, on full liquidation of all performance-based payments, for all property for which the Government elects not to require delivery under the Default clause of this contract. The Government shall be liable for no payment except as provided by the Default clause.

(k) Reservation of rights.

(1) No payment or vesting of title under this clause shall-

(i) Excuse the Contractor from performance of obligations under this contract; or

(ii) Constitute a waiver of any of the rights or remedies of the parties under the contract.

(2) The Government's rights and remedies under this clause-

(i) Shall not be exclusive, but rather shall be in addition to any other rights and remedies provided by law or this contract; and

(ii) Shall not be affected by delayed, partial, or omitted exercise of any right, remedy, power, or privilege, nor shall such exercise or any single exercise preclude or impair any further exercise under this clause or the exercise of any other right, power, or privilege of the Government.

(l) Content of Contractor's request for performance-based payment. The Contractor's request for performance-based payment shall contain the following:

(1) The name and address of the Contractor;

(2) The date of the request for performance-based payment;

(3) The contract number and/or other identifier of the contract or order under which the request is made;

(4) Such information and documentation as is required by the contract's description of the basis for payment; and

(5) A certification by a Contractor official authorized to bind the Contractor, as specified in paragraph (m) of this clause.

(m) Content of Contractor's certification. As required in paragraph (1)(5) of this clause, the Contractor shall make the following certification in each request for performance-based payment:

I certify to the best of my knowledge and belief that-

(1) This request for performance-based payment is true and correct; this request (and attachments) has been prepared from the books and records of the Contractor, in accordance with the contract and the instructions of the Contracting Officer;

(2) (Except as reported in writing on __), all payments to subcontractors and suppliers under this contract have been paid, or will be paid, currently, when due in the ordinary course of business;

(3) There are no encumbrances (except as reported in writing on ___) against the property acquired or produced for, and allocated or properly chargeable to, the contract which would affect or impair the Government's title;

(4) There has been no materially adverse change in the financial condition of the Contractor since the submission by the Contractor to the Government of the most recent written information dated ___ ; and

(5) After the making of this requested performance-based payment, the amount of all payments for each deliverable item for which performance-based payments have been requested will not exceed any limitation in the contract, and the amount of all payments under the contract will not exceed any limitation in the contract.

(End of clause)

52.252-2 Clauses Incorporated by Reference. 1998-02

Clauses Incorporated By Reference (Feb 1998)

This contract incorporates one or more clauses by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. Also, the full text of a clause may be accessed electronically at this/these address (es):

www.acquisition.gov

(End of clause)

52.252-4 Alterations in Contract. 1984-04

Alterations in Contract (Apr 1984)

Portions of this contract are altered as follows:

None

(End of clause)

52.252-6 Authorized Deviations in Clauses. 2020-11

Authorized Deviations in Clauses (Nov 2020)

(a) The use in this solicitation or contract of any Federal Acquisition Regulation (48 CFR Chapter 1) clause with an authorized deviation is indicated by the addition of "(DEVIATION)" after the date of the clause.

(b) The use in this solicitation or contract of any Defense Federal Acquisition Regulation Supplement (DFARS)(48 CFR 2) clause with an authorized deviation is indicated by the addition of "(DEVIATION)" after the name of the regulation.

(End of clause)

252.232-7013	Performance-Based Payments-Deliverable-Item Basis.	2022-12

PERFORMANCE-BASED PAYMENTS-DELIVERABLE-ITEM BASIS (DEC 2022)

(a) Performance-based payments shall form the basis for the contract financing payments provided under this contract and shall apply to Contract Line Item Numbers (CLIN(s)) 0001 through 3001. The performance-based payments schedule (Contract Attachment ___) describes the basis for payment, to include identification of the individual payment events, CLINs to which each event applies, evidence of completion, and amount of payment due upon completion of each event.

(b) In accordance with 10 U.S.C. 3802(c), the Contractor's financial statements shall be in compliance with Generally Accepted Accounting Principles in order to receive performance-based payments.

(c)(1) The Contractor shall, in addition to providing the information required by FAR 52.232-32, submit information for all payment requests using the following format:

Current performance-based payment(s) event(s) addressed by this request:
Contractor shall identify- Amount Totals
(1a) Negotiated value of all previously completed performance-based payment(s) event(s);
(1b) Negotiated value of the current performance-based payment(s) event (s);
(1c) Cumulative negotiated value of performance-based payment(s) event(s) completed to date (1a) + (1b); and
(2) Total costs incurred to date.

(2) Incurred cost is determined by the Contractor's accounting books and records, to which the Contractor shall provide access upon request of the Contracting Officer. An acceptable accounting system in accordance with DFARS 252.242-7006 is not required for reporting of incurred costs under this clause. If the Contractor's accounting system is not capable of tracking costs on a job order basis, the Contractor shall provide a realistic approximation of the allocation of incurred costs attributable to this contract in accordance with the Contractor's accounting system. FAR 52.232-32(m) does not require certification of incurred costs.

(d) Security for financing.

(1) Title to the property described in paragraph (f) of the clause at FAR 52.232-32, Performance-Based Payments, is the preferred security for receipt of performance-based payments.

(2)(i) If the Contractor's accounting system is not capable of identifying and tracking through the build cycle the property that is allocable and properly chargeable to this contract, the Contracting Officer may consider acceptance of one or a combination of the following alternative forms of security sufficient to constitute adequate security for the performance-based payments and so specify in the contract, consistent with FAR 32.202-4:

(A) A paramount lien on assets.

(B) An irrevocable letter of credit from a federally insured financial institution.

(C) A bond from a surety, acceptable in accordance with FAR part 28.

(D) A guarantee of repayment from a person or corporation of demonstrated liquid net worth, connected by significant ownership interest to the Contractor.

(E) Title to identified Contractor assets of adequate worth.

(ii) Paragraph (f) of the clause at FAR 52.232-32 does not apply to the extent that the Contractor and the Contracting Officer agree on alternative forms of security. In the event the Contractor fails to provide adequate security, as required in this contract, no financing payment will be made under this contract. Upon receipt of adequate security, financing payments will be made, including all previous payments to which the Contractor is entitled, in accordance with the terms of the provisions for contract financing. If at any time the Contracting Officer determines that the security provided by the Contractor is insufficient, the Contractor shall promptly provide such additional security as the Contracting Officer determines necessary. In the event the Contractor fails to provide such additional security, the Contracting Officer may collect or liquidate such security that has been provided and suspend further payments to the Contractor; and the Contractor shall repay to the Government the amount of unliquidated financing payments as the Contracting Officer at his sole discretion deems repayable.

(End of clause)

252.236-7001	Contract Drawings and Specifications.	2000-08

CONTRACT DRAWINGS AND SPECIFICATIONS (AUG 2000)

(a) The Government will provide to the Contractor, without charge, one set of contract drawings and specifications, except publications incorporated into the technical provisions by reference, in electronic or paper media as chosen by the Contracting Officer.

(b) The Contractor shall-

(1) Check all drawings furnished immediately upon receipt;

(2) Compare all drawings and verify the figures before laying out the work;

(3) Promptly notify the Contracting Officer of any discrepancies;

(4) Be responsible for any errors that might have been avoided by complying with this paragraph (b); and

(5) Reproduce and print contract drawings and specifications as needed.

(c) In general--

(1) Large-scale drawings shall govern small-scale drawings; and

(2) The Contractor shall follow figures marked on drawings in preference to scale measurements.

(d) Omissions from the drawings or specifications or the misdescription of details of work that are manifestly necessary to carry out the intent of the drawings and specifications, or that are customarily performed, shall not relieve the Contractor from performing such omitted or misdescribed details of the work. The Contractor shall perform such details as if fully and correctly set forth and described in the drawings and specifications.

(e) The work shall conform to the specifications and the contract drawings identified on the following index of drawings:

Title	File	Drawing No.

(End of clause)

Commencement, Prosecution, and Completion of Work - Construction

The Contractor shall be required to (a) commence work under this contract within 10 calendar days after the date the Contractor receives the notice to proceed, (b) prosecute the work diligently, and (c) complete the entire work ready for use not later than 730 calendar days after receipt of notice to proceed. The time stated for completion shall include final cleanup of the premises.

(End of condition)

Variation in Estimated Quantity

If the quantity of a unit-priced item in this contract is an estimated quantity and the actual quantity of the unit-priced item varies more than 15 percent above or below the estimated quantity, an equitable adjustment in the contract price shall be made upon demand of either party. The equitable adjustment shall be based upon any increase or decrease in costs due solely to the variation above 115 percent or below 85 percent of the estimated quantity. If the quantity variation is such as to cause an increase in the time necessary for completion, the Contractor may request, in writing, an extension of time, to be received by the Contracting Officer within 10 days from the beginning of the delay, or within such further period as may be granted by the Contracting Officer before the date of final settlement of the contract. Upon the receipt of a written request for an extension, the Contracting Officer shall ascertain the facts and make an adjustment for extending the completion date as, in the judgement of the Contracting Officer, is justified.

(End of condition)

Site Visit (Construction)

(b) An organized site visit has been scheduled for-

_____________________*TBD*___________________________________

(c) Participants will meet at-

_______________*TBD*____

PROPOSAL ORDER OF PRECEDENCE

(a) The contract includes the standard contract clauses and schedules included in the Request for Proposal. It entails

(1) the solicitation in its entirety, including all drawings, cuts, and illustrations, and any amendments, and (2) the successful offeror's accepted proposal. The contract constitutes and defines the entire agreement between the Contractor and the Government. No documentation shall be omitted which in any way bears upon the terms of that agreement.

(b) In the event of conflict or inconsistency between any of the provisions of this contract, precedence shall be given in the following order:

(1) Betterments: Any portions of the accepted proposal which both conform to and exceed the provisions of the solicitation.

(2) The provisions of the solicitation. (See also Contract Clause: SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION.)

(3) All other provisions of the accepted proposal.

(4) Any design products including, but not limited to, plans, specifications, engineering studies and analyses, shop drawings, equipment installation drawings, etc. These are "deliverables" under the contract and are not part of the contract itself. Design products must conform with all provisions of the contract, in the order of precedence herein.

(End of Condition)

TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER

1. This provision specifies the procedure for determination of time extensions for unusually severe weather in accordance with the contract clause entitled "Default: (Fixed Price Construction)". In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:

a. The weather experienced at the project site during the contract period must be found to be unusually severe, that is, more severe than the adverse weather anticipated for the project location during any given month.

b. The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the contractor.

2. The following schedule of monthly anticipated adverse weather delays is based on National Oceanic and Atmospheric Administration (NOAA) or similar data for the project location and will constitute the base line for monthly weather time evaluations. The contractor's progress schedule must reflect these anticipated adverse weather delays in all-weather dependent activities.

MONTHLY ANTICIPATED ADVERSE WEATHER DELAY WORKDAYS BASED ON (5) DAY WORKWEEK

JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
3	3	4	4	5	4	3	3	4	3	2	2

3. Upon acknowledgment of the Notice to Proceed (NTP) and continuing throughout the contract, the contractor will record on the daily CQC report, the occurrence of adverse weather and resultant impact to normally scheduled work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the contractor's scheduled workday. The number of actual adverse weather delay days shall include days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated in paragraph 2, above, the contracting officer will convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the contract clause entitled "Default (Fixed Price Construction)".

(End of Condition)

ANTITERRORISM/OPERATIONS SECURITY REQUIREMENTS, MAY 2022

X	**1. AT Level I training.** This standard language is for contractor employees with an area of performance within an Army-controlled installation, facility, or area. All contractor employees, including subcontractor employees, requiring access to Army installations, facilities, and controlled access areas shall complete AT Level I awareness training within XX calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee to the COR or to the contracting officer, if a COR is not assigned, within XX calendar days after completion of training by all employees and subcontractor personnel. AT Level I awareness training is available at the following website: http://jko.jten.mil.
X	**2. Access and general protection/security policy and procedures.** This standard language is for contractor employees with an area of performance within an Army-controlled installation, facility, or area. Contractor and all associated subcontractor employees shall provide all information required for background checks to meet installation access requirements to be accomplished by the installation Provost Marshal Office, Director of Emergency Services, or Security Office. Contractor workforce must comply with all personal identity verification requirements (CFR clause 52.204-9, Personal Identity Verification of Contract Personnel) as directed by DoD, HODA and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.
X	**2a. For contractors requiring CAC.** Before CAC issuance, the contractor employee requires, at a minimum, a favorably adjudicated National Agency Check with Inquiries (NACI) or an equivalent or higher investigation in accordance with Army Directive 2014-05. The contractor employee will be issued a CAC only if duties involve one of the following: (1) both physical access to a DoD facility and access, via logon, to DoD networks on-site or remotely; (2) remote access, via logon, to a DoD network using DoD-approved remote access procedures; or (3) physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. At the discretion of the sponsoring activity, an initial CAC may be issued based on a favorable review of the FBI fingerprint check and a successfully scheduled NACI at the Office of Personnel Management.
X	**2b. For contractors that do not require a CAC, but require access to a DoD facility or installation.** Contractor and all associated subcontractor employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (Army Directive 2014-05/AR 190-13); applicable installation, facility and area commander installation and facility access and local security policies and procedures (provided by Government representative); or, at OCONUS locations, in accordance with status-of-forces agreements and other theater regulations.

	3. AT Awareness Training for Contractor Personnel Traveling Overseas. This standard language requires U.S.-based contractor employees and associated subcontractor employees to make available and to receive Government-provided area of responsibility (AOR)-specific AT awareness training as directed by AR 525-13. Specific AOR training content is directed by the combatant commander, with the unit ATO being the local point of contact.
X	**4. Locally Developed iWATCH Army Training.** This standard language is for contractor employees with an area of performance within an Army-controlled installation, facility, or area. The contractor and all associated subcontractors shall brief all employees on the local iWATCH Army program (training standards provided by the requiring activity ATO). This locally developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within XX calendar days of contract award and within YY calendar days of new employees commencing performance, with the results reported to the COR NLT XX calendar days after contract award.
X	5. **Army Training Certification Tracking System (ATCTS) registration for contractor employees who require access to Government information systems.** All contractor employees with access to a Government info system must be registered in the ATCTS at commencement of services and must successfully complete the DoD Information Assurance Awareness prior to access to the information system and annually thereafter.
	6. For contracts that require a formal OPSEC program. The contractor shall develop an OPSEC Standing Operating Procedure (SOP)/Plan within 90 calendar days of contract award, to be reviewed and approved by the responsible Government OPSEC officer. This plan will include a process to identify critical information, where it is located, who is responsible for it, how to protect it, and why it needs to be protected. The contractor shall implement OPSEC measures as ordered by the commander. In addition, the contractor shall have an identified certified Level II OPSEC coordinator per AR 530-1.
	7. For contracts that require OPSEC Training. Per AR 530-1, Operations Security, the contractor employees must complete Level I OPSEC Awareness training. New employees must be trained within 30 calendar days of their reporting for duty and annually thereafter.
	8. For IA/IT training. All contractor employees and associated subcontractor employees must complete the DoD IA awareness training before issuance of network access and annually thereafter.

	All contractor employees working IA/IT functions must comply with DoD and Army training requirements in DoDD 8570.01, DoD 8570.01-M, and AR 25-2 within six months of appointment to IA /IT functions.
	9. For IA/IT certification. Per DoD 8570.01-M, DFARS 252.239.7001, and AR 25-2, the contractor employees supporting IA/IT functions shall be appropriately certified upon contract award. The baseline certification as stipulated in DoD 8570.01M must be completed upon contract award.
	10. For contractors authorized to accompany the force. DFARS Clause 252.225-7040, Contractor Personnel Authorized to Accompany U.S. Armed Forces Deployed Outside the United States, shall be used in solicitations and contracts that authorize contractor personnel to accompany U.S. Armed Forces deployed outside the U.S. in contingency operations; humanitarian or peacekeeping operations; or other military operations or exercises, when designated by the combatant commander. The clause discusses the following AT/OPSEC-related topics: required compliance with laws and regulations, pre-deployment requirements, required training (per combatant command guidance), and personnel data required.
	11. For contracts requiring performance or delivery in a foreign country. DFARS Clause 252.225-7043, Antiterrorism/Force Protection for Defense Contractors Outside the US, shall be used in solicitations and contracts that require performance or delivery in a foreign country. This clause applies to both contingencies and non-contingency support. The key AT requirement is for non-local national contractor personnel to comply with theater clearance requirements and allows the combatant commander to exercise oversight to ensure the contractor's compliance with combatant commander and subordinate task force commander policies and directives.
	12. For contracts that require handling or access to classified information. Contractor shall comply with FAR 52.2042, Security Requirements. This clause involves access to information classified "Confidential," "Secret," or "Top Secret" and requires contractors to comply with (1) the Security Agreement (DD Form 441), including the National Industrial Security Program Operating Manual (DoD 5220.22-M); (2) any revisions to DoD 5220.22-M, notice of which has been furnished to the contractor.
	12a. Cont. If secure telecommunication requirements apply, include clause 252.239-7016 and follow guidance at 239.74.
	12b. Cont. If covered system support requirements apply, include provision 252.239-7017 into the solicitation, clause 252.239-7018 into the contract, and follow guidance at 239.73.

	13. Controlled Unclassified Information (CUI). Include DFARS clause 252.204-7012, which requires the contractor to comply with NIST 800-171. Also include DFARS provision 252.204-2019 into solicitations. Acquisition officials should follow procedures outlined in DFARS 204.73 and verify vendors have an adequate NIST SP 800-171 summary assessment score within the Supplier Performance Risk System (within PIEE). If the score doesn't show a medium- or high-level assessment with a score of 110 or better (as described in the "NIST SP 800-171 DoD Assessment Methodology"), include clause 252.204-7020 and obtain the vendor's "system security plan" and "plan of action" for NIST SP 800-171 verification and assurance by Army security officilas.
	14. Threat Awareness Reporting Program. For all contractors with security clearances. Per AR 381-12 Threat Awareness and Reporting Program (TARP), contractor employees must receive annual TARP training by a CI agent or other trainer as specified in 2-4b.
X	**15. Escorting in classified and/or sensitive areas.** In accordance with applicable regulations, all contract personnel who do not possess the appropriate security clearance or access privileges will be escorted in areas where they may be exposed to classified information or operations, sensitive information or activities, or restricted areas.
X.	**16. Pre-screen candidates using E-Verify Program:** Contractors shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (https://www.e-verify.gov/) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form 1-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

*** END OF NARRATIVE 1

"General Decision Number: OK20260071 01/02/2026

Superseded General Decision Number: OK20250071

State: Oklahoma

Construction Type: Building
Building Construction -does not include residential construction consisting of single family homes and apartments up to and including 4 stories. (Including building projects on treatment plants)

Counties: Coal, Garvin, Haskell, Hughes, Latimer, Marshall, McCurtain, Pittsburg, Pontotoc, Pottawatomie, Pushmataha and Seminole Counties in Oklahoma.

Modification Number	Publication Date
0	01/02/2026

BOIL0592-001 01/01/2025

	Rates	Fringes
BOILERMAKER	$ 33.70	24.41

ELEC0301-006 06/01/2022

MCCURTAIN COUNTY ONLY

	Rates	Fringes
ELECTRICIAN	$ 26.60	11.92

ELEC0301-007 06/01/2022

MCCURTAIN COUNTY ONLY

	Rates	Fringes
ELECTRICIAN (Low Voltage Wiring, Alarm Installation, and Sound and Communications Systems Installation Only)	$ 26.60	11.92

ELEC0584-013 06/01/2025

COAL, HASKELL, HUGHES, LATIMER, PITTSBURG AND PUSHMATAHA COUNTIES ONLY

	Rates	Fringes
ELECTRICIAN (Low Voltage Wiring, Alarm Installation, and Sound and Communications Systems Installation Only)	$ 38.38	7%+10.20

ELEC0584-016 06/01/2025

COAL, HASKELL, HUGHES, LATIMER, PITTSBURG AND PUSHMATAHA COUNTIES ONLY

	Rates	Fringes
ELECTRICIAN	$ 38.38	7%+10.20

ELEC1141-015 05/26/2025

GARVIN, MARSHALL, PONTOTOC, POTTAWATOMIE AND SEMINOLE COUNTIES

	Rates	Fringes
ELECTRICIAN	$ 41.40	17.25+6.00

ELEC1141-019 05/26/2025

GARVIN, MARSHALL, PONTOTOC, POTTAWATOMIE AND SEMINOLE COUNTIES

	Rates	Fringes
ELECTRICIAN (Low Voltage Wiring, Alarm Installation, and Sound and Communication Systems Installation Only)	$ 24.26	15.5%+6.00

ENGI0627-018 06/01/2025

	Rates	Fringes
POWER EQUIPMENT OPERATOR:		
Group 1	$ 38.09	19.05
Group 2	$ 36.14	19.05
Group 3	$ 35.40	19.05
Group 4	$ 34.07	19.05
Group10	$ 27.25	19.05

POWER EQUIPMENT OPERATOR

GROUP 1: All Crane Type Equipment 200 ton and larger and including 400 ton capacity cranes. All Tower Cranes.

GROUP 2: All Crane Type Equipment 100 ton capacity and larger cranes, and less than 200 ton capacity.

GROUP 3: All Crane Type Equipment 50 ton capacity and larger, and less than 100 ton capacity. Crane Equipment (as rated by mfg.) 3 cu. yd. and over Guy derrick Whirley Power Driven Hole Digger (with 30' and longer mast).

GROUP 4: CRANES with Boom Incl. Jib less than 100 ft and less than 3 cu. Yd.; Overhead Monorail Crane and Bulldozer.

GROUP 10: Oiler

IRON0048-013 06/01/2025

GARVIN, MARSHALL, PONTOTOC, POTTAWATOMIE AND SEMINOLE COUNTIES ONLY

	Rates	Fringes
IRONWORKER, STRUCTURAL	$ 32.10	17.38

IRON0584-024 06/01/2024

COAL, HASKELL, HUGHES, LATIMER, MCCURTAIN, PITTSBURG AND PUSHMATAHA COUNTIES ONLY

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                                  Rates          Fringes

IRONWORKER, STRUCTURAL...........$ 30.35           16.83
----------------------------------------------------------------
 PLUM0344-015 07/01/2025

COAL, GARVIN, HUGHES, MARSHALL, PONTOTOC, POTTAWATOMIE,
PUSHMATAHA AND SEMINOLE COUNTIES


                                  Rates          Fringes

PIPEFITTER (Including
Installation of HVAC
Electrical/Temperature
Controls and HVAC Pipe
Installation)....................$ 41.00           16.03
----------------------------------------------------------------
 PLUM0344-018 07/01/2025

COAL, GARVIN, HUGHES, MARSHALL, PONTOTOC, POTTAWATOMIE,
PUSHMATAHA AND SEMINOLE COUNTIES


                                  Rates          Fringes

PLUMBER (Excludes HVAC Pipe
Installation)....................$ 41.00           16.03
----------------------------------------------------------------
 PLUM0430-015 07/01/2025

HASKELL, LATIMER, MCCURTAIN AND PITTSBURG COUNTIES

                                  Rates          Fringes

PIPEFITTER (Including
Installation of HVAC
Electrical/Temperature
Controls and HVAC Pipe
Installation)....................$ 36.65           16.03
----------------------------------------------------------------
 PLUM0430-018 07/01/2025

HASKELL, LATIMER, MCCURTAIN AND PITTSBURG COUNTIES

                                  Rates          Fringes

PLUMBER (Excludes HVAC Pipe
Installation)....................$ 36.65           16.03
----------------------------------------------------------------
 SHEE0124-013 07/01/2021

COAL, GARVIN, HUGHES, MARSHALL, PONTOTOC, POTTAWATOMIE AND
SEMINOLE COUNTIES ONLY


                                  Rates          Fringes

SHEET METAL WORKER (Including
HVAC Duct Installation)..........$ 33.38           16.36
----------------------------------------------------------------
 SHEE0270-007 06/01/2020

HASKELL, LATIMER, MCCURTAIN, PITTSBURG AND PUSHMATAHA COUNTIES
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ONLY

	Rates	Fringes
SHEET METAL WORKER (Including HVAC Duct Installation)	$ 35.49	14.60

SUOK2012-051 06/18/2012

	Rates	Fringes
BRICKLAYER	$ 20.00	0.00
CARPENTER (Acoustical Ceiling Installation Only)	$ 16.24	0.00
CARPENTER (Drywall Hanging Only)	$ 14.96	0.00
CARPENTER, Excludes Acoustical Ceiling Installation, Drywall Hanging, and Form Work	$ 16.82	2.82
CEMENT MASON/CONCRETE FINISHER	$ 14.50	0.00
FLOOR LAYER (Hardwood and Carpet)	$ 18.74	1.10
FORM WORKER	$ 16.79	3.12
GLAZIER	$ 15.27	1.46
LABORER: Common or General	$ 11.43	0.00
LABORER: Mason Tender - Brick	$ 12.00	0.00
LABORER: Mason Tender - Cement/Concrete	$ 13.04	0.00
LABORER: Pipelayer	$ 11.65	0.00
OPERATOR: Backhoe/Excavator/Trackhoe	$ 16.83	0.00
OPERATOR: Forklift	$ 18.02	0.00
PAINTER: Brush, Roller and Spray	$ 12.32	0.00
ROOFER	$ 15.00	0.29
SPRINKLER FITTER (Fire Sprinklers)	$ 21.52	6.00

WELDERS - Receive rate prescribed for craft performing operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at https://www.dol.gov/agencies/whd/government-contracts.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded. If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least $13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

a) a survey underlying a wage determination
b) an existing published wage determination
c) an initial WHD letter setting forth a position on a wage determination matter

d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION
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Section 010000 - General Requirements

Requirements

Design and construction of a new fully adhered, min. 80 mil reinforced TPO roof system to replace the existing multi-layered single membrane roof consisting of a Magnum C3 Membrane (Elvaloy) over insulation installed in 1998 over original built-up roof membrane over original insulation. New roof assembly shall consist of the TPO roof membrane, consisting of fiberglass mat face gypsum cover board over polyisocyanurate board insulation on original metal roof deck. In addition, replace existing roof drain and overflow roof drain with new sump pan assemblies. Replace existing roof drain piping and add new primary and overflow drain wall outlets per plumbing scope of work. The roof system shall be designed and installed requiring no structural modifications to the existing structure.

0001
Product Service Code : Z2QA
North American Industry Classification System (NAICS) : 238160

Option Line Item 3001
Product Service Code : Z2QA

Section 010000 - General Requirements

Requirements

Design and construction of a new fully adhered, min. 80 mil reinforced TPO roof system to replace the existing multi-layered single membrane roof consisting of a Magnum C3 Membrane (Elvaloy) over insulation installed in 1998 over original built-up roof membrane over original insulation. New roof assembly shall consist of the TPO roof membrane, consisting of fiberglass mat face gypsum cover board over polyisocyanurate board insulation on original metal roof deck. In addition, replace existing roof drain and overflow roof drain with new sump pan assemblies. Replace existing roof drain piping and add new primary and overflow drain wall outlets per plumbing scope of work. The roof system shall be designed and installed requiring no structural modifications to the existing structure.

0001
Product Service Code : Z2QA
North American Industry Classification System (NAICS) : 238160

Option Line Item 3001
Product Service Code : Z2QA

SECTION 01 00 00

GENERAL REQUIREMENTS

********* TOC PLACEHOLDER ONLY *********

-- End of Section --

SECTION 01 11 00

SUMMARY OF WORK
02/24 - SWT 03/26

PART 1 GENERAL

1.1 WORK COVERED BY CONTRACT DOCUMENTS

1.1.1 Project Description

This project will be a turnkey Design-Build (D-B) contract with all design and construction work being performed by the Contractor. The Contractor will provide all investigation, verification of site conditions and dimensions, design, research, travel, plant, materials, equipment, labor, instruments, tools, subcontractors, supervision, and management to complete all work required by these contract documents.

Requirements stated for the roof replacement in this contract are minimums. Innovative, creative, and life cycle cost effective solutions which meet or exceed these requirements are highly encouraged.

1.1.2 Location

The work is located at the Broken Bow Dam Powerhouse, North Park Drive (SH-259), Broken Bow, Oklahoma 74728, approximately as indicated. The exact location will be shown by the Contracting Officer.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. Issuance dates for references shown are the most current at the time of RFP development; however, the most current edition of each reference in effect at the time of contract award must be utilized by the selected design-build contractor. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A755/A755M	(2018; R 2024) Standard Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products
ASTM A924/A924M	(2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM C1289	(2023a) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board
ASTM C1303/C1303M	(2022) Standard Test Method for Predicting Long-Term Thermal Resistance of

Closed-Cell Foam Insulation

ASTM D471	(2016a) Standard Test Method for Rubber Property - Effect of Liquids
ASTM D522/D522M	(2017) Mandrel Bend Test of Attached Organic Coatings
ASTM D610	(2008; R 2019) Standard Practice for Evaluating Degree of Rusting on Painted Steel Surfaces
ASTM D714	(2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints
ASTM D751	(2019) Standard Test Methods for Coated Fabrics
ASTM D1149	(2018) Standard Test Method for Rubber Deterioration - Cracking in an Ozone Controlled Environment
ASTM D1204	(2014; R 2020) Linear Dimensional Changes of Nonrigid Thermoplastic Sheeting or Film at Elevated Temperature
ASTM D1654	(2008; R 2016; E 2017) Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments
ASTM D2137-11	(2018) Standard Test Methods for Rubber Property Brittleness Point of Flexible Polymers and Coated Fabrics
ASTM D4587	(2011; R 2019; E 2019) Standard Practice for Fluorescent UV-Condensation Exposures of Paint and Related Coatings
ASTM D6878/D6878M-21	(2021) Standard Specification for Thermoplastic Polyolefin-Based Sheet Roofing
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E108/UL 790	(2024) Standard Test Methods for Fire Tests of Roof Coverings
ASTM E119/UL 263	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code
ICC IEBC	(2024) International Existing Building Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-200-01	(2022; with Change 3, 2024) DoD Building Code
UFC 3-101-01	(2020; with Change 4, 2024) Architecture
UFC 3-110-03	(2012; R2020) Roofing, with Change 5
UFC 3-301-01	(2023; with Change 2, 2024) Structural Engineering

FM GLOBAL (FM)

FM 4450	(1989) Approval Standard for Class 1 Insulated Steel Deck Roofs
FM 4470	(2022) Single-Ply, Polymer-Modified Bitumen Sheet, Built-up Roof (BUR), and Liquid Applied Roof Assemblies for Use in Class 1 and Noncombustible Roof Deck Construction

UL SOLUTIONS (UL)

UL 580	(2006; Reprint Apr 2024) UL Standard for Safety Tests for Uplift Resistance of Roof Assemblies
UL 1256	(2023) Fire Test of Roof Deck Constructions
UL Bld Mat Dir	(updated continuously online) Building Materials Directory

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793	(2012) Architectural Sheet Metal Manual, 7th Edition

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Shop Drawings; G

Detailed Work Plan; G, DO

Site Condition Photos

SD-07 Certificates

Manufacturer's Roof Warranty; G

Contractor Warranty; G

1.4 SCOPE OF WORK

The work in this contract shall include design and construction of a new fully adhered, min. 80-mil reinforced TPO roof system, to replace the existing multi-layered single membrane roof. The existing roof consisting of a Magnum C3 Membrane (Elvaloy) over insulation that was installed in 1998, and over the original built-up roof membrane and original insulation.

The roof system shall be designed and installed to require no structural modifications to the existing structure. The new roof assembly shall consist of the TPO roof membrane over a 1/4-inch fiberglass mat face gypsum cover board over a polyisocyanurate board insulation on the original metal roof deck.

Roof membrane color to be white. Remove and replace existing metal flashing with new prefinished metal flashing and counterflashing per manufacturer's standard.

Remove and replace existing membrane expansion joint cover assemblies and the existing equipment curbs with new curbs.

The existing CO2 roof ventilators are to be removed and stored for replacement. There is a bid option for purchasing and installing new ventilators. the existing roof ladder shall be repainted and wall fastening repaired. There is a bid option for ladder replacement.

There is also an bid option to install new parapet guardrails to bring the roof into compliance with the code requirements for fall protection.

In addition, the Contractor shall replace existing roof drain and overflow roof drain with new sump pan assemblies. Replace existing roof drain piping and add new primary and overflow drain wall outlets per plumbing scope of work.

Design shall include a design analysis showing the existing structure can accommodate the loads applied by the new roof system without structural modifications. In addition, the design analysis shall include a test cut and inspection of the existing roof system as required by UFC 3-110-03.

Appendix A includes reference drawings of the original roof construction and sheet G-101 for general scope of work.

1.4.1 Base Bid

The bid price shall include the design, all design documentation, fabrication and installation of a fully adhered TPO roof system. Concrete repairs to existing spalled parapets and repairs associated with new roof drainage penetrations are also included in the base bid price. Also, new drainage system (primary and overflow) must be included in the base bid.

1.4.2 Option 1: Guardrail on Parapets

The bid option price shall include the procurement and installation of guardrails along the perimeter of the roof parapets at the upper and lower roofs as indicated in Appendix A.

1.4.3 Option 2: Replace Roof Access Ladder

The bid option price shall include the removal and replacement of the roof access ladder from the lower roof accessing the upper roof as indicated in Appendix A.

1.4.4 Option 3: Replace Rooftop CO2 Exhaust Fans

The bid option price shall include the removal and replacement-in-kind of six rooftop exhaust fans as indicated in Appendix A.

1.4.5 Design

1.4.5.1 Shop Drawings

Design new roof system will consist of roofing manufacturer shop drawings and fall protection data. Provide information on the existing Roof Analysis and Core Samples as described in Part 2 of Architectural.

Fall protection design must be sealed by a registered Professional Engineer. Contractor and Manufacturer developed as-built drawings, and Operations & Maintenance (O&M) manuals are required as stated in Section 01 78 00 CLOSEOUT SUBMITTALS.

1.5 REFERENCE DRAWINGS

Reference drawings of existing structures, and facilities and equipment are provided for informational purposes only and are not to be assumed as true existing as-built conditions. The Contractor is responsible for investigating and verifying actual dimensions and conditions. These drawings are provided in Appendix A - REFERENCE DRAWINGS.

1.6 SITE VISIT

An authorized Contractor representative in conjunction with the Contracting Officer and technical representatives must conduct a site visit at the Powerhouse facility.

The purpose of the site visit will be to meet with project representatives and discuss this Summary of Work. The site visit will also allow the Contractor the opportunity to examine the existing project site conditions.

1.7 VERIFICATION OF SITE CONDITIONS

Obtain all required information, measurements, dimensions, etc. necessary to field verify all existing conditions and dimensions and provide the specified Detailed Work Plan and all other necessary submittal information. If there are existing conditions that appear to affect the proposed project, inform the Contracting Officer in writing before proceeding with any design fabrication or construction processes, including ordering of materials.

The Contractor shall photograph the site and structure in connection with the project to document the initial conditions. The color photo files shall be in JPEG format and have a minimum resolution of 16 megapixels. A view location sketch indicating points of view shall be prepared, multiple view location sketches may be needed for clarity. Each photo file shall be named to indicate its location on the view location sketch.

The Contractor must upload the site condition photos to RMS as a preconstruction submittal.

1.8 DETAILED WORK PLAN REQUIREMENTS

Within 30 days after issuance of the Notice to Proceed, submit a Detailed Work Plan to the Government for review, comment, Contractor revision(s), and final Government acceptance.

The detailed work plan (DWP) must identify the personnel, equipment, and procedures that the Contractor will utilize to execute the contract. At a minimum the Work Plan must address each feature of work separately. The work plan must be in narrative format and must be in sufficient detail to tie together the design and construction activities into a comprehensive package that defines the actions in detail.

The final Government-accepted Detailed Work Plan will become a binding part of the contract. The Detailed Work Plan (DWP) requires a detailed planning effort and detailed level of documentation to define the effort proposed for the work. This effort includes step-by-step design and construction procedures.

The DWP will consist of the following unless otherwise directed by the Contracting Officer (KO):

a. DWP Preparation. Following issue of the Notice to Proceed, prepare and submit a DWP to the KO for review and approval in the format determined at a pre-design meeting (if performed). In the absence of a formal pre-design meeting, the Contracting Officer will provide guidance for the DWP format. Complete the DWP in such detail as required to tie together the design and construction activities into a comprehensive package that defines the actions in detail.

b. DWP Submittal. Submit the Work Plan in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

1.9 OCCUPANCY OF PREMISES

The Powerhouse buildings will be occupied during performance of work under this Contract. Avoid hindering employee and visitor parking and normal operations performed at this Powerhouse.

Before work is started, arrange with the Contracting Officer a sequence of procedure, means of access, space for storage of materials and equipment, and use of approaches, corridors, and stairways.

During construction, install barricades and construction fencing to prevent non-construction personnel to enter the construction area.

Signage must be installed to warn employees and visitors of hazards that exist as a result of the Contractor's activities. For additional information regarding building occupancy refer to Section 01 14 00 WORK RESTRICTIONS.

1.10 EXISTING WORK

Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

1.11 ROOF WARRANTIES

Provide roof system material and workmanship warranties (Manufacturer and Installation Contractor) meeting specified requirements. Provide revision or amendment to standard membrane manufacturer warranty as required to comply with the specified requirements.

Contractor must provide a minimum manufacturer warranty with a no dollar limit that covers the watertightness of the full system and having a minimum life of 20-years.

Submit Sample Warranty Certificates during the pre-construction phase to prove all warranty requirements will be achieved.

Contractor must include a detailed list from the roof membrane manufacturer listing all of the insulation and other products and accessories to be provided and covered under the warranty.

List products in the applicable wind uplift and fire rating classification listings, unless approved otherwise by the Contracting Officer.

1.11.1 Manufacturer's Roof Warranty

Provide the roof membrane manufacturer's no dollar limit, 20-year-life warranty covering all materials and installation workmanship. The warranty must include all flashing, insulation, and accessories necessary for a watertight roof commencing at time of Government's acceptance of the roof work. The warranty must state that:

a. If within the warranty period the roof system, as installed for its intended use in the normal climatic and environmental conditions of the facility, becomes non-watertight, shows evidence of moisture intrusion within the assembly, splits, tears, cracks, delaminates, separates at the seams, shrinks to the point of bridging or tenting membrane at transitions, or shows evidence of excessive weathering due to defective materials or installation workmanship, the repair or replacement of the defective and damaged materials of the roof system assembly and correction of defective workmanship must be the responsibility of the roof membrane manufacturer. The roof membrane manufacturer is responsible for all costs associated with the repair or replacement work.

b. When the manufacturer or his approved applicator fails to perform the
repairs within 72 hours of notification, emergency temporary repairs performed by others does not void the warranty.

1.11.2 Contractor Warranty

The Contractor must warrant that the installed workmanship of the newly installed roofing system is free from defects and errors and omissions for a period of two years.

The Contractor Warranty shall cover the roof membrane, flashing, insulation, accessories, attachments, and sheet metal installations integral for a complete watertight roof system assembly. Write the warranty to include the Government as the benefactor.

The Contractor is responsible for correction of defective workmanship and replacement of damaged or affected materials in a timely manner. The Contractor is responsible for all costs associated with the repair or replacement work.

1.11.3 Continuance of Warranty

Contractor must approve and proceed with the repair or replacement work that becomes necessary within the warranty period. Warranty actions must proceed in a manner to restore the integrity of the roof system assembly in a timely mannerand must maintain the validity of the roof membrane manufacturer warranty for the remainder of the stated warranty period.

PART 2 PRODUCTS

2.1 DELIVERABLES

All documents submitted to the Government are products of this contract and shall be the property of the Government. The Government may duplicate, use, and disclose in any manner and for any purpose all documents delivered under this contract. Include this requirement in all contracts with subcontractors and suppliers.

2.2 ARCHITECTURAL

Remove the existing single-ply membrane over asphalt built-up roof, associated miscellaneous flashings, associated miscellaneous roof penetration materials, associated wood blocking, existing metal flashings, and existing roof insulation back to the existing steel roof deck. In addition, remove existing roof ventilators and associated curbs and reinstall existing roof ventilators with new curbs and flashing. Refer to Scope of Work drawing Sheet G-101 in Appendix A. See Option 3 for replacing ventilators.

Prior to removal of the existing roof system, a detailed analysis of the existing roof deck, including inspection with non-destructive or destructive inspection methods (as necessary) along with testing shall be performed per the requirements of UFC 3-110-03 Roofing, Paragraph 6-4.1, Analysis of Existing System. Refer to Sheet G-101 in Appendix A for recommended core locations.

If inspection and testing uncovers ACM (Asbestos Containing Materials), notify the Contracting Officer immediately to determine the course of action for removal and disposal.

The new architectural roof components will consist of a new fully adhered, reinforced thermoplastic polyolefin (TPO) roof system, minimum of 80 mil thickness, white color, over 1/4-inch fiberglass mat faced gypsum cover board, over minimum R-25 polyisocyanurate board insulation, in two layers with joints staggered. Existing roof decks have an integral slope of 1/4-inch per foot, however the final roof slope must be minimum 1/2-inch per foot slope to comply with UFC 3-110-03 Roofing, 6-4.4 Slope requirements for reroof projects, which will be accomplished with tapered insulation. Provide tapered crickets as required to maintain positive slope toward drains.

Install manufacturer's standard detailing for flashing, roof penetrations, drains, curbs, etc. Install fire-retardant treated wood blocking as required for a complete system. Wrap TPO over existing expansion joint curbs, allowing for expansion/contraction per manufacturer's standard details. Wrap TPO membrane over new 1/4-inch fiberglass mat faced gypsum cover board over existing concrete parapet walls and install new prefinished galvalume metal coping as a parapet cap. Prior to installation of new TPO roofing system, inspect existing concrete parapet repairs for soundness and make additional repairs as required. In addition, repair a damaged portion of the parapet on the North East corner of the upper roof prior to installation of the TPO membrane. Refer to Scope of Work drawing Sheet G-101 in Appendix A.

Refer to Mechanical Scope of Work description for additional information about the roof drainage system Scope of Work and Option 3 to replace rooftop exhaust fans.

Remove existing roof hatch, associated curb, and exterior ladder extension and replace with new steel curb and associated blocking and painted steel roof hatch 30 inches by 36 inches with accompanying safety railing.

Install new interior ladder-mounted extendable safety post. Relocate existing equipment attached to ladder extension to another location in coordination with Powerhouse staff. Reroute existing conduit and wiring away from roof hatch curb and install proper flashing at any new roof penetrations.

Remove coating from existing roof ladder and recoat with rust-inhibitive primer and 2 coat industrial coating. Existing ladder attachments into concrete wall are pulling out in some locations where the concrete is breaking away. DOR must include the repair of wall and the reattachment with new fasteners in the design drawings. See Option 2 for replacing the ladder.

Add Guardrails as described in Option 1.

Patch and repair any areas adjacent to the roof installation affected by the installation, but not directly part of the installation.

Roof associated with the entry canopy on the west end of the building is not part of this contract.

Design and construction should be performed in accordance with ASTM D751, ASTM D6878/D6878M-21, ASTM D2137-11, ASTM D1204, ASTM D1149, ASTM D471, ASTM A755/A755M, ASTM A924/A924M, ASTM C920, ASTM D1654, ASTM D4587, ASTM D522/D522M, ASTM D610, ASTM C1289, ASTM C1303/C1303M, ASTM E84, ASTM E119/UL 263, ASTM E108/UL 790, FM 4450, and FM 4470.

Design in accordance with UFC 1-200-01, UFC 3-101-01, UFC 3-110-03, UFC 3-301-01, ASTM D714, SMACNA 1793, UL 580, UL 1256 and UL Bld Mat Dir.

2.3 MECHANICAL

All existing roof mounted equipment and penetrations type shall be re-flashed with new flashing, boots and sealant material. This would include all HVAC equipment curbs, vent piping, roof drains, etc. Roof drain strainers shall be replaced, drain bodies and piping must also be replaced.

For the base option, the existing exhaust fans must be reused and kept inside the Powerhouse. Contractor must coordinate with Powerhouse staff for storage. Provide new curbs for all exhaust fans. Finished curb elevation is 12 inches above the new insulation level.

For the roof drainage systems, remove and replace existing primary and overflow drain assemblies and piping with code compliant drain assemblies and piping at upper and lower roofs based on code requirements and roof drainage calculations as follows:

a. Separate lower roof overflow drainage piping from primary drainage system and divert primary drainage piping (currently grouped together in a single pipe) into two piping systems, one to east wall of lower roof to terminate into a stainless-steel finished downspout nozzle (Lamb's Tongue) with stainless steel perforated hinged cover. The primary drainage piping must be routed on the exterior wall (east wall) and drop to grade on a splash block.

b. Lower roof overflow drainage system must be tied into a single pipe sloped per code requirement to exit the building through the existing wall penetration above the entry canopy on west wall of the building and sized for vertical wall and half of upper roof drainage load.

c. Separate upper roof overflow drainage piping from primary drainage system and divert primary drainage piping (currently grouped together in a single pipe) into two piping systems, one each on the north and south sides upper roof sloped to terminate through the east wall of upper roof exterior, dropping cast iron piping down to splashblock.

d. Divert north and south sides of upper roof overflow drainage system to the north and south walls of upper roof to terminate into separate stainless-steel downspout nozzles at each grouping of drains for three nozzles on the north wall and three on the south wall. Ensure potential overflow drainage capacity on the south side of the upper roof is included in the calculation for the lower roof drainage.

e. The interior drainage piping for both systems must be insulated PVC.

If option for replacement of ventilators is awarded, new fans must be ECM soft start motors with integral disconnect switch and speed controller. Ventilators must match existing fans in voltage, location, Hp, RPM, CFM, etc.

Work under this contract must not diminish the function of existing mechanical systems. Installed roof penetrations must be water-tight. The contractor must provide manufacture supplied data for products associated with the roof penetrations in the roof penetration submittal. Products must be installed per manufacture's recommendations. All details of the design must be shown on the drawing submittals.

2.4 STRUCTURAL

All existing structural roof framing systems, components, and connections shall be evaluated for capability to safely support all loads from the new roof system without exceeding the allowable stress, or serviceability requirements of existing members.

If damage of existing metal deck or existing steel framing is found, then notify the Contracting Officer.

The structural analysis shall be performed in accordance with UFC 3-301-01 by a licensed Structural Engineer. Submit sealed and signed verification calculations for government-approval to be included in the design submittals, see Section 01 33 16.00 10 DESIGN AFTER AWARD.

The replacement roof's design dead, live, or snow load, including snow drift effects shall not exceed more than 5 percent of existing loads on any existing gravity carrying structural element per ICC IEBC. The component and cladding wind pressures shall be calculated in accordance with UFC 3-301-01 and submitted in accordance with Section 01 33 16.00 10 DESIGN AFTER AWARD.

Contractor shall provide calculations confirming gravity loads, including snow drift effects, are not increased by more than 5 percent over the existing condition as part of the detailed work plan.

If any existing gravity load-carrying structural element for which an alteration causes an increase of more than 5 percent, the element shall be replaced or altered as needed to carry the gravity loads required by the current International Building Code (ICC IBC) for new structures.

Existing spalled concrete parapets are to be repaired prior to the new roof installation. Penetrations from revised drainage system in the existing metal roof deck or through the existing concrete walls will be submitted for review. Locations of new penetrations in existing concrete walls should be verified with the location of existing wall reinforcement.

If option to replace roof ladder (Option 2) is selected, then anchorage design shall be submitted in accordance with Section 01 33 16.00 10 DESIGN AFTER AWARD.

PART 3 EXECUTION

3.1 SAFETY

All work shall be performed in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

3.2 APPLICABLE CRITERIA

The Contractor shall design and construct, deliver, install, and test a TPO roof system with appurtenances as designed, or per TPO manufacturer instructions.

3.3 CONTRACTOR STAGING AREA

The Government will designate a staging area for Contractor use. The Contractor shall provide all necessary facilities and utilities. The Government will provide no facilities. The Contractor shall obliterate all facilities (if any are placed) and return the area to its pre-contract state (including grading and turfing as necessary) before the contract is deemed to be complete by the Government.

3.4 PROTECTION OF GOVERNMENT PROPERTY

In addition to all contract requirements to protect Government property, the Contractor shall take particular care to protect the existing building and equipment from damage as a result of this construction. The Contractor shall not drill holes or otherwise alter or deface any existing structure outside of approved design such as new penetrations for roof drainage or holes required for guardrail and roof access ladder installation unless requested in writing with proposed repair methods and approved in writing by the Contracting Officer.

3.5 CLEAN UP

After completion of construction, the Contractor shall clean all Contractor-generated and construction debris from Government property.

-- End of Section --

SECTION 01 14 00

WORK RESTRICTIONS
11/22, CHG 5: 05/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel

1.3 CONTRACTOR ACCESS AND USE OF PREMISES

1.3.1 Activity Regulations

Ensure that Contractor personnel employed on the Activity become familiar with and obey Activity regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE) in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Ensure all Contractor equipment, including delivery vehicles, are clearly identified with their company name.

1.3.1.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.3.1.2 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer will identify designated smoking areas.

1.3.2 Working Hours

Regular Powerhouse working hours will consist of an 10.5 hour day, between 6 a.m. and 4:30 p.m., Monday through Thursday, excluding Federal Holidays.

1.3.3 Work Outside Regular Hours

Work outside regular working hours requires Contracting Officer approval. Make application 15 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours. During periods of darkness, the different parts of the work must be lighted in accordance with requirements of EM 385-1-1. Make utility cutovers after normal working hours or on Saturdays, Sundays, and Government holidays unless directed otherwise.

1.3.4 Occupied Buildings

Work will be in an existing building which is occupied.

The existing buildings and their contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer.

The Government will remove and/or relocate other Government property in the areas of the buildings scheduled to receive work.

1.3.5 Utility Cutovers and Interruptions

a. Make utility cutovers and interruptions after normal working hours or on Saturdays, Sundays, and Government holidays. Conform to procedures required in paragraph WORK OUTSIDE REGULAR HOURS.

b. Ensure that new utility lines are complete, except for the connection, before interrupting existing service.

c. Interruption to water, sanitary sewer, storm sewer, telephone service, electric service, air conditioning, heating, fire alarm, and compressed air are considered utility cutovers pursuant to the paragraph WORK OUTSIDE REGULAR HOURS.

d. Operation of Station Utilities: Do not operate nor disturb the setting of control devices in the station utilities system, including water, sewer, electrical, and steam services. The Government will operate the control devices as required for normal conduct of the work. Notify the Contracting Officer giving reasonable advance notice when such operation is required.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES
11/20, CHG 5: 02/25 - SWT 03/26

PART 1 GENERAL

1.1 CONTRACT COST BREAKDOWN

The Contractor must furnish within 30 days after the date of Notice to Proceed, and prior to the submission of its first partial payment estimate, a breakdown of its single job pay item or items which will be reviewed by the Contracting Officer as to propriety of distribution of the total cost to the various accounts. Any unbalanced items as between early and late payment items or other discrepancies will be revised by the Contracting Officer to agree with a reasonable cost of the work included in the various items. This Contract cost breakdown will then be utilized as the basis for progress payments to the Contractor.

1.2 SINGLE JOB PAYMENT ITEMS

Payment items for the work of this Contract for which Contract job payments will be made are listed in the PROPOSAL SCHEDULE and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, are included in the listed job item most closely associated with the work involved. The job price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

1.2.1 Base Bid

1.2.1.1 CLIN No. 0001 - Upper Roof Area Design

1.2.1.1.1 Payment

Payment will be made for costs associated with the design of the new roof on the upper roof area of the Powerhouse.

1.2.1.1.2 Unit of Measure

Unit of measure: job.

1.2.1.2 CLIN No. 0002 - Upper Roof Area Construction

1.2.1.2.1 Payment

Payment will be made for costs associated with the construction of the new roof on the upper roof area of the Powerhouse.

1.2.1.2.2 Unit of Measure

Unit of measure: job.1.2.1.3 CLIN No. 0003 - Lower Roof Area Design

1.2.1.3.1 Payment

Payment will be made for costs associated with the design of the new roof on the lower roof area of the Powerhouse.

1.2.1.3.2 Unit of Measure

Unit of measure: job.

1.2.1.4 No. 0004 - Lower Roof Area Construction

1.2.1.4.1 Payment

Payment will be made for costs associated with the construction of the new roof on the lower roof area of the Powerhouse.

1.2.1.4.2 Unit of Measure

Unit of measure: job.

1.2.2 Option 1

1.2.2.1 CLIN No. 1001 - Guardrail on Parapets Design

1.2.2.1.1 Payment

Payment will be made for costs associated with the design of the guardrails on Powerhouse parapets.

1.2.2.1.2 Unit of Measure

Unit of measure: job.

1.2.2.2 CLIN No. 1002 - Guardrail on Parapets Construction

1.2.2.2.1 Payment

Payment will be made for costs associated with the construction of the guardrails on the Powerhouse parapets.

1.2.2.2.2 Unit of Measure

Unit of measure: job.

1.2.3 Option 2

1.2.3.1 CLIN No. 2001 - Replace Roof Access Hatch and Ladder System to Upper Roof Design

1.2.3.1.1 Payment

Payment will be made for costs associated with the design of the replacement roof access hatch and ladder system on the upper building.

1.2.3.1.2 Unit of Measure

Unit of measure: job.

1.2.3.2 CLIN No. 2002 - Replace Roof Access Hatch and Ladder System to Upper Roof Construction

1.2.3.2.1 Payment

Payment will be made for costs associated with the construction of the replacement roof access hatch and ladder system on the lower building.

1.2.3.2.2 Unit of Measure

Unit of measure: job.

1.2.4 Option 3

1.2.4.1 CLIN No. 3001 - Replace Exhaust Fans

1.2.4.1.1 Payment

Payment will be made for costs associated with the replacement of the roof exhaust fans on the upper building.

1.2.4.1.2 Unit of Measure

Unit of measure: job.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS
11/20, CHG 5: 02/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

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EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Progress and Completion Pictures

Superintendent Qualifications; G

Red Zone Meeting Checklist; G

1.3 PROGRESS AND COMPLETION PICTURES

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600x1200x24 bit true color minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of 20 digital photographs each week throughout the entire project from a minimum of ten different viewpoints selected by the Contractor unless otherwise directed by the Contracting Officer. Submit with the monthly invoice one set of digital photographs in the agency construction management system of record, cumulative of all photos to date. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Also provide the view location sketch with the submission in the agency construction management system of record as a digital file. Include a date designator in file names. Photographs provided are for unrestricted use by the Government.

1.4 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability, during the entire period of performance under this contract. Provide other insurance coverage as required by Oklahoma State law.

1.5 SUPERVISION

1.5.1 Superintendent Qualifications

Provide project superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

1.5.2 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the Contract, then that individual must also have fluent English communication skills.

1.5.3 Duties

The project superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Red Zone meetings, partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.5.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time or excess costs or damages by the Contractor.

1.6 PRECONSTRUCTION CONFERENCE

Upon completion of design and design acceptance by the government, prior to commencing any work at the site, coordinate with the Contracting Officer a time and place to meet for the Preconstruction Conference. The purpose of this conference is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, base-access, outage requests, hot work permits, schedule requirements, quality control, schedule of prices or earned value report, shop drawings, submittals, cybersecurity, prosecution of the work, government acceptance, final inspections and contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing.

1.6.1 Attendees

Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and major subcontractors.

1.7 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start planning the turnover process at the Pre-Construction Conference meeting with a discussion of the Red Zone process and convene at regularly scheduled Red Zone Meetings beginning at approximately 75 percent of project completion. Include the following in the facility Turnover effort:

1.7.1 Red Zone Meeting Checklist

a. The Contracting Officer (KO) will provide the Contractor a copy of the Red Zone Checklist template. Attachment A presents an example Red Zone Meeting template.

b. Prior to 75 percent completion, modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. Submit the modified Red Zone Checklist to the Contracting Officer. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary.

1.7.2 Meetings

a. Conduct regular Red Zone Meetings beginning at approximately 75 percent project completion and until the project is complete.

b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.

c. Using the Red Zone Checklist as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.

d. During the Red Zone Meetings discuss with the COTR any upcoming activities that require Government involvement.

e. Maintain the Red Zone Checklist by documenting the actual completion dates as work is completed and update the Red Zone Checklist with revised planned completion dates as necessary to match progress. Distribute copies of the current Red Zone Checklist to attendees at each Red Zone Meeting.

1.8 PARTNERING

Host the partnering session within 45 calendar days of the Notice to Proceed. To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the Powerplant operations, Contractor, key subcontractors and the Designer of Record (DOR), regardless of whether the contract is a design-build or a design-bid-build, are required to participate in the Partnering process.

1.8.1 Team-Led (Informal) Partnering

a. The Contracting Officer will coordinate the initial Team-Led (Informal) Partnering Session with key personnel of the project team, including Contractor and Government personnel. The Partnering Session will be co-led by the Government Construction Manager and Contractor's Project Manager.

b. The Initial Team-led Partnering session may be held concurrently with the meeting. Partnering sessions will be held at a location mutually agreed to by the Contracting Officer and the Contractor, typically at a conference room on-base or at the Contractor's temporary trailer.

c. The Initial Team-Led Partnering Session will be conducted and facilitated using electronic media (a video and accompanying forms) provided by Contracting Officer.

d. The Partners will determine the frequency of the follow-on sessions.

e. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.

1.9 MOBILIZATION

Mobilize to the jobsite within 60 calendar days of the Notice to Proceed. Mobilize is defined as having equipment AND having a physical presence of at least one person from the contractor's team on the jobsite.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 30 00

ATTACHMENT A -

EXAMPLE RED ZONE MEETING

TEMPLATE

G/C	#	ACTIVITY DESCRIPTION	ACTION OFFICER	DUE DATE	STATUS			ACTUAL COMPLETION DATE	COMMENTS
					Working Complete	Urgent			
G+C	1	Above Ceiling Inspection (conducted, deficiencies cleared, completed)							
G+C	2	Above Ceiling Inspection (conducted, deficiencies cleared, completed)							
G+C	3	Above Ceiling Inspection (conducted, deficiencies cleared, completed)							
G+C	4	Above Ceiling Inspection (conducted, deficiencies cleared, completed)							
G+C	5	Above Ceiling Inspection (conducted, deficiencies cleared, completed)							
G+C	6	Above Ceiling Inspection (conducted, deficiencies cleared, completed)							
G+C	7	CQX/QA Mechanical & Electrical Systems Testing (conducted, verified)							
G+C	8	Commissioning (preliminary, functional & final testing of systems completed, pre-commissioning plan submitted, approved, team identified, coordinated w/CX reps, test records, reports submitted)							
C	9	Valve Tags, Charts, Diagrams (approved tags, posted properly)							

G+C	10	Communications (installation of phone, cable, fiber optics, testing, coordinate punch-out w/DPW/BCE Communications or other contractors/vendors, i.e. GCI, elevator)							
C	11	Excess Materials (inventory, spare parts, tools, quantities as spec'd, DD250, itemized list, where located/stored)							
G+C	12	Government or Contractor Furnished Equipment (date of delivery, installation, POC's)							
C	13	Landscaping/Turfing (coordinate w/DPW/BCE landscaper, utilities, irrigation system, maintenance requirements)							
C	14	Exterior (parking lots, pavement, striping, handicap signs, wheel stops, light standards, vehicle plug-in, sidewalks, fencing, bollards-safety markings, dumpster enclosures, transformer enclosure, locks)							
G or C	15	Signage (interior, exterior, directory, facility name and number, billboard)							
C	16	Keys/Tags/Box/Construction Cores (numbers req'd as spec'd, POC for turnover, key control, location, construction cores, coordination w/DPW/BCE locksmith, bldg manager, sign shops)							

C	17	Final Cleaning (spec'd waxing of floors, access controlled, bldg secured)							
C	18	Installed Property Equipment List (mechanical and electrical)							
G	19	Draft/Preliminary/Interim DD Form 1354 (coordinate w/DPW/BCE real property)							
G	20	DD Form 1354I (w/installed property list, noted deficiencies & correction dates at final inspection) *note* - hand delivered							
G	21	Submittals Closeout (outstanding, delinquent, transfer to DPW/BCE)							
C	22	Submittal Register Cleared							
G+C	23	Warranty Conference (Corps schedules w/contractor input, contractor provides booklet w/all-inclusive warranties, procedures, response order of priority, POC's phone #'s, attendees to include bldg manager, service desk, reserved room large enough for attendees)							
G+C	24	Warranty follow-up Inspections (4, 9 month & 1-yr inspections, warranty call log/procedures, contractor response times, etc.)							
C	25	Equipment Warranty Tags (type submitted &							

		approved, correct warranty date posted)							
C+G	26	As-builts (red line preliminary, final, CADD files submitted, approved, required # of copies, format as spec'd, CD's, etc.)							
C	27	CQC completion inspection and Punch List (submitted w/scheduled dates for completion of item, coordinate w/QAR. No pre-final inspection scheduled w/o this.)							
G+C	24	Warranty follow-up Inspections (4, 9 month & 1-yr inspections, warranty call log/procedures, contractor response times, etc.)							
C	25	Equipment Warranty Tags (type submitted & approved, correct warranty date posted)							
C+G	26	As-builts (red line preliminary, final, CADD files submitted, approved, required # of copies, format as spec'd, CD's, etc.)							
C	27	CQC completion inspection and Punch List (submitted w/scheduled dates for completion of item, coordinate w/QAR. No pre-final inspection scheduled w/o this.)							
C	28	Master Issue and Deficiency Log (cleared or with correction suspense dates)							
G	29	Final Quantities (verified if applicable)							

C,G	30	Snow Plow Crew (coordinate POC for "Site Familiarization Drive Through")							
G	31	Pre-final Inspection (CQC punch list received, phased by facilities, floors, mechanical rooms, etc., to include facility exterior prior to snow fall, do not schedule on early Monday or Late Friday)							
C+G	32	Beneficial Occupancy Date (BOD)							
G	33	Final Inspection (w/DD Form 1354 for signature)							
C+G	34	Final Utility Meter Readings (disconnection of meters, coordinate w/CQC, QAR, CDE/DPW utilities)							
C	35	Demobilization (winter, spring)							
C+G	36	Final Payment (withholdings/retainage /liquidated damages assessed)							
G	37	User Move-In Date (partial acceptance)							
G	38	Ribbon Cutting (if applicable)							
C+G	39	Final Payrolls(closed out)							
G	40	Release of Claims							
G+C	41	Contractor Performance Evaluation (DD2626)							
G+ AE	42	AE Performance Evaluation (DD2631)							
G+C	43	After Action Review [(AAR) (Government action, Lessons Learned, scheduled, who attends)]							

		CRITICAL PATH ISSUES:							

LEGEND: G = Government Action C = Contractor Action

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ADDITIONAL ADMINISTRATIVE REQUIREMENTS
10/19, CHG 1: 02/23 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 1110-1-8	(2021) Engineering and Design -- Construction Equipment Ownership and Operating Expense Schedule

1.2 ADDITIONAL ADMINISTRATIVE REQUIREMENTS

The conditions listed below are additional administrative requirements incorporated into this Section.

1.2.1 Contractor Supply and Use of Electronic Software for Processing Construction Wage Rate Requirements Statute Certified Labor Payrolls

a. The Contractor must use a commercially-available electronic system to process and submit certified payrolls electronically to the Government. The requirements for preparing, processing and providing certified labor payrolls are established by the Construction Wage Rate Requirements statute.

b. The Contractor must be responsible for obtaining and providing for all access, licenses, and other services required to provide for receipt, processing, certifying, electronically transmitting to the Government, and storing weekly payrolls and other data required for the Contractor to comply with the Construction Wage Rate Requirements statute. When the Contractor uses an electronic payroll system, the electronic payroll service must be used by the Contractor to prepare, process, and maintain the relevant payrolls and basic records during all work under this construction contract and the electronic payroll service must be capable of preserving these payrolls and related basic records for the required 3 years after contract completion. The Contractor must obtain and provide electronic system access to the Government, as required to comply with the Construction Wage Rate Requirements over the duration of this construction contract. The access must include electronic review access by the Government contract administration office to the electronic payroll processing system used by the Contractor.

c. The Contractor's provision and use of an electronic payroll processing system must meet the following basic functional criteria:

(1) Commercially available;

(2) Compliant with appropriate Construction Wage Rate Requirements statute payroll provisions in the Federal Acquisition Regulation

(FAR);

(3) Able to accommodate the required numbers of employees and subcontractors planned to be employed under the contract;

(4) No payroll data must be entered in RMS 3.0 CM;

(5) Demonstrated security of data and data entry rights;

(6) Ability to produce Contractor-certified electronic versions of weekly payroll data;

(7) Ability to identify erroneous entries and track the date/time of all versions of the certified Construction Wage Rate Requirements statute payrolls submitted to the government over the life of the contract; and

(8) Capable of generating a durable record copy, that is, a CD or DVD and PDF file record of data from the system database at end of the contract closeout. This durable record copy of data from the electronic payroll processing system must be provided to the Government during contract closeout.

d. All Contractor-incurred costs related to the Contractor's provision and use of an electronic payroll processing service must be included in the Contractor's price for the overall work under the contract. The costs for compliance with the Construction Wage Rate Requirements statute by using electronic payroll processing services must not be a separately bid or reimbursed item under this contract.

1.2.2 Equipment Ownership and Operating Expense Schedule

a. This requirement does not apply to terminations. See UAI 5152.249-9000, Basis for Settlement of Proposals, and FAR part 49.

b. Allowable costs for construction and marine plant and equipment in sound workable condition owned or controlled and furnished by a Contractor or subcontractor at any tier must be based on actual cost data for each piece of equipment or groups of similar serial and series for which the Government can determine both ownership and operating costs from the Contractor's accounting records. When both ownership and operating costs cannot be determined for any piece of equipment or groups of similar serial or series equipment from the Contractor's accounting records, costs for that equipment must be based upon the applicable provisions of EP 1110-1-8 Construction Equipment Ownership and Operating Expense Schedule, Region VI. Working conditions must be considered to be average for determining equipment rates using the schedule unless specified otherwise by the contracting officer. For equipment not included in the schedule, rates for comparable pieces of equipment may be used or a rate may be developed using the formula provided in the schedule. For forward pricing, the schedule in effect at the time of negotiations must apply. For retroactive pricing, the schedule in effect at the time the work was performed must apply.

c. Equipment rental costs are allowable, subject to the provisions of FAR 31.105(d)(ii) and FAR 31.205-36. Rates for equipment rented from an organization under common control, lease-purchase arrangements, and sale-leaseback arrangements, will be determined using the schedule, except that actual rates will be used for equipment leased from an organization under common control that has an established practice of leasing the same or similar equipment to unaffiliated lessees.

d. When actual equipment costs are proposed and the total amount of the pricing action exceeds the simplified acquisition threshold (SAT), the contracting officer must request the Contractor to submit either certified cost or pricing data, or partial/limited data, as appropriate. The data must be submitted on Standard Form 1411, Contract Pricing Proposal Cover Sheet.

1.2.3 Payment for Materials Delivered Off-Site

a. Pursuant to FAR 52.232-5, Payments Under Fixed Price Construction Contracts, materials delivered to the Contractor at locations other than the site of the work may be taken into consideration in making payments if included in payment estimates and if all the conditions of the General Provisions are fulfilled. Payment for items delivered to locations other than the work site must be limited to:

(1) Materials required by the technical provisions; or

(2) Materials that have been fabricated to the point where they are identifiable to an item of work required under this contract; or

(3) Items specifically listed below.

b. Payment for materials delivered off-site must be made only after receipt of paid invoices listing the value of material and labor incorporated in the items along with a canceled check showing the Prime Contractor's title to the items delivered off site. Payment for materials delivered off-site must be limited to the following items:

(1) AS AUTHORIZED BY THE CONTRACTING OFFICER.

1.2.4 Required Insurance (FAR 28.307-2)

The Contractor must procure and maintain during the entire period of his performance under this contract the following minimum insurance:

a. Workmen's Compensation and Employer's Liability Insurance in compliance with applicable state statutes, with a minimum liability coverage of $100,000.

b. Comprehensive General Liability Insurance for bodily injury in the minimum limits of $500,000 per occurrence. No property damage liability insurance is required.

c. Comprehensive Automobile Liability Insurance covering the operation of all automobiles used in connection with the performance of the contract in the minimum limits of $200,000 per person and $500,000 per occurrence for bodily injury and $25,000 per occurrence for property damage.

The Contractor agrees to insert the substance of this clause, including this paragraph, in all subcontracts hereunder.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION 01 32 01.00 31

SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT
03/22, CHG 10: 03/26 - SWT 04/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

AACE 29R-03	(2011) Forensic Schedule Analysis
AACE 52R-06	(2006) Time Impact Analysis - As Applied in Construction
AACE 84R-13	(2015) Planning and Accounting for Adverse Weather

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

AACE 67-17	(2017) Schedule Delay Analysis

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1-1-11	(2017) Administration -- PROJECT SCHEDULE

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Project Scheduler Qualifications: G

Project Schedule; G

Recovery Schedule; G

1.3 PROJECT SCHEDULER QUALIFICATIONS - DESIGNATED AUTHORIZED REPRESENTATIVE (DAR)

Designate an authorized representative (DAR)to be responsible for the preparation of the schedule and all required updating and production of reports. The DAR must have no other responsibilities involving the project unless approved by the Contracting Officer. The DAR must have a minimum of 2-years' experience scheduling construction projects and must have prepared and maintained at least three previous construction schedules similar in size and complexity to this contract, using Primavera P6 that meets the requirements of this specification. DAR must have a comprehensive knowledge of CPM scheduling principles and applications. DAR must be the same person or employee of a company approved under paragraph 1.2 "Submittals". A DAR change must be submitted and approved, per paragraph 1.2 "Submittals", before new DAR takes over responsibilities from previous DAR. DAR to be responsible for the preparation of the schedule, all required updating, preparing of reports, and all requirements of this specification. DAR must have experience in the use of the scheduling software that meets the requirements of this specification. DAR must provide certificate of scheduling software training completed from certified training company. USACE reserves the right to have DAR removed/replaced when schedule management, expertise, and performance are determined to be not meeting the requirement set forth in this specification. If so, requested by USACE, DAR must come to site or USACE office for any meeting.

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

SOFTWARE TYPES:

a. Primavera

b. Microsoft Project

c. Microsoft Excel

If Microsoft Project or Microsoft Excel is used, manual inputting of activities into RMS must be required. This includes all codes, cost loading, and other items listed (that can be manually inputted). If using Primavera, then a SDEF must be created and inputted into RMS only when USACE approves SDEF, not SDEF must be loaded into RMS without USACE approval. SDEF must be created from within the scheduling software. Using a third party software for creating SDEF is prohibited.

2.1.1 Government Default Software

The Government will use Primavera P6. Ensure exported schedule files are compatible with the version of P6 used by the Government.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required SDEF file must be created and supported by the scheduling software manufacturer.

2.1.3 Primavera

If Primavera P6 is selected for use, provide the ".xer" export file in a version of P6 importable, uncorrupted, and loadable by the Government system. If xer file provided is incapable or does not meet this requirement and cannot load correctly (examples: CLINs not balancing when loaded on Government computer or contains POBS tables) USACE will return for corrections and Contractor must revise and resend for approval. Verify at the SEKO meeting which version of the P6 is in use by the Government. Export the schedule in a version of P6 no newer than that used by the Government.

2.1.4 Other Than Primavera

If the Contractor chooses software other than Primavera P6, the software must be compliant with this specification. The software routine used to create the required SDEF file must be created and supported by the software manufacturer. In addition, the Contractor must provide for the Government's use two licenses, two computers, and software training for two Government employees by a certified training firm, which that cost must include any expenses required for travel, meals, transportation, and overnight accommodations. These computers will be stand-alone and not connected to Government network. Computers and licenses will be returned to the Contractor at project completion.

2.1.5 Conversion from One Scheduling Software to Another

The scheduling software utilized to create the schedule will be the one used for the project. Creating the schedule in one scheduling software and conversing to another scheduling software is prohibited. Also, conversion utilizing spreadsheet or word documentation software is prohibited.

PART 3 EXECUTION

3.1 Scheduling Expectations Kickoff Meeting (SEKO)

In conjunction with the post award conference or pre construction conference the Government and the Contractor will hold a separate and distinct Scheduling Expectations Kickoff Meeting(SEKO). The intent of this meeting is for the Government to review with the contractor the requirements of the scheduling specification, communicate the Government’s expectations regarding schedule development and management, and answer any questions the Contractor may have regarding the schedule requirements.

3.2 GENERAL REQUIREMENTS

Prepare Project Schedule for approval as a Bar Chart Schedule, or Project Schedule, as specified herein, pursuant to the Contract Clause, SCHEDULE FOR CONSTRUCTION CONTRACTS. Provide submittal (transmitted thru RMS via 4025) for a cost-loaded network diagram (Gantt/bar chart) for approval no later than 30 calendar days after NTP date. Show in the schedule the sequence in which the Contractor proposes to perform the work and dates on which the Contractor contemplates starting and completing all schedule activities. Schedule layout must be in sequential order from NTP to End Project. The scheduling of the entire project, including the design and construction sequences, is required. The scheduling of construction, design and construction is the responsibility of the Contractor. Contractor management personnel must actively participate in its development. Subcontractors and suppliers, Designers, Subcontractors and suppliers working on the project must also contribute in developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project-monitoring tool. Dates based on calendar days, not working days. Software used must provide equal settings in all schedule submissions to the Government.

Contracting Officer will have the right to have Contractor provide additional detail to schedule as deemed warranted besides following list.

a. If mobilization activity is used, a demobilization activity must be in the schedule. The combined total between mobilization and mobilization must be split between mobilization (60%) and demobilization (40%).

b. Once Original Durations are approved, they cannot be changed.

c. Original durations must not be over 20 calendar days. Only exceptions are Summary, Hammock, and Level of Effort activities.

d. NTP must be a start milestone, first activity in schedule with no activity ahead of it, have no predecessors. The first activity of the schedule must be called "NTP".

e. End Project activity must be based on contract completion date. Use "project must end by" or equivalent in software. Do not use any other constraints. Do not place activity constraint on End Project. End Project activity must be the only activity without a successor.

f. Project Activity Codes must be Project Level, not Global or EPS level.

g. Project Calendars must be Project Level, not Global or Resource level.

h. Project Schedule Default for defining Activity Duration Types must be set to "Fixed Duration & Units", except for Milestones.

i. Project Schedule Default for defining Percent Complete Types must be set to "Physical".

j. Project Schedule Options for defining Critical Activities must be set to "Longest Path".

k. Project Schedule Option for defining Scheduling Progressed Activities must be set to "Retained Logic".

l. Project Schedule Option for defining Expected Finish, the check box will not be checked.

m. Project Schedule Option for Compute Total Float as must be set to "Finish Float".

n. Project Admin Preference for Hours per Time Period must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days. Using any other hr/day besides 8 hours is prohibited.

o. Project cost loading using a single lump sum resource. The Price/Unit must be $1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" selected (CHECK BOX). Contractor must create project resource that does not match any resource ID and resource name in the USACE P6 database that has been dedicated to other Contractors.

p. Project Activity ID's must not exceed 10 characters. Alpha, spaces, or numbers are allowed. All other characters are prohibited.

q. Project Activity Names/Descriptions will be no more than 30 characters, including spaces. Activity Names/Description will not be duplicated; each must be unique. Abbreviations must be used to get within the 30 character limit. Use "Notebook" tab in P6 for explanation of Activity Names/Descriptions when abbreviated or for further description.

r. Calendars must be 5 day or 7 day. No 1, 2, 3, 4, or 6 day calendars or hour calendars are allowed unless approved by Government and have 8 hour work days assigned to them.

s. Provide a milestone for "RED ZONE MEETING"

t. Using a SS, FF, or FS relationship in unison as a predecessor or successor to an activity is prohibited.

u. Lags must not be more than 5 working days or 7 calendar days.

v. Start to Finish (SF) relationships, along with Leads (negative lags) are prohibited.

w. No constrained dates are allowed in the schedule other than those specified herein.

x. Using "Expected Finish", "Suspend", or "Resume" as part of the activity Status is prohibited.

y. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

z. Using BOD, Beneficial Occupancy, Substantial Completion, Turn Over, or any other description related to Owner acceptance as an activity and/or descriptions is prohibited.

aa. The Contractor must not schedule meetings, Government reviews, or responses during the last two weeks of December or other official federal Government holidays (including Friday after Thanksgiving) and must exclude such dates and periods from any durations specified for Government actions.

ab. The schedule must include mandatory tasks and/or activities along with any additional mandatory tasks and/or activities requested by Contractor or USACE in the Project Schedule and Periodic Monthly Schedule updates.

ac. The schedule must include milestones along with any additional milestones requested by Contractor or USACE in the Project Schedule and Periodic Monthly Schedule updates.

3.2.1 Project Schedule Submittal Activity Breakdown

Provide a submittal activity for administrative, design, construction, or close-out activity that requires a submittal (even if submittal is for information only).Submittal activities are not allowed to be cost loaded. A typical procurement sequence will include the string of activities: Prep, Submit, Government Review, Approve, Procure, Fabricate, Deliver, and Install. The Deliver and Install activities will be clarified during schedule submittal review if that cost will represent stored materials or materials that will be installed immediately once on site or stored on site. Contractor must follow FAR requirements for any stored materials for payments. The original duration for Submit must not include any days for Prep. Original duration for "Submit" and "Approval" activity will be one calendar day. Government Review must be based on contract requirements. If contract requirements does not give original duration for Government review in work or calendar days, then duration must be 30 calendar days.

3.2.2 Holiday Periods for Government Reviews and Actions

The Contractor must not schedule meetings, Government reviews, or responses during the last two weeks of December or other official Federal Government holidays (including Friday after Thanksgiving) and must exclude such dates and periods from any durations specified for Government actions.

3.2.3 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project Schedule. AACE 84R-13 provides methodologies for techniques in planning for adverse weather. The preferred methodology is the use of a weather calendar. If a method other than a weather calendar is proposed, discuss the reason during the SEKO meeting.

3.2.4 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in ER 1-1-11. Contractor must create the schedule using the following codes for each individual project. Contractor must define each activity code as a PROJECT activity code. This exact structure is mandatory. All Activity Codes must be developed and assigned to activities as detailed herein.

The SDEF format is as follows:

Assign Code to activity	Field	Activity Code	Length
	1	WRKP	3
YES	2	RESP	4
	3	AREA	4
	4	MODF	6
YES	5	BIDI	6
	6	PHAS	2
	7	CATW	1
YES	8	FOW	30

*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements. Activity Code shown must be exactly used in the software. Feature of Work activity code must be no more than 25 characters and description cannot be more than 30 characters to be able to provide complete description in both the schedule and SDEF.

3.2.5 Workers Per Day Coding (WRKP)

Assign Workers Per Day code (WRKP) for all field construction or direct work activities, if directed by the Contracting Officer. Workers Per Day is based on the average number of workers (crew size) expected each day to perform a task for the duration of that activity. This code must be used when Contractor is required to provide a resource (manloaded) loaded schedule. Activities may have more than one WRKP code so as to document different number of resources assigned to that activity. Using RESP or any other activity code and/or resource/expenses to provide number of resources assign to an activity is prohibited. WRKP codes must be numeric-alpha.
Example would be Code Value 3M, 3=CREW SIZE, M=Mason. Workers Per Day codes must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Workers Per Day codes into RMS is prohibited. This code must not be used as a means of tracking modifications to the contract or any other use except for above.

3.2.6 Responsible Party Coding (RESP)

Assign responsibility code (RESP) for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the activity.

a. Activities coded with a Government "GOVT" Responsible Party code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State Regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements. Government "GOVT" Responsible Party code must not be used for any activity that has costs associated with it. For activities that require Government approval and must have cost associated with them, the Contractor must use Approval By The Government "ABTG" for those activities.

b. Activities cannot have more than one Responsible Party code. Examples of acceptable Responsible Party code values for (RESP) are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE). Assign Responsible Party code for all activities. Unacceptable code values are abbreviations of the names of prime contractors or subcontractors. RESP codes must be alpha; numeric is not allowed for RESP codes. Responsible Party codes must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Responsible Party codes into RMS is prohibited.

3.2.7 Area of Work Coding (AREA)

Assign Area of Work code (AREA) to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space. Examples of Area of Work codes include different areas within a floor of a building, different floors within a building, and different buildings within a complex of buildings. Not all activities are required to be Area of Work coded. A lack of Area of Work coding indicates the activity is not resource or space constrained. Each activity using an Area of Work code must be identified within a single project area and have only one Area of Work code. AREA codes must be alpha and/or numeric or a combination of both. Area of Work codes must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Area of Work codes into RMS is prohibited.

3.2.8 Modification Number Coding(MODF)

Assign a Modification Number code (MODF) to any activity or sequence of activities added to the schedule as a result of an executed Contract Modification, when approved by Contracting Officer. Key all Modification Number code values to the Government's modification numbering system. An activity can have only one Modification Number code. Modification Number code (MODF) must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Modification Number code (MODF) into RMS is prohibited. This code must not be used as a means of tracking pending modifications to the contract or any other use except for above.

3.2.9 Bid Item Coding (BIDI)

Assign a Bid Item code (BIDI) to all activities using the Contract Line Item Number (CLIN) Schedule to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI code. Bid Item codes (BIDI) must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Bid Item (BIDI) codes into RMS is prohibited.

3.2.10 Phase of Work Coding (PHAS)

Assign Phase of Work code (PHAS) to all activities. Examples of phase of work coding are design phase, procurement phase, and construction phase. Each activity can have only one Phase of Work code. Phase of Work codes must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Phase of Work Codes into RMS is prohibited.

a. Phase of Work code proposed fast track design and construction phases proposed to allow filtering and organizing the schedule by fast track design and construction packages.

b. If the contract specifies phasing with separately defined performance periods, identify a Phase of Work code to allow filtering and organizing the schedule accordingly.

3.2.11 Category of Work Coding (CATW)

Assign Category of Work code (CATW) to all Activities based upon the predetermined codes and descriptions in RMS dependent on Tulsa District Library. Each activity must have only one Category of Work code. Category of Work codes must be imported through the SDEF conversion of the schedule into RMS. Manual inputting of Category of Work codes into RMS is prohibited.

The Tulsa District CATW codes are as follows:

Field	Category Codes	Length	Description
1	1	1	GENERAL
2	A	1	ARCHITECTURAL
3	B	1	PAINTING
4	C	1	CIVIL
5	D	1	DESIGN
6	E	1	ELECTRICAL
7	F	1	FIRE EXTINGUISH
8	G	1	GENERAL CONTRACTOR
9	H	1	HAZARDOUS/TOXIC
10	I	1	DRYWALL/STUDS
11	J	1	FLOORING
12	K	1	LANDSCAPE
13	L	1	MECHANICAL
14	M	1	PLUMBING
15	N	1	DIRTWORK/ASPHALT
16	O	1	ROOFING
17	P	1	STRUCTURAL
18	Q	1	SAFETY
19	R	1	ADMINISTRATIVE

3.2.12 Feature of Work Coding (FOW)

Assign a Feature of Work code to appropriate activities based on the Definable Feature of Work to which the activity belongs based on the approved CQC Plan Features of Work. Feature of Work codes must be created as part of the CQC Plan and inserted into the schedule. Definable Features of Work are defined in Section 01 45 00.00 QUALITY CONTROL and are required as a minimum. The Contractor's has the option to create and provide additional Features of Work codes. These additional codes must be included in the CQC Plan for approval. For new Feature of Work codes required after an executed modification, the Contractor must revise Feature of Work codes as part of CQC Plan and resubmit for approval before inserting into the schedule's Feature of Work code listing. An activity can have only one Feature of Work code. Every activity must have a Feature of Work code. Feature of Work codes (FOW) must be imported through the

SDEF conversion of the schedule into RMS. Manual inputting of Features of Work codes into RMS is prohibited. Feature of Work activity code must be no more than 25 characters and description cannot be more than 30 characters to be able to provide complete description in both the schedule and SDEF.

3.2.13 Failure to Achieve Progress

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the Contracting Officer may require a written recovery plan for approval. In either case, the plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra work days, etc.

3.2.14 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the Recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.2.15 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to all other requirements of this specification and section and FAR 52.236-15 "Schedules for Construction Contracts". Also comply with Paragraphs "Recovery Schedule Submission".

3.2.16 Recovery Schedule Submission

If the Contractor falls behind his approved schedule (behind the LS/LF cash flow curve of more than 20 calendar days of negative total float), or performs the work in such a manner that the network diagram and mathematical analysis no longer indicate reasonable logic and duration for completion of the work by the current contract completion date, as determined by the Contracting Officer, the Contractor must promptly provide a supplemental Recovery Schedule to regain the original schedule and show project will complete the work by the current contract completion date to show when project will be completed passed the current contract required completion date, as approved by the Contracting Officer. Recovery schedule must be transmitted via 4025 within the time frame the Contracting Officer has given, with all documentation and requirements per specifications. The Recovery schedule must be resource loaded, utilizing paragraphs 3.3.7 "Standard Activity Coding Dictionary",3.3.7.1 "Workers Per Day Coding (WRKP)", and 3.3.7.2 "Responsible Party Coding (RESP)", with crew size and productivity for each remaining activity, and indicating overtime, weekend work, and/or double shifts needed to either regain the schedule (Recovery) or show when project will be completed, in accordance with FAR 52.236-15, without additional cost to the Government. The Recovery schedule once submitted and accepted and approved may replace the original approved schedule as the official contract schedule. If required by Contracting Officer, the original approved schedule must be updated monthly, in addition to the Recovery schedule, and monitored by the Contractor and Contracting Officer to determine the effect of the Recovery progress for comparison to determine has the Recovery Schedule progressed to regain the rate of progress for timely completion to meet contract required completion date.

The Contractor must not artificially improve his progress by revising the schedule logic restraints, using calendars other then 5 or 7 day, or shortening future work activity durations. The Contractor may improve his progress by performing sequential work activities concurrently or by performing activities more quickly than planned, but such improvements must be indicated on the Recovery Schedule and must not be recorded on the official schedule until they have actually been achieved by the Contractor. The additional resources required to improve the progress must be evident on the work site.

Failure of the Contractor to perform work and maintain progress in accordance with the supplemental Recovery schedule may result in an interim and final unsatisfactory performance rating and/or may result in corrective action by the Contracting Officer in accordance with FAR 52.236-15.

After Recovery submittal is accepted and approved and before any manual entry or SDEF is loaded into RMS, the Contractor must update the Recovery Schedule with actual data and present the backup schedule file and SDEF file within three (3) working days to USACE for review. Once that schedule is reviewed and approved and notification from USACE is given, Contractor will then manually insert or load SDEF into RMS. Recovery Schedule must meet all requirements of this specification.

3.2.17 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental Recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.2.18 Weekly Progress Meetings

Conduct a weekly progress meeting with the Government for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming two weeks. Provide a weekly update report, with current actual start and actual finish dates, remaining durations, and percent completes on activities that has have been status between weekly progress meetings utilizing layouts part of the scheduling software. Use the current approved schedule update for the purposes of this meeting and for the production and review of reports. At the weekly progress meeting, address the status of RFIs, RFPs and Submittals. The Contractor must provide a copy of the agenda for weekly progress meeting at least 24 hours before the meeting begins. The Contractor must provide post-meeting minutes within forty-eight hours of the conclusion of the meeting.

3.2.19 Ownership of Float

The Activity float available in the schedule at any time must not be considered for the exclusive use of either the Government or the Contractor including activity and/or project float. Activity float is the number of work days that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the contract completion date milestone.

3.3 APPROVED PROJECT SCHEDULE (BAR CHART)

Use the approved Project Bar Chart Schedule to measure the progress of the work and to aid in evaluating time extensions. The Bar Chart will provide the basis for all progress payments. If the Contractor fails to submit any Bar Chart Schedules within the time prescribed, the Contracting Officer may withhold approval of progress payments until the Contractor submits the required Bar Chart Schedule. Contractor must provide a schedule update every thirty calendar days from the last approved and accepted update even if no pay application is presented. This update must be transmitted thru RMS via 4025. If using P6, SDEF must not be loaded until USACE give approval to do so.

3.4 DEFAULT TERMS

Failure of the Contractor to comply with the requirements of the Contracting Officer must be grounds for a determination, by the Contracting Officer, that the Contractor is not prosecuting the work with sufficient diligence to ensure completion within the time specified in the contract. Upon making this determination, the Contracting Officer may terminate the Contractor's right to proceed with the work, or any separable part of it, in accordance with the default terms of the contract.

3.5 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each monthly update period and therefore the amount of each progress payment. Lack of an approved monthly update, or qualified scheduling personnel, will result in the inability of the Contracting Officer to evaluate contract earned value for the purposes of payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Based on USACE requirements, Contractor can present a separate Pay Application for bonds before the Preliminary schedule is approved, however no additional Pay Applications will be presented and/or processed until an acceptable and approved schedule submittal is in place. Before any Pay Application and/or schedule update files are transmitted through RMS, the SDEF that was created from the accepted and approved schedule submittal or updated monthly schedule must be loaded into RMS and accepted. Match the progress payment "Pay Period Thru" date to match the schedule data date.

A separate "bond" Pay Application can be presented by Contractor, however USACE will require that a "bond" SDEF be created and approved before such Pay Application can be processed. Contractor must get requirements from USACE.

3.5.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.

3.5.2 Cost Loading of Submittals

No costs are to be assigned to activities for the preparation, review, or approval of submittals, except as described under paragraph Cost Loading of Closeout Activities, paragraph As-Built Drawings and Record Drawings, paragraph Record Construction Contract Specifications submittals, paragraph O&M MANUALS, and paragraph Correction of Punch List.

3.5.3 As-Built Drawings and Record Drawings

If there is no separate contract line item (CLIN) for as-built or final record drawings, cost load the "Approval of as-built and final record drawings" activity based on not less than 1 percent of the present contract value. Activity will be declared 100 percent complete upon the Government's approval of all As-Built and Final Record drawings.

3.5.4 Record Construction Contract Specifications

Cost load the "Approval of the record construction contract specifications" activity not less than 1 percent of the present contract value. Activity will be declared 100 percent complete upon the Government's approval.

3.5.5 O & M Manuals

If there is no separate contract line item (CLIN) for O & M manuals, cost load the "Approval of O & M manuals" activity based on 1 percent of the present contract value. Activity will be declared 100 percent complete upon the Government's approval of all O & M manuals.

3.5.6 Corrections of Punch List

Cost load the "Correction of punch list / Gov't final inspection" activity(ies) total not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and corrections of all punch list work identified during Government pre-final inspection(s).

3.5.7 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the Preliminary, Initial, Recovery, and/or Periodic Schedule, executed modification activities to be added to schedule, and will result in rejection of payment requests until compliance is met.

Until the Project, Recovery, or Periodic Schedule Update Schedules or executed modification activities to be added to schedule are approved, no Pay Applications will be processed or approved for payment. If Pay Application is presented before acceptance and approval of any schedule submittal, the Pay Application will be returned with no action taken. See 3.2 "BASIS FOR PAYMENT AND COST LOADING" for process given on bonds.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent project schedule revisions or updates, the Contracting Officer may withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made or reject Pay Application with no further action until such corrections are made.

In the case where the Periodic Monthly Schedule Update shows the project is 20 calendar days or more in total negative total float, the Contracting officer may withhold 10 percent in retainage.

3.5.7.1 Submission Requirements

Submit the following items for the Project submittal, Recovery submittals, and every Periodic Schedule Update throughout the life of the project. The following items will be submitted individually and not provided in a .zip file.

3.5.7.2 Electronic Scheduling Data

Provide electronic scheduling data containing the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Project, Recovery, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings. The file naming convention is to be determined at SEKO meeting.

3.5.7.3 Narrative Report

Provide a Narrative Report with the Project, Recovery, and each Periodic Update of the project schedule, as the basis of the progress payment request and/or each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for any problems, either current or potential, which are revealed through that analysis. The narrative report must not be used to provide comments or clarifications to a schedule submittal. Include the following information as minimum in the Narrative Report:

a. Must be addressed to the PE/CM for the project.

b. Statement declaring who is the Designated Authorized Representative (DAR) for Schedule.

c. Compliance with paragraphs 1.1 "References"

d. Specifically reference, on an activity-by-activity basis, all changes made since the previous period and relate each change to documented, approved schedule changes.

e. Identify and discuss the work scheduled to start in the next update period.

f. Describe activities along the longest path in addition to activities on the next float path after the longest path.

g. A description of current and anticipated problem areas or delaying factors and their impact and an explanation of corrective actions taken or required to be taken. Identify the party responsible for delay, if applicable.

h. Identify and explain why activities based on their calculated late dates should have either started or finished during the update period but did not.

i. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.

j. Identify and discuss out-of-sequence work.

k. If the longest path changes from a monthly update to the next, provide an explanation of the factor driving the change.

l. Respond to USACE Schedule Review Comments from the previous update(s).

3.5.7.4 Schedule Reports

The format, filtering, organizing and sorting for each schedule report must be as directed by the Contracting Officer. Typically, reports must contain Activity Numbers, Activity Description, Original Duration, Remaining Duration, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Total Float, Actual Start Date, Actual Finish Date, and Percent Complete. Layouts are acceptable. Utilize or create reports with the schedule software. Provide the reports electronically in .pdf format. The following lists typical reports that will be requested:

3.5.7.4.1 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date. This report must reflect the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments along with agreed to percentages from Contractors Pay Request Worksheet (CPRW) presented for USACE approval and finalized at Periodic Schedule Update (PSU) Meeting) or Monthly Progress (MPM) Meeting. Group activities by CLIN number and sort by activity number. This report must also provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. The printed report must contain the following columns for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings this period, Total Quantity, Quantity to Date, and Percent Complete (based on cost).

3.5.7.5 Network Diagram and Longest/Critical Path Bar (Gantt)Charts

A Network Diagram and Longest/Critical Bar (Gantt) Chart is required for the Project, Recovery, and Periodic Schedule Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.7.5.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram along with the columns as described per paragraph 3.5.3 "Schedule Reports".

3.5.7.5.2 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram along with the columns as described per paragraph 3.5.3 "Schedule Reports".

3.5.7.5.3 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.7.5.4 Longest/Critical Path

Show all activities on the longest/critical path. Clearly show the longest/critical path, using RED. The critical path is defined as the longest path.

3.5.7.5.5 Grouping and Sorting

Group and sort activities as directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.7.5.6 Banding

Organize activities using the WBS or as otherwise directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.7.6 Longest/Critical Path Bar (Gantt) Chart

The Longest/Critical Path Bar Chart is required for the Preliminary, Initial, Recovery, and Periodic Schedule Updates. The two main float paths must be depicted and displayed with the order and interdependence of activities and the sequence in which the work is to be accomplished. Show all activities on the longest path. Clearly show the longest path, using RED. Each float path must be separate and in stair step fashion. Do not use WBS format.

3.5.7.7 Requests for Time Extensions

AACE 67-17 provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in AACE 67-17, the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occurring. Also prepare a time impact analysis for each Government request for proposal (RFP).

3.5.7.7.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date or interim completion date(s). Multiple impacts must be evaluated chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.5.7.7.2 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the project schedule.

3.5.7.7.3 Time Impact Analysis (Prospective Analysis)

Submit requests for time extensions based on a prospective analysis if the work involved or the impact identified has not already occurred. Where the impact is ongoing a prospective analysis may be considered. However, the Government reserves the right to require a retrospective analysis be prepared after the impact has ended. Prepare a time impact analysis for approval by the Contracting Officer based on industry standard AACE 52R-06. Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

3.5.7.7.4 Forensic Schedule Analysis (Retrospective Analysis)

Submit requests for time extensions based on a retrospective analysis to see if the work involved or the impact identified has already occurred. The analysis must account for the actual performance of both the impacted work and all other contract work in the schedule.
If a methodology is chosen from AACE 29R-03, the method must adhere to the principles identified in AACE 67-17. If there is a conflict with the methodology chosen from AACE 29R-03 and AACE 67-17, AACE 67-17 will govern. Choice of methodology should be discussed prior to submission to Government.

3.5.7.7.5 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis. The proposed fragnet must consist of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. Typically, report must contain Activity ID, Activity Description, Current Schedule, Data Date, Start Date, Finish Date, Duration in Calendar Days, Predecessor(s), Successor(s), Relationship Type, Constraint, and Calendar. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the project schedule.

3.5.7.7.6 Submission Requirements for Time Extensions

Submit a justification for each request for a change in the contract completion date based upon the most recent approved updated project schedule at the time of the impact. Such a request must be in accordance with the requirements of other appropriate Contract Clauses and must include, as a minimum:

a. List of affected activities, with their associated project schedule activity number (3.8.5 Fragmentary Network (Fragnet).

b. Brief explanation of the causes of the change (narrative).

c. An analysis of the overall impact of the changes proposed (part of narrative).

d. A sub-network of the affected area (detailed fragnet breakdown per 3.8.4).

e. .pdf and backup file schedule at time of impact (using data date of start of impact).

f. .pdf and Backup file of schedule after inserting impact activities (using same data date of start of impact).

3.5.7.7.7 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay will not be a cause for an extension to the performance period, completion date, or any interim milestone date.

3.5.7.7.8 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless the Contractor actually performed work in accordance with that Early Completion Schedule. The Contractor must show that an early completion was achievable had it not been for the impact.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21 - SWT 03/26

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor and the Designer of Record (DOR), if applicable, are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

Design submittals, design substantiation submittals and extensions of design submittals.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and Safety Data Sheets(SDS)concerning impedances, hazards and safety precautions.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 Design-Build Submittal Classifications

1.4.2.1 Designer of Record Approved (DA)

Designer of Record (DOR) approval is required for extensions of design; critical materials; any variations from the Solicitation, the Accepted Proposal, or the completed design; equipment whose compatibility with the entire system must be checked; and other items as designated by the Contracting Officer. Provide the Government with the number of copies designated hereinafter of all DOR approved submittals. The Government may review any or all Designer of Record approved submittals for conformance with the Solicitation, the Accepted Proposal, and the completed design. The Government will review all submittals designated as varying from the Solicitation or Accepted Proposal, as described below. Provide design submittals in accordance with Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD). Generally, list design submittals under SD-05 Design Data.

1.4.2.2 Government Conformance Review of Design (CR)

The Government will review all intermediate and final design submittals for conformance with the technical requirements of the Solicitation. Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD) covers the design submittal and review process in detail. Review will be only for conformance with the applicable codes, standards, and contract requirements. Design data includes the design documents described in Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD).

1.4.2.3 Designer of Record Approved/Government Conformance Review (DA/CR)

1.4.2.3.1 Variations from the Accepted Design

DOR approval and the Government's concurrence are required for any proposed variation from the accepted design that still complies with the contract before the Contractor is authorized to proceed with material acquisition or installation. If necessary to facilitate the project schedule, before official submission to the Government, the Contractor and the DOR may discuss with the Contracting Officer's Representative a submittal proposing a variation. However, the Government reserves the right to review the submittal before providing an opinion. In any case, the Government will not formally agree to or provide a preliminary opinion on any variation without the DOR's approval or recommended approval. The Government reserves the right to reject any design, variation that may affect furniture, furnishings, equipment selections, or operational decisions that were made, based on the reviewed and concurred design.

1.4.2.3.2 Substitutions

Unless prohibited or otherwise provided for elsewhere in the contract, where the Accepted Proposal named products, systems, materials or equipment by manufacturer, brand name, model number, or other specific identification, and the Contractor desires to substitute a manufacturer or model after award, submit a requested substitution for Government concurrence. Include substantiation, through identifying information and the DOR's approval, that the substitute meets the contract requirements and that it is equal in function, performance, quality, and salient features to that in the accepted contract proposal. If the contract otherwise prohibits substitutions of equal named products, systems, materials or equipment by manufacturer, brand name, model number or other specific identification, the request is considered a "variation" to the contract. Variations are discussed below in paragraphs: "DESIGNER OF RECORD APPROVED/GOVERNMENT APPROVED" and VARIATIONS.

1.4.2.4 Designer of Record Approved/Government Approved (DA/GA)

In addition to the above-stated requirements for proposed variations to the accepted design, both DOR and Government Approval and, where applicable, a contract modification are required before the Contractor is authorized to proceed with material acquisition or installation for any proposed variation to the contract (the Solicitation or the Accepted Proposal), that constitutes a change to the contract terms. The Government reserves the right to accept or reject any such proposed variation.

1.4.3 For Information Only

Submittals not requiring Government approval will be for information only. For Design-build construction all submittals not requiring DOR or

Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.4.4 Sustainability Reporting Submittals (S)

Submittals for Guiding Principle Validation (GPV) or Third Party Certification (TPC) are indicated with an "S" designation. These submittals are for information only and for use as specified in Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING.

Schedule submittals for these items throughout the course of construction as provided; do not wait until closeout.

1.5 PREPARATION

1.5.1 Transmittal Form

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms are included in the RMS CM software that the Contractor is required to use for this contract. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Submittal Format

1.5.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.

b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format and native electronic format.

1.5.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 2 inches on the right-hand side of each sheet for the Government disposition stamp.

1.5.2.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.2.4 Format of SD-04 Samples

1.5.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

a. Sample of Equipment or Device: Full size.

b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.

c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.

d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.

e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.

f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.

g. Sample Panel: 4 by 4 feet.

h. Sample Installation: 100 square feet.

1.5.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.2.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3 Source Drawings for Shop Drawings

1.5.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.4 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. Compile the submittal file as a single, complete document, to include the Transmittal Form described within, and also separately attach the native files which were used to create PDF. The attached files should include the original digital files used to create the submittal. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the DOD SAFE Web Application located at the following website: https://safe.apps.mil/.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit two sets of administrative submittals.

1.6.2 Number of SD-04 Samples

a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.

b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.

c. Submit one sample installation, where directed.

d. Submit one sample of nonsolid materials.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe. For Design-Build construction, the Government will retain one copy of information-only submittals.

1.8 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is presented in Section 01 33 01 SUBMITTAL REGISTER

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM). The Government will provide the initial submittal register in electronic format with the following fields completed, to the extent that will be required by the Government during subsequent usage.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Design-Build Submittal Register

The Designer of Record develops a complete list of submittals during design and identify required submittals in the specifications, and use the list to prepare the Submittal Register. The list may not be all inclusive and additional submittals may be required by other parts of the contract. Complete the submittal register and submit it to the Contracting Officer for approval within 30 calendar days after Notice to Proceed. The approved submittal register will serve as a scheduling document for submittals and will be used to control submittal actions throughout the contract period. Coordinate the submit dates and need dates with dates in the Contractor prepared progress schedule. Submit monthly or until all submittals have been satisfactorily completed, updates to the submittal register showing the Contractor action codes and actual dates with Government action codes. Revise the submittal register when the progress schedule is revised and submit both for approval.

1.8.3 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.4 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.5 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.6 Action Codes

1.8.6.1 Contractor Action Codes

DESIGN BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS – The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA
BLANK	Submittal is For Information Only(FIO)	BLANK	FIO
DA	Submittal requires Designer of Record Approval	D	DA
CR	Submittal requires Government Conformance Review	C	CR
DA/CR	Submittal requires Designer of Record Approval and Government Conformance Review	R	DA/CR
DA/GA	Submittal requires Designer of Record Approval and Government Approval	A	DA/GA

1.8.7 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of 30 calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. Allow an additional 14 calendar days for review and approval of submittals for refrigeration and HVAC control systems.

a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 30 calendar days after the date of submission.

1.10.1 Government Reviewed Design

The Government will review design submittals for conformance with the technical requirements of the Solicitation. Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD) covers the design submittal and review process in detail. Government review is required for variations from the completed design. Review will be only for conformance with the contract requirements. Included are only those construction submittals for which the DOR's design documents do not include enough detail to ascertain contract compliance. The Government may, but is not required to, review extensions of design such as structural steel or reinforcement shop drawings.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

a. Note the date on which the submittal was received.

b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.

c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. One copy of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.

b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.

c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.

d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only thatthe general method of construction, materials, detailing, and other information are satisfactory. the design, general method of construction, materials, detailing, and other information appear to meet the Solicitation and Accepted Proposal.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained with in each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

______NAME OF CONTRACTOR ______ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION 01 33 01

SUBMITTAL REGISTER

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
BROKEN BOW ROOF REPLACEMENT - DESIGN BUILD

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 11 00	SD-01 Preconstruction Submittals															
			Shop Drawings	1.4.5.1	G													
			Detailed Work Plan	1.8	G	DO												
			Site Condition Photos	1.7														
			SD-07 Certificates															
			Manufacturer's Roof Warranty	1.11.1	G													
			Contractor Warranty	1.11.2	G													
		01 14 00	SD-01 Preconstruction Submittals															
			List of Contact Personnel	1.3.1.1														
		01 30 00	SD-01 Preconstruction Submittals															
			Progress and Completion	1.3														
			Pictures															
			Superintendent Qualifications	1.5.1	G													
			Red Zone Meeting Checklist	1.7.1	G													
		01 32 01.00 10	SD-01 Preconstruction Submittals															
			Project Scheduler Qualifications	1.3	G													
			Preliminary Project Schedule	3.4.1	G													
			Initial Project Schedule	3.4.2	G													
			Periodic Schedule Monthly	3.4.4	G													
			Update															
			Recovery Schedule	3.4.3	G													
			Weekly Progress Meeting	3.7	G													
			Minutes															
		01 33 00	SD-01 Preconstruction Submittals															
			Submittal Register	1.8	G													
		01 33 16.00 10	SD-01 Preconstruction Submittals															

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
BROKEN BOW ROOF REPLACEMENT - DESIGN BUILD

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION / GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	CONTRACTOR: SCHEDULE DATES APPROVAL NEEDED BY	CONTRACTOR: SCHEDULE DATES MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	CONTRACTOR ACTION DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	APPROVING AUTHORITY DATE RCD FROM OTH REVIEWER	APPROVING AUTHORITY ACTION CODE	APPROVING AUTHORITY DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 33 16.00 10	Design Quality Control Plan	1.5.1	G												
			Initial Design Conference	1.5.2.3													
			Preconstruction Conference	1.5.2.4													
			DCM Procedures	3.4.1	G												
			Submittal Register	3.7.4	G												
			SD-02 Shop Drawings														
			Small Scale Site Plan	1.5.2.2	G DO												
			Sketch Elevations	1.5.2.2	G DO												
			Proposed Materials Of	1.5.2.2	G DO												
			Construction														
			SD-05 Design Data														
			Design and Code Checklists	1.7.1	C												
			Interim Design Submittals	3.2.1.2	C												
			Conference Documentation	3.6.3													
			Final Design Submittals	3.2.2	R												
			Design Complete Documents	3.8	C												
			Concept Design Submittals	3.2.1.1	C												
		01 33 29	SD-01 Preconstruction Submittals														
			Preliminary High Performance	1.5.3.2	G												
			and Sustainable Building Checklist														
			Sustainability Action Plan	1.4.1	G												
			Preliminary Sustainability	1.5.3.2	G												
			eNotebook														
			SD-05 Design Data														
			Interim Design High Performance	1.5.3.2	G												
			and Sustainable Building Checklist														

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 33 29	Interim Design Sustainability eNotebook	1.5.3.2	G													
			Final Design High Performance and Sustainable Building Checklist	1.5.3.2	G													
			Final Design Sustainability eNotebook	1.5.3.2	G													
			SD-11 Closeout Submittals															
			Final High Performance and Sustainable Building Checklist	1.5.3.2	G													
			Final Sustainability eNotebook	1.5.3.2	G													
		01 35 26	SD-01 Preconstruction Submittals															
			Accident Prevention Plan (APP) Design	1.8.1	G													
			Accident Prevention Plan (APP) Construction	1.8.2	G													
			SD-06 Test Reports															
			Accident Reports	1.12.2	G													
			LHE Inspection Reports	1.12.3														
			Monthly Exposure Reports	1.5	G													
			SD-07 Certificates															
			Activity Hazard Analysis (AHA)	1.9	G													
			Certificate of Compliance	1.12.4														
			Hot Work Permit	1.13														
			Standard Lift Plan	1.12.4.2	G													
			Critical Lift Plan	1.12.4.2	G													
		01 45 00.00	SD-01 Preconstruction Submittals															

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 45 00.00	Contractor Quality Control (CQC) Plan	1.5.2	G													
			Additional Requirements For Design Quality Control (DQC) Plan	1.5.2.2	G													
			SD-05 Design Data															
			Design Quality Control	1.12.2														
			Discipline-Specific Checklists	1.5.2.2														
			SD-06 Test Reports															
			Verification Statement	1.12.3														
		01 50 00	SD-01 Preconstruction Submittals															
			Construction Site Plan	1.3	G													
			Traffic Control Plan	3.3.1	G													
			Haul Road Plan	2.2.1	G													
			Contractor Computer Cybersecurity Compliance Statements	1.6.1.4	G													
			Contractor Temporary Network Cybersecurity Compliance Statements	1.6.6	G													
			SD-06 Test Reports															
			Backflow Preventer Tests	3.4														
			SD-07 Certificates															
			Backflow Tester	1.4.1														
			Backflow Preventers	1.4														
		01 57 19	SD-01 Preconstruction Submittals															
			Preconstruction Survey	1.5.1														

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19	Regulatory Notifications	1.5.2													
			Employee Training Records	1.5.4													
			Environmental Protection Plan	1.6	G												
			Dirt and Dust Control Plan	1.6.9.1													
			Solid Waste Management Permit	1.9													
			Spill Prevention Control And	3.8.2	G												
			Countermeasure (SPCC) Plan														
			Waste Determination	3.5.1	G												
			Documentation														
			SD-06 Test Reports														
			Monthly Solid Waste Disposal	1.9.1	G												
			Report														
			SD-07 Certificates														
			Employee Training Records	1.5.4	G												
			SD-11 Closeout Submittals														
			Regulatory Notifications	1.5.2	G												
			Assembled Employee Training	1.5.4	G												
			Records														
			Solid Waste Management Permit	1.9	G												
			Project Solid Waste Disposal	3.5.2.1	G												
			Documentation Report														
			Sales Documentation	3.5.2.1	G												
		01 58 00	SD-02 Shop Drawings														
			Sign Legend Orders	1.4.1	G												
		01 74 19	SD-01 Preconstruction Submittals														

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 74 19	Construction Waste Management	1.6	G												
			Plan														
			SD-06 Test Reports														
			Quarterly Reports	1.8.2													
			Annual Report	1.8.3													
			SD-11 Closeout Submittals														
			Final Construction Waste	1.9	S												
			Diversion Report														
		01 78 00	SD-03 Product Data														
			Warranty Management Plan	1.8.1													
			Warranty Tags	1.8.5													
			Spare Parts Data	1.6													
			SD-08 Manufacturer's Instructions														
			Instructions	1.8.1													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.7	G												
			Manuals														
			SD-11 Closeout Submittals														
			As-Built Drawings	3.1	G												
			Record Drawings	3.3	G												
			As-Built Record of Equipment	3.6													
			and Materials														
			Warranted Equipment and	1.8.1													
			Materials														
			Final Approved Shop Drawings	3.4	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 00	Construction Contract	3.5	G												
			Specifications														
			Certification of EPA Designated	2.2	G												
			Items														
			Certification Of USDA Designated	2.3	G												
			Items														
			Interim DD FORM 1354	3.9.1	G												
			List Of Installed Building	3.9.2													
			Equipment Items For DD Form														
			1354														
		01 78 23	SD-10 Operation and Maintenance														
			Data														
			Facility Data Workbook	1.4	G												
			Training Plan	3.1.1	G												
			Training Outline	3.1.3	G												
			Training Content	3.1.2	G												
			Operation And Maintenance	3.2.1	G												
			Manual, Progress Submittal														
			Operation And Maintenance	3.2.2	G												
			Manual, Prefinal Submittal														
			Operation And Maintenance	3.2.3	G												
			Manual, Final Submittal														
			SD-11 Closeout Submittals														
			Training Video Recording	3.1.4	G												
			Validation of Training Completion	3.1.6	G												
			Training Plan	3.1.1	G												

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 23	Record Drawings And Utility Systems	1.6.6.7	G													
		02 41 00	SD-01 Preconstruction Submittals															
			Demolition and Deconstruction Plan	1.2.2	G													
			Existing Conditions	1.9														
			SD-07 Certificates															
			Notification	1.6	G													
			SD-11 Closeout Submittals															
			Receipts	3.2.4														

SECTION 01 33 16.00 10

DESIGN DATA (DESIGN AFTER AWARD)
05/16 - SWT 03/26

PART 1 GENERAL

1.1 SUMMARY

After award, develop the accepted proposal into the completed design, as described herein. Use a collaborative, integrated design process for all stages of project delivery with comprehensive performance goals for site development, energy, water, material selection, indoor environmental quality, and waste diversion. Ensure incorporation of these goals in project delivery. Consider all stages of the building lifecycle, including deconstruction, rehabilitation, re-purposing, or demolition.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 19005-3	(2012) Document Management -- Electronic Document File Format for Long-Term Preservation -- Part 3: Use of ISO 32000-1 with Support for Embedded Files (PDF/A-3)
ISO 32000-1	(2008) Document Management -- Portable Document Format -- Part 1: PDF 1.7

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-345-700	(2026) Engineering and Design -- Design Analysis, Drawings, Specifications, and Minimum Submittal Requirements
ERDC/ITL TR-19-6	(2019) A/E/C Graphics Standard, Release 2.1
ERDC/ITL TR-19-7	(2019) A/E/C CAD Standard - Release 6.1

1.3 DEFINITIONS

1.3.1 Designer of Record (DOR)

Professional Registered members of the Contractor’s Design-Build team that check, approve, sign, date, and certify, prior to submitting the deliverables to the Government, that the D-B design submittals comply with the contract requirements.

The DOR's stamp, sign, and date each design drawing and other design deliverables under their responsible discipline at each design submittal stage. The DOR(s) are responsible for maintaining the integrity of the design and for compliance with the contract requirements through construction and documentation of the as-built condition by coordination, review and approval of extensions of design, material, equipment and other construction submittals, review and approval or disapproval of requested deviations to the accepted design or to the contract, coordination with the Government of the above activities, and by performing other typical professional design responsibilities.

1.3.2 Government Furnished Material (GFM)

Government material that may be incorporated into, or attached to, an end item to be delivered under a contract or which may be consumed in the performance of a contract. It includes, but is not limited to, raw and processed material, parts, components, assemblies, and small tools and supplies.

1.3.3 Model Element

A self-contained graphical element with a unique identification that is used to populate a model, and whose behavior and properties are defined by facility/site data and software processes. Model elements can represent a physical entity, such as a pump, a concrete wall, or a utility vault and range from the simple to the complex and can be custom modified.

1.3.4 USACE Minimum Modeling Matrix (M3)

The USACE Minimum Modeling Matrix (M3) describes the minimum modeling and data requirements by defining the level of development (LOD) and element grade.

1.3.5 Facility Data

Non-graphical data attached to surface and subsurface components for both building and site model elements that describe various facility characteristics such as parametric values that drive physical sizes, material definitions (e.g. wood, metal), manufacturer data, industry standards (e.g. AISC steel properties), location, and project identification numbers. Facility data can also define supplementary physical entities that are not shown graphically in the model, such as the system of a duct, hardware on a door, content of conduit, site surface, alignment, levee, channel or transformer properties.

1.3.6 USACE CAD/BIM Technology Center

The USACE CAD/BIM Technology Center hosts all standard content for USACE. This content can be accessed through the CAD/BIM Technology Center website, https://cadbimcenter.erdc.dren.mil/.

1.4 ORDER OF PRECEDENCE

In the event of a conflict or inconsistency between any of the requirements within the Contract, precedence is applied:

a. Any portions of the accepted proposal which both conform to and exceed the requirements of the solicitation.

b. The provisions of the solicitation.

c. All other provisions of the accepted proposal.

d. Any design products including, but not limited to, plans, specifications, engineering studies and analyses, shop drawings, and equipment installation drawings. These are "deliverables" under the contract are not part of the contract itself. Design products must conform to all provisions of the contract, in the order of precedence.

1.5 PRECONSTRUCTION ACTIVITIES

1.5.1 Design Quality Control Plan

Submit a Design Quality Control Plan in accordance with Section 01 45 00.00 QUALITY CONTROL before design may proceed.

1.5.2 Meetings and Conferences

1.5.2.1 Post Award Conference

The Government will conduct a post award conference at the project site, as soon as possible after Contract award, coordinated with issuance of the notice to proceed (NTP). Participation by the Contractor and major subcontractor representatives is mandatory. All designers need not attend this first meeting. The government will provide an agenda, meeting goals, meeting place, and meeting time to participants prior to the meeting.

As a minimum the following will be addressed during the conference: determination and introduction of contact person and their authorities; contract administration requirements; discussion of expected project progress processes; and coordination of subsequent meeting.

a. The government will introduce the Government project delivery team members, facility users, facility command representatives, and installation representatives.

b. Introduce key personal, major subcontractors and other needed staff.

c. Define expectations and duties of each participant.

d. Develop a meeting roster with complete contact information including name, office, project role, phone, mailing and physical address, and e-mail address for distribution to all participants. Also, provide minutes of the meeting to all participants.

1.5.2.2 Schematic Design Submittal (5 percent Review)

The submittal consists of single-line type drawings showing the design team's interpretation of the User's stated needs. This is accomplished through presentation of different layout schemes to the User to review and select the one that best meets their functional needs. The submittal shall consist of the following unless otherwise specified:

a. Small Scale Site Plan indicating siting of building, parking, existing utility locations and ingress/egress to site and/or building.

b. Sketch Elevations to show scale and mass of the facility including story heights.

c. Proposed Materials of Construction.

1.5.2.3 Initial Design Conference

After Contract award, conduct the initial design conference, and provide a record of the meeting. All Designers of Record must participate in the conference. The primary purpose of the meeting is to make sure any needs are assigned and due dates established, as well as points of contact identified. The initial design conference may be scheduled and conducted at the project installation after the Post Award Conference and prior to initiation of significant preliminary design development, although it is recommended that the partnering process be initiated at the time of or before the initial design conference. Limit any design work conducted after award and prior to this conference to site work.

1.5.2.4 Pre-Construction Conference

Before starting any construction activities, jointly conduct an administrative conference with the Government to discuss any outstanding requirements and to review local installation requirements. It is possible there will be multiple Pre-Construction Conferences based on the configuration of the design packages. Provide minutes of the meeting(s) to all participants.

1.6 SUBMITTALS

Each submittal includes an associated approval level designation as defined in the following table:

Approval Level Designation	Definition
G	Government approval
no designation	for information only
D	Designer of Record approval
C	Government Conformance Review of Design
R	Designer of Record Approval and Government Conformance Review
A	Designer of Record Approval and Government Approval
S	inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING

When used, a designation following the approval level designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Design Quality Control Plan; G

Initial Design Conference

Preconstruction Conference

DCM Procedures; G

Submittal Register; G

SD-02 Shop Drawings

Small Scale Site Plan; G, DO

Sketch Elevations; G, DO

Proposed Materials Of Construction; G, DO

SD-05 Design Data

Design and Code Checklists; C

Interim Design Submittals; C

Conference Documentation

Final Design Submittals; R

Design Complete Documents; C

Concept Design Submittals; C

1.7 DESIGN QUALITY CONTROL

1.7.1 Design And Code Checklists

Develop and utilize appropriate discipline-specific checklists during the design and quality control of each submittal. Submit these completed checklists with each design submittal, as applicable, as part of the project documentation. See Section 01 45 00.00 QUALITY CONTROL and paragraph FIRE PROTECTION AND LIFE SAFETY CODE REVIEW for a sample Fire Protection and Life Safety Code Review checklist.

1.8 DELIVERY, STORAGE, AND HANDLING

1.8.1 Electronic Design Submittal

Provide identical copies of discs for approval, for each submittal required. Provide submittal files on electronic storage media in compliance with the quality requirements identified in this specification.

1.8.1.1 Malicious Content

Scan all electronic files for malicious viruses using commercially available scanning program that is routinely updated to identify and remove current virus threats.

1.8.1.2 Storage Media

Provide project data on disc-based (DVD±R/RW) media. Provide the full submittal on one single disc whenever possible. When separation of the submittal is required separate deliverables onto separate media. Document any media divisions in the PxP for approval by the Contracting Officer.

a. Directly print identification of contents onto storage media. Do not provide adhesived labels. Include the name of the submittal, project, project location, Contract number, Designer of Record firm/Prime Contractor company's name, title of submission, and security classification (in accordance with the applicable security classification labeling regulations) on the label. If multiple discs are provided, clearly document the contents of each disc on the label.

b. Include the name and contact information of the individual who produced the final data disc to ensure that any problems with the data or media can be easily resolved.

c. When browsed on any computer, the disc displays the following folders and their associated content:

(1) Submittal files (containing all submittal data)

(2) All supporting documents associated with the submittal

(3) Readme containing one TXT, PDF, or HTML file with general use information, organizational instructions, and basic preparer contact information.

1.8.2 PDF File Packaging

Utilize PDF file format in accordance with ISO 32000-1 and ISO 19005-3. Provide files from original sources, text-searchable, and saved in "Standard" (uncompressed) resolution whenever possible.

1.8.2.1 Bookmarking

a. Bookmark drawing submittal PDF sets to include one Parent Bookmark per Discipline and one Child Bookmark per sheet within each Discipline. Format Parent Bookmarks as "Discipline" (e.g. Architectural). Format Child Bookmarks as "Sheet ID Sheet Title" (e.g. A-101 First Floor Plan).

b. Bookmark specification submittal PDF sets using the SpecsIntact Print Processing PDF Print/Publish feature, combining processed sections into one PDF document. Insert the Submittal Register into the file where specified by Section 01 33 00 SUBMITTAL PROCEDURES and bookmark.

c. Bookmark design analysis and calculation submittal PDF sets to include one Parent Bookmark per design analysis section and one Child Bookmark per major paragraph per section. Format Parent Bookmarks as "Section" (e.g. Architectural). Format Child Bookmarks as "major paragraph designation Sheet Title" (e.g. 2.1 Primary Facility Functions).

1.8.2.2 Hyperlinking

Hyperlink all reference annotation symbology (e.g. section cut symbology, detail callout symbology, elevation callout symbology) to the sheet referenced by the annotation.

1.8.3 Encryption

Encrypt deliverable data as directed by Resident Office Engineer. Document the encryption in the PxP.

PART 2 PRODUCTS

2.1 DESIGN DRAWINGS

From advanced model files, produce design drawings that describe the scope of the Contract for all required submittals including all interim and final deliverables.

2.1.1 Electronic Drawing Files

Provide electronic drawing files in PDF format for each project drawing in the design set.

2.1.2 Drawing Index

Provide an index of drawings sheet as part of the drawing set, and an electronic table of all drawings submitted. Include the electronic file name, the sheet reference number, the sheet number, and the sheet title containing the data for each drawing.

2.1.3 Shop Drawings Used as Design Drawings

Design drawings may be prepared similar to shop drawings to minimize construction submittals after the Design Complete Submittals. Prepare and submit with the design drawings, appropriate connection, fabrication, layout, and product specific drawings.

2.1.3.1 Drawing Format For Shop Drawings Used as Design Drawings

Use the Contractor-originated drawings as the basis for the record drawings. Conform shop drawings included as design documents with the same drawing requirements such as drawing format, sheet size, layering, lettering, and title block used in design drawings.

2.1.3.2 Identification of Shop Drawings Used as Design Drawings

Indicate which shop drawings are being submitted as design drawings in the transmittal letter.

2.1.4 Seal on Documents

Sign, date and seal all Contractor-originated design drawings by the registered architect or the registered engineer of the respective discipline. This is the seal of the Designer of Record for that drawing. Application of the electronic seal and signature accepts responsibility for the work shown thereon.

2.2 SPECIFICATIONS

Provide design specifications in accordance with ER 1110-345-700 (Attachment A of this Section).

2.2.1 Specification Deliverable

Submit a bundled specification package in PDF format for each design package. As a minimum, bookmark each specification section in the bundled package. Also, submit the source files, in the processing system format, used to create the PDF.

2.3 DESIGN ANALYSIS

Prepare, organize, and present a design analysis in accordance with the ER 1110-345-700 (Attachment A of this Section).

2.3.1 Design Requirements and Provisions

Include subparts for each major design discipline and basic project design requirements for each discipline that justify and validate design decisions to include, but not limited to: life cycle cost effectiveness.

2.3.1.1 Civil

Include soil analysis and survey data, site design, site improvements, planting and landscaping, paving, grading and drainage, water, waste-water and soil treatment, contaminant containment, utilities systems analysis and design, and provisions for airfields, ports and railroads, if required.

2.3.1.2 Architectural

Include space allowance, functional layout, unique features, interior design ,furniture planning, signage, accessibility, security, air barriers, energy conservation and sustainable design to include site analysis focusing on orientation, space-mass composition, materials used and details with respect to image, safety, maintenance and cost effectiveness and historical context.

2.3.1.3 Structural

Include foundation, structural, and seismic analysis and design.

2.3.1.4 Mechanical

Include heating, ventilation and air conditioning systems, refrigeration, plumbing, elevators and cranes, energy conservation, pollution control, noise and vibration control, heating and chilled water distribution, gas distribution, fuel storage and dispensing, and process systems design.

2.3.1.5 Electrical

Include power generation, transmission and distribution systems, lighting (interior and exterior), voice and video communications, intrusion detection, utilities monitoring control systems (UMCS), cathodic protection, lightning and static electricity protection systems analysis and design, aviation lighting, and electromagnetic protection

2.3.1.6 Fire Protection and Life Safety

Include building construction, exit requirements, fire extinguishing systems, fire protection water supplies, surge analysis, and alarm and detection systems analysis and design.

2.3.1.7 Physical Security

Include fencing, vaults, protective lighting, security systems, locks, arms rooms, controlled substances, entrances, guard facilities, classified material, patrol roads, clear zones, restricted areas, surveillance and penetration resistance.

2.3.2 Operations and Maintenance (O&M) Provisions

Identify design provisions made to enhance and to reduce the cost of operating and maintaining the facility when completed. Identify any special safety considerations or occupational health related considerations that may affect operation and maintenance activities as a result of the final design.

2.3.3 Design Analysis Packaging

2.3.3.1 Assembly and Identification

Assemble design analysis in a single volume with a table of contents if possible. Include a cover page in the basis of design for each discipline indicating the project title and locations, contract number, table of contents, and tabbed separations or bookmarks for quick reference. At a minimum tab or bookmark for each discipline.

2.3.4 Calculations

Place the signature and seal of the designer of record responsible for the work on the cover page of the calculations for the respective design discipline.

PART 3 EXECUTION

3.1 DESIGN SUBMITTALS

Include all deliverable products and associated support documents described in Part 2 of this specification with each design submittal.

3.2 DESIGN SUBMITTALS PHASES

The stages of design submittals described below define requirements with respect to process and content. Determine how to best plan and execute the design and review process for the project, within the parameters listed below. As a minimum, provide at least one concept (35%) design submittal, one interim (65%) design submittal, at least one final (95%) design submittal before construction of a design package may proceed, and at least one 100% Design Complete submittal that documents the accepted design.

In addition to the requirements below, ensure design submittals meet the requirements for the phases as defined in the ER 1110-345-700, as applicable. Notify the Contracting Officer in the case of conflict between the requirements stated here and the ER 1110-345-700.

3.2.1 Conceptual and Interim Design Submittals

3.2.1.1 Concept Design Submittals

A Design Analysis as outlined in ER 1110-345-700, including but not limited to a written narrative to define the project and basis of design, functional and engineering criteria, preliminary design analysis computations, concept drawings, a specifications table of contents, and preliminary estimated costs.

Schematics/Sketches per ER 1110-345-700 are single-line drawings to scale representing 5 to 10 percent project design development. They include but are not limited to basic site plan (including orientation and contours), floor plan (with overall dimensions), gross area, two elevations and a section.

Include all discipline requirements outlined in ER 1110-345-700, as applicable.

3.2.1.2 Interim Design Submittals

After acceptance of the concept design package, revise the conceptual design package to incorporate the comments generated and resolved, perform and document a back-check review and submit the interim design package that represents a complete package with all design disciplines. This is not necessarily a hold point for the design process; the Contractor may designate the interim design submittal(s) as a snapshot and proceed with design development at its own risk.

3.2.1.3 Interim Design Development Management

Maintain a fully functional configuration management system as described herein to track design revisions, regardless of whether or not there is a need for a formal interim design development review.

3.2.1.4 Over-the-Shoulder Progress Review

To facilitate a streamlined design-build process, the Government and the Contractor may agree to one-on-one review or small group reviews, on-line, or at the Contractor's design offices or other agreed location, when practicable to the parties. Coordinate such reviews to minimize or eliminate disruptions to the design process. Due to limits on project funding, utilize the maximum virtual teaming methods. Facilitate these reviews with electronic format data transfer and collaboration. Through the partnering process, find ways to facilitate the quality assurance process and to facilitate meeting or bettering the design-build schedule.

3.2.1.5 Interim Design Development Review Waiver

The Government may agree to shorten or waive the formal interim design development review period for design package(s) if an effective, mutually agreeable partnering procedure is established and implemented for regular (e.g., weekly) over-the shoulder review. During the course of the procedure, keep the Government reviewers fully informed of the progress, contents, design intent, design documentation, and other pertinent factors of the design package.

3.2.2 Final Design Submittals

After acceptance of the interim design package, revise the design package to incorporate the comments generated and resolved, perform and document a back-check review and submit the final design package.

Refer to the paragraph FINAL DESIGN REQUIREMENTS below for submittal details.

3.2.3 Design Complete Submittals

After the final design submission and review conference for a design package, revise the design package to incorporate the comments generated and resolved in the final review conferences, perform and document a back-check review and submit the final, design complete documents, which represents released for construction documents.

3.3 DESIGN PLATFORM AND FILE FORMATS

Design the project using the systems and platforms defined below:

3.3.1 CAD

3.3.1.1 Native CAD Authoring Content

All content produced through CAD authoring software outside of any object/element based BIM or CIM platform must be compliant with ERDC/ITL TR-19-6 and ERDC/ITL TR-19-7. Autodesk AutoCAD Template Files Most recent version at the time of Contract award. Download form the CAD/BIM Technology Center website as part of the A/E/C Work Structure.

3.4 DESIGN CONFIGURATION MANAGEMENT (DCM)

3.4.1 Procedures

Develop and maintain effective, DCM procedures to control and track all revisions to the design documents subsequent to the Interim Design Submission and continuing through submission of the As-Built documents. After the final design is accepted, this process provides control of and documents revisions to the accepted design (See Special Contract Requirement: Deviating From the Accepted Design). Submit the DCM procedures within the Design Quality Control Plan.

a. Include authorities and concurrences in the DCM system to authorize revisions, including documentation as to why the revision is required.

b. An internal system may be used with interactive Government concurrences or the Government's "Dr Checks Design Review and Checking System" may be used.

c. Make the DCM data available to the Government reviewers at all times.

3.4.2 Tracking Design Review Comments

Although an internal system for overall design configuration management is allowed, use the DrChecks Design Review and Checking System to initiate, respond to, resolve and track Government design review comments.

The Government will set up the project in DrChecks. Throughout the design process parties enter, track, and back-check comments using the DrChecks system. Designers of Record annotate comments timely and specifically to indicate exactly the action to be taken or why the action is not required. After the design review conference and prior to the next design submittal for the package, the DORs annotate those comments that require DOR action or design revision to show how and where it has been addressed in the design documents. These procedures are part of the required design configuration management plan. Flag comments considered critical by the conference participants.

3.4.2.1 DrChecks Initial Account Set-Up

Identify a contact person within the office to act as the administrator for all Contractor personnel, including subcontractors, that will be accessing the PROJNET Dr Checks system. Through the Contracting Officer, coordinate with the Project Manager and the District PROJNET administrator for system access, system instruction and comment process instructions.

PROJNET contains an introductory file and other tutorial material that can be accessed once user accounts have been established. Upon log in, select Portals/User Documentation.

3.4.2.2 DrChecks Review Comments

Annotate and resolve all comments prior to the next submittal. Include the DrChecks comments and responses in the design analysis for record in the next design submittal for the package.

a. Upon review of comments prior to the design review conference, the DOR(s) evaluate the comments. Include exactly what action will be taken or why action is not required.

b. After the review conference, the DOR(s) formally respond to each applicable comment in DrChecks a second time, prior to the next submittal, clearly indicating what action was taken and what drawing/spec/analysis changed. Designers of Record are encouraged to directly contact reviewers to discuss and agree to the formal comment responses rather than relying only on DrChecks and review meetings to discuss comments. With the next design submittal, reviewers will back-check answers to the comments against the new submittal, in addition to reviewing additional design work.

c. Clearly annotate in DrChecks those comments that require effort outside the requirements of the contract. Do not proceed with work outside the contract until a modification to the contract is properly executed.

3.5 DISCIPLINE DESIGN REQUIREMENTS

Provide interim design deliverables that comply with requirements of ER 1110-345-700.

3.6 INTERIM DESIGN REQUIREMENTS

At least one interim design submittal, review and review conference is required for each design package (except that the Contractor may, upon Government approval, skip the interim design submission and proceed directly to final design of the sitework and utilities package). Additional interim design conferences or over-the-shoulder reviews may be scheduled, as needed, to assure continued Government concurrence with the design work. Include the interim submittal review periods and conferences in the Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT and indicate in periodic schedule updates what part of the design work is at what percentage of completion. See also paragraph INTERIM DESIGN DEVELOPMENT REVIEW WAIVER for a waiver to the formal interim design review.

3.6.1 Submission Review

After receipt of an Interim Design submission, the Government requires 30 calendar days after receipt of the submission to review and comment on the interim design submittal. For smaller design packages, especially those that involve only one or a few separate design disciplines, the parties may agree on a shorter review period or alternative review methods (e.g., over-the -shoulder or electronic file sharing), through the partnering process.

a. For each interim design review submittal, the Contracting Officer will furnish a single consolidated, validated set of comments from the various design sections and from other concerned agencies involved in the review process using the DrChecks Design Review and Checking System. The review will be for conformance with the technical requirements of the Contract.

b. The Government reserves the right to reject design document submittals if comments are deemed significant.

c. Furnish disposition of all comments, in writing, through DrChecks. If there are technical disagreements with any comments, clearly outline, with justification, the reasons for disagreement and noncompliance within five calendar days after receipt of these comments.

d. The Contractor is cautioned that if it believes the action required by any comment exceeds the requirements of this contract, that it should take no action and notify the Contracting Officer in writing immediately.

3.6.2 Interim Review Conference

Hold an Interim Review conference for each design submittal at either the installation or as agreed upon as part of the partnering process. Attendees include, at a minimum, the DOR(s) involved in development of the design submittal. Schedule the conference to take place the week after the receipt of the comments. Notify the Contracting Officer of any comments that with concurrence would require further design development.

For smaller fast-track packages that involve only a few reviewers, the parties may agree to alternative conferencing methods, such as teleconferencing, or televideo, where available, as determined through Partnering.

3.6.3 Conference Documentation

3.6.3.1 Minutes and Comment Process

Provide meeting minutes within two work days after the conference adjourns, and enter final resolution of all comments into DrChecks. Include copies of comments, annotated with comment action agreed on, with the minutes.

a. Resolve issues remaining open after the conference adjourns by immediate follow-on action to close the issue within 30 calendar days.

b. Incorporate comments as agreed upon during the conference.

3.6.3.2 Availability

In order to facilitate the Government code and contract conformance reviews, identify, track resolution of, and maintain all comments and action items generated during the design review process. Make this available to the designers and reviewers prior to the subsequent design reviews.

3.7 FINAL DESIGN REQUIREMENTS

Provide final design submittals that consist of 100 percent complete drawings, specifications, submittal register, design analyses for Government review and acceptance.

a. Include any permits required by the contract for each package submitted.

b. In order to expedite the final design review, prior to the conference, ensure that the design configuration management data and all review comment resolutions are up-to-date.

c. Perform independent technical reviews and back-checks of previous comment resolutions, as required by Section 01 45 00.00 QUALITY CONTROL.

3.7.1 Design Drawings

Submit drawings complete with all contract requirements incorporated into the documents to provide a 100 percent design for each package submitted. In addition to all native Advanced Modeling files, provide separate electronic files in a PDF format.

3.7.1.1 Geo-Referenced Data

Capture geo-referenced coordinates of all changes that will be made to the existing site (facility footprint, utility line installations and alterations, roads, parking areas, etc) as a result of this contract.

Close-out requirements at the as-built stage, require final geo-referenced GIS Database of the new facility along with all exterior modifications. The Government will incorporate this data set into the Installation's GIS Masterplan or Enterprise GIS System. See also, Section 01 78 00 CLOSEOUT SUBMITTALS.

3.7.2 Design Analysis

Provide a design analysis with calculations necessary to validate and support all design work submitted. Expand and advance calculations and information presented in the interim design stage to the current level of design. The responsible DOR(s) stamp, sign and date the design analysis.

3.7.3 Specifications

Provide specifications 100 percent complete and in final form.

3.7.4 Submittal Register

Provide an updated, cumulative submittal register with each design package that identifies the design and construction submittals required by that design package.

3.8 DESIGN COMPLETE CONSTRUCTION DOCUMENT REQUIREMENTS

After the Final Design Submission and Review Conference, revise the design documents for the design package to incorporate the comments generated and resolved in the final review conference. Perform and document a back-check review and submit the final, design complete documents. The deliverable includes all documentation and supporting design analysis in final form, as well as the final review comments, disposition and the back-check. As part of the quality assurance process, the Government may perform a review of the released for construction documentation. Promptly correct any errors or omissions found during the Government review.

3.9 ACCEPTANCE AND RELEASE FOR CONSTRUCTION

After acceptance of the Design Complete Construction Document(s) the Contracting Officer will allow construction to start for that design package.

Government review and acceptance of design submittals is for contract conformance only and does not relieve the Contractor from responsibility to fully adhere to the requirements of the contract, including the Contractor's accepted proposal, or limit the Contractor's responsibility of design as prescribed under Special Contract Requirement: "Responsibility of the Contractor for Design" or limit the Government's rights under the terms of the contract. The Government reserves the right to rescind inadvertent acceptance of design submittals containing contract deviations not separately and expressly identified in the submittal for Government consideration and approval.

-- End of Section --

ATTACHMENT A -

SECTION 01 33 16.00 10

ER 1110-345-700 - Engineering and Design

Design Analysis, Drawings, Specifications, and Minimum Submittal Requirements

Department of the Army
U.S. Army Corps of Engineers
Washington, DC
4 February 2026

Engineer Regulation* 1110-345-700

Effective 4 March 2026

CEEC

Engineering and Design
Design Analysis, Drawings, Specifications, and Minimum Submittal Requirements

FOR THE COMMANDER:

DAMON A. DELAROSA
COL, EN
Chief of Staff

Purpose. This engineer regulation provides U.S. Army Corps of Engineers-wide consistency in criteria and requirements for developing design analysis, drawings, and specifications necessary for facility project design, estimating, approval, contracting, and construction.

Applicability. This regulation applies to all U.S. Army Corps of Engineers elements, major subordinate commands, district commands, technical centers, laboratories, and field operating activities. This regulation applies to all facility design and construction efforts within the Directorate of Military Programs regardless of funding or location. This includes, but is not limited to, Military Construction regardless of service; host nation-funded design and construction in outside the continental United States locations; sustainment, restoration, and modernization regardless of service; and vertical design and construction undertaken according to ER 1140-1-211.

Distribution Statement. Approved for public release; distribution is unlimited.

Proponent and Exception Authority. The proponent of this regulation is the Headquarters, U.S. Army Corps of Engineers, Directorate of Engineering and Construction. The proponent has the authority to approve exceptions or waivers to this regulation that are consistent with controlling law and regulations. Only the proponent of a publication or form may modify it by officially revising or rescinding it.

*This regulation supersedes ER 1110-345-700, dated 30 May 1997.

Summary of Change

ER 1110-345-700
Design Analysis, Drawings, Specifications, and Minimum Submittal Requirements

This revision, dated 4 February 2026:

- Resolves and deconflicts overlapping terms and definitions for Basis of Design, Design Analysis, Owner's Project Requirements, and stakeholder requirements between Department of Defense, U.S. Army Corps of Engineers, and non-governmental commissioning and sustainability authority documents.
- Identifies the multiple purposes and functions of the design analysis and its component parts.
- Outlines the design analysis development process and clarifies the format, content, and use of its components.
- Prescribes the requirements for the preparation and approval of drawings.
- Prescribes the requirements for the preparation of specifications.
- Provides recommended minimum submittal requirements for facility design.
- Incorporates policy related to signature on design documents that was previously included in ER 1110-1-8152.
- Incorporates all policy from ER 1110-1-8155.
- Updates the list of references.
- Adds reference to Whole Building Design Guide for design analysis sample content and templates.
- Adds the Glossary of Terms.

Contents

Appendixes

Table List

Chapter 1
Introduction

1–1. Purpose

This engineer regulation provides U.S. Army Corps of Engineers-wide consistency in criteria and requirements for developing design analysis, drawings, and specifications necessary for facility project design, estimating, approval, contracting, and construction.

1–2. Distribution statement

Approved for public release; distribution is unlimited.

1–3. References

See Appendix A.

1–4. Records management (recordkeeping) requirements

The records management requirement for all record numbers, associated forms, and reports required by this publication are addressed in the Army Records Retention Schedule. Detailed information for all related record numbers is located on the U.S. Army Corps of Engineers (USACE) Records Management Site https://usace.dps.mil/sites/INTRA-CIOG6/SitePages/Records-Management.aspx. If any record numbers, forms, and reports are not current, addressed, and/or published correctly, see DA Pam 25-403 for guidance.

1–5. Associated publications

This section contains no entries.

1–6. Design analysis

A design analysis (DA) is a written document developed by the designer of record (DOR) that describes the project at initiation and is updated at each subsequent phase. It presents facts to demonstrate that the concept of the project is understood and that the design is based on sound engineering principles and judgement. It documents the history and criteria for the project, including any criteria designated by the stakeholder, correspondence, codes, references, meeting minutes, and research. The DA addresses the entire facility and site design. The justification for each major design decision must be clearly stated. The DA documents must conform to the stakeholder requirements in EM 5-1-11. Coordinate the DA with the development of the Project Management Plan (PMP). Refer to Chapter 2 for detailed DA requirements.

a. Sequence of development. Initial preparation of the DA follows the development of the PMP and is updated at each project phase. It is prepared according to AR 420-1 and applicable Unified Facilities Criteria (UFC). The initial preparation of the DA may be referred to as the basis of design (BOD) as it establishes the design requirements for the project. Subsequent submittals will build on these design requirements to document how the design complies with the established criteria. The DA is a living document during project design and construction. It must be updated as appropriate to reflect design progress or changes in project status, conditions, or requirements.

b. *Contract document status*. The DA informs and documents the work of the DOR. The DA is not considered a contract document for solicitation of construction services; however, it may be included as an informational attachment when appropriate. The DA, if completed, may be used as part of contract documents in solicitations for Architect-Engineer (A-E) services.

c. *Proprietary product or brand-specific information*. Proprietary product or brand-specific information is prohibited in all documents included in a solicitation unless a justification and approval (J&A) has been prepared and authorized. If any portions of the DA are included in the solicitation, then all proprietary or brand-specific information that is not covered by a J&A must be removed prior to its inclusion in the solicitation. Proprietary information may be included in updates to the DA provided to the contractor after contract award.

1–7. Drawings

A drawing is a static graphic representation of design content formatted as a two-dimensional sheet. Refer to Chapter 3 for detailed drawing requirements.

a. *Preparation*. Prepare drawings according to the latest version of the Architecture/Engineering/Construction (A/E/C) Graphics Standards (ERDC/ITL SR-23-1) and A/E/C Computer-Aided Design (CAD) Standard (ERDC/ITL TR-19-7).

b. *Standard drawings*. Standard drawings are drawings that are not created for a specific project and are available to be adapted with project-specific information. They are not contract documents or project drawings in terms of this regulation. They include the following drawing types:

(1) Adapt-build drawings provided by a Center of Standardization (COS).

(2) Non-government standard drawings.

c. *Drawing types*. Project drawings include the following drawing types:

(1) Parametric drawings and sketches (see AR 420-1).

(2) Concept drawings (see AR 420-1).

(3) Design drawings.

(4) Construction drawings, also known as solicitation or contract documents.

(5) Amendment drawings.

(6) Conformed drawings.

(7) Modification drawings.

(8) Shop drawings.

(9) Record and as-built drawings.

d. Drawings prepared by the contractor. This regulation does not govern drawings prepared by the contractor in fulfillment of construction responsibilities such as shop drawings, record drawings, or as-built drawings.

1–8. Specifications

A specification is a written document describing in detail the scope of work, materials to be used, and quality of workmanship for work to be placed under contract. Refer to Chapter 4 for detailed specification requirements.

a. Preparation. Prepare specifications according to Unified Facilities Supplements (UFS) 1-300-02 and Chapter 4.

b. SpecsIntact. The use of SpecsIntact and the Unified Facilities Guide Specifications (UFGS) database is mandatory for production of all project specifications, except for overseas projects designed to host nation standards.

1–9. Minimum submittal requirements by phase

Refer to Chapter 5 for minimum submittal requirements by phases. Project teams may modify the minimum submittal requirements to align with project scopes. Project teams must record the minimum submittal requirements by phase in the Quality Management Plan (QMP) as part of the PMP, per EM 5-1-11.

Chapter 2
Design Analysis Requirements

2–1. General

This chapter prescribes the requirements for the preparation and approval of the DA. It applies to DAs prepared at all phases of design. For detailed requirements related to quality control reviews of the DA, refer to ER 1110-3-12.

a. For design-bid-build (D-B-B) projects, prepare a DA for the following submittals for all applicable projects: parametric design (5% to 15%), concept design (35%), interim design (65%), final design (95%), corrected final (100%), and ready to advertise (RTA).

b. For design-build (D-B) request for proposals (RFP) prepare a DA for the following submittals for all applicable projects: draft RFP, final RFP, corrected final RFP, and RTA RFP.

c. This regulation does not describe submittals by the D-B construction contractor.

d. Refer to AR 420-1 for more information on design codes.

e. Develop the DA in coordination with the installation and stakeholders.

f. Prepare the DA in a format appropriate for:

(1) Review, approval, and recordkeeping purposes.

(2) Coordination of design modifications during construction.

(3) Incorporation of stakeholder requirements documentation from the PMP.

(4) Use of Owner's Project Requirements (OPR) documentation for commissioning and sustainability purposes.

(5) Operations and maintenance (O&M) purposes.

(6) Post-occupancy evaluation.

g. The information provided in this chapter is geared toward typical military construction (MILCON) projects and can be adapted for all other project types, or as needed by project teams to align with project scopes.

2–2. Organization and content

The DA is organized into three parts: Part 1 – General Description, Part 2 – Design Requirements and Provisions, and Part 3 – O&M Provisions.

a. *Part 1: General description.* Organize Part 1 to facilitate coordination with stakeholder requirements documentation in the PMP and with OPR for commissioning. Include the Table 2-1**.** components in Part 1 of the DA as well.

Table 2-1
General description design analysis components

Component	Description
Purpose	State the functional purpose.
Authorization and directives	Include applicable design directives. Describe the ownership of the facility, site, and installation, and describe the stakeholder structure identifying decision-making responsibilities and the authority having jurisdiction.
Scope of work	Include the Department of Defense “DD” Form 1391, current working estimate (CWE), planning charrette documents, and other governing scoping documents, when available, as attachments in the appendices. Include the record of decision or other decision document for hazardous, toxic, and radioactive waste (HTRW) projects.
General criteria	List of overarching criteria that apply to the facility and site design. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
Project description	Include the functional objective, performance requirements, personnel and equipment, facility operational requirements, constructability, master plan compliance, economic summary, and accessibility.

b. *Part 2: Design requirements and provisions*. Include subparts for each major design discipline and basic project design requirements with justifications for the design decisions, assumptions, feature and system narratives, and calculations applicable to the project.

(1) *Environmental*. Include the Table 2-2 components in the Environmental DA as applicable to the scope of work.

Table 2-2
Environmental design analysis components

Component	Description
Environmental impact assessment	Include an environmental impact assessment checklist covering air, water, and noise effects from the project and construction.
Health and Safety	Describe requirements for worker health and safety specific to a project or location.
Hazardous, toxic, and radioactive waste (HTRW)	Provide HTRW assessment, remediation cleanup, and action levels. Include all applicable regulatory, chemical sampling/analytical, safety and occupational health, geotechnical, cost engineering, and process engineering provisions and criteria required by EM 200-1-2.
Transportation and disposal	Provide transportation and disposal regulation requirements.
Chemical sampling/analysis	Describe quality control for chemical sampling/analysis.

Component	Description
Wetlands	Provide wetlands determination (tidal and non-tidal), special wildlife, plant, and endangered species considerations.
Water protection	Provide groundwater, waterway, and floodplain protection assessment.
Pollution prevention	Provide pollution prevention control requirements and design measures to be implemented (such as construction site sediment and erosion control requirements by federal, state, and local governments).
Permits	Describe required hazardous material management, natural and cultural resources, and environmental permits.

(2) *Geotechnical*. Include the geotechnical data report, regional and site-specific geology, groundwater elevations, geotechnical data, seismic design parameters, geotechnical design considerations, foundation and earthwork recommendations, soil infiltration rates, pavement design, soil corrosivity and conductivity, geotechnical engineering analysis, and special site considerations.

(3) *Civil*. Include survey data, site design, antiterrorism site requirements, site improvements, pavement design, grading and drainage, low-impact development, water, wastewater, permits, contaminant containment, utilities systems analysis and design, and provisions for airfields, ports, and railroads, if required.

(4) *Landscape*. Include planting and landscaping, hardscape, irrigation systems, site furnishings, natural resource mitigation, and recreational site features.

(5) *Structural*. Include overall discussion on applicable codes and standards, building structure, structural materials, structural load path, loading criteria (gravity loads, lateral loads, and environmental loads), risk category, foundation design criteria, retaining walls, identification of any deferred design elements, and any special design requirements such as seismic, tornado, hardened structure design, vibration, progressive collapse, and conveying equipment. Include analysis of all code-compliant major structural materials, including at least one option where mass timber is a substantial structural component, and a justification for the structural system selected.

(6) *Architectural*. Include the Table 2-3 components in the Architectural DA.

Table 2-3
Architectural design analysis components

Component	Description
General Criteria	List all criteria that apply to architectural design. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
Scope of work	Summarize the architectural program or scope of work, listing the overall square footage, the functional requirements of the facility, and a tabulation of rooms with square footages of each space.
Type of construction	Describe the type of construction selected and justify its use relative to building permanency, life cycle cost, functionality, and fire resistance.

Component	Description
Gross floor area calculations	Provide a tabulated breakdown of the net and gross programmed areas to confirm project scope and statutory criteria compliance. Provide a supplemental drawing keyed to the area tabulation.
Accessibility	Describe accessibility features included in the project. If a partial or full exclusion to accessibility requirements has been or is being pursued, indicate proper documentation and the status of the exclusion.
Architectural compatibility	Identify the design guidelines that pertain to this project and describe how the proposed design incorporates these guidelines. Discuss the approach to achieving architectural compatibility with other surrounding architecture according to the installation exterior architectural guidelines. Exterior color boards are required for all projects. For all other service projects, refer to service-specific facilities standards and applicable installation facilities standards.
Roof system selection	Indicate the construction of the roof, roof membrane selection, substrate, roof slope, and roof drainage system.
Energy conservation	Identify the American Society of Heating, Refrigerating, and Air-conditioning Engineers (ASHRAE) Standard 90.1 climate zone and describe the types of insulation to be provided including specific “U” values for the wall, roof, and floor construction. Provide a description of all architectural energy conserving and generating features, including any passive solar systems. Discuss air barrier components and testing requirements. Provide a moisture vapor analysis.
Security requirements	Describe the physical security or hardening requirements such as controlled access, sensitive compartmented information facility (SCIF), and secure room requirements that will be used in the design.
Antiterrorism	Describe the occupancy of the facility, if the facility is within a controlled perimeter, and what the standoff distances will be. Include sketches as required to depict the site of the project and standoff distances. Include a summary of how the facility meets each of the applicable Standards in UFC 4-010-01 and geographic combatant commander antiterrorism construction standards. Outline any special requirements, including any requirement for hardening of the facility.
Architectural acoustics	Identify design team members responsible for the acoustical engineering. Provide a detailed identification of conditions, materials, or features that will impact the acoustic design of the project. Describe testing, mock-up, commissioning, and quality control processes.
Sustainable design	Include the architectural narrative related to sustainable design features in the “Sustainable Design” chapter. Describe the overall sustainability and energy performance of the project, with the architectural portion leading the process of compliance.
Doors and windows	Indicate the types of doors and windows selected for the project and explain the basis for their selection. Indicate special door requirements such as sound transmission class (STC) ratings, cipher locks, and special window requirements such as outdoor-indoor transmission class ratings.
Demolition or deconstruction	Describe the extent of any architectural demolition or deconstruction and items to be salvaged.

Component	Description
Special construction features	Describe the special construction features built into the facility.
Commissioning	Identify architectural systems scheduled for commissioning as required by the commissioning plan.

(7) *Interior design*. Include the Table 2-4 components in the Interior Design DA.

Table 2-4
Interior design analysis components

Component	Description
General criteria	List all criteria that apply to interior design. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
Space utilization	Describe the BOD as it relates to the user's requirements and building design.
Wayfinding	Describe the BOD and wayfinding elements used as it relates to the user's requirements and building design. Describe architectural, building-related finish, and signage strategies used in the overall wayfinding plan.
Structural interior design (SID)	Describe the BOD, building-related finishes, and furniture floor plan development and features as it relates to the user's requirements and building design. Include topics that relate to installation standards, life safety, aesthetics, durability, and maintenance. A SID binder must be provided as an appendix. Reference 2–2.d(6) for detailed SID binder requirements.
Furniture, fixtures, and equipment (FF&E)	Describe the design objectives as they relate to the selection of furniture, finishes, and colors; the furniture plan development and features; and how project-specific requirements are met. Identify the procurement method and format of FF&E package government furnished government installed (GFGI), contractor furnished contractor installated (CFCI), government furnished contractor installed (GFCI) or a combination. Include design decisions made to fully coordinate the SID and the FF&E design, including function, safety and ergonomic considerations, durability, sustainability, and aesthetics. As required, the FF&E package must be provided as an appendix. Reference 2–2.d(7) for detailed FF&E package requirements.

(8) *Fire protection and life safety*. Projects requiring the services of a qualified fire protection engineer (QFPE) must have a fire protection DA that includes the components listed in Table 2-5. Provide both the code requirement and the proposed design as part of any building code or life safety analysis.

Table 2-5
Fire protection design analysis components

Component	Description
General criteria	List all criteria that apply to fire protection. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
Specific hazards	Identify hazardous areas (such as chemicals, fuels, and ordnance) with maximum allowable quantities allowed and quantities that will be stored, processes, and special hazards or features requiring special fire protection considerations, such as radio frequency shielded rooms, secured rooms, computer rooms, and commercial kitchen appliances. Provide relevant information pertaining to the hazards and how they are protected.
Summary of fire protection features	Provide a summary of the active and passive features of fire protection. Provide a description and identify the location of new and existing fire extinguishing systems, detection systems, fire alarm systems, or fire pumps to be provided or existing systems to remain or be modified.
Summary of existing conditions	Provide a summary of existing conditions impacting the project, such as existing detection/suppression systems or existing building construction features.
Summary of design enhancements	Identify items in excess of the contract, criteria, or code requirements.
Building code analysis	Include the following information on occupancy classification; mixed use and occupancy separation, height and area calculations (area per floor and total); type of construction; required building separation or exposure protection; rating of structural components; description of construction; whether rated floor and roof assemblies are restrained or unrestrained; and interior fire or smoke rated wall/partition requirements, fire rating of each floor, ceiling system, and roofing system when applicable. Discuss if and how the proximity to, and classification of, adjacent structures factors into the analysis. A summary fire protection and Life Safety Code ® review must be provided as an appendix. Reference 2–2.d(9) for detailed requirements.
Life safety code analysis	Identify occupancy classification; total occupant load; capacity of mean of egress, number of exits; type of exits; exit travel distance; total exit width; common path of travel; illumination of means of egress, emergency lighting, marking of means of egress classification of interior finishes; location of fire-rated walls and partitions; and opening protection, special hazard protection, and other applicable provisions of National Fire Protection Association (NFPA) 101. A summary fire protection and life safety code review must be provided as an appendix. Reference 2–2.d(9) for detailed requirements.
Water supply analysis	Provide a summary of the data obtained from the water flow test and provide a determination of the adequacy of the water supply (even for facilities without sprinkler protection), along with sketches of the water distribution system. If fire flow demands cannot be met, cite the deficiencies and recommend design alternatives/solutions to correct the problem of an insufficient water supply (such as fire pump(s) or water storage tank(s)).

Component	Description
Fire pump(s)	Provide information regarding project site reliable power. Provide fire pump(s) flow and pressure requirements, locations, construction enclosure, fire pump, and fire pump controller maintenance access.
Water storage tank	Indicate water storage requirements including capacity, type of storage tank, and storage tank monitoring.
Adequacy of water supply for fire protection	• Pre-award design services: If the water supply analysis determines that the water supply cannot support the anticipated hydraulic fire flow or fire sprinkler demand, contact the District Fire Protection Engineer (DFPE), as defined in UFC 3-600-01 as soon as possible. Provide appropriate supporting calculations and propose design options or alternatives for consideration. • Post-award design/construction services: Design assumptions must be based on the water supply data cited in the solicitation. This data must be used as the basis for fire flow and sprinkler design even if flow testing performed by the QFPE or sprinkler subcontractor reveals more favorable results. If flow testing performed by the QFPE or sprinkler contractor reveals flow/pressure less than that specified, immediately submit a request for information (RFI) citing the concern. Provide supporting information and calculations to substantiate the claim and request clarification or direction.
Fire suppression system	Describe the area(s) that will be protected, the hazard classification of these area(s), and the type of system protecting these area(s). For sprinkler systems, describe the design density, demand area, and hose stream allowance to be specified for each different area. Describe the method for connecting the suppression system to the fire alarm system, as well as the method of annunciating the systems, and a description of power disconnects or pre-alarms, that are required. Where appropriate, provide sketches representing the water distribution system, sprinkler demand areas, and show hydraulic reference points for the hydraulic sprinkler calculations.
Fire alarm/mass notification system(s)	Describe what type of initiation devices and notification appliances will be provided for all areas. Identify areas that may have challenging features that will make it difficult to achieve intelligibility requirements. Provide information for connecting to the base-wide fire reporting system and the base-wide mass notification system. Provide drawings or sketches. Provide detailed information on existing fire detection and suppression systems for existing buildings (examples: type of systems; area of coverage; make and model of equipment; and why system is or is not being replaced). For fire alarm systems, provide the following information (at a minimum): number of spare zones and spare spaces for modules, capacity of control panel(s), list of existing fire alarm zones, list of outputs, number of audio/visual (AV) circuits, and standby battery capacity. Indicate the working order of each system (condition or status).

(9) *Mechanical*. Include heating, ventilation, and air conditioning (HVAC) systems; dehumidification and humidification systems; special ventilation or exhaust systems; plumbing systems; interior roof drainage; special piping systems (such as acid waste, chemical, compressed air, vacuum, medical gases, and lubrication); gas distribution; exterior mechanical utilities (such as chilled water, hot water, and steam); design temperatures; utility rates; refrigeration; air pollution control; noise and vibration control; fuel storage and dispensing; radon mitigation; controls; utilities monitoring and control system (UMCS); mechanical utility metering; and process systems design. System descriptions must include rationale and justification for equipment and system selection and information about materials and sizes. Include the Table 2-6**.** components in the Mechanical DA as well.

Table 2-6
Mechanical design analysis requirements

Component	Description
General criteria	List all criteria that apply to mechanical design. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
Exterior mechanical utilities	Describe exterior mechanical utilities such as hot and chilled water distribution, steam distribution, condensate recovery, and gas and fuel distribution and storage including capacities.
Existing conditions	Describe existing conditions for renovation or rehabilitation projects.
Climatic design conditions	Provide climatic design conditions including, but not limited to, design temperatures and humidities, degree days, and climate zone.
Interior design conditions	Provide interior design conditions including, but not limited to, building population including male to female ratio; design temperatures and humidities; ventilation and exhaust; hazardous pollutant sources and target concentrations; personnel occupant capacity and operating schedule information; and internal heat gain information such as from lighting, equipment, and personnel heat release.
Heat transmission values	Provide heat transmission values for building envelope components.
Alternative systems	Describe alternative mechanical system and energy-conserving and recovery features analyzed. Describe considerations for fuel sources, energy performance, life cycle cost analyses, passive design, sustainability, and environmental impact considered during life cycle cost analyses (LCCA). Provide a final energy optimization and LCCA report as an appendix. Reference 2–2.d(15) for detailed requirements.
Mechanical systems	Describe design mechanical systems including all major features and equipment, how the systems will work, capacities, connection to utilities, antiterrorism measures, seismic design measures, redundance and standby systems, vibration and noise control, system expandability and flexibility, enhancements to support operations and maintenance, and associated control systems.
Special provisions	Provide special provisions for balancing HVAC systems.

Component	Description
Control systems	Describe integration of mechanical control systems with building control systems and UMCSs.

(10) *Electrical*. Include power generation, backup power system, generators, transmission and distribution systems, lighting (interior and exterior), wiring methods, intrusion detection, cathodic protection, grounding, lightning and static electricity protection systems analysis and design, aviation lighting, metering, and electromagnetic protection.

(11) *Telecommunications*. Include overall discussion on interior and exterior telecommunication systems and services, to include telecommunication rooms, network and voice equipment, cabling and cabling infrastructure, network classifications, audio communications, video communications, outlets and terminations, pathways, grounding and bonding, cable television system, and calculations.

(12) *Conveying equipment*. Include the Table 2-7 components in the Conveying Equipment DA.

Table 2-7
Conveying equipment design analysis requirements

Component	Description
General criteria	List all criteria that apply to conveying equipment design. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
Type of conveying equipment	Describe type of conveying equipment provided in facility design including elevators, lifts, crane, hoists, etc. Describe the functional requirements resulting in the selection of the type of equipment, required capacity, rated speed, etc.

(13) *Physical security*. Include access control systems, fencing, vaults, protective lighting, security systems, locks, arms rooms, entrances, guard facilities, classified material storage requirements per AR 380-5, patrol roads, clear zones, restricted areas, surveillance, and penetration resistance.

(14) *Cybersecurity*.

(a) The design and installation of the following systems must be in accordance with UFC 4-010-06, UFS 4-010-06, and use UFGS 25 05 11 as the basis for project specifications:

1. Networked Facility Related Control System (FRCS) Control Systems
2. Networked non-FRCS Control Systems
3. Networked low-voltage systems that will integrate to a control system

(b) Additional cybersecurity requirements may be added to meet specific project needs provided those requirements align with the “designer” categorization defined in UFS 4-010-06. Implementation of cybersecurity controls or requirements beyond those that create a functional or documentation requirement for the system are not the responsibility of the designer or construction contractor.

(c) Include scope of design, control systems, system owner (SO) information, authorizing official, communication transport, known authority to operate (ATO) status if any, how the confidentiality, integrity, and availability (C-I-A) was derived, the level of control system work, and connection to base wide platforms. Include a high-level/notional diagram showing major components and connectivity. Cybersecurity design documentation must meet the requirements of UFC 4-010-06 and other applicable supplements or manuals.

(15) *Sustainability*. Include project sustainability requirements according to UFC 1-200-02 and a narrative for how the project is meeting all requirements. Include the Table 2-8 components in the Sustainability DA as well.

Table 2-8
Sustainability design analysis components

Component	Description
General criteria	List all criteria that apply to sustainability. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.
High Performance Sustainable Buildings (HPSB) Guiding Principles	If the project meets the threshold established in UFC 1-200-02 for HPSB Guiding Principles compliance tracking and reporting, include the checklist and a detailed narrative for how the project is achieving HPSB Guiding Principles.
Third-party certification (TPC)	If the project meets the threshold established in UFC 1-200-02 for third-party certification, include the checklist and a detailed narrative for how the project is achieving the required TPC.
Sustainable design elements	Provide narrative related to federal building performance standards, decarbonization approaches, use of sustainable materials, and all supporting energy models and LCCA.

(16) *Commissioning*. Include project commissioning requirements to meet requirements of UFC 1-200-02 and ER 1110-345-723. Include the Table 2-9 components in the Commissioning DA as well.

Table 2-9
Commissioning design analysis components

Component	Description
General criteria	List all criteria that apply to Commissioning. Refer to UFC 1-200-01 for applicability of the latest UFCs. Refer to the appropriate UFCs for applicable standard/code versions.

Component	Description
Third-party certification	Describe the third-party certification requirements related to commissioning.
List of systems	Provide a list of systems to be commissioned.
Commissioning leadership structure	Identify the Government Commissioning Specialist (CxG) and Design Phase Commissioning Specialist (CxD). Indicate whether a third-party contract will be used to hire or supplement the CxG role. Identify the commissioning authority for leadership in energy and environmental design Leadership in Energy and Environmental Design (LEED) purposes.

(17) *Cost and economic analysis*. Include CWE, program budget limitations, analysis of potential economic impacts and risks, and cost schedule risk analysis (CSRA).

c. Part 3: Operations and maintenance provisions. Describe the design provisions made to enhance and to reduce the cost of operating and maintaining the facility. This section may be omitted if not applicable to the DOR's scope during design. During construction, the construction contractor will include this as applicable and per UFGS 01 33 16.00 10, which is outside the scope of this regulation.

(1) Include the O&M design intentions for the major design disciplines covered in Part 2.

(2) Format this part of the DA so that it can be used separately to supplement required project closeout records, or to form the basis for assembling a facility user's manual.

d. Appendices. Include appendices as appropriate for design calculations, data, analysis, reports, meeting minutes, supplemental information, and the engineering considerations and instructions for field personnel (ECIFP). All calculations must be legible, orderly and easily understood. For computer generated calculations, provide the software program name, version and source used to produce each computer output or report. The following items must be included as appendices to the DA.

(1) DD Form 1391 and design directives.

(2) Cost estimate and risk register.

(3) Geotechnical and environmental data reports.

(4) Civil calculations.

(5) Architectural BOD cutsheets and calculations. Include cut sheets of pertinent items specific to the project used as a BOD for coordination purposes. Provide the architectural calculations outlined in Table 2-10.

Table 2-10
Architectural calculation and documentation requirements

Calculation	Description
Net and gross area	Provide net and gross area calculations in line with UFC 3-101-01.
Scupper, gutter and downspout sizing	Provide calculations indicating size, type, and number of scuppers, gutters, and downspouts for adequate roof drainage in line with UFC 3-110-03.
Cutsheets	Provide BOD cut sheets for all architectural features that have a significant impact to the overall building design.

(6) Structural interior design (SID) binder. Provide a physical and digital SID binder. The physical binder must display actual interior finish color samples proposed for the project; the digital binder must include high resolution images of finish samples. Include the Table 2-11**.** components in the SID binder as well.

Table 2-11
Structural interior design requirements

Component	Description
Cover sheet	Include the following information on the cover page: • Structural interior design • Design stage • Date • Design firm • Project title • Project number • Project location • Volume number
Table of contents	Provide a table of contents.
Color boards	Provide physical and digital interior and exterior finish samples proposed for the project. Samples should be identified and keyed to the item numbers included in the finish legend on the contract documents.

(7) Furniture, fixtures, and equipment (FF&E) package.

(a) USACE standard FF&E nomenclature. Utilize the Standard FF&E Nomenclature document when tagging and cataloging FF&E items regardless of procurement method.

(b) Government furnished; government installed (GFGI) furniture packages. FF&E is procured outside of the construction contract by the government.

1. *Huntsville Center Furnishings Program*. All Army MILCON, Sustainment Restoration and Modernization (SRM), Initial Issue and Replacement Unaccompanied Housing (UH) FF&E procurements are centrally administered through the Huntsville Center (HNC), per Installation Management Command

(IMCOM) narrative funding guidance. The HNC Furnishings Program supports all Department of Defense (DoD) components. Refer to the HNC Furnishings Program MRSI Website for furniture construction standards, additional information, and contacts. For a list of approved items for procurement through HNC, refer to HNC Centralized Furnishings Program – Ancillary Items Authorized for Procurement Matrix.

2. *Agency specific GFGI FF&E package*. Refer to agency specific guidance for FF&E package requirements.

3. *Format.* Provide a physical and digital copy of the FF&E binder. The physical binder must display actual FF&E samples proposed for the project; the digital binder must include high resolution images of finish samples. Include the Table 2-12 components in the FF&E package as well.

Table 2-12
Furniture, fixtures, and equipment package requirements

Component	Description
Cover sheet	Include the following information on the cover page: • Furniture, fixtures, and equipment • Design stage • Date • Design firm • Project title • Project number • Project location • Volume number
Table of contents	Provide a table of contents.
Point of contact list	Provide a points of contacts (POC) list to include appropriate project team members, user contacts, interior design representatives, contractors and installers involved in the project. Include each contact's name, address, phone number, email, and job description.
Itemized furniture cost estimate	Provide an itemized cost estimate of furniture keyed to the plans and the data sheets. Organize the estimate by furniture item code and item name so the estimate can also be used as the item code legend. Base the estimate on applicable pricing. The estimate must include percentage allowances for general contingency, shipping, inflation, and installation costs, listed as separate line items. Consult with the government interior designer for any additional costs to be included. Installation and freight quotes from vendors should be used in lieu of a percentage allowance when available.
Item code legend	Provide a consolidated list of all FF&E items in the package with the item code and name of each item. Organize item codes by product category such as workstations, seating, and tables.

Component	Description
Item installation list	The item code legend may be expanded for use as an item installation list. Indicate quantity per room, model number, and manufacturer of each furniture item.
Data sheets	Prepare one data sheet for each item specified in the design. The data sheet identifies all information required to order each item. Ideally, information is presented on one page; however, a second page may be used for additional detailed specifications. Indicate open market justifications and any other critical procurement information. Organize data sheets by product category and key to the item code legend. Include the following on the data sheet: • Item code and name • General Services Administration (GSA) Schedule Information: Federal Supply Category (FSC) group and part number and special item number (SIN); GSA contract number and expiration date; and GSA contractor name, ordering address, phone number, and email or website • Federal Stock information (if applicable) • Product specification information including manufacturer's item name, series, model number, dimensions, and description including minimum quality standards, construction materials and methods, configuration, features, options, warranty, and critical dimensions • Finish and fabric information coded to the finish samples • Illustration that is close to the actual item specified or it must be noted as representative or similar • Quantity by room number and name • Total quantity of items • Special instructions for ordering such as packaging information, mounting height, and installation coordination notes • Interior design source contact, whether it be the A-E interior designer or the government interior designer • Two alternates for each item to ensure multiple manufacturers can meet the BOD
Furniture and finishes material samples	Provide finish and fabric samples keyed to the item codes and names used on the data sheets and the furniture plans. Use photographs or color photocopies of materials or fabrics to show large overall patterns in conjunction with samples to show the true colors. Samples must be large enough to show a complete pattern or design where practical.
Drawings	Include furniture plans, systems furniture plans, and artwork placements plans in the FF&E package. Reference paragraph 3–12 for detailed interior design drawing requirements.

(c) Contractor furnished; contractor installed (CFCI) furniture packages. FF&E is provided as a part of the construction contract within performance specifications.

1. *FF&E criteria*. Provide non-proprietary, project-specific salient characteristics for all items specified in the FF&E package.

2. *Format.* Provide a physical copy of the FF&E finish samples only. The physical samples must display actual FF&E samples proposed for the project. Samples may be provided in a binder or as loose samples. Refer to Table 2-12 for coversheet and furniture and finish sample requirements.

(8) Structural calculations.

(9) Fire protection and life safety. Provide the life safety and fire protection calculations and code analysis outlined in Table 2-13**.** and in line with UFC 3-600-01.

Table 2-13
Fire protection and life safety calculation and documentation requirements

Component	Description
Fire flow data	Provide fire flow test results in line with UFC 3-600-01.
Hydraulic demand analysis	Using computer program-generated hydraulic calculations, calculate the "anticipated" demand of a facility to validate the adequacy of the available water supply or to establish the minimum water supply required. Refer to UFC 3-600-01 for hazard classifications and design criteria determination. The proposed piping layout must accompany the hydraulic sprinkler calculations included with the fire protection calculations submittals. Plot the available water supply versus the hydraulic demand on the Q1.85 hydraulic graph paper.
Summary fire protection and life safety code review	The summary fire protection and life safety code review includes both the building code analysis and life safety code analysis. Reference UFC 3-600-01 for all requirements of summary fire protection and life safety code review. Utilize the fire protection and life safety code review template in the Whole Building Design Guide (WBDG) resource library (https://wbdg.org/ffc/army-coe/resource-library) for compliance with UFC requirements. The summary fire protection and life safety code review must also appear on the fire protection and life safety drawings. Reference paragraph 3–13 for detailed requirements.

(10) Mechanical and plumbing. Methodology, assumptions, conditions, and data sources must be apparent or explicitly stated for all calculations outlined in Table 2-14.

Table 2-14
Mechanical and plumbing calculation and documentation requirements

Component	Description
Calculations	Include calculations for sizing for all equipment items and distribution systems, distribution system pressure drop calculations, building envelope heat transmission values, pipe stress, liquid and piping material expansion, ventilation and exhaust, building air balance and pressurization, and noise and vibration analyses.
Plumbing fixture calculations	Provide calculations that indicate the required number of plumbing fixtures for the facility for all people in line with UFC 3-420-01. The number of fixtures must be based on occupancy type and number of building occupants as calculated for NFPA 101.

Component	Description
Computerized heating and cooling load calculations	Provide computerized heating and cooling load calculations with input and outputs organized so that each space, zone, system, item of equipment, and building component is correlated with identifiers on plans and easily identifiable. Bookmark each input and output report.
Psychrometric calculations	Proved psychrometric calculations for each air-conditioning process showing all process points on psychrometric charts. Provide system schematics indicating state point dry-bulb and wet-bulb temperatures (or humidity ratios) for the outdoor, mixed, supply, and return airflow streams. Also demonstrate capability to condition ventilation air and maintain space relative humidity design conditions over the full range of cooling loads including part-load.
Floor plan diagram	Provide a floor plan diagram showing and identifying HVAC zones.
Air flow diagrams	Provide air flow diagrams showing supply, return, relief, exhaust, ventilation, and transfer flow throughout the system including air flow volumes, flow direction, and equipment labels.
Cutsheets	Provide BOD cut sheets for all mechanical and plumbing equipment to support calculations.

(11) Electrical BOD cutsheets and calculations.

(12) Telecommunication calculations.

(13) Cybersecurity.

(14) Sustainability. Provide third-party certification checklists and high performance and sustainable building (HPSB) tracking and reporting checklists in line with UFC 1-200-02. USACE-specific HPSB compliance tracking checklists can be found on WBDG at https://www.wbdg.org/dod/sustainability/tracking-reporting.

(15) Energy optimization and LCCA. Provide energy and water optimization and LCCA reports and energy compliance analyses in line with UFC 1-200-02 and ER 1110-1-8173.

(16) Commissioning OPR and BOD. Provide OPR and BOD Commissioning (Cx) in line with ER 1110-345-723.

(17) Design phase commissioning plan.

(18) Value engineering (VE) study. Include a record copy of any VE studies conducted as part of the design process. This documentation should include a summary of all proposal items and if proposals were accepted and incorporated in the final design.

(19) Project correspondence. Include digital copies of all significant project correspondence, which would include design directions, decisions and approval of the design directions and design; correspondence associated with scope, schedule, or

budget changes; and critical information from project stakeholders that influence the design.

(20) Engineering considerations and instructions for field personnel. The DOR must prepare an ECIFP as a transition document from engineering to construction. The document must provide insight and background necessary for construction personnel to conduct quality assurance (QA) oversight, review submittals, and resolve minor construction problems without compromising the design intent. The document should not duplicate extensive information included in the DA. The document will be augmented by briefings and instructional sessions as required. A sample ECIFP is available in the WBDG Resource Library. The ECIFP must outline the following:

(a) Engineering considerations used to make design decisions.

(b) Communicating unique, critical, and/or atypical aspects of the design and contract documents.

(c) Discussions on the intent and why specific design decisions were made.

(d) Why certain systems or materials were selected.

(e) Any features requiring special attention (such as long lead items, extensions of design, extraordinary quality specifications, unique testing, and permits).

(f) Critical submittals for review and identification of reviewer.

(g) Document milestones for subject matter expert or DOR participation in field oversight.

(21) Advanced modeling project execution plan (PxP). The DOR must prepare an advanced modeling PxP. Download the latest template from the USACE CAD/BIM Technology Center website.

2–3. Preparation

a. Format. All DA deliverables must be in portable document format (PDF) unless otherwise specified. PDF files must be text searchable, and documents must include a table of contents with page numbers. Include bookmarks for all major sections in the table of contents, at minimum.

b. Classified material. Prepare the DA as an unclassified document with proper references to sources of classified material.

c. Design calculations. Design calculations will be computed following all code and criteria requirements and checked for accuracy following the QMP per ER 1110-3-12.

d. *Use of existing DA.* If a standard design or other design is being site adapted, the DA for the new project will include appropriate material from the existing DA modified to incorporate site adaptations and other essential requirements.

2–4. Review and approval

Review and approval of DA will coincide with the review and approval of the project drawings and specifications. For detailed requirements, refer to ER 1110-3-12.

a. For D-B-B projects, review and approval steps will include:

(1) Parametric design phase (5% to 15%). The general summary statement will comply with AR 420-1.

(2) Concept design phase (35%). The DA will cover the progress of all the design disciplines (refer to Part 2).

(3) Interim design phase (65%). All parts of the DA will continue to be developed from the concept design phase.

(4) Final design phase (95%). All parts of the DA will be complete. Approval of final design drawings and specifications will be done concurrently with the review and approval of the final DA.

(5) Corrected final design phase (100%). Back check all comments from the final design and incorporate revisions as necessary to address all previous review comments.

(6) RTA. All review comments will be resolved and incorporated into the RTA DA and kept for recordkeeping purposes.

(7) Drawing modifications. When modifications of project drawings are authorized, the changed conditions may be added to the DA and the revision date or dates noted.

b. For D-B RFP (based on parametric or concept design), the review and approve process will include the following documents:

(1) Draft RFP. Initial DA covering the progress of all the design disciplines (refer to Part 2).

(2) Final RFP. All parts of the DA will be complete. Approval of final RFP concept design drawings (Division 00 and Division 01 specifications) will be done concurrently with the review and approval of the final DA.

(3) Corrected final RFP. Back check all comments from the final RFP and incorporate revisions as necessary to address all previous review comments.

(4) RTA RFP. All review comments will be resolved and incorporated into the RTA DA and kept for recordkeeping purposes.

2–5. Disposition and reference copy

The final DA with revisions and the as-awarded drawings will be transferred to the using service after contract award. A reference copy of the DA will be retained by the executing District and by the design agency, if different, for use in adapting the project design to another site or in evaluating lessons learned.

Chapter 3
Drawing Requirements

3–1. General

This chapter prescribes the requirements for the preparation and approval of drawings. It applies to design drawings prepared at all phases of design and construction, other than drawings prepared by the contractor during construction, such as shop drawings, record drawings, or as-built drawings, which are outside the scope of this regulation. For detailed requirements related to quality control (QC) reviews of drawings, refer to ER 1110-3-12.

3–2. Project drawings

a. Project acquisition and delivery method. The following guidance follows design phases for D-B-B execution. For D-B RFP preparation, the final RFP documents are based on approved concept or parametric designs; therefore, the project drawings should follow the guidance for parametric or concept design outlined below to the level appropriate for D-B. Concept drawings will be attached to UFGS 01 11 00 for D-B RFPs.

b. Parametric design – 5% to 15%. Parametric design incorporates the following information:

(1) Preliminary sketches of the site plan showing project features such as proposed buildings, roads, and parking.

(2) Predesign level functional relationship diagrams showing the functional space arrangements and required program.

(3) Review of existing geotechnical and survey data to identify probable utility connection points and identification of unusual requirements that will significantly influence the cost.

(4) Permanent stormwater management best management practices.

(5) Site inventory and analysis.

c. Concept design drawings – 35%. Concept designs are used to define the functional, technical, architectural, and engineering aspects of a project and to help verify project costs to provide a firm basis on which to initiate the final project design. Concept design drawings typically include the following information:

(1) Fire protection plan.

(a) Provide code compliance summary sheets.

(b) Provide building site code compliance plans.

(c) Life safety floor plan. At a minimum, identify building areas having different occupancy and hazard classifications and identify egress travel requirements.

(d) Fire suppression plans. At a minimum, provide floor plans and identify hazard classifications. Where a facility has multiple hazard classifications, differentiate each classification area by border or hatching. Identify areas to be protected with special fire suppression systems.

(e) Fire alarm and mass notification plans. At minimum, provide plans that include notification and detection areas of protection to indicate required coverage.

(2) Geotechnical investigation results, boring data, subsurface soil classification and profiles, allowable soil bearing capacity, groundwater elevations, etc.

(3) Project site plans showing existing and proposed buildings, roads, parking, landscape planting masses, contours, the utilities in the immediate vicinity of the project, and applicable removal and demolition.

(4) Foundation plans and preliminary framing plans.

(5) Diagrammatic air barrier plans and sections indicating the location of the air barrier envelope on all building surfaces.

(6) Preliminary building floor plans, reflected ceiling plans, roof plans, building sections, wall sections, and exterior elevations showing the overall building dimensions and heights, functional layout, space configuration and form, and exterior materials, including walls, glazing, soffits, and roofs.

(7) Enlarged floor plans of specific areas where more detail is required to explain design relationships or function.

(8) Preliminary schedules of windows, doors, and interior and exterior finishes.

(9) Perspective/isometric renderings to convey campus relations and clearly depict the massing of individual buildings.

(10) Preliminary FF&E floor plans indicating basic furniture arrangements and primary equipment items to convey basic requirements and confirm room geometry is compatible with FF&E requirements.

(11) Schematic systems plans for fire protection, fire alarms, security systems, plumbing, mechanical, electrical, and telecommunications, showing the proposed locations of all equipment and distribution systems and proposed riser diagrams.

(12) Preliminary electrical and telecommunications site plans depicting exterior electrical and telecom services (existing and new), site-related systems, exterior lighting, major equipment items and electrical pathways to/from the building(s),

electronic security systems, intrusion detection systems, antennas, grounding, and lightning protection.

d. Interim design drawings – 65%. Interim design drawings will continue to develop and refine the approved drawings from concept design and add all necessary drawings for a complete construction contract. In addition to the above drawings, the interim design will include the following:

(1) Site details as required for pavement, reinforcement, water, sanitary sewer, stormwater, and signage.

(2) Framing elevations, sections, details, diagrams, and schedules as applicable for foundations, roof framing, walls, and floor framing with sizing of members indicated. Include typical sections showing the methods of framing, typical structural details, and general structural notes including applicable codes used for design. Design loads, including wind component and cladding pressures, identification of delegated design components, and other pertinent information required by International Building Code (IBC) must be noted on the drawings and specified for each facility. Indicate the locations of any expansion/contraction joints in the structure and slabs and provide details.

(3) Building details for continuous air barrier, fire ratings, penetrations, exterior and interior elements, assemblies, detailed wall sections and casework sections.

(4) Provide additional enlarged plans and sections as required.

(5) Room finish plans, interior finish schedules, and interior elevations.

(6) Developed systems plans, diagrams, details, and schedules for fire protection, fire alarm, security systems, plumbing, mechanical, electrical, and telecommunications. Include power calculations.

(7) Fire protection plans (update from Concept Design Submittal). Refer to paragraph 3–13).

e. Final design drawings – 95%.

(1) Final design drawings will be prepared from the approved interim design and will fully develop and expand on all previous drawings for a complete design. Include details for construction of fire protection, firm alarm, security systems, plumbing, mechanical, electrical, structural, and telecommunications, and seismic control for nonstructural components. Include isometric views, sequences of operation, control drawings, and point schedules. When standard design drawings are used, additional sheets will be incorporated as appropriate.

(2) Final design drawings, together with a complete DA, construction specifications, and a CWE covering all technical, architectural, and engineering details are the basis for construction contracting. The drawings will be sufficient in detail, in

combination with the construction specifications, to provide for fair and competitive bids from contractors, and to provide for the construction of the project without additional drawings, except for shop drawings or as may be required to deal with unforeseen conditions encountered during construction.

f. Corrected final design drawings – 100%. Corrected final design drawings will be prepared from the final design drawings and will incorporate all necessary revisions to back check all review comments.

g. Ready to advertise. RTA drawings will be prepared from the corrected final design drawings (100%). This is the final set of drawings that will be used by the contracting officer to advertise the project.

h. As-awarded drawings. Sometimes referred to as “conformed set.” During the solicitation process, a requirement to issue an amended drawing may be identified. At the time of contract award, a conformed set will be prepared and issued by contracting that reflects the scope of work agreed on by the contractor and the government in the executed contract. The as-awarded drawings may reflect additions or deletions realized through scope changes negotiated prior to contract award. These are the official drawings used in the final contract and will be digitally signed.

i. Consolidated set for construction. Depending on the complexity of the bid options executed, there may be a need for the DOR to develop a consolidated set for construction, issued for information only, to provide clarity to the construction contractor. This set of drawings will consolidate the base bid and executed options to avoid confusion during construction. This may involve removing sheets or other partial content to reflect bid options not exercised or combining information from the base bid and option sheets to create a concise and clear plan for construction. There will be no scope changes, modifications, or deviations from the as-awarded drawings.

j. Modifications. Modifications to construction contracts may require modifications to the contract drawings in a clear and concise manner following the A/E/C Graphics Standard (ERDC/ITL SR-23-1).

k. Shop drawings. These are drawings submitted by a contractor, manufacturer, vendor, or others that show in detail the proposed fabrication and assembly of specific building components, or that show the installation details (such as form, fit, and attachment) of materials or equipment. Preparation, approval, and transmittal of shop drawings are outside the scope of this regulation.

l. As-built drawings. These are drawings submitted by a contractor as part of the completion records transferred to the using service on completion of the project. Preparation, approval, and transmittal of as-built drawings are outside the scope of this regulation.

3–3. Drawing preparation

a. Required work. Drawings will delineate the required work clearly and adequately to the level appropriate for the submittal phase as described in paragraph 3–2.

b. Drafting standards and practices. Format and organization, control data blocks, title blocks, drawing conventions, schedules, and standard details must conform to the most recent version of the A/E/C CAD Standard (ERDC/ITL TR-19-7), A/E/C Graphics Standard (ERDC/ITL SR-23-1), and USACE Advanced Modeling Building Information Modeling (BIM)/Civil Information Modeling (CIM) Object Standard (ERDC/LAB TR-21-4).

3–4. General sheets

Cover sheets must comply with the requirements in Chapter 6 concerning signatures of design documents by registered USACE and contractor professionals. General drawing requirements are outlined in Table 3-1.

Table 3-1
General drawing requirements

Drawing	Description
Cover sheet	Project drawings will have a cover sheet with the project name, project location, vicinity map, project rendering, design agency logo and identification, contract information, project number, fiscal year, and signature block area.
Sheet index	All drawing sets will have a sheet index. For small projects, the sheet index may go on the cover sheet. For larger projects, the sheet index should be moved to a separate sheet.
Abbreviations, general notes, and legends	A sheet or sheets with abbreviations, general notes, and legends should follow the index sheet or may be combined with the index sheet. Include definitions of abbreviations used; legends for materials, mechanical, and electrical symbols; a graphic illustration of details and cross-section reference indicators; and other information as required for that set of drawings. For larger projects, it may be appropriate to have discipline-specific abbreviations, general notes, and legends included at the beginning of discipline drawing sheets.
Code compliance summary sheet	The code compliance summary sheet(s) must be prepared by the QFPE and must be included as “general sheets” immediately following the title sheets. At a minimum, include the following information: 1. Building code summary. Identify each of the following elements in the building code summary: • Classification of occupancy • Allowed vs. provided type of construction • Basic allowable heights and areas vs. actual heights and areas

Drawing	Description
	• Allowable vs. provided height or area increases per floor and total • Calculations supporting height and area modifications/increases • Required vs. provided exterior exposure protection • Required vs. provided interior fire-rated occupancy separations • Required vs. provided internal fire area separations • United States required vs. host nation required vs. provided for projects (outside of the continental United Stats) only 2. Life Safety Code summary. Identify the following elements in the Life Safety Code summary: • Classification of occupancy of each room, area, or compartment (on the drawings or in tabular form) • Occupant load factor(s) and total calculated load • Required vs. provided number of exits • Required vs. provided capacity of means of egress • Required vs. provided arrangement of means of egress including remoteness of exits, horizontal exits, travel distance, common path of travel, and dead-end corridor lengths; when suites are used, indicate type, location, area, and arrangement • Required vs. provided accessible means of egress • Required vs. provided discharge from exits
Code compliance site plan	The building code site plan must be included as "general sheets" immediately following the title sheets co-located with the code compliance summary sheet. Identify the following elements on the building code site plan: • Line of encroachment identifying minimum separation distances from adjacent buildings and assumed property lines of the new construction and of the adjacent structures • Building perimeter used for frontage increases • Exit discharge paths • Fire department vehicle access to building • Fire lane width, marking and locations, approach roads and turn radius, and location • Intended fire department main entrance to building • Location of fire department connections • Fire hydrants, post indicator valve or valves, and their connected water distribution mains serving the building • Fire pump room • Water storage tanks • Hazardous material spill containment tanks • Backflow prevention assembly or assemblies serving water-based fire protection systems (if located outside of building)
Life safety plans	The life safety plans must be prepared by the QFPE and must be included as "general sheets" immediately following the title sheets and code compliance summary sheets. Scale the floor plans so the entire footprint fits on a single sheet if information is clearly legible, and the scale is no

Drawing	Description
	smaller than 1/16-inch (1:200). At a minimum, include the following information: • Location and rating of fire walls, fire barriers, fire partitions, smoke barriers, and smoke partitions (both horizontal and vertical). Barriers requiring fire resistance rated supporting construction must be specifically identified for coordination with the structural design partition locations with fire-rated partitions and horizontal exits identified. • Building areas having different occupancy and hazard classifications. • Room numbers, corresponding occupancy classification, and calculated occupant load. Include the occupant load of each room, area, or compartment (on the drawings or in tabular form). Similar occupancies can be grouped together for occupant load calculations. • Capacity and number of occupants using each major means of egress component (such as stairs, stair doors, exterior doors, and assembly exit doors). • Rooms or areas requiring special life safety or fire protection features. • Location of hazardous materials storage, handling and use that exceed the maximum allowable quantities. • Egress travel requirements (such as travel distances, common paths of travel, and dead-end corridors). • Structural fireproofing locations and associated ratings. • When required, fire extinguisher cabinet and surface-mounted fire extinguisher locations. • When required, fire extinguisher type/quantity table identifying the total number and type of extinguishers required. • Location of primary fire alarm/mass notification control panel.

3–5. Discipline drawing sheets

Discipline drawing sheets follow the general sheets in the order and with the naming and numbering sequences in the A/E/C CAD Standard and must include:

a. General notes. Individual discipline drawing sheets will include general notes, keynotes, legends, north arrow, and key plans as applicable for a clear and concise set of drawings.

b. Demolition drawings. For projects that include removal and demolition, all disciplines will include applicable demolition drawings clearly defining the scope of demolition and removal.

c. Room numbering. Every room will be assigned a unique number according to the A/E/C CAD Standard. Room numbering must be consistent throughout all discipline drawings.

3–6. Survey/mapping drawings

Survey/mapping drawing requirements are outlined in Table 3-2.

Table 3-2
Survey/Mapping drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general survey/mapping notes, and other information as required.
Topographic survey	Provide a topographic survey including all planimetric features within the survey limits including, but not limited to, buildings, sidewalks, roadways, parking areas, trees, ground covers, road culverts, fences, drainage structures, fire hydrants, water valves, visible utility lines, boxes, and signs including electric, phone, cable, gas, and water. The survey should also show the property boundary.

3–7. Geotechnical drawings

Geotechnical drawing requirements are outlined in Table 3-3.

Table 3-3
Geotechnical drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general geotechnical notes, and other information as required. Include a note that references the specification section for any foundation or subgrade areas that require special treatment.
Geotechnical boring plan and logs	Include a general site plan showing the locations of the geotechnical explorations like boreholes or test pits. Place the exploration logs, including all relevant data, onto sheets. Provide all relevant information from the geotechnical report on the drawings (for example, geophysics results if appropriate).

3–8. Civil drawings

Civil drawing requirements are outlined in Table 3-4.

Table 3-4
Civil drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general civil notes, and other information as required.
Access and haul route	Include an aerial image or location plan showing the haul route(s), project site, and staging areas.

Drawing	Description
General site plan	Show the new proposed features such as building(s), paved areas, utilities with sizes, demarcations for privatized utilities, hydrants, valves, fences, security systems, stormwater management features, accessible path of travel, setbacks, and antiterrorism standoff distances, as applicable.
Third-party certification site plan	Show the site boundary to comply with TPC requirements (this may be different from the construction limits boundary), total site area, square footages for hardscape and site vegetation, location of recycling, bicycle storage, location of building entry, etc.
Existing conditions site plan	Show the existing site conditions and identify the sources for the base survey data. Include existing site features, utilities, and topography.
Enlarged plans	Include enlarged plans to show areas in more detail where appropriate and include a key plan.
Demolition plans	Include plans showing the demolition and removal of site features, utilities, and roadways, as applicable.
Roadway plans	For projects with roadways, include roadway plans in coordination with the installation. Include enlarged plans and sections showing the roadway profiles.
Grading plans	Show all existing and new grading.
Storm drainage plans	Show all storm drainage features.
Utility plans	Show all existing and new utilities with profiles and inverts.
Sediment and erosion control plans	Provide sediment and erosion control plans as needed.
Pavement plans	Provide pavement plans in coordination with the geotechnical engineer showing the pavement types, joint layouts, striping, and signage.
Profiles	Provide sanitary sewer and storm drain profiles.
Details	Provide details that include the following: • Pavement sections, joint details, and pavement reinforcement details • Utility details • Special site feature details like retaining walls, exterior stairways, sport courts, fences, etc. • Exterior site signage details like sign mounting details, accessible parking space signs, roadway signage, etc. • Roadway cross sections

3–9. Landscape drawings

Landscape drawing requirements are outlined in Table 3-5.

Table 3-5
Landscape drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general landscape notes, and other information as required.
Landscape plan schedule	Indicate the planting types, to include the container size, quantity, and spacing. Indicate types of ground treatment with square footage areas, as applicable.
Landscape site plans	Provide landscape site plans that include the following: • General site plan • Ground treatment plan • Planting plan to include the locations of trees and shrubs • Tree preservation/removal plan
Enlarged plans	Include enlarged plans to show areas in more detail where appropriate.
Details	Include necessary planting details for trees, shrubs, ground cover, and planting soil preparation. Include details of unique site features such as seat walls, landscape edging, material transition details, and site furnishings.

3–10. Structural drawings

Structural drawing requirements are outlined in Table 3-6.

Table 3-6
Structural drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general structural notes, and other information as required. Include standard general structural notes sheets, to include the minimum requirements found in the IBC and UFCs. These note sheets must include all required building design loads and structural system information as required by the IBC. Include a list of delegated design submittal items that will be designed and coordinated during construction, such as roofing and siding attachments, seismic bracing or equipment, and other items not designed by the structural DOR. Include information on special inspections and testing.
Overall plans and loading plans	Include overall plans showing the structure at each level with overall dimensions and include loading plans showing applied loads used in design (for example, live loads, unbalanced snow loading, wind load diagrams, and any special load considerations). Additional loading diagrams are required for all delegated engineered systems (for example, contractor-designed components and cladding, loading diagrams, and snowdrift loading).
Foundation plans	Include all necessary foundation information, including general plan notes, symbols, schedules, section callouts, foundation tags, grids, dimensions, and key plans.

Drawing	Description
Floor and roof plans	Include framing information, callouts, span directions, bearing elevations, lateral force-resisting system locations, general plan notes, symbols, tags and schedules, section callouts, grids, dimensions, and key plans.
Framing elevations	Provide specific elevations to show unique and typical conditions for framing (such as braced frame elevations, moment frame elevations, and multistory framing considerations). Include elevations, dimensions, framing sizes, opening sizes where applicable, detailing requirements, section callouts, notes, schedules, and information that has not been previously relayed.
Framing and roof framing sections	Include overall or isolated sections to relay information not previously shown or required to illustrate framing complexities or unique conditions. Include elevations, dimensions, framing sizes, detailing requirements, section callouts, notes, schedules, and information that has not been previously relayed.
Enlarged plans	Include enlarged plan views where required due to plan view constraints, amount of information to be relayed, and complexity of construction in the area in detail.
Foundation, framing, and roof framing details	Provide standard construction details based on material type (typical details) and specific details from referenced plan views, elevation views, and section views. Include the following as applicable: • Details of ancillary architectural features and associated reinforcing requirements including, but not limited to, canopies, facades, overhangs, openings, non-load bearing walls, retaining walls, and all connections to structure • Details of compatible connections for delegated design features including, but not limited to, railings, curtainwalls, and pre-engineered components • Details of attachments and supports for large mechanical and electrical units; coordinate with mechanical and electrical for attachment details for light units and equipment
Schedules	Provide structural schedules when required based on type of construction, number of scheduled structural components, and applicability of providing schedules on plan views or other drawing sheets. Examples could include column schedules, beam schedules, and reinforcing schedules.

3–11. Architectural drawings

Architectural drawing requirements are outlined in Table 3-7.

Table 3-7
Architectural drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general architectural notes, and other information as required.

Drawing	Description
Air barrier diagrams and calculations	Provide schematic air barrier plans and building sections indicating the volume enclosed by the air barrier and any building areas excluded from the air barrier. Schematic details or notes will clearly indicate the location of the air barrier within the building wall and roof assemblies. Provide calculations for the surface area of the air barrier enclosure.
Floor plans	Include floor plans that show overall building dimensions in two directions as well as dimensions for all building offsets and openings. Overall dimensions will be coordinated with civil and structural drawings so that each discipline shows the same overall building dimensions. Indicate wall type locations on floor plans.
Reflected ceiling plans	Provide reflected ceiling plans matching the floor plans. Indicate ceiling heights above the finish floor, soffits, lighting fixtures, suspended equipment, mechanical features, and other items as necessary for coordination of disciplines. Indicate the starting points of suspended ceiling grids or patterns.
Roof plans	Roof plans will show ridges, valleys, slopes, drainage elements (such as gutters and downspouts), parapets, crickets, roof penetrations in coordination with disciplines (such as mechanical and plumbing), applicable roof elements (such as roof hatches), rooftop fall arrest systems, snow guards, and any roof-mounted equipment.
Building exterior elevations	Exterior elevations will show actual finished grade lines coordinated with civil grades. Building heights according to code requirements will be shown. Exterior finish materials and elements will be called out with keynotes and indicate the location of control joints and expansion joints. Exterior doors, louvers, and windows will be tagged.
Building sections	Building sections will show finished floor elevations, parapet coping and soffit elevations, high slope roof peaks or roof heights according to applicable code provisions, lowest adjacent grade elevations according to applicable code provisions, interior roof, and floor structure in sufficient detail to verify interior and interstitial clearances and grid lines. Show major mechanical, plumbing, electrical, telecommunications, fire protection, and structural systems.
Exterior wall sections	Exterior wall sections will be drawn at large enough scale and in sufficient detail to indicate floor-to-floor heights, soffits, fenestrations, material changes, and interior finish elements. Assemblies will be tagged, and detail callouts will be added at typical and unique conditions such as footings, wall-to-roof transitions, opening heads and sills, and material transitions.
Enlarged plans	Provide enlarged plans of toilet rooms, pantries and kitchen areas, typical units, and areas with significant casework as required to fully describe the design intent and to key relevant interior elevations or details. Each enlarged plan must indicate dimensions to the nearest grid line, existing building elements, or face of structure, in two directions.
Stairs and vertical circulation	Provide enlarged floor plans and sections for stair and vertical circulation drawn at large enough scale to show sufficient detail.
Interior elevations	Provide elevations of interior elements sufficient to define the design intent fully and adequately. Coordinate with interior drawings to avoid duplication of information.

Drawing	Description
Details	Interior and exterior building elements must be adequately detailed. Place exterior details and interior details on separate sheets if possible. Exterior details can include footings; material transitions and terminations; interior and exterior corners; penetrations; expansion joints; drainage; roof and canopy details; and opening head, jamb, and sill details. Exterior details will sufficiently show the building envelope systems and transition details that clearly note the moisture barrier, continuous air barrier, and flashings. Interior details can include vertical circulation details, ceiling details, continuity of fire-rated partitions or shafts, and floor transitions.
Assembly types	Interior and exterior assembly types will be shown as a wall section at adequate scale to indicate all elements of construction. Each unique assembly, including distinctions in fire resistance, will have a unique assembly type indicator. Assemblies include interior and exterior walls, floors, ceilings, and roof assemblies. Each partition section will show a partition thickness dimension. Indicate any required ratings such as fire or smoke resistance and STC ratings. Include references to testing sources for any required ratings.
Schedules	Provide schedules when required based on applicability of providing schedules on plan views or other drawing sheets. Schedules and item or element numbers and identifiers will follow the requirements of the A/E/C Graphics Standard. Include schedules for the following items: • Window schedules. A tabular schedule of windows will be included on the drawings. • Door and louver schedules. A tabular schedule of doors and louvers will be included on the drawings. • Exterior finish and material schedules. Provide a comprehensive exterior finish and color schedule, indicating selections for all exterior materials. Locate this schedule on the finish schedule sheet or on the sheet with the exterior building elevations. • Interior finish and material schedules. For projects without separate interior drawings, this schedule may go in the architectural sheets. See requirements in paragraph 3–12.

3–12. Interior design drawings

Interior design drawing requirements are outlined in Table 3-8.

Table 3-8
Interior design drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general interior design notes, and other information as required.

Drawing	Description
FF&E plans	Provide FF&E plans to indicate the locations of all furniture, equipment, and accessories. FF&E plans must include composite and enlarged area plans; area plans must have key plans. Identify these items with an item code that is keyed to the FF&E package and the finish color boards. Each drawing sheet must include a legend listing all furniture item codes and names. Provide to scale as required by scope of work. Typically, FF&E plans will be the same scale as the architectural drawings. Some projects may require furniture plans for individual rooms or areas to show furniture in sufficient detail for installation. Examples of this include enlarged plans for systems furniture or individual room drawings where typical room configurations are repeated throughout a project.
Systems furniture	Provide large-scale plans, elevations, and isometrics showing typical workstation configurations that clearly identify major workstation components to include, but not limited to, panels, storage, worksurfaces, accessories (such as monitor arms and keyboard trays), and task lighting. Include the location of all power, voice, and data outlets and indicate the height on panels by note or symbol.
Artwork placement plans	If the artwork locations and item codes cannot be clearly shown on the FF&E plans, provide separate artwork placement plans. Ensure that mounting heights and special installation instructions are indicated on the plans and on the data sheets.
Element plans	Include floor plans indicating interior elements to be procured and installed by the construction contract including corner guards, wall guards, markerboards, tackboards, window blinds, mop rack and shelf, mailboxes, permanently fixed kitchen equipment, etc. Include an interior elements schedule on the elements plan sheet(s) indicating the type, description, and specification section. Coordinate with the architectural enlarged plans to avoid duplication of information.
Room finish plans	Include floor plans indicating the finishes, to include floor finishes, material transitions, floor patterns, and a finish legend.
Enlarged floor plans	Provide enlarged plans of toilet rooms, pantries and kitchen areas, typical units, and areas with significant casework as required to fully describe the design intent and for key relevant interior elevations or details. Coordinate with the architectural enlarged plans to avoid duplication of information. Include schedules as appropriate for interior elements and accessories tagged on these enlarged plans.
Interior elevations	Provide interior elevations of walls and rooms with built-in features, casework, special equipment, visual display components, finishes/trim or walls with multiple finishes, or complex finishes and trim details.
Signage plans	Include floor plans showing the locations of building signage. Include a schedule indicating the sign type, copy, room number, and location.
Details	Interior elements should be adequately detailed, to include the following: mounting heights, restroom details, interior element details, furniture details, casework details, building signage details, and finish details.

Drawing	Description
Schedules	Provide schedules when required based on applicability of providing schedules on plan views or other drawing sheets as referenced above. Include schedules for the following items: • Finish and material schedules. A tabular schedule of interior finishes and materials will be included on the drawings. Finish and material schedules should identify by room number the finish and material to be used for the floor, to include the base, the walls (including wainscoting and trim), and the ceiling. • Finish materials legend. Describe all finishes and materials listed in the schedules, to include the specification section, item code, description, sustainability features or TPC credits it contributes to, and remarks. If a J&A is in place for interior finishes, then the name of the manufacturer and color used as the BOD may be listed in coordination with the contracting officer.

3–13. Fire protection drawings

Fire protection drawing requirements are outlined in Table 3-9.

Table 3-9
Fire protection drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general fire protection notes, and other information as required.
Fire protection plans	Include floor plans showing the following information. Scale the floor plan so the entire footprint fits on a single sheet if information is clearly legible, and the scale is no smaller than 1:200 or 1/16-inch. Where a building has multiple hazard classifications or areas protected with special fire suppression systems, differentiate each area by border or hatching. Information pertaining to electronic control/release systems may be shown on the fire alarm/mass notification systems drawings specified below. 1. Fire sprinkler systems. Include the following information: • Locations of sprinkler riser room • Fire department connections • Post indicator valves • Isolation control valves • Sprinkler branch lines or feed main piping if a specific routing is required, such as single feed to computer room or elevator equipment room and hoistway • Location of control panels used for release of pre-action or deluge systems • Fire pump and associated equipment 2. Gaseous fire extinguishing systems. Include the following information: • Outline of area/hazard to be protected • Location of storage cylinders and releasing panel

Drawing	Description
	• System initiating devices (such as manual releases and automatic detection devices) • Notification appliances • Main/reserve transfer switches • Control devices (such as dampers, shunt trip breakers for computer equipment shutdown, and air-conditioning units to be shut down) and electromagnetic door hold-open devices
Fire suppression details	Provide fire suppression detail sheets that include the following information: 1. Fire sprinkler systems. • Enlarged plan view of sprinkler riser room showing sprinkler risers, control valves, backflow prevention device, and service entrance (supply) manifold drawn to scale. • Cross-sectional elevations of sprinkler and standpipe risers. • Enlarged plan view of fire pumps and piping arrangement, jockey pump, and associated controllers and equipment drawn to scale. • Cross-sectional elevations of fire pump supply and discharge piping arrangement. • Releasing system riser diagram for pre-action or deluge sprinkler systems. Identify zones, circuit inputs, and circuit outputs necessary for controls, including interconnection with building fire alarm control panel. 2. Gaseous fire extinguishing systems. • Releasing system riser diagram identifying zones, circuit inputs and circuit outputs necessary for controls, including interconnection with building fire alarm control panel. • Elevation view of storage cylinders and manifold. • Isometric detail drawing of agent distribution piping including storage cylinder manifold and discharge nozzles. • Sequence of operation matrix. See NFPA 72 for a sample.
Fire alarm/mass notification (FAMN) system plans	Include floor plans identifying the location of field-installed components and interconnected devices. Plans may identify fire suppression control/release system information identified above. At a minimum, identify the location of the following information: 1. Control panel(s). 2. Notification appliance circuit extender panels. 3. Radio transmitter or master box. 4. Line and low voltage surge arrestors. 5. Initiating devices (including duct smoke detectors). In lieu of locating devices on the plans, and as authorized by the DFPE, provide the following information on the plans: • Ambient sound pressure levels and audible design sound pressure levels • Area boarders or other means to identify differing acoustically distinguishable spaces

Drawing	Description
	• Area borders to indicate the type of detection system, initiating devices, notification appliances, and releasing service or area borders for detection and notification zones • Rooms and spaces that will have visible notification and those where visible notification will not be provided • Rooms and spaces that will have initiating devices and the design performance requirements for those devices 6. Supplemental equipment interfaced with the fire alarm system such as voice evacuation panels, electromagnetic door holders, delayed-egress or access-controlled doors, and elevator system components. 7. Supplemental fire suppression equipment control panels such as fire pump controllers and fire suppression control/release panels.
FAMN details	Detail sheets may identify fire suppression control/release system information identified above.
Riser diagram	Provide a riser diagram showing hierarchy, arrangement, and zoning of the system. Identify typical circuits, interconnections, and interlocks necessary for associated controls. Do not identify every field device individually, such as smoke and heat detectors. Identify required line and low voltage surge arrestors. Interface with security systems for required delayed-egress or access-controlled doors. Identify interface with fire suppression control/release panels.
Sequence of operation matrix	Provide a sequence of operation matrix. See NFPA 72 for sample figure A.14.6.1.

3–14. Plumbing drawings

Plumbing drawing requirements are outlined in Table 3-10.

Table 3-10
Plumbing drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general plumbing notes, and other information as required. Include general notes relating to compliance with manufacturer recommendations and requirements, applicable codes and standards, permit responsibility, and coordination between trades. Include notes that relate to installation requirements that generally apply throughout the drawings, such as sealing penetrations, general elevation of pipes, or diagrammatic nature of the drawings.
Site plans	Provide site plans for exterior equipment and distribution systems coordinated with civil site plans. Show connection points to existing systems and locations of tanks.

Drawing	Description
Floor plans	Include floor plans indicating the location of all equipment and fixtures, schematic pipe layout for domestic water, sanitary waste, vent, and natural gas indicating the pipe sizes and locations of access panels for isolation valves and cleanouts. In addition, floor plans should include the following: • Routing of all piping distribution systems • Unique identifiers for each item of equipment • Separate water system drawings from sanitary and vent drawings • Other piping systems such as fuel, gas, lubricant, or compressed air on separate drawings where necessary to clearly demonstrate the layout and sizes • Anticipated grading of piping systems for drainage for systems requiring the ability to drain • Means of access, including ceiling or wall panels, catwalks, ladders, platforms, or access doors • Locations of valves, service taps, and pits • Invert elevations for connection to exterior utilities
Sections and elevations	Include building sections and elevations as necessary to indicate the routing of pipes in coordination with other disciplines to avoid conflicts in congested spaces such as mechanical rooms, laundry rooms, areas above ceilings, and typical units. Demonstrate adequate clearance and maintenance access between plumbing systems and other building components. The number and location of elevations and sections must be sufficient for constructability without further design work by the DOR or construction contractor. Label all equipment, devices, and distribution systems and show sizes for all piping.
Enlarged plans	Include enlarged plan views where required due to plan view constraints, amount of information to be relayed, and complexity of the area in detail. Include enlarged mechanical room plans. Label all equipment, devices, and distributions systems. Provide enlarged plans for restroom or shower areas, at a minimum, with separate plans for water and sanitary and vent systems.
Details	Provide standard details based on design (typical details) and specific details from referenced plan views and sections. The number and types of details must be sufficient for constructability and installation without further design by the DOR or construction contractor. At a minimum, show the following: structural supports for equipment; roof mounting details to avoid water leakage and limit air intrusion; pits; drain terminations; vent stacks through roofs; water hammer arrestors; equipment connection details; sinks and drains; building entrance piping and penetrations; cleanouts; water heating system schematics showing all equipment and devices; pumps and appurtenances; lift station and force mains; meters; and exterior tanks.

Drawing	Description
Schedules	Provide equipment schedules with all necessary performance characteristics and features for equipment selection by the construction contractor. Include efficiency and electrical information. Performance characteristics must be minimums or maximums for proper system operation. Include a plumbing fixture schedule indicating cold water, hot water, vent, and sanitary connection sizes. Include additional schedules as applicable, such as pumps (such as ejector, booster, or sump pumps), interceptors, water treatment equipment, backflow preventer schedule, hot water heat pump schedule, recirculation pump for domestic hot water, water heaters, hot water generator (heat exchanger), storage tank schedule, chilled water control valve, thermostatic mixing valve, control valves, expansion tank schedule, compressed air equipment, and water heater point schedule.
Control diagrams	Provide plumbing controls drawings as described in mechanical drawings, to include water heating systems or other systems, equipment, and meters that integrate with the building control systems, meter data management systems, or utility monitoring and controls systems. Plumbing control drawings may be collocated with mechanical controls drawings. Include a schematic diagram showing the hot water controls with information about the sequence of operations in coordination with the requirements of the installation. Include the required interface with control systems, meter data management system.
Isometric diagrams	Provide isometric riser diagrams showing plumbing system piping without other building components, to include domestic system, sanitary system, vents, and natural gas. Show the location of shutoff valves, water hammer arrestors, vent through roof penetrations, building entrance, plumbing system equipment, and cleanouts. Label all equipment and piping and show sizes for all piping.

3–15. Mechanical drawings

Mechanical drawing requirements are outlined in Table 3-11.

Table 3-11
Mechanical drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general mechanical notes, and other information as required. Include general notes relating to compliance with manufacturer recommendations and requirements, applicable codes and standards, permit responsibility, and coordination between trades. Include notes that relate to installation requirements that generally apply throughout the drawings, such as sealing penetrations, general elevation of pipes, or diagrammatic nature of the drawings.
Site plans	Provide site plans for exterior equipment and distribution systems coordinated with civil site plans including below and above grade steam, condensate, chilled water, and hot water systems. Include profile drawings.

Drawing	Description
HVAC Floor plans	Include floor plans that show HVAC distribution layout as well as supply/return air distribution device layout and identify or outline individual HVAC zones. Indicate primary equipment such as chillers or refrigeration compressors, boilers, pumps, condensers cooling towers, air handling units, fans, hoods, terminal devices, unitary equipment, access panels for equipment maintenance, required clearances, and other items of major equipment required for the facility. Provide ductwork plans on separate drawings from hydronic system plans. In addition, floor plans must include the following: • Routing and size of all duct work, hydronic lines, and condensate lines for HVAC systems. Separate ductwork and piping plans. • Unique identifiers for each item of equipment and label all devices. • Pressure classification of ductwork. • Anticipated grading of piping systems, as applicable. • Locations of expansion loops, expansion joints, joints, guides, and anchors. • Locations for control devices such as thermostats and humidistats, temperature and humidity sensors, building differential pressure sensors, emergency shutdown switch, and duct pressure sensors. Indicate the terminal equipment each thermostat or humidistat is associated with. • Means of access, including ceiling panels, catwalks, ladders, platforms, and access doors. Show the location and size of access panels in floors, walls, and ceilings. When mechanical rooms have interior access in lieu of exterior access, show the route for replacement of the largest item of equipment including space for a dolly, lift, crane, or similar transport aids. • Locations of isolation valves, manual volume dampers, fire and smoke dampers, door louvers, and turning vanes.
Roof plans	Provide a roof plan showing mechanical penetrations and equipment and distribution systems on the roof.
Sections and elevations	Include building sections and elevations as necessary to indicate the routing of pipes and ducts in coordination with other disciplines to avoid conflicts in congested spaces such as mechanical rooms, laundry rooms, areas above ceilings, typical units, etc. Demonstrate adequate clearance and maintenance access between mechanical systems and other building components. Provide at least one section in the mechanical room. The number and location of elevations and sections must be sufficient for constructability without further design work by the DOR or construction contractor. Label all equipment, devices, and distribution systems and show sizes for all ductwork and piping.
Enlarged plans	Include enlarged plan views where required due to plan view constraints, amount of information to be relayed, and complexity of the area in detail. Include an enlarged plan of the mechanical room showing all equipment, including HVAC, plumbing, and fire suppression system equipment to demonstrate sufficient clearance between equipment and distribution systems. Show the location of control panels and variable frequency drives. Show maintenance clearances around equipment and label all equipment, devices, and distribution systems.

Drawing	Description
Details	Provide standard details based on design (typical details), and specific details from referenced plan views and sections. The number and types of details must be sufficient for constructability and installation without further design by the DOR or construction contractor. At a minimum, show the following: structural supports for equipment (including connection to building structure), penetrations, fire dampers, roof mounting details to avoid water leakage and limit air intrusion, pits, condensate drain details (including depth of water traps and slope from drain pan), air vent and drain details, expansion loops, expansion joins, anchors, make-up water to hydronic systems (including all associated devices), equipment connection details, and hydronic system schematics (including flow direction arrows) showing all equipment and devices. Include an air balance diagram for kitchens and a capture hood design showing capture velocities and distances.
Schedules	Provide equipment schedules with all necessary performance characteristics and features for equipment selection by the construction contractor and to demonstrate compliance with project requirements. Such equipment includes, but is not limited to, dedicated outside air systems, fan coil units, exhaust fans, louvers, air separators, expansion tanks, unit heaters, terminal devices, air distribution devices, plate heat exchangers, split-system indoor units, split-system outdoor units, air-cooled chillers, boilers, hydronic control valves, and pumps. Include efficiency and electrical information. Performance characteristics must be minimums or maximums for proper system operation, as calculated, and not based solely on a specific manufacturer’s products. Include a design condition schedule, ventilation schedule, vibration isolator schedule, and duct construction and leak testing table showing duct pressure classes, seal classes, leakage classes, and leak test pressures.
Isometric view	Provide isometric views (three dimensional) of the mechanical systems in mechanical rooms showing ductwork, piping, and equipment for designs that use BIM and three-dimensional modeling.

Drawing	Description
HVAC Control System	Provide control drawings that include every mode of operation, sequence of operation interlock, safety, and control devices required to fully describe system operation for all equipment and systems. HVAC control system drawings must comply with UFC 3-410-02 and all applicable supplements or manuals. • Include a legend for all symbology • Provide a schematic for each unique item of equipment or system showing all associated devices with all devices labeled • Show failure positions of dampers and valves • Include written sequences of operation for each unique item of equipment or system • Provide points schedules indicating points names; network information; setpoints; control and operating ranges; overrides; alarms; and trending, monitoring, and control capabilities • Include a diagram of the system architecture and provide alarm handling, system scheduling, and system integration requirements, including integration with other building control systems (such as lighting and metering) and UMCSs • Provide ladder diagrams where necessary to detail equipment interlocks and interfaces • Provide control valve, control damper, zone sensor, and occupancy schedules

3–16. Electrical drawings

a. Provide adequate plans including demolition; existing conditions; and new work, legends, details, and diagrams to clearly define the work to be accomplished. Coordinate construction drawings and specifications; show information only once to avoid conflicts.

(1) Reference DoD supplemental technical documents located at https://www.wbdg.org/dod/supp-tech-documents for background information (supplemental documentation, best practices, sample calculations, etc.).

(2) Provide a general note at the beginning of the electrical drawings clarifying the work to be accomplished. The following note is required unless directed or approved by the government: "ELECTRICAL WORK AND MATERIAL IS NEW AND PROVIDED BY THE CONTRACTOR UNLESS INDICATED OTHERWISE."

(3) Follow the NFPA 70 metric designations (mm) and trade sizes (in) for conduit size designations.

(4) Indicate the quantity and size of all conduits and cables required in multiple conduit and cable runs.

b. Electrical drawing requirements are outlined in Table 3-12.

Table 3-12
Electrical drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general mechanical notes, and other information as required.
Electrical site plans	Show utility point of common coupling to the installation power and telecommunications systems on the site plan. Provide explicit direction on the method of entering existing manholes. Provide all details including composition of duct banks and depth and configurations of the duct banks. Depict the existing and new exterior electrical service, site-related systems, major equipment items, and electrical pathways to and from the building(s). Locate exterior lighting on the plans. Provide schematic diagrams of exterior systems such as cathodic protection, grounding, and lightning protection. Coordinate with other utilities. 1. Electrical site plans must be separate and distinct from other utility site plans. Electrical and civil site plans may be combined when a project requires minor utility work. 2. Provide the orientation of electrical site drawings consistently with the civil drawings. In addition, provide the orientation of partial building or site plans identical to the orientation of the larger plan from which the partial was taken. Indicate the exact title of each detail, partial plan, or elevation as identified on the cross-referenced sheet.

Drawing	Description
	3. For overhead distribution, use a separate symbol for each individual circuit; define each circuit by voltage level as well as number, size, and type of conductors. Coordinate guying and conductor sag information shown on the drawings with that shown in the specifications. 4. Indicate overhead distribution pole details on the drawings. Identify the pole number assignment consistent with installation standards on the site plan and require a pole identifier number to be installed on the pole matching the installation standard.
Pole details	Indicate overhead distribution pole details on the drawings. 1. Format. • Naval Facilities Engineering Systems Command (NAVFAC) pole details are available in PDF format and in CAD format on the WBDG website at https://www.wbdg.org/dod/ufgs/ufgs-33-71-01. • Provide details in situations where an applicable pole detail has not been developed. Designer developed details must contain the same level of detail equivalent to the NAVFAC pole details and include material requirements. • Review the information contained in the NAVFAC pole details OH-1.1 through OH-1.5a for examples of how to show overhead distribution work. Do not describe proposed work by referencing sketch numbers instead of pole detail designation symbols. Do not use pole detail designation symbols to describe existing facilities to be removed. To maintain the integrity of the pole details, do not modify pole details; include any required exceptions or modifications as supplemental information with the pole detail designation symbols. When using pole details, place a note referencing the pole detail designation symbols (like the following) on the drawings: “XFB, 15FR3-N are pole detail designation symbols. Refer to Sketches OH-1.1 through OH-41 on Sheets [TBD] for an explanation of the use and description of equipment provided by these symbols.” 2. Conductor sag. Indicate conductor initial sag values. Provide initial sag values at ambient temperatures in 18 degrees F (10 degrees C) increments for a temperature range, which includes the outside summer and winter design temperature values. Clearly indicate each different calculated ruling span on the plans and provide initial sag for one span in the calculated ruling span. 3. Fusing. Provide the appropriate symbol and detail indicating the use of backup current-limiting fuses with the device being protected. Indicate the fuse type and ampere rating as well as the voltage rating and current designation of the backup current-limiting fuse.

Drawing	Description
Transformer details	Indicate transformer details on the drawings. Transformer details are available in a PDF format and an CAD format on the WBDG website at https://www.wbdg.org/dod/ufgs/ufgs-26-12-19. Provide the following transformer descriptive information: • Transformer type (such as pad-mounted, pole mounted, station type, or unit-sub) • Kilovolt-Ampere (kVA), single or three phase • Voltage ratings per Institute of Electrical and Electronics Engineers (IEEE) C57.12.00 (such as 11.5kV – 208Y/120 volts) • Primary and secondary connection (when using single-phase units for three-phase service, specifically indicate how the units will be connected such as connect delta-wye grounded for 208Y/120-volt secondary service) Include the following information for surge arresters and fused cutouts: • Surge arrester kV rating • Cutout kV, continuous ampere, and interrupting ampere rating • Fuse link type and ampere rating
Underground distribution	Profiles may be required for duct bank runs. Discuss profile requirements with the electrical reviewer. Indicate structure (manhole and handhole) tops, duct bank elevations, slopes, and diameters. Coordinate structure numbers with plan sheets. Show and label all crossing utility lines, both existing and new. If depths of existing utilities are unknown, indicate the horizontal location of the utility and indicate the vertical location with a line representing the anticipated range of elevations where the utility will be found in the field. Indicate the method of new utility installation routing above or below conflicts. Notes can be used to identify parts. Provide cable/duct bank information indicating cable identification, description, conduit size, and remarks for any special instructions. Where four or more conduits will be installed, provide in schedule format. For smaller applications, schedule or notational conveyance is acceptable. Provide manhole foldout details or exploded views for all multiple-circuit primary systems and all primary systems requiring splices. Indicate the entrance of all conduits and the routing of all conductors in the manholes. Manhole details are available in PDF format and in CAD format on the WBDG website at https://www.wbdg.org/dod/ufgs/ufgs-33-71-02.
Demolition plans	Provide demolition plans separate and distinct from new work plans, except where only minor demolition work is required. Clearly show what will be demolished, at an appropriate scale. Indicate the beginning and ending points of circuit removals. For modifications or additions to existing equipment, provide the manufacturer’s name and other pertinent manufacturer’s identification (such as serial number, model number, style, and any other manufacturer’s identifying markings). Provide a sequence of demolition as required. Include known requirements for continuous operation and/or limited shutdowns. Identify these in the special scheduling paragraphs of the specifications. Indicate the quantity of lighting ballasts that contain polychlorinated biphenyls and the quantity of lamps that contain mercury.

Drawing	Description
Lighting plans and details	Do not show lighting and power on the same floor plan, unless the scale of the plan is 1:50 (1/4 in = 1 ft – 0 in) or larger. Provide a photometric layout plan as an appendix to the BOD. The photometric layout plan may also be a “for information only” drawing but may not be a contract drawing. Provide luminaire (lighting fixture) details, a separate luminaire schedule, controls, and control strategies for each space. Details and a luminaire schedule are available in PDF format and in CAD format on the WBDG website at https://www.wbdg.org/dod/ufgs/ufgs-26-51-00. To maintain the integrity of the details, do not modify details; make any required exceptions or modifications in the remark’s column of the luminaires schedule and not on the details themselves. Provide applicable luminaires type symbol(s) with each luminaire sketch/detail. When using luminaire(s) not included in the database, detail the luminaire(s) on the drawings providing the following minimum information: luminaire type (such as high bay, fluorescent, industrial, downlight, roadway type, or floodlight); physical construction including housing material and fabrication method, description of lens, reflector, refractor; electrical data including number of lamps, lamp type, ballast data, operating voltage; mounting (surface, suspended, flush) and mounting height; and special characteristics such as wet label, specific hazardous classification, or air handling.
Power plans	Show all power requirements and points of connections. Indicate the connection between the exterior electrical systems with the building systems, including indication of electrical characteristics, voltage, phase, conductor size, etc. Indicate the locations of electrical rooms and primary pathways inside facilities. Specifically identify each piece of equipment such as transformers, switchgear, motor control centers, and panelboards including HVAC and mechanical equipment (such as unit heater No. 1, unit heater No. 2). Include required clearances for maintenance in line with applicable codes. Identify physical areas that may be subject to damage as defined in UFC 3-501-01 and indicate the appropriate raceway type for each one.
Grounding plan	Provide grounding plans and details at an appropriate scale.
Roof plan	When roof mounted equipment, including HVAC equipment, cannot be adequately shown on the power plan, provide an appropriately scaled roof plan.
Lightning protection plan	Provide a lightning protection plan and details at an appropriate scale. Indicate the locations and number of system components required. Show air terminal installation details, roof and wall penetration details, and details to show concealed components of the system. Coordinate roof and wall penetrations with other disciplines to ensure that the integrity of the facility envelope is not compromised.
Hazardous location plan	Provide on the drawings the boundaries and classifications of all hazardous locations in line with NFPA 70.

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<td>Power one-line/riser diagrams</td>
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Provide a power one-line (single-line) diagram for:

Medium-voltage distribution systems, including substations and switching stations
Systems involving generation, either low voltage or medium voltage
Facility switchgear, switchboards, motor control centers, and main distribution panels (MDPs)

The one-line diagram must show all components (including metering and protective relaying). Indicate sizes of bus, feeders, and conduits. Show connections of transformers, current transformers, potential transformers, and capacitors on the one-line diagram using the proper symbol. Show potential and current transformer ratios. Indicate relay quantity and function (overcurrent, voltage, differential) using the American National Standards Institute (ANSI) designation numbers.
On most facility-related projects, it is acceptable to combine the one-line diagram with a riser diagram. The one-line diagram would begin with the medium voltage system and continue through the transformer up to and including the main breaker and feeder breakers within the MDP. Sub-panels beyond the MDP may be shown in the riser diagram format.
Include the following on one-line diagrams:

Indicate kV ratings for surge arresters, and kV and ampere ratings for cutouts. Indicate the fuse link type and ampere rating. For capacitors, indicate Kilovolt-Ampere Reactive Power (kVAR) per unit, number of units per bank, voltage (voltage rating of units, not the system voltage), phase (for example, three-phase or single-phase units), fuse size, and fuse type.
Show the following on the one-line diagram when a transformer is indicated, as applicable: primary switches, wye or delta connection, loadbreak elbows, lightning arresters, kVA rating, rated voltage (primary and secondary), transformer identification number, industry standard impedance, meter type, current transformer and potential transformer sizes, and fuse sizes. Show all pertinent transformer information on the one-line diagram as opposed to the specifications. Items that are common to all transformers can be indicated by notes on the one-line diagram if a typical detail drawing is provided.
Show the following on the one-line diagram when pad-mounted switchgear is indicated: spare ways (cubicles), protective devices, loadbreak elbows, and switch identification number.
Show the following on the one-line diagram when a new primary is indicated: in-line splices in manholes, normally open points, number and sizes of phase, neutral and ground cables, and conduit sizes.
If there is demolition involved or work is to be done to existing equipment, provide an existing one-line diagram showing the current arrangement of the gear and then show a new one-line diagram indicating by line weights what is existing or new.

Ensure that information shown on the one-line diagram is not duplicated elsewhere in the construction package, as this will likely cause conflicts if changes are necessary. Indicate on the electrical legend the exact nomenclature used to indicate conductor and conduit sizing. Provide a schedule for feeder runs. Provide medium voltage one-line diagrams for stations and distribution systems that have a geographic affiliation to the actual constructed distribution system.
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Drawing	Description
Fire alarm riser diagram	If required, fire alarm riser diagrams, including a mass notification system, will be provided by the fire protection engineer.
Special systems riser diagrams	Provide riser diagrams for other support systems, which may include electronic security, intrusion detection, access control, video surveillance, and lighting.
Enlarged plans	Include enlarged plan views where required due to plan view constraints, amount of information to be relayed, and complexity of the area in detail. Include enlarged electrical room plans.
Schematics and diagrams	Diagrams of high- and low-voltage interior electrical distribution will show features such as auto-transfer switches, emergency generators, emergency systems, and major subpanels. Provide control schematics, including motor and lighting controls and specialized controls.
Details	Include details for all electrical equipment, lighting calculations, light fixtures, exit signs, mounting details, occupancy sensors, exterior lighting control connection detail, daylighting and occupancy sensor diagram, system ground detail, main electrical grounding bar, panel details, arc flash warning label, ground rod test well details, duct bank details, power trench details, hand hole, etc.
Schedules and elevations	Provide elevations, sections, and details to clearly identify space constraints, unique conditions, design intent, or to ensure a specific method is implemented. Include schedules as applicable, such as lighting fixtures, lighting control scenarios and strategies, wire, cable and raceway, switchboard, motor control center, and panelboard schedules. 1. Panelboards. Provide schedules for all panelboards. Provide the panelboard schedule reflecting the actual circuit breaker and bus arrangement. Include the following: • Panelboard designation and location (such as room number). • Voltage, phase, frequency, number of poles, and minimum interrupting rating. • Main amperes indicating main breakers or lugs only. • Surface or flush mounting. • Circuit number, wire size, breaker trip, connected load, and identification of load associated with each branch or feeder (for example, do not merely indicate “Lighting,” but rather “Lighting, Room 102” for the directory marking). • Total connected load (calculated load) including demand factors of all circuits. • Any special breaker requirements such as ground fault circuit interrupter, arc fault circuit interrupter, switch duty, adjustable trip, and 100% rated. *Note*: Many manufacturers require minimum 400A panel frames for 100% breakers to be used. • Neutral bus size (100% or 200%). Additionally, consider other conductor sizing factors specific to a given circuit such as anticipated ambient temperature on the assumed routing path, whether the load is nonlinear, and the number of current-carrying conductors for purposes of re-sizing should circuits be consolidated into a common raceway during construction.

Drawing	Description
	Show all circuiting (identifying conduit and wiring back to specific panels but not identifying the exact routing required during construction) on the design drawings exactly as they will be installed. 2. Switchboards and switchgear. Provide plan and elevation or isometric drawings for switchboards and switchgear, showing compartments, their intended use, and instruments, relays, and controls. Clearly show contents of all sections including whether breakers are individually or group mounted. Indicate that switchboards and switchgear are mounted on 4 in (100 mm) elevated concrete pads. Coordinate design of pad with a structural engineer. 3. Motor control centers (MCCs). Provide plan and elevation or isometric drawings for MCCs identifying compartments. Provide a schedule listing each compartment. Include on the schedule (for each compartment) the description of load, load in amperes, load in horsepower, National Electric Manufacturers Association (NEMA) size and type of starter, breaker size, conductor and conduit size, control devices, and other special requirements. • Indicate, on plans or in specifications, the enclosure type, bus rating, bus material, bus bracing, NEMA class and wiring type, service voltage, control voltage and source, and top or bottom feed. • Indicate on the drawings that MCCs are mounted on 4 in (100 mm) elevated concrete pads. Coordinate design of pad with a structural engineer. • Provide elevation of control panels, indicating front panel devices such as indicator lights, pushbuttons, gauges, and switches.

3–17. Telecommunications drawings

Telecommunications drawing requirements are outlined in Table 3-13.

Table 3-13
Telecommunications drawing requirements

Drawing	Description
Abbreviations, general notes, and legends	Include definitions of abbreviations used, legends showing graphic illustration symbols used, general telecommunications notes, and other information as required.
Site plans	Depict the existing and new exterior telecommunication services, site-related systems, major equipment items, and pathways to and from the building(s). Indicate the telecommunications pathway improvements required from the installation-identified connection point to the site. Provide schematic diagrams of exterior systems such as grounding, electronic security systems, intrusion detection systems, and ground- or building-mounted antennas.

Drawing	Description
Floor plans	Provide a plan for the interior distribution systems, including access control, closed circuit television and cable television systems, AV systems, proposed location of all major items of telecommunications equipment, telecommunication connections, and cable raceways. Include required clearances for maintenance. Indicate pathways for cables, conduit, and equipment installed by the government or others (for example, antenna systems).
Enlarged plans	Include enlarged plan views where required due to plan view constraints, amount of information to be relayed, and complexity of the area in detail. Include enlarged telecommunication room plans and operating equipment spaces for control of secure areas within the facility, including exterior sally ports.
Diagrams	Include interior distribution control system diagrams, outside plant telecommunications one-line diagrams, telecommunication riser diagrams, and cable television diagrams.
Details	Include details for all telecommunication equipment and outlets, security equipment, cable trays, duct banks, penetrations, conduit entry into building, vaults, support details, and grounding.
Schedules	Include schedules as applicable, such as cables and cable trays, either on the floor plans or separate sheets as appropriate.
Control drawings	Include the interactive control and operations of all systems.

3–18. Utility monitoring and control system drawings

UMCS drawings for all UMCS Front Ends and for integration of building control systems with a UMCS must comply with UFC 3-470-01 and all applicable supplements or manuals.

3–19. Review and approval of project drawings

All reviews must be conducted in line with ER 1110-3-12. District and Center Chiefs of Engineering or their deputies will approve all appropriate in-house design drawings per Chapter 6.

Chapter 4
Specifications Requirements

4–1. General

This chapter prescribes the requirements for preparing specifications. It applies to specifications prepared at all phases of the design and construction process, and by both in-house teams and A-E contractors. Refer to UFS 1-300-02 for specifications formatting requirements.

4–2. Unified Facilities Guide Specifications

UFGS are a set of standardized, comprehensive specifications used for DoD design and construction. These specifications provide common requirements across DoD for safety, sustainability, durability, and functionality and refer to the applicable consensus building codes, DoD-defined requirements, and statutory and regulatory requirements. UFGS promotes consistency, quality, and efficiency across various projects by providing a standardized framework for a project team to use and edit for their project.

a. UFGS contain designer notes providing guidance on use of the specifications and the coordination required with the other project specification sections and the project drawings. Many UFGS contain “tailoring options” that allow SpecsIntact to globally delete products or requirements with minimal effort. Additionally, the guide specifications use brackets to identify alternative text and fill-in blanks for selection by designers.

b. The UFGS, combined with the use of SpecsIntact automated processing, improve project specification production, uniformity, consistency, and overall quality according to DoD policy. Uniformity and consistency of project specifications aid contractors in preparing their bids, improve quality of construction, and reduce cost to DoD stakeholders.

c. Use of UFGS is mandatory for developing project specifications. Use the latest version of UFGS per UFC 1-200-01.

d. UFGS are developed and maintained per MIL-STD-3007.

(1) Proposals for new criteria and UFGS with potential DoD-wide application are encouraged for submission. Proposals for technical or editorial changes to existing criteria and UFGS that are necessary or desirable for general application or to reflect local availability of materials and construction practice are also encouraged. Submit such proposals to DoD electronically as a criteria change request (CCR) in the WBDG website.

(2) UFGS are revised and reissued periodically to incorporate CCR, lessons learned, and technological advances. UFGS are published quarterly in February, May, August, and November.

4–3. SpecsIntact

SpecsIntact is a software program developed in partnership between the National Aeronautics and Space Administration (NASA) and the DoD, for use in producing USACE project specifications and maintaining guide specifications. The software provides specification automation to users and incorporates a wide range of quality control features.

a. Using SpecsIntact is mandatory for production of all project specifications, except for overseas projects that are designed to host nation standards. Maximum efficiency and quality are obtained when project specifications are prepared using SpecsIntact and the latest UFGS edited to suit the specific requirements of projects.

b. Refer to the eLearning modules and Web-based Help on the SpecsIntact website for instructions on its use.

4–4. Project specifications

a. General requirements. Ensure that high-quality and concise specifications are prepared, that the preparation of project specifications is fully coordinated with agency construction and contracting representatives, and that the project specifications comply with format and content requirements according to UFS 1-300-02. As needed, designate a lead specification engineer to oversee and coordinate the preparation of project specifications and to ensure compliance with these requirements. A specifications engineer has knowledge and experience in developing construction contract documents and project specifications.

b. Use of existing project specifications. To ensure that the latest UFGS is used, using sections from previous design project specifications is prohibited.

c. Tailor and edit the UFGS to fit specific project requirements. Preserve the intent of and wording of UFGS to the extent practicable as they incorporate public laws, federal mandates, DoD policy, industry coordination, and lessons learned.

d. Specifications development during project phases.

(1) Project specifications, when combined with the project drawings, must provide a comprehensive set of construction documents that can be bid fairly, competitively, and executed without change, except as necessary to resolve unforeseen conditions or changes made during construction. See ER 415-1-11 for guidance on biddability, constructability, operability, environmental, and sustainability (BCOES) reviews. Identify and resolve unusual design or contract administration problems and assure that project specifications comply with technical policy established by Headquarters, U.S. Army Corps of Engineers (HQUSACE).

(2) Close coordination between the specifications engineer, specifiers, and the designers is important throughout all design phases to produce complete and accurate project specifications. Specifications engineering and designing are distinct professional

functions that must be performed during specifications development. A person responsible for the specifications engineering function on a project may also handle some design tasks. However, one person may focus solely on specifications engineering while others handle the design functions. The specifications engineer is responsible for establishing local processes for developing and tracking project specifications.

(3) Specifiers should assist designers in identifying UFGS sections to use in the project, operating the SpecsIntact software, and incorporating a designer's technical requirements into the project specifications.

(4) Designers are responsible for designing the technical project features and for the technical content of the project specifications for those features. Specifiers are responsible for formatting all project specification sections and for ensuring that proper and non-contradictory contract language is used throughout. Additionally, specifiers determine the project-specific information that must be inserted into the non-technical provisions and the Division 01 general requirements sections.

(5) Designers will prepare technical specification sections for which no UFGS exists. When a new specification section must be developed for a particular project, the designer will provide the technical information and technical requirements to be included in the specification section. The specifier will work with the designer to ensure that the section contains proper language and is properly formatted per UFS 1-300-02.

(6) Project bid schedules (also known as pricing schedules) will be prepared in close coordination with contracting, Counsel, project management, design, cost engineering, and construction. For Civil Works projects, the lump sum and unit-priced items defined for incorporation in the bid schedule must be consistent with the work breakdown structure. Conform bid schedules to USACE guidance and all aspects of Federal Acquisition Regulation (FAR) Subpart 36.207. Contracting is responsible for incorporating the bid schedule into the contract documents.

(7) As part of the routine QA/QC process, Specifiers should perform quality checks (for example, SpecsIntact reports, visual scan of pages for errors, verification of specification inserts such as the submittal register, test requirement list) on project specifications prior to advertisement.

(8) Appropriate design staff should make field trips during the construction phase of projects to identify specifications and contract administration problems to be avoided in future project specifications. Implement corrective action to resolve problems identified during all project phases that could have been prevented by improved specifications, such as recommend changes in UFGS.

(9) Contact the Resident and Area Engineers during the design process and solicit their input, particularly for Division 01 sections.

e. Specifications prepared by architect-engineer firms. Include the requirement to use SpecsIntact for production of project specifications in all procurement of A-E design services. Give preference to A-E firms that utilize Construction Specifications Institute (CSI) certified construction specifiers.

(1) Provide the A-E copies of regulations, manuals, engineer technical letters, and other information not available on TECHINFO and the WBDG website.

(2) Clearly specify in the A-E scope of work who will prepare the Division 01 sections. If the A-E is responsible, provide guidance and agency-unique information to A-E firms for incorporation into Division 01 sections.

f. Construction documents format. Format project specifications per UFS 1-300-02.

(1) Specification section numbering will follow CSI MasterFormat® (latest edition). If CSI MasterFormat lists a section number and title, that is the number and title to use. If either the section number or title changes, update both to maintain a unique and consistent set.

(2) CSI MasterFormat classifies the fifth-level numbering as user-defined for internal agency use only. The UFGS uses a fifth-level section numbering scheme to designate agency-specific sections; fifth-level 10 indicates a USACE-specific section. Refer to Table 4–1 for fifth-level numbers assigned to USACE Districts for local masters.

Table 4–1
U.S. Army Corps of Engineers District fifth levels

District	5th Level	District	5th Level
CEHNC (Huntsville)	01	CENWP (Portland)	25
CELRB (Buffalo)	02	CENWS (Seattle)	27
CELRC (Chicago)	03	CENWW (Walla Walla)	28
CELRE (Detroit)	04	CEPOA (Alaska)	29
CELRH (Huntington)	05	CEPOF (Far East)	31
CELRL (Louisville)	06	CEPOH (Honolulu)	32
CELRL-PM-R (Army Reserve)	48	CEPOJ (Japan)	33
CELRN (Nashville)	07	CESAA (Caribbean)	51
CELRP (Pittsburgh)	08	CESAC (Charleston)	34
CEMVK (Vicksburg)	09	CESAJ (Jacksonville)	35
CEMVM (Memphis)	11	CESAM (Mobile)	36
CEMVN (New Orleans)	12	CESAS (Savannah)	37
CEMVP (St. Paul)	13	CESAW (Wilmington)	38
CEMVR (Rock Island)	14	CESPA (Albuquerque)	39

District	5th Level
CEMVS (St. Louis)	15
CENAB (Baltimore)	16
CENAE (New England)	17
CENAN (New York)	18
CENAO (Norfolk)	19
CENAP (Philadelphia)	21
CENAU (Europe)	22
CENWK (Kansas City)	23
CENWO (Omaha)	24

District	5th Level
CESPK (Sacramento)	41
CESPL (Los Angeles)	42
CESPN (San Francisco)	43
CESWE (Expeditionary)	52
CESWF (Fort Worth)	44
CESWG (Galveston)	45
CESWL (Little Rock)	46
CESWT (Tulsa)	47
CESWM (Middle East)	49

g. Reference publications.

(1) Describe materials, workmanship, and equipment, where possible, by referring to industry and government standards generally known to the construction community, citing the type, class, or other designation necessary to identify fully the item required. The reference approval date and the dates of any applicable amendment and revisions must be included in the solicitation (FAR Subpart 11.201a). Reference standards should not be used to describe minor, non-critical items (such as incidental fasteners) when any commercially available product of that nature may be adequate. To the maximum extent practicable, references will be to nationally recognized industry and technical society specifications and standards. If industry documents are unavailable or unsuitable, refer to the applicable commercial item descriptions.

(2) Publications referenced in project specifications need to be no later than the editions cited in the current notice for the corresponding UFGS. Publications not readily available to bidders, such as locally developed policy or guidance, should not be referenced, but if referenced, will be furnished with the solicitation (FAR Subpart 11.201b). Per DoD direction, do not use federal specifications (FED-SPEC), federal standards (FED-STD), military specifications (MIL-SPEC), and military standards (MIL-STD) in contracts unless exempted by HQUSACE. These publications, when cited in UFGS, are approved for use. FED-SPEC, FED-STD, and commercial item descriptions are available on the GSA website. MIL-SPEC and MIL-STD can be located through the Defense Standardization Program website.

h. Submittals. Construction submittals such as shop drawings, samples, test reports, certificates, and manufacturer’s instructions are not required for non-critical items of relatively low value when the cost of making the submittal exceeds the benefit to the project (see ER 415-1-10). Avoid such submittal requirements for small projects. Keep submittals requiring government approval to a minimum.

i. Testing. Ordinarily, testing is the contractor’s responsibility under the Contractor Quality Control (CQC) provisions of the specifications (see ER 1180-1-6). Specifications that designate testing as the contractor’s responsibility will not override the Contracting

Officer’s right to conduct confirmation testing, QA testing, or to observe the contractor’s testing. Keep government testing at government expense (outside of the tests performed by the contractor under the CQC procedures) to a minimum and specify it only when necessary to assure the quality of critical construction. SpecsIntact produces a test/requirements list that includes all the test requirements cited in the text, along with the sections and subparts. Provide this report to the Resident Engineer when required.

j. *Warranties.* Warranty requirements extending beyond the normal one-year construction warranty period, or such other period required by UFGS will be specified only for materials, equipment, or systems for which longer warranties are normally provided in the industry. Evaluate the increased cost of the extended warranties and the costs of administering and enforcing such warranties prior to their specification.

k. *New materials and methods.* Designers are encouraged to consider using new or innovative materials, equipment, systems, or methods in designs when evidence shows that their use is in the best interest of the government in terms of value, lower life cycle costs, and quality of construction. Manufacturers are to prove the merits of their product by certified, independent laboratory results, evidence of satisfactory installation under conditions like those anticipated for the proposed construction, and compliance with appropriate industry standards, if they exist. Refer to UFS 1-300-02 for specifying new items. Consider submitting new and innovative items to the DoD as a CCR on the WBDG website. The CCR will inform the DoD about the innovation, enabling the evaluation and implementation of change to the criteria.

l. *Brand names and proprietary items.* Specifying items peculiar to one manufacturer (closed proprietary), either by brand name or by peculiar characteristic, is prohibited unless specially justified and approved (FAR, Subpart 11.105). The use of brand name or equal (open proprietary) descriptions also requires J&A (DFAR 206.302-1(c)). When such descriptions are used, include the salient features of the item specified on which equality can be determined (FAR, Subpart 11.104, Subpart 11.107, and Subpart 36.202(c)).

m. Discipline-specific specification requirements.

(1) *Facility-related control system cybersecurity specifications*. Provide separate specifications for each FRCS and cybersecurity impact level. Begin all FRCS cybersecurity specification with “25 05 11” for the first three levels of numbers and use fourth-level numbering to differentiate specifications as described in UFC 4-010-06 and UFS 4-010-06.

(2) *Contractor furnished contractor installed furniture*. Include specification UFGS 12 50 00.13 10 or a comparable specification in the contract documents when the FF&E is contractor furnished contractor installed. Specifications must include non-proprietary, project-specific, salient characteristics for furniture items. An FF&E package is required as an attachment to UFGS 12 50 00.13 10 but may not be required for other comparable specifications. Reference 2–2.d(7) for detailed FF&E package

requirements. When including an FF&E attachment, omit unit, extended, or shipping costs on data sheets.

Chapter 5
Recommended Minimum Submittal Requirements for Military Vertical Design

5–1. General

This list of requirements represents typical minimum submittal content for MILCON projects but can be adapted for all other project types. Project teams are to review this information and validate this list against the specific scope and project requirements to generate project-specific submittal requirements. Coordinate the lists with project requirements, AR 420-1, all relevant ERs, UFCs, and other applicable criteria. The submittal phases outline requirements for D-B-B execution. Notations are added to help adapt the list for D-B RFP preparation.

5–2. Parametric design – 5% to 15% submittal

a. *Drawings.* Refer to Chapter 3 for detailed drawing requirements.

b. *Specifications.* Not required at this phase.

c. *Design analysis.* Refer to Chapter 2 for detailed DA requirements and for the OPR that corresponds to “Stakeholder Requirements” in the PMP.

5–3. Concept design – 35% submittal

AR 420-1, paragraph 4-40 concept design (code 2) states: “The design will establish all basic features, materials, construction methods, facility systems, fire plans, and related costs of the facility.” The 35% design effort establishes that the facility can be constructed according to applicable codes and criteria, the user program, and the DD Form 1391 requirements. The overall size, shape, and configuration of the building(s) are defined. The requirements for a concept or 35% submittal include, but are not limited to:

a. *Drawings.* Refer to Chapter 3 for detailed drawing requirements. A project delivery team must tailor minimum content to match the project scope.

b. *Advanced modeling.* Include the following documents and files and the USACE Advanced Modeling submittal checklist for the advanced modeling submittal. Refer to EM 1110-1-2909.

(1) *Advanced modeling project execution plan.* An electronic copy of the most current approved version of the project advanced modeling PxP. For in-house design projects, this will be included as an appendix to the DA. For projects designed by A-Es, this will be a required submittal.

(2) *Electronic files.* An electronic list (.txt file or similar) of all submitted electronic files, including a description, directory, and file name for each file submitted. Identify which files were produced from the model and facility data. For all sheet files, include a list of the sheet titles and sheet numbers.

(3) *Advanced modeling submittal checklist.* Complete the USACE BIM/CIM advanced modeling submittal checklist and include it with each submittal. Download the latest checklist from the USACE CAD/BIM Technology Center website (https://cadbimcenter.erdc.dren.mil/).

(4) *Advanced modeling files.* Provide all native advanced modeling files associated with the production of the contract drawings and associated as-modeled drawings.

(5) *Quality control reports.* Provide electronic PDFs of all QC reports and checklists used to verify full compliance with the contract requirements and standards.

(6) *Computer-aided design exports of building information modeling, generated sheets, and drawings.* Provide supplemental two-dimensional CAD exports from the project BIM model as needed to demonstrate compliance with contract requirements.

c. Specifications. Edited table of contents. (For draft D-B RFP submittal requirements, refer to 65% submittal for specification requirements).

d. Design analysis. Refer to Chapter 2 for detailed DA requirements.

(1) Initial acquisition strategy.

(2) (For D-B-B) Updated CWE (class 3 or better), CSRA (if applicable), ENG Form 6196, and ENG Form 3086.

(3) BOD Cx format covering commissioned systems.

(4) OPR format covering commissioned systems.

(5) Geotechnical data report.

(6) Initial site survey including identification of all major site utilities.

5–4. Interim design – 65% submittal

a. Drawings. Update and develop the drawings from the 35% submittal and add new 65% drawings as required by each discipline. Incorporate revisions as necessary to address previous review comments. Refer to Chapter 3 for detailed drawing requirements.

b. Advanced modeling. See paragraph 5–3.b for detailed requirements.

(1) Advanced modeling PxP.

(2) Electronic files.

(3) Advanced modeling submittal checklist.

(4) Advanced modeling files.

(5) QC reports.

(6) CAD exports of BIM-generated sheets and drawings.

c. Specifications. Refer to Chapter 4.

(1) Updated table of contents.

(2) Edit specification sections and include any necessary attachments. Leave edits visible for review. For Draft D-B RFP, include all Division 01 specifications and attachments. Example of attachments to the UFGS 01 11 00 include: scope of work, concept drawings, FF&E, communication information, environmental information, fire flow test results, photos of existing conditions, and as-built drawings. Level of development may vary between disciplines.

(3) Bid schedule indicating bid items, units of measure, and estimated quantities.

d. Design analysis. Updated from 35% submittal. Refer to Chapter 2 for detailed DA requirements.

(1) Final acquisition strategy.

(2) Updated CWE (class 2 or better).

(3) VE study.

(4) Outline of ECIFP.

(5) Final geotechnical data report.

5–5. Final design – 95% submittal

a. Drawings. Complete the drawings from the previous submittal. Incorporate revisions as necessary to address all previous review comments. Refer to Chapter 3 for detailed drawing requirements.

b. Advanced modeling. See paragraph 5–3.b for detailed requirements.

(1) Advanced modeling PxP.

(2) Electronic files.

(3) Advanced modeling submittal checklist.

(4) Advanced modeling files.

(5) QC reports.

(6) CAD exports of BIM-generated sheets and drawings.

c. *Specifications.* Final edited specifications, including all required attachments. Refer to Chapter 4.

(1) Coordinate with contracting to prepare the Division 00 specifications to be included in this submittal. Contract clauses, instructions to bidders, and selection criteria are established.

(2) All final specifications will incorporate necessary revisions to address review comments. Edits no longer need to be visible for final review.

(3) All errors have been resolved, and all brackets have been edited.

(4) Updated bid schedule.

d. *Design analysis.* Updated from the 65% submittal. Refer to Chapter 2 for detailed DA requirements.

(1) Updated acquisition strategy if it has changed.

(2) Updated CWE (Class 1).

(3) VE study documentation, if applicable, complete and certified.

(4) Final ECIFP.

5–6. Corrected final – 100% submittal

a. *Drawings.* Corrected final drawings incorporating any required revisions from the 95% (or Final D-B RFP) submittal to allow all reviewers to back check and close remaining comments. Refer to Chapter 3 for detailed drawing requirements.

b. *Advanced modeling.* See paragraph 5–3.b.

(1) Advanced modeling PxP.

(2) Electronic files.

(3) Advanced modeling submittal checklist.

(4) Advanced modeling files.

(5) QC reports.

(6) CAD exports of BIM-generated sheets and drawings.

c. *Specifications.* Corrected final specifications with all comments addressed. Refer to Chapter 4.

(1) Final division 00 Specifications.

(2) Final bid schedule.

d. Design analysis. Corrected final DA updated to document any revisions made to the final design, including any final updates to the acquisition strategy, CWE, VE Study, or ECIFP. Refer to Chapter 2 for detailed DA requirements.

5–7. Ready to advertise (Ready to advertise/Ready to advertise design-build request for proposal)

a. Drawings. Final drawings that resolve all previous review comments, including all appropriate signatures. Refer to Chapter 3 for detailed drawing requirements.

b. Advanced modeling. See paragraph 5–3.b for detailed requirements.

(1) Advanced modeling PxP.

(2) Electronic files.

(3) Advanced modeling submittal checklist.

(4) Advanced modeling files.

(5) QC Reports.

(6) CAD exports of BIM-generated sheets and drawings.

c. Specifications. Final edited specifications. Coordinate with the project Specifications Engineer on additional documents required for the RTA submittal, such as:

(1) Signed ENG Form 6055 or waiver.

(2) Signed liquidated damages/construction memo.

(3) (D-B-B only) Signed district quality control/A-E quality control certifications from each discipline.

(4) DrChecks screenshot showing all review comments as closed.

(5) Report with DrChecks BCOES review comments.

(6) Report with DrChecks DQC review comments.

(7) Final bid schedule.

(8) (D-B-B Only) Attachment 00 72 00-A index of drawings.

(9) BCOES certification form.

(10) (D-B-B in-house only) Design authentication form.

(11) Certification needed prior to advertisement, if applicable (for example, see BCOES Certification per ER 415-1-11, Appendix 1).

(12) Real estate certification form.

(13) Scope certification form.

(14) Other documents on request (for example, J&As).

d. Design analysis. Final DA that resolves all previous review comments. Refer to Chapter 2 for detailed DA requirements.

Chapter 6
Signatures on design documents

6–1. Signature of design documents by registered U.S. Army Corps of Engineers and contractor professionals

a. USACE District and Center Chiefs of Engineering, District Chiefs of Construction, or their equivalent will sign and indicate their professional registration on appropriate design documents, permit applications, and certifications. When administratively requested by state or local authorities, although not legally required to do so, a currently registered USACE professional at a USACE Center or District may also stamp or seal documents, or portions of documents, prepared by in-house USACE personnel.

b. A-E contractors will prepare or review and approve design documents, permit applications, or certifications as required by the A-E contract. If the work requirements and the A-E contract necessitate signature and a stamp or seal for specific document sections for specific engineering disciplines such as structural, mechanical, electrical, or fire protection designs, then the currently registered A-E professional(s) overseeing the production of those documents will sign and stamp or seal the documents associated with that specialty. The complete set of design documents will be signed and stamped or sealed by the currently registered designer(s) of record.

c. The D-B contractor's designer(s) of record will prepare or review and approve their work by affixing their signature and stamp or seal on design documents, permit applications, or certifications as required by the construction contract and applicable state laws or regulations.

6–2. Review, approval, and signature on design documents and indication of registration

a. District and Center Chiefs of Engineering or their deputies will review and approve all appropriate design documents and associated certifications, as well as all appropriate permit applications prepared by in-house USACE personnel. Their approval and professional registration will be indicated by their signature or stamp and date on appropriate design documents and permit applications. District and Center Chiefs of Construction (or their equivalent) or their deputies will sign and date USACE certifications required during or after construction. Districts and Centers are encouraged to contact HQUSACE for guidance concerning unusual situations.

b. The responsible registered professional's signature must be followed by "P.E." (Professional Engineer), "R.A." (Registered Architect), or other appropriate designation indicating that the signer is a currently registered professional. At the discretion of the District or Center engineering and construction leadership, all documents may be sealed or stamped rather than using the P.E. or R.A. designation.

c. Individuals signing per paragraph 6–1 of this regulation are required to do so within the scope of their employment. Documents to be submitted to federal, state, or local authorities that contain initials, signatures, or seals (or other indication of registration such as P.E.) will contain a statement that the documents are executed according to this regulation. For example, the cover sheet for the project plans and specifications will include a statement such as the following: “This project was designed by the (name of District) District of the U.S. Army Corps of Engineers. The initials or signatures and registration designations of individuals appear on these project documents within the scope of their employment as required by ER 1110-345-700.”

d. The purpose of this requirement is to establish a clear, written record to be used in case of litigation, indicating that USACE employees who sign documents are doing so within the scope of their employment and specific, written authority and thus are not personally liable.

6–3. Signing and sealing or stamping of Architect-Engineer or design-build contract deliverables

A-E and D-B contracts will require the A-E or D-B contractor to sign and stamp or seal and date at least one set of electronic design documents (including DA, drawings, and specifications), permit applications, or certifications.

6–4. Changing in-house Architect-Engineer or design-build contract designs

a. In-house designs.

(1) If a design document, permit application, or certification created by USACE in-house designers is changed by someone other than the original professional, a clear record of internal responsibility for the change must be maintained. Accordingly, when a change is prepared by someone other than the original professional, before the change is implemented, a written record will be made describing the change and the reason for making the change, showing the date, signature, and title of the individual making the change.

(2) Significant changes, such as changes impacting the design intent or details of implementation, will be made only in consultation with the original designer and with the written concurrence of the original designer. If the original designer is no longer employed by USACE, then the professional who is duly designated to succeed or assume the original designer’s role must be consulted appropriately, and this professional will provide the necessary review and written concurrence of the significant change to the original design.

(3) In the event of a conflict between the original professional and acceptance of a management-directed change, the resolution must be made by the supervising technical professionals and documented to establish clear internal responsibility for the change.

b. Architect-engineer contractor designs.

(1) Changes to A-E contractor-prepared design documents, permit applications, and certifications should not be made by anyone other than the original designers. If a document prepared under an A-E contract is changed by someone other than the original A-E designers, then the A-E may successfully argue that it is relieved of some responsibility for the original design, and the government may have difficulty enforcing A-E liability under FAR Subpart 36.608.

(2) If a document created by an A-E contractor is changed by someone other than the original professional, a clear record of internal responsibility for the change must be maintained. Accordingly, when a change is prepared by someone other than the original professional, before the change is implemented, a written record must be made describing the change and the reason for making the change, showing the date, signature, and title of the individual making the change.

(3) Significant changes, such as changes impacting the design intent or details of implementation, will be made only in consultation with the original designer and with the original designer's written concurrence.

c. Design-bid contractor designs. Changes to D-B contractor-prepared design documents, permit applications, and certifications should not be made by anyone other than the original designers.

Appendix A
References

Section I

Required Publications

Unless otherwise indicated, all Army and USACE publications are available at https://armypubs.army.mil and https://www.publications.usace.army.mil. DoD publications are available on the Executive Services Directorate website at https://www.esd.whs.mil. Federal Acquisition Regulations (FARs) are available at https://www.acquisition.gov/browse/index/far. National Fire Protection Association (NFPA) publications are available at https://www.nfpa.org/for-professionals/codes-and-standards/list-of-codes-and-standards. Unified Facilities Criteria (UFC) and Unified Facilities Supplements (UFS) publications are available on the Whole Building Design Guide website at https://www.wbdg.org/dod/ufc.

AR 380-5
Army Information Security Program

AR 420-1
Facilities Management

ASHRAE Guideline 0
The Commissioning Process

ASHRAE Standard 90.1
Energy Standard for Sites and Buildings Except Low-Rise Residential Buildings

ASHRAE Standard 202
Commissioning Process for Buildings and Systems
(Available at https://www.ashrae.org/)

CSI MasterFormat® 2020 Edition
(Available at csiresources.org)

DA Pam 25-403
Army Guide to Recordkeeping

DFAR 206.302-1(c)
Application for brand-name descriptions
(Available at https://www.acquisition.gov/dfars/206.302-1-only-one-responsible-source-and-no-other-supplies-or-services-will-satisfy-agency-requirements.?searchTerms=206.302-1)

EM 5-1-11
Project Delivery Business Process

EM 200-1-2
Environmental Quality Technical Project Planning Process

EM 1110-1-2909
Geospatial Data and Systems

ER 415-1-10
Contractor Submittal Procedures

ER 415-1-11
Biddability, Constructability, Operability, Environmental, and Sustainability (BCOES) Reviews

ER 1110-1-8152
Professional Registration and Signature on Design Documents

ER 1110-1-8155
Specifications

ER 1110-3-12
Quality Management

ER 1110-345-723
Total Building Commissioning Procedures

ER 1140-1-211
Support for Others: Reimbursable Services

ER 1180-1-6
Construction Quality Management

ERDC/ITL SR-23-1
A/E/C Graphic Standards
(Available at https://cadbimcenter.erdc.dren.mil/)

ERDC/ITL TR-19-7
A/E/C Computer-Aided Design (CAD) Standard
(Available at https://cadbimcenter.erdc.dren.mil/)

ERDC/LAB TR 21-4
USACE Advanced Modeling BIM/CIM Object Standard
(Available at https://cadbimcenter.erdc.dren.mil/BIM)

FAR Subpart 11.104
Use of brand name or equal purchase descriptions

FAR Subpart 11.105
Items peculiar to one manufacturer

FAR Subpart 11.107
Solicitation provision

FAR Subpart 11.201 (a and b)
Identification and availability of specifications

FAR Subpart 36.102
Definitions

FAR Subpart 36.202(c)
Specifications

FAR Subpart 36.207
Pricing fixed-price construction contracts

FAR Subpart 36.608
Liability for Government costs resulting from design errors or deficiencies

Federal Specifications, Federal Standards, and Commercial Item Descriptions
(Available at https://fedspecs.gsa.gov/s/)

HNC Furniture Program MRSI Website
(Available at https://mrsi.erdc.dren.mil/furniture/)

IEEE C57.12.00
IEEE Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers(Available at https://standards.ieee.org/ieee/C57.12.00/6962/)

High Performance and Sustainable Building (HPSB) Guiding Principles(Available at https://www.wbdg.org/FFC/FED/hpsb_guidance.pdf)

International Building Code (IBC)
(Available at https://codes.iccsafe.org)

Military Specifications and Military Standards
(Available at https://www.dsp.dla.mil)

MIL-STD-3007
Standard Practice Unified Facilities Criteria, Facilities Criteria and Unified Facilities Guide Specifications
(Available at https://www.wbdg.org/dod/fedmil/mil-std-3007)

NFPA 70
National Electrical Code (NEC)

NFPA 72
National Fire Alarm and Signaling Code

NFPA 101
Life Safety Code®

SpecsIntact
eLearning Modules and Web-Based Help
(Available at https://SpecsIntact.wbdg.org)

UFC 1-200-01
DoD Building Code

UFC 1-200-02
High Performance and Sustainable Building Requirements

UFC 3-101-01
Architecture

UFC 3-410-02
Direct Digital Control for HVAC and Other Building Control Systems

UFC 3-420-01
Plumbing Systems

UFC 3-470-01
Utility Monitoring and Control System (UMCS) Front End and Integration

UFC 3-501-01
Electrical Engineering

UFC 3-600-01
Fire Protection Engineering for Facilities

UFC 4-010-01
DoD Minimum Antiterrorism Standards for Buildings

UFC 4-010-06
Cybersecurity of Facility-Related Control Systems (FRCS)

UFGS 01 11 00
Summary of Work

UFGS 01 33 16.00 10
Design Data (Design After Award)

UFGS 12 50 00.13 10
Furniture and Furniture Installation

UFS 1-300-02
Unified Facilities Guide Specifications (UFGS) Format Standard

USACE Technical Information-Facilities Design Website
(Available at https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/)

WBDG Resource Library
(Available at https://wbdg.org/ffc/army-coe/resource-library)

Section II

Prescribed Forms

DD Form 1391
FY__ Military Construction Project Data

ENG Form 3086
Current Working Estimate for Budget Purposes

ENG Form 6055
Contract Requirements Package Security Review Cover Sheet

ENG Form 6196
Designated Department of Defense Construction Agent (DCA) Assessment

Glossary of Terms

Section I

Acronym List

Term	**Definition**
A/E/C	Architecture/Engineering/Construction
A-E	Architect-Engineer
ANSI	American National Standards Institute
AR	Army Regulation
ASHRAE	American Society of Heating, Refrigerating, and Air-Conditioning Engineers
ATO	Authority to Operate
AV	Audio/Visual
BCOES	Biddability, Constructability, Operability, Environmental and Sustainability
BIM	Building Information Modeling
BOD	Basis of Design
CAD	Computer-Aided Design
CCR	Criteria Change Request
CFCI	Contractor furnished, contractor installed
C-I-A	Confidentiality, Integrity, and Availability
CIM	Civil Information Modeling
CQC	Contractor Quality Control
COS	Center of Standardization
CSI	Construction Specifications Institute
CSRA	Cost Schedule Risk Analysis
CWE	Current Working Estimate
Cx	Commissioning
CxD	Commissioning Specialist for the Design Phase
CxG	Commissioning Specialist for the Government
DA	Design Analysis
D-B	Design-Bid
D-B-B	Design-Bid-Build
DFAR	Defense Federal Acquisition Regulation
DFPE	District Fire Protection Engineer
DoD	Department of Defense
DOR	Designer of Record
ECIFP	Engineering Considerations and Instructions for Field Personnel
EM	Engineer Manual
ER	Engineer Regulation

Term	Definition
ERDC/ITL	Engineer Research and Development Center/Information Technology Laboratory
FAMN	Fire Alarm/Mass Notification
FAR	Federal Acquisition Regulation
FED-SPEC	Federal Specifications
FED-STD	Federal Standards
FF&E	Furniture, Fixtures, and Equipment
FRCS	Facility Related Control System
FSC	Federal Supply Category
GFCI	Government furnished, contractor installed
GFGI	Government furnished, government installed.
GSA	General Services Administration
HNC	Huntsville Engineering and Support Center
HPSB	High Performance and Sustainable Building
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HTRW	Hazardous, Toxic, and Radioactive Waste
HVAC	Heating, Ventilation, and Air Conditioning
IBC	International Building Code
IEEE	Institute of Electrical and Electronics Engineers
IMCOM	Installation Management Command
J&A	Justification and Approval
kVA	Kilovolt-Ampere
kVAR	Kilovolt-Ampere Reactive Power
LCCA	Life Cycle Cost Analysis
LEED	Leadership in Energy and Environmental Design
MCC	Motor Control Center
MDP	Main Distribution Panel
MILCON	Military Construction
MIL-SPEC	Military Specifications
MIL-STD	Military Standards
NASA	National Air and Space Administration
NAVFAC	Naval Facilities Engineering Systems Command
NEMA	National Electric Manufacturers Association
NFPA	National Fire Protection Association
O&M	Operations and Maintenance
OPR	Owner’s Project Requirements
P.E.	Professional Engineer
PDF	Portable Document Format
PMP	Project Management Plan
POC	Point of Contact

Term	Definition
PxP	Project Execution Plan
QA	Quality Assurance
QC	Quality Control
QFPE	Qualified Fire Protection Engineer
QMP	Quality Management Plan
R.A.	Registered Architect
RFI	Request for Information
RFP	Request for Proposal
RTA	Ready to Advertise
SCIF	Sensitive Compartmented Information Facility
SID	Structural Interior Design
SIN	Special Item Number
SO	System Owner
SRM	Sustainment Restoration and Modernization
STC	Sound Transmission Class
TPC	Third-Party Certification
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
UH	Unaccompanied Housing
UMCS	Utility Monitoring and Control System
USACE	U.S. Army Corps of Engineers
VE	Value Engineering
WBDG	Whole Building Design Guide

Section II

Terms

Advanced Modeling Project Execution Plan
The governing document that establishes how to execute, monitor, and control projects. The plan is the main communication vehicle to ensure that everyone is aware of and knows the project objectives and how they will be accomplished. Project objectives are derived from the mission needs statement, and an integrated project team assists in developing the PxP. The plan is a living document and should be updated to describe current and future processes and procedures such as integrating safety into the design process. Updates are common as a project moves through critical decision phases.

Basis of Design for Commissioning
A stand-alone subset or sections of the design analysis that address commissioned systems and elements. Where ER 1110-345-723, ASHRAE Standard 202, or ASHRAE Guideline 0 use the terms “Basis of Design” or “BOD,” those references will be understood as referring to BOD Cx.

CSI MasterFormat
Industry standard for organizing construction project information to create a consistent, logical format for specifications, costs data and other project details using a hierarchical system of 50 divisions.

Design Analysis
The document that contains written material covering general parameters, functional and technical requirements, design objectives, design assumptions, and provides design calculations applicable to a project’s design.

Design-Bid-Build
Per FAR 36.102, a project acquisition and delivery method where design and construction are sequential, and construction is contracted separately. For USACE in-house D-B-B, the executing District is DOR. For A-E contracted D-B-B, the A-E is DOR.

Design-Build
Per FAR 36.102, a project acquisition and delivery method that combines design and construction in a single contract with one contractor.

Designer of Record
The registered professional responsible for the technical correctness of the design, supervises the design team, and who signs and seals the design documents.

Drawing
A static graphic representation of design content formatted as a two-dimensional sheet.

Engineering Considerations and Instructions for Field Personnel
An information bridge between designers, design-build preparers, and construction personnel for both design-build and design-bid-build projects. The ECIFP conveys risk elements identified during the design, design assumptions of critical project features, and initial project coordination of the project delivery team to USACE construction field personnel.

Justification and Approval
A document required to justify and obtain appropriate level approvals to contract without providing for full and open competition as required by the FAR.

Project Management Plan
Per EM 5-1-11, “A formal, approved, living document used to define requirements and expected outcomes and guide project execution and control.”

Owner's Project Requirements
A stand-alone subset of the DA that addresses commissioned systems and elements. Per ER 1110-345-723, it is a "written document that details the project requirements and the expectations of how the building and its systems must be used and operated. These include project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information."

Reference Standards
Documents that contain requirements set by authority, custom, or general consensus and are established as accepted criteria. Trade associations, professional societies, standards-writing organizations, governments, and institutional organizations such as the American National Standards Institute (ANSI) and ASTM International publish reference standards. The UFGS and project specifications refer to these documents to define qualitative and performance requirements for materials, equipment, systems, test methods, and workmanship.

Request for Proposal
A request for proposal is the procurement document normally used to procure D-B projects. An RFP for a D-B contract should state project requirements, criteria, and evaluation factors. It provides the information as well as framework necessary for offerors to submit proposals.

Specifications Engineer
An Architect, Engineer or Technician who is assigned primary responsibility for overseeing the collective preparation of project specifications/construction documents on an individual project and has the responsibility for overall coordination of the final RTA package for the individual project.

Specifier
An Architect, Engineer, or Technician who is assigned the preparation of project specifications and coordination of the specifications with the other construction documents. The Specifier works under the direction of a Specifications Engineer or Designer.

SpecsIntact
A software program, developed in partnership between NASA and DoD, mandated for use in producing USACE project specifications and maintaining guide specifications. The software provides specification automation to users and incorporates a wide range of quality control features.

Stakeholder Requirements
Per EM 5-1-11, Stakeholder Requirements "outline project objectives, key deliverables, decision points, and constraints related to the project. Assumptions made during the development of the scope should also be documented."

Submittal

The complete package of documents that is delivered at completion of a design milestone as identified in the PMP, or as required during the design process. Submittals can be for internal or external use.

Third-Party Certification
Per UFC 1-200-02, third-party certification is “the generic term for a third-party product that provides either certification of the third-party’s specific product or a validation program by the third-party.”

SECTION 01 33 29

SUSTAINABILITY REQUIREMENTS AND REPORTING
02/21, CHG 1: 08/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. For references without a year, use the current published version.

COUNCIL ON ENVIRONMENTAL QUALITY (CEQ) (WHITE HOUSE)

HPSB Guiding Principles	Guiding Principles for Sustainable Federal Buildings and Determining Compliance with the Guiding Principles for Sustainable Federal Buildings

INTERNATIONAL CODE COUNCIL (ICC)

ICC IgCC	(2018) International Green Construction Code

U.S. DEPARTMENT OF AGRICULTURE (USDA)

FSRIA 9002	Farm Security and Rural Investment Act Section 9002 (USDA BioPreferred Program)

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-200-02	(2020; Change 2, 2022) High Performance and Sustainable Building Requirements
UFC 3-410-01	(2013; with Change 9, 2024) Heating, Ventilating, and Air Conditioning Systems
UFC 3-600-01	(2016; with Change 6, 2021) Fire Protection Engineering for Facilities

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star	Energy Star Energy Efficiency Labeling System (FEMP)
FEMP	Federal Energy Management Program (FEMP) Energy-Efficient Product Procurement

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 247	Comprehensive Procurement Guideline for Products Containing Recovered Materials

1.2 SUMMARY

This section includes requirements for Sustainability documentation and reporting submittals per the federally mandated High Performance and Sustainable Building (HPSB) or HPSB "Guiding Principles" (GP), in accordance with UFC 1-200-02 High Performance and Sustainable Building Requirements, and other identified requirements.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government:

SD-01 Preconstruction Submittals

Preliminary High Performance and Sustainable Building Checklist; G

Sustainability Action Plan; G

Preliminary Sustainability eNotebook; G

SD-05 Design Data

Interim Design High Performance and Sustainable Building Checklist; G

Interim Design Sustainability eNotebook; G

Final Design High Performance and Sustainable Building Checklist; G

Final Design Sustainability eNotebook; G

SD-11 Closeout Submittals

Final High Performance and Sustainable Building Checklist; G

Final Sustainability eNotebook; G

1.4 GUIDING PRINCIPLES VALIDATION (GPV)

Guiding Principles Validation (GPV) is the project delivery team's self-assessment of UFC 1-200-02 compliance requirements, which includes meeting Federally-required tracking and reporting requirements, in order to report a sustainable Federal building. The use of a third-party certification is an additional and separate requirement.

Provide the following sustainability activities and documentation to verify achievement of HPSB Guiding Principles Validation (GPV):

a. Analysis of each Guiding Principle Requirement and how project complies. Include final government approved narrative(s) in the HPSB Checklist submittal. Multiple checklists indicate multiple buildings that require individual HPSB Checklist tracking.

b. No changes to the HPSB Checklist are allowed without approval from the Contracting Officer, in accordance with Section 01 33 00 SUBMITTAL REQUIREMENTS. Immediately bring to the attention of the Contracting Officer any project changes that impact meeting the approved HPSB Guiding Principles Requirements for this project. Demonstrate the change will not increase the life-cycle cost and maintains or improves the building performance.

c. Documentation of all work required to incorporate the applicable HPSB Guiding Principles requirements indicated on the HPSB Checklist and in this contract, including all "S" submittals.

d. Sustainability Action Plan.

e. Design and construction related documentation for the project Sustainability eNotebook and keep updated with regularly-scheduled Construction Quality Control Meetings. Include design and construction related documentation containing the following components:

(1) HPSB Checklist(s)

(2) Sustainability Action Plan

(3) Documentation illustrating HPSB Guiding Principles Requirements compliance, including "S" submittals

1.4.1 Sustainability Action Plan

Include the following information in the Sustainability Action Plan:

a. Analysis of each HPSB Guiding Principles Requirement and how project will comply. Final government approved narrative(s) must be included in the HPSB Checklist submittal.

b. Name and contact information for: Contractor's Point of Contact (POC) ensuring sustainability goals are accomplished and documentation is assembled. For TPC that include on-site visit by third party representative, provide list of required attendees.

c. Indoor Air Quality plan.

1.4.2 Calculations

Provide all design data, calculations, product data, labels and product certifications required in this specification to demonstrate compliance with the HPSB Guiding Principles Requirements.

1.5 SUSTAINABILITY SUBMITTALS

Provide HPSB Checklist and other documentation in the Sustainability eNotebook to indicate compliance with the sustainability requirements of the project.

1.5.1 High Performance Sustainable Building (HPSB) Checklist

Provide construction documentation that provides proof of, and supports compliance with, the completed HPSB Checklist.

1.5.1.1 HPSB Checklist Submittals

Submit updated HPSB Checklist with each Sustainability eNotebook submittal. Include the final HPSB Checklist(s) with the interim DD1354 Real Property Record Submittal.

1.5.2 "S" Submittals for Sustainability Documentation

"S" submittals are the sustainability documentation requirements cited in the various sections of this contract. Submit the GPV sustainability documentation required in this section as "S" submittals in all affected UFGS Sections.

a. Highlight GPV compliance data in "S" submittal.

b. Add "S" submittals to the Sustainability eNotebook only after submittal approval, and bookmark them as required in paragraph SUSTAINABILITY ENOTEBOOK below.

c. Ensure all approved "S" submittals are included in each Sustainability eNotebook submittal.

1.5.3 Sustainability eNotebook

The Sustainability eNotebook is an electronic organizational file that serves as a repository for all required sustainability submittals. To support documentation of compliance with an approved HPSB checklist, provide and maintain a comprehensive and current Sustainability eNotebook. Include all required data in Sustainability eNotebook, to support full compliance with the HPSB Guiding Principles Requirements, including:

a. HPSB checklist

b. Sustainability Action Plan

c. Calculations

d. Labels

e. "S" submittals

1.5.3.1 Sustainability eNotebook Format

Provide Sustainability eNotebook in the form of an Adobe PDF file; bookmark each HPSB Guiding Principles Requirement and sub-bookmark at each document. Match format to HPSB Guiding Principles numbering system indicated herein. Maintain up-to-date information, such as spreadsheets, templates, with each current submittals.

Contracting Officer may deduct from the monthly progress payment accordingly if Sustainability eNotebook information is not current and on track per project goals.

1.5.3.2 Sustainability eNotebook Submittal Schedule

Provide Sustainability eNotebook Submittals at the following milestones of the project:

a. Preliminary Sustainability eNotebook

b. Submit preliminary Sustainability eNotebook with updated Preliminary High Performance and Sustainable Building Checklist at the first post award meeting in accordance with Section 01 30 00 ADMINISTRATIVE REQUIREMENTS.

c. Interim Design Sustainability eNotebook

d. Submit updated Sustainability eNotebook with updated Interim Design High Performance and Sustainable Building Checklist with the 100 percent design. If issues relating to achieving the sustainability goals of the project are subsequently identified, identify reasons and mitigation from DOR, and resubmit to the Contracting Officer for approval.

e. Final Design Sustainability eNotebook

f. Submit updated Sustainability eNotebook with updated Final Design High Performance and Sustainable Building Checklist with the final design. If issues relating to achieving the sustainability goals of the project are subsequently identified, identify reasons and mitigation from DOR, and resubmit to the Contracting Officer for approval.

g. Construction Quality Control Meetings.

h. Provide up-to-date GP documentation in the Sustainability eNotebook for each meeting.

i. Final Sustainability eNotebook

j. Submit updated Sustainability eNotebook with updated Final High Performance and Sustainable Building Checklist within 7 days of the completion of the proejct. Final progress payment retainage may be held by Contracting Officer until Final Sustainability construction phase documentation is complete.

1.6 DOCUMENTATION REQUIREMENTS

a. Incorporate each of the following HPSB Guiding Principles requirements into project and provide documentation that proves compliance with each listed requirement. Items below are organized by HPSB Guiding Principles. For life-cycle cost analysis requirements, one document with all analyses is acceptable, with Contracting Officer approval.

b. For each of the following paragraphs that require the use of products listed on Government-required websites, provide documentation of the process used to select products, or process used to determine why listed products do not meet project performance requirements.

1.6.1 Integrated Design Process

For the submittal documentation below, demonstrate compliance with UFC 1-200-02.

1.6.1.1 Design Submittal Documentation

a. List the sustainability integrated design team, and a description of their roles in all stages of a project's planning and delivery:

(1) Include Contractor's Sustainability Coordinators; Architecture and Engineering disciplines involved on the project, and the DOR in charge of the overall project and each discipline; Construction Subcontractors and the company representatives that

align with each architectural and engineering discipline, Planning, Public Works, Environmental Specialist and other appropriate installation personnel.

(2) Describe their roles and responsibilities and plan-of-action for how each team member will be involved to achieve the project sustainability requirements, and how the Contractor will coordinate with Government personnel.

(3) Maintain an up-to-date list with descriptions throughout the project.

b. Provide narratives that:

(1) Indicate each performance goal cited in UFC 1-200-02 paragraph "Integrated Design Process", along with other comprehensive design goals and ensures incorporation and integration of these goals throughout the planning, design, construction, and life cycle of the building..

(2) Demonstrate integration of the goals into design and construction.

(3) Demonstrate collaboration with other providers, such as Commissioning Provider.

1.6.2 Optimize Energy Performance

For the submittal documentation below, demonstrate compliance with UFC 1-200-02.

1.6.2.1 Design Submittal Documentation

a. Narrative that provides a summary of:

(1) The decision-making process for considering and implementing passive strategies prior to designing active and mechanical systems.

(2) The decision-making process leading to the selection of at least three energy-efficient solutions (for each system contributing to the energy footprint of the building) to be analyzed; and the selected design solution(s)

(3) The specific energy standard and version utilized; and the software used in the analysis

(4) The calculated energy consumption and energy use intensity (EUI in kBTU/sf/yr) of the baseline building and the proposed design alternatives

b. A minimum of the following energy modeling files and summaries for the baseline and proposed alternatives:

(1) Input, schedules and libraries; and output

(2) Calculated energy use by energy type

(3) Calculated energy use by building system

c. The life-cycle cost analysis input and output files for the baseline and the proposed alternatives

1.6.2.2 Construction Submittal Documentation

Provide revised energy modeling for actual system constructed.

1.6.3 Energy Efficient Products

Provide only energy-using products that are Energy Star Certified by the Environmental Protection Agency (EPA) or that meet or exceed the efficiency standard designated by the Federal Energy Management Program (FEMP). Where Energy Star or FEMP requirements have not been established, provide most efficient products that are life-cycle cost-effective. Provide only energy using products that meet FEMP requirements for low standby power consumption. Energy efficient products can be found at: https://www.energystar.gov/.

For construction submittal documentation, provide proof that product is labeled energy efficient and complies with the cited requirements.

1.6.4 Building-level Power Metering

Provide building-level meters for electricity, natural gas, and steam where applicable.

1.6.4.1 Design Submittal Documentation

Provide design drawings that highlight meter locations on the site.

1.6.4.2 Construction Submittal Documentation

Provide documentation of meter communication with the base-wide meter monitoring system, and manufacturer's data validating compatibility with base-wide system and component advanced meter requirements.

1.6.5 Ventilation and Thermal Comfort

For the submittal documentation below, demonstrate compliance with UFC 1-200-02 and UFC 3-410-01.

1.6.5.1 Design Submittal Documentation

Provide design drawings and calculations that demonstrate HVAC systems and the building envelope have been designed to meet the requirements.

1.6.6 Daylighting

For the submittal documentation below, demonstrate compliance with UFC 1-200-02.

1.6.6.1 Design Submittal Documentation

a. Provide floor plans and elevations that demonstrates all required illumination levels, including automated and daylighting controls, and daylighting where people regularly occupy and gather.

b. Provide design analysis delineating requirements, to include compliant reflective surface locations and shading devices (where applicable).

1.6.7 Moisture Control

Provide the following:

1.6.7.1 Design Submittal Documentation

Provide drawings of building envelope details, HVAC humidity controls, and design calculations.

1.6.7.2 Construction Submittal Documentation

Ensure construction materials are separated and protected in accordance with other sections in this contract document, with adequate humidity controls during construction. In accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA, includes plan for ongoing building moisture control.

1.6.8 Reduce Volatile Organic Compounds (VOC) (Low-Emitting Materials)

Meet the requirements of Table 3-1 at the end of this specification.

For Construction submittal documentation, provide certifications or labels that demonstrate compliance with cited requirements, based on the attached TABLE 3-1.

1.6.9 Indoor Air Quality During Construction

Prior to construction, create indoor air quality plan. Develop and implement an IAQ construction management plan during construction and flush building air before occupancy.

Provide documentation showing that after construction ends and prior to occupancy, HVAC filters were replaced and area air was flushed out in accordance with the cited standard.

1.6.10 Recycled Content

Comply with 40 CFR 247. Refer to:

https://www.epa.gov/smm/comprehensive-procurement-guideline-cpg-program

for assistance identifying products cited in 40 CFR 247. Selected products must comply with non-proprietary requirements of the Federal Acquisition Regulation and must meet performance requirements.

1.6.10.1 Construction Submittal Documentation

a. Provide manufacturers' documents stating the recycled content by material, or written justification for claiming one of the exceptions allowed on the cited website.

b. Substitutions: Submit for Government approval for proposed alternative products or systems that provide equivalent performance and appearance and have greater contribution to project recycled content requirements. For all such proposed substitutions, submit with the Sustainability Action Plan accompanied by product data demonstrating equivalence.

c. In order to complete compliance with FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items, refer to submittal requirement for recycled/recovered material content in Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.11 Bio-Based Products

Provide products and materials composed of the highest percentage of bio-based materials (including rapidly renewable resources and certified sustainably harvested products), consistent with FSRIA 9002 USDA BioPreferred Program, to the maximum extent possible without jeopardizing the intended end use or detracting from the overall quality delivered to the end user and when available at a reasonable cost. Use only supplies and materials of a type and quality that conform to applicable specifications and standards.

Comply with FSRIA 9002 USDA BioPreferred Program. Refer to www.biopreferred.gov for the product categories and BioPreferred Catalog. Selected products must comply with non-proprietary requirements of the Federal Acquisition Regulation and must meet performance requirements. Provide the following documentation:

a. USDA BioPreferred label for each product; for bio-based products used on project but not listed with BioPreferred program, provide bio-based content and percentage.

b. In order to complete compliance with FAR 52.223-1 Biobased Product Certification, refer to submittal requirement for biobased products in Section 01 78 00 CLOSEOUT SUBMITTALS, paragraphs CERTIFICATION OF EPA DESIGNATED ITEMS and CERTIFICATION OF USDA DESIGNATED ITEMS.

1.6.12 Waste Material Management (Recycling – Design)

For the submittal documentation below, demonstrate compliance with UFC 1-200-02.

For design submittal documentation, provide drawing showing an appropriately sized and placed dedicated storage area for recyclables.

1.6.13 Waste Material Management (Recycling – Construction)

Divert demolition and construction debris in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

1.6.14 Address Risk

For design submittal documentation, provide narrative of decisions for design associated with scoped requirements.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 SUSTAINABILITY COORDINATION

Provide sustainability focus and coordination at all meetings to achieve sustainability goals. Coordinate meeting requirements with other UFGS

Sections meeting requirements in this project. Ensure the designated sustainability professional responsible for GP documentation participates in these meetings to coordinate documentation completion. Review GP sustainability requirements, HPSB Checklist documentation, Sustainability Action Plan, and completeness status of Sustainability eNotebook at the following meetings:

a. Pre-Construction Conference

b. Construction Quality Control Meetings

c. Refer to Section 01 30 00 ADMINISTRATIVE PROCEDURES for Post Award Meetings.

d. Post Award Meeting

e. Design Quality Assurance Meetings

f. Design Complete Review Meetings

g. Conduct review no later than 60 days after final design complete submission and identify any outstanding issues that affect correct completion of all documentation requirements, and actions that will achieve requirements. Conduct corrective actions.

h. Facility Turnover Meetings

i. Conduct review no later than 60 days before final turnover and identify any outstanding issues that affect correct completion of all documentation, and actions that will achieve requirements. Conduct corrective actions prior to turnover, to ensure all requirements are achieved.

3.2 TABLE 3-1 VOLATILE ORGANIC COMPOUNDS (VOC) (LOW EMITTING MATERIALS) REQUIREMENTS

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Adhesives and Sealants	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	Adhesives (carpet, resilient, wood flooring; base cove; ceramic tile; drywall and panel; primers) Sealants (acoustical; firestop; HVAC Air duct; primers) Caulks	**SCAQMD Rule 1168** (Use "other" category for HVAC duct sealant) (for firestop adhesive, UFC 3-600-01 overrides conflicting requirements)
			Aerosol adhesives	**Section 3 of Green Seal Standard GS-36** (except: cleaners, solvent cements, and primers used with plastic piping and conduit in plumbing, fire suppression, and electrical systems; HVAC air duct sealants when the application space air temp is less than 40 F (4.5 C).

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Paints and Coatings	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	Flat and nonflat, nonflat high-gloss, specialty, basement specialty, fire-resistive, floor, low-solids, rust preventative, wood, reflective wall coatings; concrete/masonry sealers; primers; sealers; undercoaters; shellacs (clear and opaque); stains; varnishes; conjugated oil varnish; lacquer; clear brushing lacquer	**Green Seal Standard GS-11**

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	EMISSIONS REQUIREMENTS
Paints and Coatings	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	Concrete curing compounds; dry fog, faux finishing, graphic arts (sign paints), industrial maintenance, mastic texture, metallic pigmented, multicolor, recycled coatings; pretreatment wash primers, reactive penetrating sealers; specialty primers, wood preservatives, and zinc primers	**California Air Resources Board (CARB) Suggested Control Measure for Architectural Coatings** or **SCAQMD Rule 1113r**
Paints and Coatings	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	High-temperature coatings; stone consolidants; swimming-pool coatings; tub- and tile-refining coatings; and waterproofing membranes	**California Air Resources Board (CARB) Suggested Control Measure for Architectural Coatings**

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	EMISSIONS REQUIREMENTS
Floor Covering Materials	For carpet, all locations: **CDPH/EHLB/Standard Method V1.1 (California Section 01350)** or label for **Section 9** of **CDPH/EHLB/Standard Method V1.1 (California Section 01350)**		none	none
Insulation	**CDPH/EHLB/Standard method V1.1** (California Section 01350) (Use "office" or "classroom" space limits for all applications)		none	none

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Composite Wood, Wood Structural Panel, and Agrifiber Products, no added urea-formaldehyde resins including laminating adhesives for composite wood and agrifiber assemblies - particleboard, medium density fiberboard (MDF), wheatboard, strawboard, panel substrates, door cores	Third-party certification (approved by CARB) of **California Air Resource Board's (CARB) regulation**, Airborne Toxic Control Measure to Reduce Formaldehyde Emissions from Composite Wood Products	or	none	**CDPH/EHLB/Standard method V1.1** (California Section 01350) (Use "office" or "classroom" space limits for all applications) (except: Structural panel components such as plywood, particle board, wafer board, and oriented strand board identified as "EXPOSURE 1," "EXTERIOR," or "HUD-APPROVED" are considered acceptable for interior use.)
Office Furniture Systems and Seating installed prior to occupancy	**ANSI/BIFMA X7.1** **ANSI/BIFMA X7.1:** (95-percent of installed office furniture system workstations and seating units) **Section 7.6.2 of ANSI/BIFMA e3** (50-percent of office furniture system workstations and seating units)		none	none

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	EMISSIONS REQUIREMENTS
Ceiling and Wall assemblies and systems including: acoustical treatments; ceiling panels and tiles; tackable wall panels and coverings; wall coverings; wall and ceiling paneling and planking	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)		none	none

-- End of Section --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
05/24, CHG 1: 05/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ANSI/ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2020) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1	(2014; R 2019) American National Standard for Industrial Head Protection

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022; ERTA 22-1) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1926	Safety and Health Regulations for Construction
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP) Design; G

Accident Prevention Plan (APP) Construction; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Hot Work Permit

Standard Lift Plan; G

Critical Lift Plan; G

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth day of each month for the previous month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment. Submit the hours via the agency construction management system of record.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and the following federal, state, and local laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

   (1) Are designated on an ENG Form 6282.

   (2) Meet minimum training and experience requirements identified in EM 385-1-1.

(3) Execute roles and responsibilities identified in EM 385-1-1.

1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1 for this project.

1.7.4 Competent Person for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of EM 385-1-1 and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1.

1.7.5 Preconstruction Conference

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.

c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan (APP) Design

Provide a site-specific Accident Prevention Plan (APP), including Activity Hazard Analyses (AHA), in accordance with EM 385-1-1, ENG Form 6293, for the design team to follow during site visits and investigations. For subsequent visits, update the plan if there are changes in the personnel who will be attending, or the tasks to be performed. Submit the APP for review and acceptance by the Government at least 15 calendar days prior to the start of the design field work after being given Notice to Proceed. Field work must not begin until the design APP is accepted by the Contracting Officer. Prior to the start of construction incorporate the Design APP into the Construction APP so that one site specific APP exists for the project and submit to the Contracting Officer for acceptance.

If the design scope includes borings or other subsurface investigations, include in the APP the type of field investigation and verification techniques, such as visual, local utility locating service scanning and third party subcontractor scanning, potholing, or hand digging within two feet of a known utility that will be required. Mark underground utilities before starting any ground-disturbing actions. Notify the Contracting Officer 15 days prior to the start of soil borings or sub-surface investigations.

1.8.2 Accident Prevention Plan (APP) Construction

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 30 calendar days after the Notice to Proceed and not less than 10 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

1.8.3 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.8.4 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including the following:

1.8.4.1 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION and referenced sources.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table
Accident Reporting Required Timeline

Accident Type	**Notify KO or COR**	**Complete Final Accident Report on ENG 3394 and provide to KO or COR**
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over $600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.

b. Near Misses: For Army projects, report all "Near Misses" to the Contracting Officer, using local accident reporting procedures, within 24 hours. The Contracting Officer will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

1.12.4.1 Certificate of Compliance

Provide a Certificate of Compliance for LHE utilized to perform work under this Contract in accordance with Section 16-7 of the EM 385-1-1. Post completed certificate on the crane.

1.12.4.2 Pre-Lift Plans

Develop either a Standard Lift Plan (SLP) or Critical Lift Plan (CLP) as required in accordance with EM 385-1-1. Utilize the Standard Lift Plan ENG Form 6203 for routine crane lifts and use the Critical Lift Plan ENR Form 6213 for non-routine crane lifts requiring more detailed planning and additional safety precautions, as defined by Section 16-2j of the EM 385-1-1. Submit lift plans to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e., welding or cutting) or operating other flame-producing/spark producing devices, from the local Fire Department. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. Contractors are required to meet all criteria before a permit is issued. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of 1 hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency local Fire Department phone number. Report any fire, no matter how small, to the responsible local Fire Department immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1.

1.14 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with EM 385-1-1, 29 CFR 1926, 29 CFR 1910, and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.14.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.15 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

a. Secure outside equipment and materials and place materials that could be damaged in protected areas.

b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.

c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

a. Head Protection that meets ANSI/ISEA Z89.1

b. Long Pants

c. Appropriate Safety Footwear

d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR Part 1910.1000). If material(s) that can be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations, stop portions of work that would disturb the material and notify the Contracting Officer immediately. Within 30 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages in sufficient time as to not result in impacts or delays to the project schedule. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1. In accordance with EM 385-1-1, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Powerhouse representative, and the Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1.

3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

3.6 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1.

3.6.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.

3.6.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State and Local requirements applicable to where work is being performed.

3.6.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.6.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.6.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
05/24

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AACE INTERNATIONAL (AACE)
726 East Park Avenue #180
Fairmont, WV 26554 USA
Ph: 304-296-8444
Fax: 304-291-5728
Internet: https://web.aacei.org/

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2311 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: http://www.ahrinet.org

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300
Email: customercare@asce.org
Internet: https://www.asce.org/

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)
180 Technology Parkway NW
Peachtree Corners, GA 30092
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: https://www.ashrae.org/

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929

E-mail: customerservice@assp.org
Internet: https://www.assp.org/

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: https://www.awwa.org/

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: https://www.astm.org/

COUNCIL ON ENVIRONMENTAL QUALITY (CEQ) (WHITE HOUSE)
722 Jackson Place
Washington DC 20506
Internet: https://www.whitehouse.gov/administration/eop/ceq

FM GLOBAL (FM)
270 Central Avenue
Johnston, RI 02919-4949
Ph: 401-275-3000
Fax: 401-275-3029
Internet: https://www.fmglobal.com/

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 and 501 Hoes Lane
Piscataway, NJ 08854-4141
Ph: 732-981-0060 or 800-701-4333
Fax: 732-981-9667
E-mail: onlinesupport@ieee.org
Internet: https://www.ieee.org/

INTERNATIONAL CODE COUNCIL (ICC)
200 Massachusetts Ave, NW Suite 250
Washington, DC 20001
Ph: 800-786-4452 or 888-422-7233
Fax: 202-783-2348
E-mail: order@iccsafe.org
Internet: https://www.iccsafe.org/

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)
ISO Central Secretariat
BIBC II
Chemin de Blandonnet 8
CP 401 - 1214 Vernier, Geneva
Switzerland
Ph: 41-22-749-01-11
E-mail: central@iso.ch
Internet: https://www.iso.org

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)
1101 Wilson Blvd, Suite 1425

Arlington, VA 22209-1762
Ph: 703-525-1695
Fax: 703-528-2148
Internet: https://safetyequipment.org/

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
Ph: 800-344-3555
Fax: 800-593-6372
Internet: https://www.nfpa.org

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)
4201 Lafayette Center Drive
Chantilly, VA 20151-1219
Ph: 703-803-2980
Fax: 703-803-3732
Internet: https://www.smacna.org/

U.S. ARMY CORPS OF ENGINEERS (USACE)
CRD-C DOCUMENTS available on Internet: https://www.wbdg.org/army
THE CAD/BIM TECHNOLOGY CENTER FOR FACILITIES, INFRASTRUCTURE, AND ENVIRONMENT (USACE)on Internet:
https://cadbimcenter.erdc.dren.mil/
Order Other Documents from:
Official Publications of the Headquarters, USACE
E-mail: hqpublications@usace.army.mil
Internet: http://www.publications.usace.army.mil/
or
https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/

U.S. DEFENSE LOGISTICS AGENCY (DLA)
Andrew T. McNamara Building
8725 John J. Kingman Road
Fort Belvoir, VA 22060-6221
Ph: 877-352-2255
E-mail: dlacontactcenter@dla.mil
Internet: http://www.dla.mil

U.S. DEPARTMENT OF AGRICULTURE (USDA)
Order AMS Publications from:
AGRICULTURAL MARKETING SERVICE (AMS)
Seed Regulatory and Testing Branch
801 Summit Crossing Place, Suite C
Gastonia, NC 28054-2193
Ph: 704-810-8884
E-mail: PA@usda.gov
Internet: https://www.ams.usda.gov/
Order Other Publications from:
USDA Rural Development
Rural Utilities Service
STOP 1510, Rm 5135
1400 Independence Avenue SW
Washington, DC 20250-1510
Phone: (202) 720-9540
Internet:
https://www.rd.usda.gov/about-rd/agencies/rural-utilities-service

U.S. DEPARTMENT OF DEFENSE (DOD)
Order DOD Documents from:
Room 3A750-The Pentagon
1400 Defense Pentagon
Washington, DC 20301-1400
Ph: 703-571-3343
Fax: 215-697-1462
E-mail: customerservice@ntis.gov
Internet: https://www.ntis.gov/
Obtain Military Specifications, Standards and Related Publications from:
Acquisition Streamlining and Standardization Information System (ASSIST)
Department of Defense Single Stock Point (DODSSP)
Document Automation and Production Service (DAPS)
Building 4/D
700 Robbins Avenue
Philadelphia, PA 19111-5094
Ph: 215-697-6396 - for account/password issues
Internet: https://assist.dla.mil/online/start/; account registration required
Obtain Unified Facilities Criteria (UFC) from:
Whole Building Design Guide (WBDG)
National Institute of Building Sciences (NIBS)
1090 Vermont Avenue NW, Suite 700
Washington, DC 20005
Ph: 202-289-7800
Fax: 202-289-1092
Internet:
https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc

U.S. DEPARTMENT OF ENERGY (DOE)
1000 Independence Avenue Southwest
Washington, D.C. 20585
Ph: 202-586-5000
Fax: 202-586-4403
E-mail: The.Secretary@hq.doe.gov
Internet: https://www.energy.gov/

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)
1200 New Jersey Ave., SE
Washington, DC 20590
Ph: 202-366-4000
E-mail: ExecSecretariat.FHWA@dot.gov
Internet: https://highways.dot.gov/
Order from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272

Internet: https://www.archives.gov/
Order documents from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

UL SOLUTIONS (UL)
333 Pfingsten Road
Northbrook, IL 60062
Ph: 877-854-3577 or 360-817-5500
E-mail: CustomerExperienceCenter@ul.com
Internet: https://www.ul.com/
UL Directories available through IHS at https://accuristech.com/

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION 01 45 00.00

QUALITY CONTROL
08/23, CHG 3: 08/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E329	(2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements
ER 1110-3-12	(2021) Military Engineering and Design Quality Management

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Pricing Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

Additional Requirements For Design Quality Control (DQC) Plan; G

SD-05 Design Data

Design Quality Control

Discipline-Specific Checklists

SD-06 Test Reports

Verification Statement

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all design and construction operations, both onsite and offsite, and must be keyed to the proposed design and construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the Post Award Conference, before start of design or construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, design activities, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor, the Architect/Engineer (A/E), and the Government. Provide a copy of the signed minutes to all attendees and incorporate into the relevant QC Plan. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

a. Waste Management Plan.

b. Procedures for noise and acoustics management.

c. Environmental Protection Plan.

d. Environmental regulatory requirements.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, the QC Specialists (if used), and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

a. Review the minutes of the previous meeting.

b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.

c. Review the status of submittals and Request For Information (RFIs).

d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY PROGRESS MEETINGS in Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT and all quality control documentation required for that work.

e. Review Testing Plan and Log including status of tests performed since last QC Meeting.

f. Resolve QC and production problems. Discuss status of pending change orders.

g. Address items that may require revising the QC Plan.

h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.

i. Review environmental requirements and procedures.

j. Review Environmental Management Plan.

k. Review Waste Management Plan.

l. Review the status of training completion.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 30 days after receipt of the Notice to Proceed, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. The Government will consider an interim plan for the first 5 days of operation. Design and Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work. Design work is not permitted to start prior to approval of a Design Quality Control Plan.

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.

b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.

c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction

Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.

d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.

e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.

f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, Design Quality Control Manager, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists or quality control representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.

g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, designers of record, consultants, architect-engineers (AE), offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.

h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.

i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.

j. Procedures to complete design and construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.

k. Reporting procedures, including proposed reporting formats.

l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.

m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Provide separate DFOWs in the Network Analysis Schedule for each design development stage and submittal package. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination and Mutual Understanding Meeting.

n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.

o. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.

p. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.

q. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

1.5.2.2 Additional Requirements for Design Quality Control (DQC) Plan

The following additional requirements apply to the DQC Plan:

a. Submit and maintain a DQC Plan as an effective quality control program which assures that all services required by this Contract are performed and provided in a manner that meets professional architectural and engineering quality standards. As a minimum, all documents must be technically reviewed by competent, independent reviewers identified in the DQC Plan. The same element that produced the product may not perform the Independent Technical Review (ITR) and/or Agency Technical Review (ATR). Correct errors and deficiencies in the design documents prior to submitting them to the Government.

b. Include the design schedule in the master project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific Contract period. Schedule must include sufficient detail to identify all major design tasks, including those that control the flow of work. Include review and correction periods associated with each item. Schedule must be a forward planning as well as a project monitoring tool. The schedule reflects calendar days and not dates for each activity. If the schedule is changed, submit a revised schedule reflecting the change within 7 calendar days.

c. Include in the DQC Plan the discipline-specific checklists to be used during the design and quality control of each submittal. Submit at each design phase as part of the project documentation these completed discipline-specific checklists. ER 1110-3-12 provides some useful information in developing checklists.

d. Implement the DQC Plan by a Design Quality Control Manager who has the responsibility of being cognizant of and assuring that all documents on the project have been coordinated and who coordinates work with and reports to the Contracting Officer via the QC Manager. Notify the Contracting Officer, in writing, of the name of the individual and the name of an alternate person assigned to the position.

e. Provide Quality Control Documentation procedures such as QC review sets and QC comments to demonstrate that cross checking of all engineering disciple's design drawings and specifications has taken place. The QC review documentation must exhibit a checking process of the design documents for completeness, accuracy, and constructability.

1.5.3 Acceptance of the Quality Control (QC) and Design Quality Control (DQC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan, or interim plan applicable to the particular feature of work to be started, and Design Quality Control Plan is required prior to the start of design and construction. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work. The Contracting Officer reserves the right to require changes in the QC and DQC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC and DQC organization at any time to verify the submitted qualifications. All QC and DQC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), QC Manager, a Design Quality Manager, and enough qualified personnel to ensure safety and Contract compliance. The SSHO reports directly to a senior project (or corporate) official independent from the QC Manager. The SSHO will also serve as a member of the CQC Staff Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly. The CQC staff always maintains a presence at the site during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules, and all other project documentation to the CQC organization. The CQC organization is responsible for always maintaining these documents and records at the site, except as otherwise acceptable to the Contracting Officer.

1.6.2 Quality Control (QC) Manager

1.6.2.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The QC Manager must be employed by the Prime Contractor. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the Coordination and Mutual Understanding Meeting, perform the three phases of control except for those phases of control designated to be performed by QC Specialists (if employed for this work), perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by the QC Specialists (if employed for this work), testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities.

1.6.2.2 Qualifications

The QC Manager must be an individual with a minimum of 10 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least 4 years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.2.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager Alternate QC Manager, Superintendent, QC Specialist, and DQCM must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

1.6.3 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.4 Design Quality Control (DQC) Manager

The DQC Manager must be a member of the QC organization, must coordinate actions with the QC Manager, and must not be subordinate to the Project Superintendent or the Project Manager.

1.6.4.1 Qualifications

a. Must be a person who has verifiable engineering or architectural design experience and is a registered professional engineer or architect.

1.6.4.2 Responsibilities

a. Be responsible for the design integrity, professional design standards, and all design services required.

1.6.5 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6.6 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who may perform production related duties but must be allowed sufficient time to perform their assigned quality control duties. These individuals or specialized technical companies are employees of the Prime or subcontractor including Designer of Record (DOR). A single person can cover more than one area provided that the single person is qualified to perform quality control activities in each designated and that workload allows. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix

1. Area	**2-1. Qualification** **2-2. Experience**
Roofing Manufacturer's Technical Representative	2-1. Installation and testing of roofing systems 2-2. 5 years experience minimum

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least 2 business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by the QC Specialists (if employed for this work), the Project Superintendent, and the foreman responsible for the DFOW. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFOW:

a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the prepatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

b. Review the Contract drawings.

c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.

d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.

e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.

f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.

g. Review coordination of product/material delivery to designated prepared areas to execute the work.

h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.

i. Check to assure that all materials and equipment have been tested, submitted, and approved.

j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOW. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.

k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

l. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.

m. Discuss schedule and execution of the initial control phase and confirmation or construction quality compliance.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with the QC Specialists (if employed for this work), the Project Superintendent, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFOW:

a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.

b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.

c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

d. Resolve any workmanship issues.

e. Ensure that testing is performed by the approved laboratory.

f. Check work procedures for compliance with the APP and the appropriate

AHA to ensure that applicable safety requirements are met.

g. Review project specific work plans (i.e., HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.

b. Maintain the quality of workmanship required.

c. Ensure that testing is performed by the approved laboratory.

d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.

e. Do not build upon nor conceal non-conforming work.

f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least 2 weeks prior to the start of the preparatory and initial phases.

1.8.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.

b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.

c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.

d. The Contractor is responsible for including those items identified by the Contracting Officer.

e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.

f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) and DESIGN QUALITY CONTROL (DQC) PLAN.

g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site or within 200 miles. Perform the following activities and record and provide the following data:

a. Verify that testing procedures comply with Contract requirements.

b. Verify that facilities and testing equipment are available and comply with testing standards.

c. Check test instrument calibration data against certified standards.

d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.

e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Laboratory Accreditation Authorities

All testing laboratories must be validated by the USACE Material Testing Center (MTC) for the tests to be performed. Information on the USACE MTC with web-links to both a list of validated testing laboratories and for the laboratory inspection request for can be found at:

https://mtc.erdc.dren.mil

1.9.2 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

1.9.2.1 Capability Recheck

If the selected laboratory fails the capability check, the Contractor will be assessed a charge of $10,000 to reimburse the Government for each succeeding recheck of the laboratory or the checking of a subsequently selected laboratory. Such costs will be deducted from the Contract amount due the Contractor.

1.9.2.2 Onsite Laboratory

The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

1.9.3 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager.

1.10 COMPLETION INSPECTIONS

1.10.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.10.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred by the Contracting Officer in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.10.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

Maintain current and complete records of on-site and off-site QC program operations and activities.

Contact the Contracting Officer for sample forms or print from RMS-QCS as needed. Prior to commencing work on construction, the Contractor must obtain a copy set of the current report forms. The report forms will consist of the Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Contractor Production Report, Contractor Production Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Testing Plan and Log, and Rework Items List. Unless otherwise provided by the Contracting Officer, Contractor may use the forms provided as related material located at www.wbdg.org/dod/ufgs/ufgs-01-45-00.

1.12.1 Construction Documentation

Prepare a Construction Production Report and a Contractor Quality Control Report for each day that work is performed. Attach the Construction Production Report to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities.
Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

a. The name and area of responsibility of the Contractors and any subcontractors.

b. Operating plant/equipment with hours worked, idle, or down for repair.

c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.

d. Control phase activities performed. Preparatory, and Initial phase

Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.

e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.

f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.

g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.

h. Offsite surveillance activities, including actions taken.

i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.

j. Instructions given/received and conflicts in plans and specifications.

k. Provide documentation of design quality control activities. For independent design reviews, provide, as a minimum, identification of the Independent Technical Review (ITR) and/or Agency Technical Review (ATR) team and their review comments, responses and the record of resolution of the comments.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Reports from the Quality Control (QC) Specialist(s)

If QC Specialist are employed for this work, document inspection results on a QC specialist report prepared each day work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

1.12.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.

b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.

c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.

d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES.

e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.

f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.

g. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.6 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.12.7 As-Built Drawings

The QC Manager must ensure the as-built drawings, required by Section 01 78 00 CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document commences with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the Contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager or QC Specialist assigned to an area of responsibility must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/23 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

Administration
Finances
Quality Control
Submittal Monitoring
Scheduling
Closeout
Import/Export of Data

The Contractor must be prepared to transition from the Resident Management System (RMS) to the new Construction Management Platform (CMP) at any time during the period of performance of this contract, once the USACE enterprise implements this change. The Contractor's obligation to transition from RMS to the CMP system is a requirement of this contract. Therefore, additional time or cost increase to complete this transition will not be considered. In the event of the transition, RMS project information will be migrated to CMP by the government and available in the CMP system. After the transition, the Contractor must utilize the CMP system to perform all contract administration functions previously required in RMS, including all areas listed above. After the transition, “RMS” and “CMP” will be synonymous for all “Resident Management System” and “RMS” references in this contract’s specification. To facilitate the transition from RMS to CMP, online training for the CMP system will be made available to the Contractor with enough time to train staff, prior to the transition occurring.

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some documents such as correspondence, including pay requests and payrolls, may be required in paper format with original signatures. Paper documents will govern in the event of discrepancy with the electronic version. Contracting Officer will determine which documents require submission of paper copies.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Priced Construction Contracts; Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 00.00 QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a web based application. Download, install and be able to utilize the latest version of RMS within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (https://rms.usace.army.mil). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.6 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.6.1 Administration

1.6.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.6.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within seven calendar days of receiving access to the Contract in RMS or within seven calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.6.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.6.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates in accordance with Section 01 45 00.00 QUALITY CONTROL.

1.6.1.5 Reports

Track the status of the project using the reports available in RMS. The value of these reports reflects the quality of the data input. These reports include but are not limited to the Progress Payment Request worksheet, Quality Control (QC) comments, Deficiency Tracking Log, Submittal Register Status, and Three-Phase Control worksheets in accordance with Sections 01 20 00 PRICE AND PAYMENT PROCEDURES, Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT, 01 33 00 SUBMITTAL PROCEDURES, and 01 45 00.00 QUALITY CONTROL.

1.6.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response. Provide status at weekly progress meeting in accordance with Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT.

1.6.2 Finances

1.6.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule in accordance with Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT. The sum of pay activities equals the total Contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.6.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the Contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data, either electronically or by hard copy.

1.6.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Link each deficiency and QC test to a pay activity. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Ensure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments. Provide within seven calendar days of Government acceptance of the CQC Plan.

1.6.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report.

1.6.3.2 Deficiency Tracking

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.6.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS, and upload meeting minutes of the meetings. Worksheets for the three-phase control meetings are generated within RMS.

1.6.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.6.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.6.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.6.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager. Prepare activity hazard analysis in accordance with Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS.

1.6.4 Submittal Management

Enter all current submittal register data and information into within the time frame directed by the Contracting Officer. The Contractor will import the information shown in the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES unless otherwise directed by the Contracting Officer. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the Contract.

Enter all submittals as listed in the approved design's submittal register within 14 calendar days of design approval for construction.

Contractor may request submittals to be added to RMS throughout the lifecycle of the project. Approval/disapproval of Contractor requests is at the discretion of the Contracting Officer. Submittals entry into RMS must be updated daily or as practical for performance of the Contract, as approved by the Contracting Officer.

1.6.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 01.00 31 SMALL CIVIL PROJECT CONSTRUCTION PROGRESS SCHEDULES - BARCHART FORMAT.

1.6.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.7 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.8 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within seven calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
11/20, CHG 3: 08/24 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511	(2017; R 2021) Reduced-Pressure Principle Backflow Prevention Assembly

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 241	(2022; ERTA 22-1) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD	(2023) Manual on Uniform Traffic Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

SD-06 Test Reports

Backflow Preventer Tests

SD-07 Certificates

Backflow Tester Certification

Backflow Preventers Certificate of Full Approval

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 BACKFLOW PREVENTERS CERTIFICATE

1.4.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with a company participating in other phases of this Contract.

1.4.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.5 CONDITION OF READINESS (COR) LEVELS FOR WEATHER

The Condition of Readiness (COR) is based on the weather forecast and the potential for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Emergency Plan requirements and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain

"Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.

b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.

c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.

d. Condition ONE. (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.6 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.6.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.6.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.6.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.6.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.6.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at:

https://www.wbdg.org/dod/ufgs/ufgs-01-50-00.

Each Statement must be signed by a cybersecurity representative for the relevant company.

1.6.2 Temporary IP Networks

Temporary Contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.6.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.6.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. If there is a Government agency responsible for network access, identify that agency.

1.6.4 Temporary Wireless IP Networks

1.6.4.1 Contractor Provided Wi-Fi Network

Provide standalone Wi-Fi network for authorized Government personnel. Network access must be available throughout the jobsite and for the entire Contract period. The Government will use the Wi-Fi service for coordination in the field and related activities. Do not attach or connect the temporary Wi-Fi network to Government networks or systems unrelated to construction activities.

Password information must adhere to requirements outlined in Paragraph PASSWORDS AND PASSPHRASES. Provide password and Wi-Fi network name (SSID) to the Contracting Officer for Government personnel access. Wi-Fi performance must be capable of handling bandwidth to support Government and Contractor activities, including drawing and model access onsite. Provide a point of contact for reporting Wi-Fi outages.

1.6.4.2 Temporary Wireless IP Networks Security

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.6.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.6.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at:

https://www.wbdg.org/dod/ufgs/ufgs-01-50-00.

Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as specified in Section 01 58 00 PROJECT IDENTIFICATION and Section 01 58 01 PROJECT SIGNAGE GUIDANCE. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract in accordance with EM 385-1-1. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control, although optional, must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and haul roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work.

To block public view of the construction, enclose the project work area and Contractor lay-down area with a 8 ft high chain link fence and gates with brown, UV light resistant, plastic fabric mesh netting (similar to tennis court or other screening).

2.3.1 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 48 inches high and maximum mesh size of 2 inches. Fencing must extend from the grade to a minimum of 48 inches above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 8 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged cast iron, ductile iron, bronze, or brass mounted gate valve and strainer, 304 stainless steel or bronze, internal parts.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Construction Contract employees must park privately owned vehicles in an area designated by the Contracting Officer. Employee parking must not interfere with existing and established parking requirements of the Government installation.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

3.2.2 Payment for Utility Services

a. The Government will make all reasonably required utilities available from existing outlets and supplies, as specified in the Contract. Unless otherwise provided in the Contract, the amount of each utility service consumed will be charged to or paid at prevailing rates charged to the Government or, where the utility is produced by the Government, at reasonable rates determined by the Contracting Officer. Carefully conserve utilities furnished without charge.

b. Reasonable amounts of electricity and potable water will be made available without charge.

c. The point at which the Government will deliver such utilities or services and the quantity available is must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing backflow-preventing devices on connections to domestic water lines; and make disconnections.

d. The Contractor must provide their own utilities for the Contractor Administrative Field Trailer.

3.2.3 Meters and Temporary Connections

Provide and maintain necessary temporary connections, distribution lines, and meter bases (if needed, and the Government will provide meters) required to measure the amount of each utility used for the purpose of determining charges.

3.2.4 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities in accordance with EM 385-1-1. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or station sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Coordinate with the Contracting Officer and follow station regulations and procedures when discharging into the station sanitary sewer system. Maintain these conveniences at all times. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.5 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.2.6 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.3 TRAFFIC PROVISIONS

3.3.1 Maintenance of Traffic

a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Make all notifications and obtain all permits required for modification to traffic movements outside Station's jurisdiction. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.

b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.

c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, Life Safety Signage, overhead protection, and other items that may be required by the authority having jurisdiction.

d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.

3.3.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. All personnel working in roadways will wear brightly-colored vests or other high visibility apparel in accordance with EM 385-1-1. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.3.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.3.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

3.4 REDUCED PRESSURE BACKFLOW PREVENTERS

Provide an approved reduced pressure backflow prevention assembly at each location where the Contractor taps into the Government potable water supply.

Perform backflow preventer tests using test equipment, procedures, and certification forms conforming to those outlined in the latest edition of the Manual of Cross-Connection Control published by the FCCCHR Manual. Test and tag each reduced pressure backflow preventer upon initial installation (prior to continued water use) and monthly thereafter. Tag must contain the following information: make, model, serial number, dates of tests, results, maintenance performed, and signature of tester. Record test results on certification forms conforming to requirements cited earlier in this paragraph.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.5.1 Contractor Administrative Field Trailer

Contractor must provide and maintain an administrative field trailer within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

The adminstrative field trailer must be a minimum 12 feet in width, 16 feet in length and have a minimum of 7 feet headroom. Equip the trailer with approved electrical wiring, at least one double convenience outlet and the required switches and fuses to provide 110-120 volt power. Provide a work table with stool, desk with chair, two additional chairs, and one legal size file cabinet that can be locked. The trailer must be waterproof, supplied with a heater, have a minimum of two doors, electric lights, a telephone, a battery-operated smoke detector alarm, a sufficient number of adjustable windows for adequate light and ventilation, and a supply of approved drinking water. Provide electrical service, potable water and approved sanitary facilities. Screen the windows and doors and provide the doors with deadbolt type locking devices or a padlock and heavy-duty hasp bolted to the door. Door hinge pins must be non-removable. Arrange the windows to open and to be securely fastened from the inside. Protect glass panels in windows by bars or heavy mesh screens to prevent easy access. In warm weather, provide air conditioning capable of maintaining the office at 50 percent relative humidity and a room temperature 20 degrees F below the outside temperature when the outside temperature is 95 degrees F. Unless otherwise directed by the Contracting Officer, remove the trailer from the site upon completion and acceptance of the work.

3.5.2 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Include plastic strip inserts, colored green or brown, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.5.3 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.5.4 Appearance of Trailers

a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Installation property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.

b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.5.5 Safety Systems

Protect the integrity of all installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.6 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.6.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 PLANT COMMUNICATIONS

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

3.7 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.8 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.9 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

-- End of Section --

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS
08/22, CHG 2: 08/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1053	Respirable Crystalline Silica
29 CFR 1910.1200	Hazard Communication
29 CFR 1926.1153	Respirable Crystalline Silica
40 CFR 50	National Primary and Secondary Ambient Air Quality Standards
40 CFR 60	Standards of Performance for New Stationary Sources
40 CFR 63	National Emission Standards for Hazardous Air Pollutants for Source Categories
40 CFR 64	Compliance Assurance Monitoring
40 CFR 82	Protection of Stratospheric Ozone
40 CFR 112	Oil Pollution Prevention
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 261.7	Residues of Hazardous Waste in Empty Containers
40 CFR 262.11	Hazardous Waste Determination and Recordkeeping
40 CFR 268	Land Disposal Restrictions
40 CFR 273	Standards for Universal Waste Management

40 CFR 279	Standards for the Management of Used Oil
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 355	Emergency Planning and Notification
40 CFR 403	General Pretreatment Regulations for Existing and New Sources of Pollution
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink:

https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink:

https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances.

1.2.2 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.3 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.4 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.5 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.6 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with 40 CFR 261.

1.2.7 Hazardous Materials

Hazardous material is any material that: Is defined in 49 CFR 171, listed in 49 CFR 172, regulated as a hazardous material in accordance with 49 CFR 173; or requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.1200; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D. Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this section for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.8 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibits a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D, or meets a state, local, or host nation definition of a hazardous waste.

1.2.9 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with other state and local requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in other state regulations or local ordinances; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.10 Regulated Waste

Regulated waste are solid wastes that have specific additional federal, state, or local controls for handling, storage, or disposal.

1.2.11 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.12 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.12.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 2.5-inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.12.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.12.3 Material Not Regulated As Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.12.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 261.

1.2.12.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of "lead contaminated" or "lead based paint contaminated" may be included as recyclable only if the metal is handled and transported in accordance with regulations, and the facility receiving the material is permitted and authorized to accept and process lead contaminated materials. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

1.2.12.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.12.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.12.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.13 Wastewater

Wastewater is the aqueous liquid waste generated by a project, to include both domestic wastewater (sewage) and industrial wastewater.

1.2.13.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.14 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

1.2.15 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.16 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Survey

Regulatory Notifications

Employee Training Records

Environmental Protection Plan; G

Dirt and Dust Control Plan

Solid Waste Management Permit

Spill Prevention Control And Countermeasure (SPCC) Plan; G

Waste Determination Documentation; G

SD-06 Test Reports

Monthly Solid Waste Disposal Report; G

SD-07 Certificates

Employee Training Records; G

SD-11 Closeout Submittals

Regulatory Notifications; G

Assembled Employee Training Records; G

Solid Waste Management Permit; G

Project Solid Waste Disposal Documentation Report; G

Sales Documentation; G

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

1.4.1 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 QUALITY ASSURANCE

1.5.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.5.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with federal, state and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of regulatory notifications to the Contracting Officer at least 14 days prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.5.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP) or equipment local requirement. Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

1.5.4 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet Oklahoma State requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States, and endangered species and their habitat that are known to be in the area.

1.5.5 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. FAR 52.242-14 Suspension of Work provides that a suspension, delay, or interruption of work due to the fault or negligence of the Contractor allows for no adjustments to the contract for time extensions or equitable adjustments. In addition to a suspension of work, the Contracting Officer may use additional authorities under the contract or law.

1.6 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken.

Submit the EPP within 30 days after the Notice to Proceed and not less than 10 days before the preconstruction meeting. Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of any applicable federal, state, and local environmental protection laws and regulations. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version onsite.

The EPP includes, but is not limited to, the following elements:

1.6.1 General Overview and Purpose

1.6.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as spill control plan, solid waste management plan, wastewater management plan, contaminant prevention plan, a historical, archaeological, cultural resources, biological resources and wetlands plan, traffic control plan and the non-hazardous solid waste disposal plan.

1.6.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

1.6.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

1.6.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.6.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.6.2 General Site Information

1.6.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.6.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

Show where any fuels, hazardous substances, solvents, or lubricants will be stored. Provide a spill plan to address any releases of those materials.

1.6.3 Management of Natural Resources

a. Land resources

b. Tree protection

c. Replacement of damaged landscape features

d. Temporary construction

e. Stream crossings

f. Fish and wildlife resources

g. Wetland areas

1.6.4 Protection of Historical and Archaeological Resources

a. Objectives

b. Methods

1.6.5 Stormwater Management and Control

a. Ground cover

b. Erodible soils

c. Temporary measures

(1) Structural Practices

(2) Temporary and permanent stabilization

d. Effective selection, implementation and maintenance of Best Management Practices (BMPs).

e. Stormwater Pollution Prevention Plan (SWPPP).

1.6.6 Protection of the Environment from Waste Derived from Contractor Operations

This item consists of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan when within an installation. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan as applicable.

As a minimum, include the following:

a. List of the types of hazardous wastes expected to be generated

b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated

c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications

d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)

e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)

f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions (40 CFR 268)

g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar

h. Used oil management procedures in accordance with 40 CFR 279; Hazardous waste minimization procedures

i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.

1.6.7 Prevention of Releases to the Environment

Procedures to prevent releases to the environment

Notifications in the event of a release to the environment

1.6.8 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.6.9 Clean Air Act Compliance

1.6.9.1 Haul Route

Submit truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.6.9.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may require federal, state, or local permits under the Clean Air Act. Determine requirements based on any current installation permits and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager). Ensure required permits are obtained prior to installing and operating applicable equipment/processes.

1.6.9.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced. Comply with 40 CFR 60 Subpart IIII, 40 CFR 60 Subpart JJJJ, 40 CFR 63 Subpart ZZZZ, and local regulations as applicable. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between maintenance/testing, emergency, and non-emergency operation.

1.6.9.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants complies with 40 CFR 82 requirements. Technicians must be certified, maintain copies of certification on site, use certified equipment and log work that requires the addition or removal of refrigerant. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.6.9.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.6.9.6 Compliant Materials

Provide the Government a list of SDSs for all hazardous materials proposed for use on site. Materials must be compliant with all Clean Air Act regulations for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.7 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7 Permits and Responsibilities. Notify the Government of all equipment that may require permits or special approvals that the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7 Permits and Responsibilities.

1.8 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.9 SOLID WASTE MANAGEMENT PERMIT

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.9.1 Monthly Solid Waste Disposal Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

1.10 FACILITY HAZARDOUS WASTE GENERATOR STATUS

The Broken Bow Powerhouse is designated as a Conditionally Exempt-Small Quantity Generator. Meet the regulatory requirements of this generator designation for any work conducted within the boundaries of this Installation. Comply with provisions of federal, state, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats.

Broken Bow Powerhouse lies in McCurtain County in Oklahoma. The county is host to eight Federally protected, threatened, endangered, and candidate species. Endangered species include the Gray Bat and Ozark Big-eared Bat. Threatened species include the American Burying Beetle, Ozark Cavefish, Northern Long-eared Bat, Piping Plover, and the Red Knot. Candidate species include the Monarch Butterfly.

This project will not adversely affect or threaten the continued existence of any Federally listed threatened or endangered species. Therefore, a determination of “No Effect” was made for all Federally protected threatened, endangered, or candidate species from this activity.

The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees and critical root zones (CRZ) that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.1.3 Streams

Stream crossings must allow movement of materials or equipment without violating water pollution control standards of the federal, state, and local governments. Construction of stream crossing structures must be in compliance with all required permits including, but not limited to, Clean Water Act Section 404, and Section 401 Water Quality.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer.

3.2 PROTECTION OF CULTURAL RESOURCES

3.2.1 Archaeological Resources

If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

3.2.2 Historical Resources

Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

3.3 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with 40 CFR 64 and state air emission and performance laws and standards.

3.3.1 Preconstruction Air Permits

Notify the Air Program Manager, through the Contracting Officer, at least 6 months prior to bringing equipment, assembled or unassembled, onto the Installation, so that air permits can be secured. Necessary permitting time must be considered in regard to construction activities. Clean Air Act (CAA) permits must be obtained prior to bringing equipment, assembled or unassembled, onto the Installation.

Confirm that these permits have been obtained.

3.3.2 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Provide in accordance with paragraph PRECONSTRUCTION AIR PERMITS.

3.3.3 Burning

Burning is prohibited on the Government premises.

3.3.4 Class I and II ODS Prohibition

Class I and II ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant.

3.3.5 Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Intentional venting of refrigerants (including most Non-ODS substitute refrigerants) is prohibited per 40 CFR 82.

3.3.6 EPA Certification Requirements

Heating and air conditioning technicians must be certified through an EPA-approved program. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by environmental law.

3.3.7 Dust Control

Keep dust down at all times, including during nonworking periods. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster. Since these products contain Crystalline Silica, comply with the applicable OSHA standard, 29 CFR 1910.1053 or 29 CFR 1926.1153 for controlling exposure to Crystalline Silica Dust.

3.3.7.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with state and local visibility regulations.

3.3.7.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with 29 CFR 1910.

3.3.8 Odors

Control odors from construction activities. The odors must be in compliance with state regulations and local ordinances and may not constitute a health hazard.

3.4 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.4.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

3.4.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	cubic yards or tons, as appropriate
C&D Debris Recycled	cubic yards or tons, as appropriate
C&D Debris Composted	cubic yards or tons, as appropriate
Total C&D Debris Generated	cubic yards or tons, as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	cubic yards or tons, as appropriate

3.5 WASTE MANAGEMENT AND DISPOSAL

3.5.1 Waste Determination Documentation

Complete a Waste Determination form (provided at the pre-construction conference) for Contractor-derived wastes to be generated. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g., scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of 40 CFR 262.11 or corresponding applicable state or local regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination form. As a minimum, provide a Waste Determination form for the following waste (this listing is not exhaustive): oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.5.2 Solid Waste Management

3.5.2.1 Project Solid Waste Disposal Documentation Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation must include the receiver's tax identification number and business, EPA or state registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.5.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste.

Transport solid waste off Government property and dispose of it in compliance with 40 CFR 260, state, and local requirements for solid waste disposal. A Subtitle D RCRA permitted landfill is the minimum acceptable offsite solid waste disposal option. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Solid waste disposal offsite must comply with most stringent local, state, and federal requirements, including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with 49 CFR 173.

3.5.3 Releases/Spills of Oil and Hazardous Substances

3.5.3.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Installation Fire Department, the Installation Command Duty Officer, the Installation Environmental Office, the Contracting Officer and the state or local authority.

Submit verbal and written notifications as required by the federal (40 CFR 300.125 and 40 CFR 355), state, local regulations and instructions. Provide copies of the written notification and documentation that a verbal notification was made within 20 days. Spill response must be in accordance with 40 CFR 300 and applicable state and local regulations. Contain and clean up these spills without cost to the Government.

3.5.3.2 Clean Up

Clean up hazardous and non-hazardous waste spills. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor- responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.5.4 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.5.5 Wastewater

3.5.5.1 Disposal of Wastewater

Disposal of wastewater must be as specified below.

3.5.5.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged. Dispose of the construction-related waste water off-Government property in accordance with 40 CFR 403, state, regional, and local laws and Prior to disposal, submit a WIS to the Tulsa District Environmental Office and have the results reviewed and approved by the Contracting Officer prior to being discharged or disposed of off-Government property.

3.6 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Accident Prevention Plan (APP), in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this contract. Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with 40 CFR 261 and state and powerhouse requirements.

3.7 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds. Consult with the U.S. Department of Agriculture jurisdictional office for additional cleaning requirements.

3.8 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with EPA 40 CFR 112, and other federal, state, regional, and local laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of the United States. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. Storage of fuel on the project site must be in accordance with EPA, state, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS.

3.8.1 Used Oil Management

Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.8.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 5 inches freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 1320 gallons will be used onsite (only containers with a capacity of 55 gallons or greater are counted), provide and implement a Spill Prevention Control and Countermeasure (SPCC) plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.9 INADVERTENT DISCOVERY OF HAZARDOUS WASTES

If suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.10 SOUND INTRUSION

Make the maximum use of low-noise emission products, as certified by the EPA. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and then only during the designated times.

Keep construction activities under surveillance and control to minimize environment damage by noise. Comply with the provisions of the State of Oklahoma rules.

3.11 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

-- End of Section --

SECTION 01 58 00

PROJECT IDENTIFICATION
08/19, CHG 5: 08/22 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 310-1-6a	(2006; 2019 Change 2) Project Operation -- Sign Standards Manual, VOL 1
EP 310-1-6b	(2006) Sign Standards Manual, VOL 2, Appendices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Sign Legend Orders; G

1.3 QUALITY CONTROL

1.4 PROJECT IDENTIFICATION SIGN

1.4.1 Construction Project Signs

Furnish the construction project sign package, maintain the signs during construction, and remove the signs from the job site upon completion of the project. The construction project sign package consists of two signs: one for project identification and the other to show the on-the-job safety performance of the contractor. Ensure that the package conforms to the requirements of EP 310-1-6a and EP 310-1-6b, specifically Section 16. Submit the sign legend orders as described in Section 16 of EP 310-1-6a before erecting the signs.

Section 01 58 01 PROJECT SIGNAGE GUIDANCE presents example sign graphics with dimensions. Contractor must follow USACE policy provided in Engineering Pamphlet (EP) 310-1-6a, Sign Standards Manual, VOL 1 for guidelines on fabricating, locating, and mounting construction project signs:

(https://www.publications.usace.army.mil/Portals/76/Users/182/86/2486/EP_310-1-6a%20change%202.pdf.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

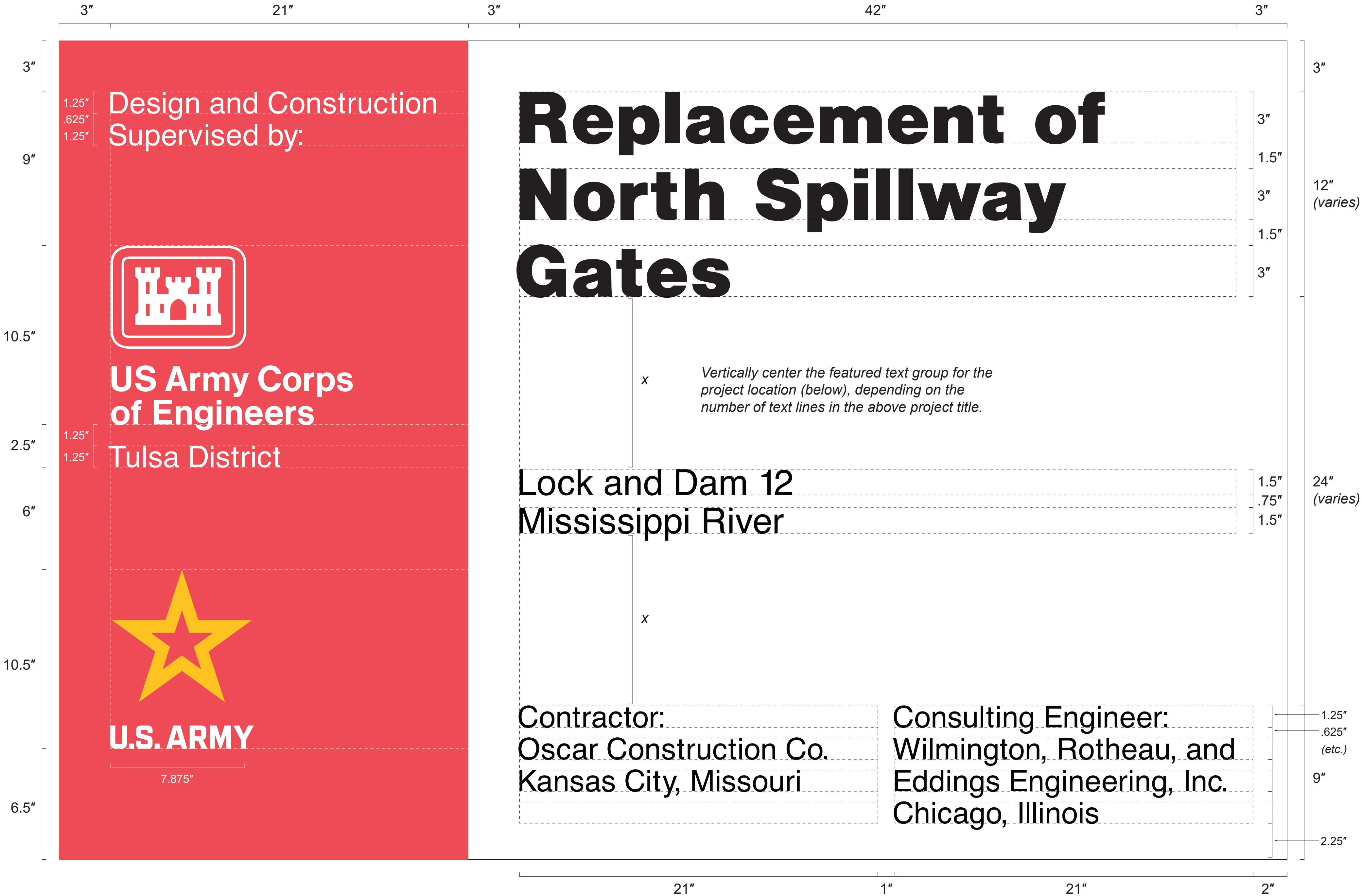
Not used.

-- End of Section --

SECTION 01 58 01

PROJECT SIGNAGE GUIDANCE

Example Graphic of Signage with Dimensions for Civil Works Project



Template designed 20230414 in Adobe Illustrator by AJ Jensen

This template is derived from the Corps *Sign Standards Manual*, (EP 310-1-6a), Section 16. Consult the *Manual* for additional guidance.

SAFETY PERFORMANCE SIGN TEMPLATE

Note:

Change 1) the name of the project and 2) the name of the primary servicing contractor. The font to be used is COEHelvetica2019.1 and may be downloaded at the NRM Gateway or by contacting the USACE National Sign Program Manager.

Ensure a 3″ margin to the left and to the right.

This sign is designed to be 4ft x 4ft, installed to the right of the Construction Identification sign.

In the text for the Name of Prime Contractor, lines 1 and 2, the logo of the contractor may stand in place of the template text, so long as the name of the contractor is clear and easy to read. If the logo needs to be taller than the space provided, it is permissible to reduce the 9″ space immediately below the “Safety Is a Job Requirement” text to 5″ and shift both the name of the project and the

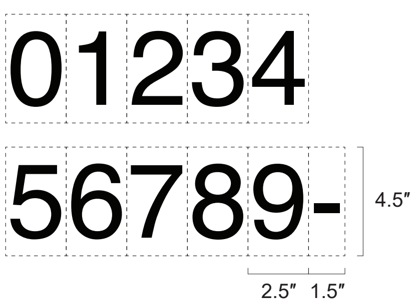
Outlines and measurement markings are provided for convenience of placement and are deleted when everything is in place. Margins and placement of all elements in this sign are standard and should not be altered.

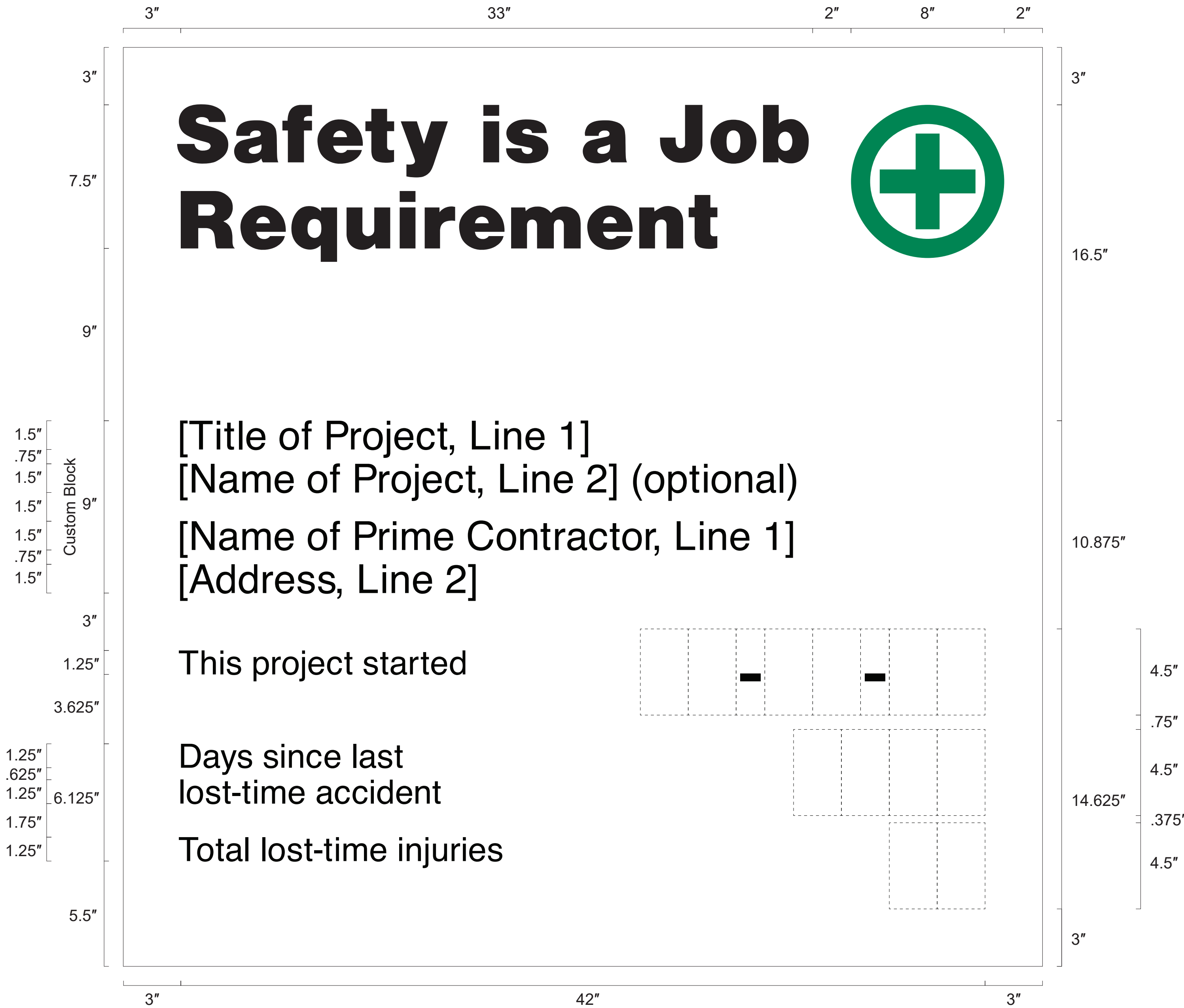
This template is derived from the Corps *Sign Standards Manual*, (EP 310-1-6a), Section 16. Consult the *Manual* for additional guidance.

The numbers below are 2.5″ x 4.5″. They are vectorized and are designed to be printed out for placement into the blank spots in the lower right area of the sign.

The outlined dashed margins below and on the actual sign are to show placement and are not to appear on the actual sign.

A preceding zero is not required for the month or day in the “This project started” line, as long as the right margin for the entire date remains 3″ in from the right edge of the sign.





Template designed 20200214 in Adobe Illustrator by AJ Jensen

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
02/19, CHG 3: 11/21 - SWT 03/26

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.1.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.1.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.1.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.1.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.1.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.1.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.1.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.1.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.1.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.2 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.3 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.3.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.3.2 Oversight

The Quality Control Manager, as specified in Section 01 45 00.00 QUALITY CONTROL, is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project.

1.3.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.3.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.3.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste steams not listed:

a. Land Clearing Debris

b. Asphalt

c. Masonry and CMU

d. Concrete

e. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)

f. Wood (nails and staples allowed)

g. Glass

h. Paper

i. Plastics (PET, HDPE,PVC,LDPE,PP,PS, Other)

j. Gypsum

k. Non-hazardous paint and paint cans

l. Carpet

m. Ceiling Tiles

n. Insulation

o. Beverage Containers

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-06 Test Reports

Quarterly Reports

Annual Report

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; S

1.5 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00.00 QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

a. Preconstruction meeting.

b. Regular Quality Control meetings.

c. Work safety meeting (if applicable).

1.6 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 30 calendar days after the Notice to Proceed. Revise and resubmit Construction Waste Management Plan as necessary, in order for construction to begin.

Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.

b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.

c. Description of the regular meetings to be held to address waste management.

d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.

e. Name of landfill and incinerator to be used.

f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.

g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.

h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.

i. Description of the means by which materials identified in item (g) above will be protected from contamination.

j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).

k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.

Distribute copies of the waste management plan to each subcontractor, Quality Control Manager, and the Contracting Officer.

1.7 RECORDS (DOCUMENTATION)

1.7.1 General

Maintain records to document the types and quantities of waste generated and diverted though re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.7.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.8 REPORTS

1.8.1 General

Maintain current construction waste diversion information on site for periodic inspection by the Contracting Officer. Include in the quarterly reports, annual reports and final reports: the project name, contract information, information for waste generated, diverted and disposed of for the current reporting period and show cumulative totals for the project. Reports must identify quantifies of waste by type and disposal method. Also include in each report, supporting documentation to include manifests, weigh tickets, receipts, and invoices specifically identifying the project and waste material type and weighted sum.

1.8.2 Quarterly Reporting

Provide cumulative reports at the end of each quarter (December, March, June, and September, corresponding with the federal fiscal year for reporting purposes). Submit quarterly reports not later than 15 calendar days after the preceding quarter has ended. Submit Quarterly Reports to the appropriate office or identified point of contact.

1.8.3 Annual Reporting

Provide a cumulative construction waste diversion report annually. Submit annual report not later than 30 calendar days after the preceding fourth quarter has ended. Provide copy of annual construction waste diversion report to the installation POC.

1.9 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide Final Construction Waste Diversion Report within 30 days follìwng the completion of the project.

1.10 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS. Separate materials by one of the following methods:

1.10.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.10.2 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.11 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.11.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the Installation. Consider the use of surplus industrial supply broker services, who match entities with reusable or repurpose industrial materials with entities with need of such materials.

1.11.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.11.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used. -- End of Section --

SECTION 01 78 00

CLOSEOUT SUBMITTALS
05/19, CHG 1: 08/21 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

ERDC/ITL TR-19-6	(2019) A/E/C Graphics Standard, Release 2.1
ERDC/ITL TR-19-7	(2019) A/E/C CAD Standard - Release 6.1

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-300-08	(2023) Criteria for Transfer and Acceptance of DoD Real Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site and/or red-lined PDF files. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.2.3 Record Model

A model reflecting approved changes during construction including red-lines, requests for information (RFI's), and contract modifications. Include updated construction phase facility/site data for components.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.4 RECORD DRAWINGS

The Government will provide pdf and or program files at the preconstruction conference that contains one set of "as-designed" electronic CAD files in the specified software and format revised to reflect all amendments and the final contract PDF drawings. The CAD files are provided to enable preparation of as-built or as-constructed drawings. If discrepancies exist between the CAD files and the contract PDF drawings, correct the CAD files to show the contract PDF drawings.

1.4.1 Variation with Contract Drawings

The electronic files provided are not part of the contract documents. If there is any discrepancy between the electronic files and the contract drawings, the contract drawings govern. The Government has no responsibility to modify any Government furnished materials (GFM) due to changes in the design that occur after award.

Evaluate the content and quality of the GFM upon receipt. If major discrepancies or omissions occur in the GFM, notify the Contracting Officer and indicate the nature of such variations.

1.4.2 Data Loss, Corruption, and Error

Transfer of GFM files may result in corrupted files resulting in data loss and errors. Use of GFM files at own risk. Verify data integrity upon receipt and request a replacement if necessary. Make any adjustment in file structure, format, or software version as needed to make GFM compatible with computer systems and/or software to meet the requirements of the contract.

1.4.3 Modeling Completeness and Quality

The Government makes no guarantee that the GFM provide the level of completeness or quality as required by the contract. Further, the Government makes no guarantee that identified variations will be corrected upon notification.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan

Warranty Tags

Spare Parts Data

SD-08 Manufacturer's Instructions

Posted Instructions

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

As-Built Record of Equipment and Materials

Warranted Equipment and Materials

Final Approved Shop Drawings; G

Construction Contract Specifications; G

Certification of EPA Designated Items; G

Certification Of USDA Designated Items; G

Interim DD FORM 1354; G

List Of Installed Building Equipment Items For DD Form 1354

1.6 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

b. At acceptance of commissioning, ensure the required stock level is supplied as indicated in subparagraph a for maintenance and repair activities through the facilities warranty period. Provision of spare parts does not relieve the Contractor of responsibilities listed under the contract guarantee provisions.

1.7 QUALITY CONTROL

Additions and corrections to the contract drawings must be equal in quality and detail to that of the originals. Line colors, line weights, lettering, layering conventions, and symbols must conform to ERDC/ITL TR-19-7.

1.8 WARRANTY MANAGEMENT

1.8.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.

c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.

d. Warranted Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:

(1) Name of item.

(2) Model and serial numbers.

(3) Location where installed.

(4) Name and phone numbers of manufacturers or suppliers.

(5) Names, addresses and telephone numbers of sources of spare parts.

(6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.

(7) Cross-reference to warranty certificates as applicable.

(8) Starting point and duration of warranty period.

(9) Summary of maintenance procedures required to continue the warranty in force.

(10) Cross-reference to specific pertinent Operation and Maintenance manuals.

(11) Organization, names and phone numbers of persons to call for warranty service.

(12) Typical response time and repair time expected for various warranted equipment.

e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.</OLI>

f. Procedure and status of tagging of equipment covered by warranties longer than one year.

g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.8.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.
<OLG>
<OLI>In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.</OLI>

<OLI>In the event sufficient funds are not available to cover the

construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.</OLI>

<OLI>Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.</OLI>
</OLG>

1.8.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.8.4 Contractor's Response to Construction Warranty Service Requirements

Following oral or written notification by the Contracting Officer, respond to construction warranty service requirements in accordance with the "Construction Warranty Service Priority List" and the three categories of priorities listed below. Submit a report on any warranty item that has been repaired during the warranty period. Include within the report the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and back charge the construction warranty payment item established.
<OLG>
<OLI>First Priority Code 1. Perform onsite inspection to evaluate situation, and determine course of action within 4 hours, initiate work within 6 hours and work continuously to completion or relief.</OLI>

<OLI>Second Priority Code 2. Perform onsite inspection to evaluate situation, and determine course of action within 8 hours, initiate work within 24 hours and work continuously to completion or relief.</OLI>

<OLI>Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.</OLI>

<OLI>The "Construction Warranty Service Priority List" is as follows:</OLI>

Code 1-Life Safety Systems
<OLI LEVEL="2">Fire suppression systems.</OLI>
<OLI LEVEL="2">Fire alarm system(s) in place in the building.</OLI>

Code 1-Air Conditioning Systems

<OLI LEVEL="2">Recreational support.</OLI>
<OLI LEVEL="2">Air conditioning leak in part of building, if causing damage.</OLI>
<OLI LEVEL="2">Air conditioning system not cooling properly.</OLI>

Code 1-Doors
<OLI LEVEL="2">Overhead doors not operational, causing a security, fire, or safety problem.</OLI>
<OLI LEVEL="2">Interior, exterior personnel doors or hardware, not functioning properly, causing a security, fire, or safety problem.</OLI>

Code 3-Doors
<OLI LEVEL="2">Overhead doors not operational.</OLI>
<OLI LEVEL="2">Interior/exterior personnel doors or hardware not functioning properly.</OLI>

Code 1-Electrical
<OLI LEVEL="2">Power failure (entire area or any building operational after 1600 hours).</OLI>
<OLI LEVEL="2">Security lights</OLI>
<OLI LEVEL="2">Smoke detectors</OLI>

Code 2-Electrical
<OLI LEVEL="2">Power failure (no power to a room or part of building).</OLI>
<OLI LEVEL="2">Receptacle and lights (in a room or part of building).</OLI>

Code 3-Electrical
<OLI LEVEL="2">Street lights.</OLI>

Code 1-Gas
<OLI LEVEL="2">Leaks and breaks.</OLI>
<OLI LEVEL="2">No gas to family housing unit or cantonment area.</OLI>

Code 1-Heat
<OLI LEVEL="2">Area power failure affecting heat.</OLI>
<OLI LEVEL="2">Heater in unit not working.</OLI>

Code 2-Kitchen Equipment
<OLI LEVEL="2">Dishwasher not operating properly.</OLI>
<OLI LEVEL="2">All other equipment hampering preparation of a meal.</OLI>

Code 1-Plumbing
<OLI LEVEL="2">Hot water heater failure.</OLI>
<OLI LEVEL="2">Leaking water supply pipes.</OLI>

Code 2-Plumbing
<OLI LEVEL="2">Flush valves not operating properly.</OLI>
<OLI LEVEL="2">Fixture drain, supply line to commode, or any water pipe leaking.</OLI>
<OLI LEVEL="2">Commode leaking at base.</OLI>

Code 3 -Plumbing
<OLI LEVEL="2">Leaky faucets.</OLI>

Code 3-Interior
<OLI LEVEL="2">Floors damaged.</OLI>
<OLI LEVEL="2">Paint chipping or peeling.</OLI>
<OLI LEVEL="2">Casework.</OLI>

Code 1-Roof Leaks
<OLI LEVEL="2">Temporary repairs will be made where major damage to property is occurring.</OLI>

Code 2-Roof Leaks
<OLI LEVEL="2">Where major damage to property is not occurring, check for location of leak during rain and complete repairs on a Code 2 basis.</OLI>

Code 2-Water (Exterior)
<OLI LEVEL="2">No water to facility.</OLI>

Code 2-Water (Hot)
<OLI LEVEL="2">No hot water in portion of building listed.</OLI>

Code 3-All other work not listed above.
</OLG>

1.8.5 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	
Address	
Telephone number	
Warranty contact	
Address	
Telephone number	

Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

PART 2 PRODUCTS

2.1 RECORD DRAWINGS

Prepare the CAD drawing files in AutoCAD Release 2013, or newer, format compatible with a Windows 7, or newer, operating system.

2.1.1 Additional Drawings

If additional drawings are required, prepare them using the specified electronic file format applying ERDC/ITL TR-19-7 and ERDC/ITL TR-19-6. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings.

2.1.1.1 Sheet Numbers and File Names

If a sheet needs to be added between two sequential sheets, append a Supplemental Drawing Designator in accordance with ERDC/ITL TR-19-7 Adding a drawing sheet, and ERDC/ITL TR-19-6 Adding or deleting drawing sheets and index sheet procedures.

2.2 CERTIFICATION OF EPA DESIGNATED ITEMS

Submit the Certification of EPA Designated Items as required by FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items and FAR 52-223-17 Affirmative Procurement of EPA designated items in Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:
<OLG>
<OLI>The product does not meet appropriate performance standards;</OLI>

<OLI>The product is not available within a reasonable time frame;</OLI>

<OLI>The product is not available competitively (from two or more sources);</OLI>

<OLI>The product is only available at an unreasonable price (compared with a comparable non-recycled content product)."</OLI>
</OLG>
Record each product used in the project that has a requirement or option of containing recycled content in accordance with SECTION 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING, noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, exemptions (a, b, c, or d, as indicated), and comments. Recycled content values may be determined by weight or volume percent, but must be

consistent throughout.

2.3 CERTIFICATION OF USDA DESIGNATED ITEMS

Submit the Certification of USDA Designated Items as required by FAR 52-223-1 Bio-based Product Certifications and FAR 52.223-2 Affirmative Procurement of Biobased Products Under Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current USDA standards for biobased materials content. The following exemptions may apply to the non-procurement of biobased content materials:

<OLG>
<OLI>The product does not meet appropriate performance standards;</OLI>

<OLI>The product is not available within a reasonable time frame;</OLI>

<OLI>The product is not available competitively (from two or more sources);</OLI>

<OLI>The product is only available at an unreasonable price (compared with a comparable bio-based content product)."</OLI>
</OLG>

Record each product used in the project that has a requirement or option of containing biobased content in accordance with SECTION 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING, noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, total value of biobased content, exemptions (a, b, c, or d, as indicated), and comments. Biobased content values may be determined by weight or volume percent, but must be consistent throughout.

2.4 PDF AS-BUILT FILES

Provide electronic PDF "plots" of all contract drawings sheets associated with the as-built drawing submittal. Compile and organize the PDF set to match the contract drawings. Bookmark and label the pages of the PDF file in accordance with Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD).

2.5 REDLINES AND MARKUPS

Provide PDFs of the current working redlines and/or markups complying with the as-builts drawing and markup requirements contained in this specification.

2.6 AS-BUILT OR ADVANCED MODELING RE-SUBMISSION REQUIREMENTS

If elements of an as-built submittal or advanced modeling package are rejected, provide the following for each re-submission, in addition to any information required in Section 01 33 00 SUBMITTAL PROCEDURES:

<OLG>
<OLI>Re-submit all components required under paragraph As-Builts or Advanced Modeling Package, including a new Advanced Modeling Submittal Checklist and updated content in response to Government comments.</OLI>

<OLI>Provide a copy of all Government review comments.</OLI>

<OLI>Provide a disposition/response to each Government review comment for a back-check of the re-submission deliverable.</OLI>
</OLG>

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site and/or red-lined PDF files.

Contractor shall submit the current As-Builts to the Contracting Officer on a monthly basis with the normal pay request.

Contrator shall submit the final As-Built Drawings within 30 days following the completion of the project.

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

<OLG>
<OLI>Use base colors of red, green, and blue. Color code for changes as follows:</OLI>

<OLI LEVEL="2">Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.</OLI>

<OLI LEVEL="2">Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.</OLI>

<OLI LEVEL="2">Additions (Green) - Added items, lettering in notes and leaders.</OLI>

<OLI>Provide a legend if colors other than the "base" colors of red, green, and blue are used.</OLI>

<OLI>Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.</OLI>

<OLI>Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.</OLI>

<OLI>Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.</OLI>

<OLI>Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.</OLI>

<OLI>For deletions, cross out all features, data and captions that relate to that revision.</OLI>

<OLI>For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.</OLI>

<OLI>Indicate one of the following when attaching a print or sketch to a markup print:</OLI>

<OLI LEVEL="2">Add an entire drawing to contract drawings</OLI>

<OLI LEVEL="2">Change the contract drawing to show changes on the drawing.</OLI>

<OLI LEVEL="2">Provided for reference only to further detail the initial design.</OLI>

<OLI>Incorporate all shop and fabrication drawings into the markup drawings.</OLI>
</OLG>
3.1.2 As-Built Drawings Content

Revise As-Built Drawings and/or red-lined PDF files in accordance with ERDC/ITL TR-19-6 and ERDC/ITL TR-19-7. Keep these working as-built markup drawings current on a weekly basis and at least one set available on the jobsite at all times. Changes from the contract drawings which are made during construction or additional information which might be uncovered in the course of construction must be accurately and neatly recorded as they occur by means of details and notes. Submit the working as-built markup drawings for approval prior to submission of each monthly pay estimate. For failure to maintain the working and final record drawings as specified herein, the Contracting Officer will withhold 10 percent of the monthly progress payment until approval of updated drawings. Show on the as-built drawings, but not limited to, the following information:
<OLG>
<OLI>The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.</OLI>

<OLI>The location and dimensions of any changes within the building structure.</OLI>

<OLI>Layout and schematic drawings of electrical circuits and piping.</OLI>

<OLI>Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.</OLI>

<OLI>Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.</OLI>

<OLI>The topography, invert elevations and grades of drainage installed or affected as part of the project construction.</OLI>

<OLI>Changes or Revisions which result from the final inspection.</OLI>

<OLI>Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.</OLI>

<OLI>If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.</OLI>

<OLI>Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.</OLI>

<OLI>Changes in location of equipment and architectural features.</OLI>

<OLI>Modifications.</OLI>

<OLI>Actual location of anchors, construction and control joints, etc., in concrete.</OLI>

<OLI>Unusual or uncharted obstructions that are encountered in the contract work area during construction.</OLI>

<OLI>Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.</OLI>
</OLG>

3.2 RECORD DRAWING FILES

If additional drawings are required, prepare them using the specified electronic file format applying ERDC/ITL TR-19-7 and ERDC/ITL TR-19-6. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings. Accomplish additions and corrections to the contract drawings using CAD files. Provide all program files and hardware necessary to prepare final PDF record drawings. The Contracting Officer will review final PDF record drawings for accuracy and return them to the Contractor for required corrections, changes, additions, and deletions.

3.2.1 Rename the CAD Drawing files

Rename the CAD Drawing files using the contract number as the Project Code field,(e.g., W91238-15-C-10A-102.DWG) as instructed in the Pre-Construction conference. Use only those renamed files for the Marked-up changes. Make all changes on the layer/level as the original item.
<OLG>
<OLI>For AutoCAD files (DWG), enter all as-built delta changes and notations on the AS-BUILT layer.</OLI>

<OLI>When final revisions have been completed, show the wording "RECORD DRAWING AS-BUILTS" followed by the name of the Contractor in letters at least 3/16 inch high on the cover sheet drawing. Date RECORD DRAWING AS-BUILTS" drawing revisions in the revision block.</OLI>

<OLI>Within 10 days after Government approval of all of the working record drawings for a phase of work, prepare the final CAD record drawings for that phase of work and submit PDF drawing files and two sets of prints for review and approval. The Government will promptly return one set of prints annotated with any necessary corrections. Within 7 days revise the CAD

files accordingly at no additional cost and submit one set of final prints for the completed phase of work to the Government. Within 10 days of substantial completion of all phases of work, submit the final record drawing package for the entire project. Submit one set of electronic CAD files, and one set of the approved working record PDF and or programfiles with two sets of prints. The CAD files must be complete in all details and identical in form and function to the CAD drawing files supplied by the Government. Prepare AutoCAD files for transmittal using e-Transmit. Make any transactions or adjustments necessary to accomplish this. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CAD system. Paper prints, drawing files and storage media submitted will become the property of the Government upon final approval. Failure to submit final record PDF drawing files, CAD files and marked prints as specified will be cause for withholding any payment due under this contract. Approval and acceptance of final record drawings must be accomplished before final payment is made.</OLI>
</OLG>

3.3 RECORD DRAWINGS

Prepare final record drawings after the completion of each definable phase of work as listed in the Contractor Quality Control Plan (such as Foundations, Utilities, or Structural Steel as appropriate for the project). Transfer the changes from the approved working as-built markup drawings to the original electronic CAD drawing files. Modify the as-built CAD drawing files to correctly show the features of the project as-built by bringing the working CAD drawing set into agreement with approved working as-built markup drawings, and adding such additional drawings as may be necessary. Refer to ERDC/ITL TR-19-6. Jointly review the working as-built markup drawings with printouts from working as-built CAD drawing PDF files for accuracy and completeness. Monthly review of working as-built CAD drawing PDF file printouts must cover all sheets revised since the previous review. These PDF drawing files are part of the permanent records of this project. Any drawings damaged or lost must be satisfactorily replaced at no expense to the Government.

Drawing revisions (include within change order price the cost to change working and final record drawings to reflect revisions) and compliance with the following procedures.
<OLG>
<OLI>Follow directions in the revision for posting descriptive changes.</OLI>

<OLI>The revision delta size must be 5/16 inch unless the area where the delta is to be placed is crowded. Use a smaller size delta for crowded areas.</OLI>

<OLI>Place a revision delta at the location of each deletion.</OLI>

<OLI>For new details or sections which are added to a drawing, place a revision delta by the detail or section title.</OLI>

<OLI>For minor changes, place a revision delta by the area changed on the drawing (each location).</OLI>

<OLI>For major changes to a drawing, place a revision delta by the title of the affected plan, section, or detail at each location.</OLI>

<OLI>For changes to schedules or drawings, place a revision delta either by the schedule heading or by the change in the schedule.</OLI>
</OLG>

3.3.1 Final Record Drawing Package

Submit the final record PDF and CAD drawings package for the entire project within 20 days of substantial completion of all phases of work. Submit one set of ANSI D size PDF and CAD files, two sets of ANSI D size prints and one set of the approved working record drawings. The package must be complete in all details and identical in form and function to the contract drawing files supplied by the Government.

3.4 FINAL APPROVED SHOP DRAWINGS

Submit final approved project shop drawings within 30 days following the completion of the project.

3.5 CONSTRUCTION CONTRACT SPECIFICATIONS

Submit final PDF file record construction contract specifications, including revisions thereto, within 30 days following the completion of the project.

3.6 AS-BUILT RECORD OF EQUIPMENT AND MATERIALS

Furnish one copy of preliminary record of equipment and materials used on the project 15 days prior to final inspection. This preliminary submittal will be reviewed and returned 2 days after final inspection with Government comments. Submit two sets of final record of equipment and materials 10 days after final inspection. Key the designations to the related area depicted on the contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

3.7 OPERATION AND MAINTENANCE MANUALS

Provide project operation and maintenance manuals as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit to the Contracting Officer for approval within 30 calendar days following the completion of the project.

3.8 CLEANUP

Leave premises "broom clean". Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site.

3.9 REAL PROPERTY RECORD

Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354.

3.9.1 Interim DD FORM 1354

Near the completion of Project, but a minimum of 60 days prior to final acceptance of the work, complete, update draft DD FORM 1354, and submit an accounting of all installed property with Interim DD FORM 1354. Include any additional assets, improvements, and alterations from the Draft DD FORM 1354.

3.9.2 Completed DD FORM 1354

For convenience, a blank fillable PDF DD FORM 1354 may be obtained at the following link:

www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf

Submit the completed List of Installed Building Equipment items for DD FORM 1354. Attach this list to the updated DD FORM 1354.

-- End of Section --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA
05/23, CHG 1: 05/25 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE GUIDELINE 1.4	(2019) Preparing Systems Manuals for Facilities

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2024) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings
ASTM E2166	(2016; R 2023) Standard Practice for Organizing and Managing Building Data

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

Facility Data Workbook; G

Training Plan; G

Training Outline; G

Training Content; G

Operation And Maintenance Manual, Progress Submittal; G

Operation And Maintenance Manual, Prefinal Submittal; G

Operation And Maintenance Manual, Final Submittal; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

Training Plan; G

Record Drawings And Utility Systems; G

1.3 MEETINGS

To assure that Operation and Maintenance Manual and Facility Data Workbook requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

1.3.1 Pre-Construction Meeting

At a minimum, discuss the following:

a. The requirement for Operation and Maintenance Manuals and Facility Data deliverables under this contract including coordination meetings

b. Processes and method of gathering Facility Data information during construction

c. Primary roles and responsibilities associated with the development and delivery of the Operation and Maintenance Manuals and Facility Data deliverables, and

d. Identify and agree upon a date and attendance list for the meetings described below:

1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Pre-Construction Meeting prior to the submission of the Operation and Maintenance Manual Progress Submittal. Meeting attendance must include the Contractor's Operation and Maintenance Manual and Facility Data Workbook Preparer, Designer of Record (DOR), Quality Control Manager, the Government's Design Manager (DM), Contracting Officer's Representative, and Government's facility data reviewer. Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for Operation and Maintenance Manual and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of Operation and Maintenance Manual.

1.3.3 Submittal Coordination Meeting

Facilitate a meeting following submission and Government review of each design or progress submittal of the Operation and Maintenance Manual and Facility Data Workbook.

a. Include personnel from the Coordination meeting and any additional personnel identified.

b. The purpose of this meeting is to demonstrate ongoing compliance with the requirements identified in this specification. Discuss Government review comments and unresolved items preventing completion and Government approval of the Operation and Maintenance Manuals and Facility Data Workbook (FDW).

c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting.

1.4 FACILITY DATA WORKBOOK

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program. As a minimum, provide Facility Data Workbook as a list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL MEDIA

Assemble Operation and Maintenance Manual into an electronically bookmarked file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete bookmarked operation and maintenance directory.

1.5.1 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

a. Building Number

b. Project Title

c. Activity and Location

d. Construction Contract Number

e. Prepared For: (Contracting Agency)

f. Prepared By: (Name, title, phone number and email address)

g. Include the disk content on the disk label

h. Date

i. Virus scanning program used

1.6 O&M MANUAL CONTENT

Organize the bookmarked Operation and Maintenance Manual into the following Parts in accordance with ASHRAE GUIDELINE 1.4, and as modified and detailed below. Word template for O&M Manual is available at:

www.wbdg.org/dod/ufgs/ufgs-01-78-23

1.6.1 Part 1: Executive Summary

Provide a summary of the information found in the O&M manual including the purpose of the manual and a description of the manual's organization.

1.6.2 Part 2: Facility Design and Construction

1.6.2.1 General Facility and Systems Description

Provide an overview of the intent for design and use of the facility. Provide a PDF of the Record Drawings prepared in accordance with 01 78 00 CLOSEOUT SUBMITTALS and bookmarked using the sheet title and sheet number. Include uncluttered 11 by 17 inches floor plans with room numbers, type or function of space, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code list and generally describe all the facility systems and any special building features (for example, HVAC Controls, Sprinkler Systems, Cranes, Elevators, and Generators). Include photographs marked up and labeled to show key operating components and the overall facility appearance.

1.6.2.2 Basis of Design

Provide a copy of the contract Basis of Design.

1.6.2.3 Contract Documents, RFP, Amendments, and Modifications

Provide the contract construction documents complete, to include specifications, drawings, Request for Proposal, amendments, and modifications.

1.6.2.4 Room Inventory of Real Property and Finishes

Provide a list of installed equipment furnished under this contract. Include all information usually listed on manufacturer's name plate. Include, as applicable, the following information for each piece of equipment installed: description of item, all dimensions, location by room number, model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Real property includes, but is not limited to, floor coverings, wall surfaces, ceiling surfaces, windows, roofing, HVAC filters, plumbing fixtures, and lighting fixtures. Submit the final list within 30 days following the completion of the project.

Include spatial data defining actual net square footage and data of each room. Also include the room finish schedule including room names and numbers. Include schedules in the construction drawings in the room inventory. Add a column to each schedule to record what was provided by the contractor during construction. Provide a PDF of room inventory. Key the designations to the related area depicted on the contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

1.6.3 Part 3: Facilities, Systems, and Assemblies Information

1.6.3.1 Organization

Bookmark information in this section using the current version of ASTM E2166 Uniformat II, UFGS numbers, and document type as outlined in the example below. Bookmark/tab each item to the third level for easy navigation of the manual.

Example as shown in Table below:

PARTS AND SUBPART NUMBERING
3.1 B20 EXTERIOR CLOSURE (System)
3.1.1 B2030 EXTERIOR DOORS (Subsystem)
3.1.1.1 B2030110 GLAZED DOORS (Component)
3.1.1.1.1 Applicable specifications List in UFGS Format
3.1.1.1.2 Manufacturer's Operations and Maintenance Data
3.1.1.1.3 Approved Submittal
3.1.1.1.4 Coordination/Shop Drawings
3.1.1.1.5 Sequence of Operation for Operating Equipment
3.1.1.1.6 Testing Equipment Information and Performance Data
3.1.1.1.7 Routine Maintenance Requirements
3.1.1.1.8 Repair Procedures
3.1.1.1.9 Emergency Procedures & Locations of Applicable Controls
3.1.1.1.10 Warranties
3.1.1.1.11 Record Drawings and Utility Systems
3.1.1.1.12 Contractor / Supplies Listing and Contact Information

1.6.3.2 Related Specifications

Reference each specification related to the subsystem in this section, and locate the actual specification section in Part 2 of the O&M Manual. List specifications in table format as shown in the below example.

UFGS Number	Specification Title	Page Spec Begins in Part 2

1.6.3.3 Manufacturer's Operations and Maintenance Data

Provide a copy of all manufacturer specifications and cutsheets. Provide text-searchable, high-quality document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Provide Uniformat II Level 3 identification for D20, D30, D40 installed equipment. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same files. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master part catalog.

1.6.3.4 Approved Submittals and Certificates

Provide a copy of all submittals documented with the required approval as applicable for each UFGS specification listed in the table outlined in applicable specifications. Include copies of SD-07 Certificates submittals documented with the required approval, SD-08 Manufacturer's Instructions submittals documented with the required approval, and SD-10 Operation and Maintenance Data submittals documents with the required approval.

1.6.3.5 Approved Coordination/Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work. Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project. Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

1.6.4 Sequence of Operation for Operating Equipment

Provide record one-line diagrams for each floor, delineating mechanical equipment location within the building. Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.4.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard. Specify if any certifications or licenses are required to operate the equipment.

1.6.4.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.4.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.6.4.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.4.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.4.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.4.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.6.4.9 Additional Requirements for Equipment Control Systems

Provide Data Package 5 and the following for all control systems:

a. Provide a narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.

b. Submit complete controls equipment schedules, full as-built sequence of operations, wiring and logic diagrams, Input/Output Tables, equipment schedules, copies of checkout tests and calibrations performed by the Contractor (not Cx tests), and all associates information.

c. Full points list. Provide a listing of rooms with the following information for each room:

    (1) Floor

    (2) Room number

    (3) Room name

    (4) Air handler unit ID

    (5) Reference drawing number

    (6) Air terminal unit tag ID

    (7) Heating or cooling valve tag ID

    (8) Minimum cfm

    (9) Maximum cfm

d. Full print out of all schedules and set points after testing and acceptance of the system.

e. Full as-built print out of software program.

f. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.4.10 Testing Equipment Information and Performance Data

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5 Routine Maintenance Requirements

1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including requirements by type of activity. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

a. Define the anticipated time required to perform each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.

b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.5.2 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

a. A table showing recommended lubricants for specific temperature ranges and applications.

b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide procedural instructions for Oil Sampling for all equipment.

c. A Lubrication Schedule showing service interval frequency.

1.6.6 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards. Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials. Specify if any certifications or licenses are required to repair the equipment.

1.6.6.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.6.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.6.3 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required specialty tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.6.4 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.6.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Identify if replacement of a subassembly, attachment or accessory requires the entire assembly to be replaced. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.6.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.6.7 Record Drawings and Utility Systems

The record drawings are the final compilation of actual conditions reflected in the as-built drawings. Provide record drawings as outlined in 01 78 00 CLOSEOUT SUBMITTALS.

Using Record Source Drawings, show and document details of the actual installation of the utility systems, annotate and highlight the Operation and Maintenance information. Provide the following drawings at a large enough scale to differentiate designated isolation units from surrounding valves and switches.

1.6.6.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.6.9 Contractor / Supplier Listing and Contact Information

Provide a list that includes the name, address, telephone number, email and website of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6.7 Part 4: Facility Operations

1.6.7.1 Completed Facility Operating Plan

Provide a plan that documents that procedures for the operation of systems and assemblies in the facility. The systems that should be included in the Operating Plan include, but are not limited to:

a. Electrical systems and equipment

b. Mechanical systems and equipment

c. Fire Protection systems and equipment

d. Control Systems and equipment

e. Architectural and Structural systems, fixtures, structures, and equipment

f. Vertical transportation such as elevators and escalators

1.6.7.2 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.7.3 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports

Compile and provide approved Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals.

1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory

1.6.7.5.1 Maintenance Schedules

Include recommended maintenance schedules for systems and equipment.

1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping

Include ongoing commissioning and optimization procedures and documentation to monitor and improve the performance of facility systems.

1.6.7.5.3 Janitorial and Cleaning Plans and Procedures

Include a copy of facility cleaning and janitorial plan with procedures and intended chemicals and equipment.

Provide environmentally friendly cleaning recommendations in accordance with ASTM E1971.

1.6.7.6 Utility Record Drawings

1.6.7.6.1 Utility Schematic Diagrams

Provide a one-line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the 5 foot connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.

1.6.7.6.2 Enlarged Connection and Cutoff Plans

Provide enlarged floor plans and provide information between the 5 foot utilities connection point and where utilities connect to facility distribution. Enlarge floor plans/elevations of the rooms where the utility enters the building and indicate on these plans the locations of the main interiors and exterior connection and cutoff points for the utilities. Also enlarge floor plans/elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems

Provide in narrative format a description of the utility metering and monitoring systems. Include locations, function, and related systems.

1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting

Procedures for usage reporting and tracking in support of establishing and monitoring utility budgets and costs, and in developing annual energy reports.

1.6.7.6.2.3 One-Line Diagrams and Meter Location of System

Provide one-line diagrams and design drawings that highlight meter locations on the site.

1.6.7.6.3 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.8 Part 5: Training

Provide a copy of training plans used for each type of equipment along with training materials used, arranged in specification sequence. Provide a copy of training records, sign-in sheets, and agendas. Include training and documentation on the updating and continued use of the O&M Manual.

1.6.9 Part 6: Cx Project Report and TAB Report

Provide the final Cx Plan and complete Cx reports with evaluation and testing forms and records for each building system. Include relevant commissioned system assemblies test reports including installers checklists of assemblies. Provide all Cx Progress Reports, issues and resolutions logs with resolution or status of each item, and a list of any open items and seasonal or additional testing required.

1.6.10 Part 7: Regulatory Requirements

Provide information describing regulatory and policies compliance requirements or provide a reference to where it is stored.

1.6.11 Part 8: Permits

Provide information requiring frequently asked questions and associated answers or provide a reference to where it is stored.

1.6.12 Part 9: Operations and Maintenance Manual Approval

Provide a signed document stating that the project O&M Manual has been reviewed and confirming agreement with the approach it presents. Include contact information for the signer for coordination of any future changes.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. O&M Data Packages are one of the components of the O&M Manual. The information required in each type of data package follows:

1.7.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.7.2 Data Package 1

a. Safety precautions and hazards

b. Cleaning recommendations

c. Maintenance and repair procedures

d. Warranty information

e. Extended warranty information

f. Contractor information

g. Spare parts and supply list

1.7.3 Data Package 2

a. Safety precautions and hazards

b. Normal operations

c. Environmental conditions

d. Lubrication data

e. Preventive maintenance plan, schedule, and procedures

f. Cleaning recommendations

g. Maintenance and repair procedures

h. Removal and replacement instructions

i. Spare parts and supply list

j. Parts identification

k. Warranty information

l. Extended warranty information

m. Contractor information

1.7.4 Data Package 3

a. Safety precautions and hazards

b. Operator prestart

c. Startup, shutdown, and post-shutdown procedures

d. Normal operations

e. Emergency operations

f. Environmental conditions

g. Operating log

h. Lubrication data

i. Preventive maintenance plan, schedule, and procedures

j. Cleaning recommendations

k. Troubleshooting guides and diagnostic techniques

l. Wiring diagrams and control diagrams

m. Maintenance and repair procedures

n. Removal and replacement instructions

o. Spare parts and supply list

p. Product submittal data

q. O&M submittal data

r. Parts identification

s. Warranty information

t. Extended warranty information

u. Testing equipment and special tool information

v. Testing and performance data

w. Contractor information

x. Field test reports

1.7.5 Data Package 4

a. Safety precautions and hazards

b. Operator prestart

c. Startup, shutdown, and post-shutdown procedures

d. Normal operations

e. Emergency operations

f. Operator service requirements

g. Environmental conditions

h. Operating log

i. Lubrication data

j. Preventive maintenance plan, schedule, and procedures

k. Cleaning recommendations

l. Troubleshooting guides and diagnostic techniques

m. Wiring diagrams and control diagrams

n. Repair procedures

o. Removal and replacement instructions

p. Spare parts and supply list

q. Repair work-hours

r. Product submittal data

s. O&M submittal data

t. Parts identification

u. Warranty information

v. Extended warranty information

w. Personnel training requirements

x. Testing equipment and special tool information

y. Testing and performance data

z. Contractor information

aa. Field test reports

1.7.6 Data Package 5

a. Safety precautions and hazards

b. Operator prestart

c. Start-up, shutdown, and post-shutdown procedures

d. Normal operations

e. Environmental conditions

f. Preventive maintenance plan, schedule, and procedures

g. Troubleshooting guides and diagnostic techniques

h. Wiring and control diagrams

i. Maintenance and repair procedures

j. Removal and replacement instructions

k. Spare parts and supply list

l. Product submittal data

m. Manufacturer's instructions

n. O&M submittal data

o. Parts identification

p. Testing equipment and special tool information

q. Warranty information

r. Extended warranty information

s. Testing and performance data

t. Contractor information

u. Field test reports

v. Additional requirements for HVAC control systems

1.7.7 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS. Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Quality Control Manager (QC) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and QC. Include within the plan the following elements:

a. Equipment included in training

b. Intended audience

c. Location of training

d. Dates of training

e. Objectives

f. Outline of the information to be presented and subjects covered including description

g. Start and finish times and duration of training on each subject

h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)

i. Instructor names and instructor qualifications for each subject

j. List of texts and other materials to be furnished by the Contractor that are required to support training

k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QC is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.

b. Relevant health and safety issues.

c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

d. Design intent.

e. Use of O&M Manual Files.

f. Review of control drawings and schematics.

g. Interactions with other systems.

h. Special maintenance and replacement sources.

i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the QCin accordance with Section 01 45 00.00 QUALITY CONTROL.

3.2 SUBMITTAL SCHEDULING

3.2.1 Operation and Maintenance Manual, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Operation and Maintenance Manual Files (Bookmarked PDF). Include the elements and portions of system construction completed up to this point. The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual Coordination Meeting.

3.2.2 Operation and Maintenance Manual, Prefinal Submittal

Submit the 100 percent submittal of the Operation and Maintenance Prefinal Submittal to the Contracting Officer for approval within 60 calendar days following the completion of the project. This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of Operation and Maintenance Progress submittal must be corrected prior to the Prefinal submission. The Prefinal Submittal must include Operation and Maintenance Manual Files (Bookmarked PDF).

3.2.3 Operation and Maintenance Manual, Final Submittal

Submit completed Operation and Maintenance Manual Files (Bookmarked PDF) within 30 days following the completion of the project. Any discrepancies discovered during the Government's review of the Prefinal submittal, including the Field Verification, must be corrected prior to the Final submission.

-- End of Section --

SECTION 02 41 00

DEMOLITION AND DECONSTRUCTION
08/22 - SWT 03/26

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K	(2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6	(2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25	(Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders; https://www.dla.mil/Portals/104/Documents/DispositionS /ddsr/docs/cylinderjointpub.pdf

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M	(2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures
MIL-STD-129	(2014; Rev R; Change 1 2018; Change 2 2019; Change 3 2023) Military Marking for Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 82	Protection of Stratospheric Ozone
49 CFR 173.301	Shipment of Compressed Gases in Cylinders and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Critical Root Zone (CRZ)

The critical root zone is the area around and under a tree where the majority of its roots are located. The CRZ is usually measured as a circle with a radius that's 1.5- foot for every inch of the tree's diameter at breast height (DBH), and a depth of 24 inches.

1.2.1.2 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.3 Deconstruction

Deconstruction is the process of taking apart a facility with the primary goal of preserving the value of all useful building materials.

1.2.1.4 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.1.5 Deconstruction Plan

Deconstruction Plan is the planned steps and processes for dismantling all or portions of a structure or assembly, to include managing sequencing activities, storage, re-installation activities, salvage and disposal mechanisms.

1.2.2 Demolition and Deconstruction Plan

Prepare a Demolition and Deconstruction Plan and submit a proposed salvage, demolition, deconstruction, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Coordinate with Waste Management Plan in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

Include statements affirming Contractor inspection of the existing roof deck and its suitability to perform as a safe working platform or if inspection reveals a safety hazard to workers, state provisions for securing the safety of the workers throughout the performance of the work. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by the Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work includes demolition, deconstruction, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 17, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove snow, dust, dirt, and debris from work areas daily.

1.3.2 Weather Protection

For portions of the building to remain, protect building interior and materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, have materials and workmen ready to provide adequate and temporary covering of exposed areas.

1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations. Prior to start of work, the Government will disconnect and seal utilities serving each area of alteration or removal upon written request from the Contractor.

1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished or deconstructed, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted. Where burning is permitted, adhere to federal, state, and local regulations.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition and Deconstruction Plan; G

Existing Conditions

SD-07 Certificates

Notification; G

SD-11 Closeout Submittals

Receipts

1.6 QUALITY ASSURANCE

Submit timely notification of demolition and deconstruction projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the State's environmental protection agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.6.1 Dust Control

Prevent the spread of dust to occupied portions of the building and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Vacuum and dust the work area daily. Sweep pavements as often as necessary to control the spread of debris that may result in foreign object damage potential to aircraft.

1.7 PROTECTION

1.7.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Notify the Contracting Officer prior to beginning such work.

1.7.2 Protection of Personnel

Before, during and after the demolition and deconstruction work continuously evaluate the condition of the site specific features being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.8 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.9 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Inspect and evaluate existing structures onsite for reuse. Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Structures

a. Remove existing structures indicated to be removed as shown in the drawings.

b. Demolish structures in a systematic manner from the top of the structure to the ground. Complete demolition work above each tier or floor before the supporting members on the lower level are disturbed. Demolish concrete and masonry walls in small sections. Remove structural framing members and lower to ground by means of derricks, platforms hoists, or other suitable methods as approved by the Contracting Officer.

c. Locate demolition and deconstruction equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location at the Powerhouse in accordance with instructions of the Contracting Officer.

3.1.3 Roofing

Remove existing roof system and associated components in their entirety down to existing roof deck. Remove built-up and/or single-ply roofing to effect the connections with new flashing or roofing. Remove gravel surfacing from existing roofing felts for a minimum distance of 18 inches back from the cut. Remove gravel without damaging felts. Cut existing felts, membrane and/or insulation along straight lines. Remove roofing system and/or insulation without damaging the roof deck. Sequence work to minimize building exposure between demolition or deconstruction and new roof materials installation.

3.1.3.1 Temporary Roofing

Install temporary roofing and flashing as necessary to maintain a watertight condition throughout the course of the work. Remove temporary work prior to installation of permanent roof system materials unless approved otherwise by the Contracting Officer. The existing deck and support structure is deteriorated where indicated, such that ability to support foot traffic and construction loads is unknown. Make provisions for worker safety during demolition, deconstruction, and installation of new materials as described in paragraphs entitled "Statements" and "Regulatory and Safety Requirements."

3.1.3.2 Reroofing

When removing the existing roofing system from the roof deck, remove only as much roofing as can be recovered by the end of the work day, unless approved otherwise by the Contracting Officer. Do not attempt to open the roof covering system in threatening weather. Reseal all openings prior to suspension of work the same day.

3.1.4 Masonry

Sawcut and remove masonry so as to prevent damage to surfaces to remain, to removed materials being salvaged and to facilitate the installation of new work. Where new masonry adjoins existing, abut or tie the new work into the existing construction as indicated. Provide square, straight edges and corners where existing masonry adjoins new work and other locations.

3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 2 inch. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete. Sawcuts in existing concrete pads must be at the nearest existing expansion joint or weakened plane joint and at full depth.

3.1.6 Structural Steel

Dismantle structural steel at field connections and in a manner that will prevent bending or damage. Salvage for recycle structural steel, steel joists, girders, angles, plates, columns and shapes. Flame-cutting torches are permitted when other methods of dismantling are not practical. Transport steel joists and girders as whole units and not dismantled. Transport structural steel shapes to a designated storage area, stacked according to size, type of member and length, and stored off the ground, protected from the weather.

3.1.7 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, steel trusses, metal gutters, roofing and siding, metal toilet partitions, toilet accessories and similar items.

Recycle scrap metal as part of demolition and deconstruction operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

3.1.8 Carpentry

Salvage for recycle lumber, millwork items, and finished boards, and sort by type and size. Chip or shred and recycle salvaged wood unfit for reuse, except stained, painted, or treated wood. Salvage windows, doors, frames, and cabinets, and similar items as whole units, complete with trim and accessories. Do not remove hardware attached to units, except for door closers. Brace the open end of door frames to prevent damage.

3.1.9 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990.

3.1.10 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.11 Mechanical Equipment and Fixtures

Disconnect mechanical hardware at the nearest connection to existing services to remain, unless otherwise noted. Disconnect mechanical equipment and fixtures at fittings. Remove service valves attached to the unit. Salvage each item of equipment and fixtures as a whole unit; listed, indexed, tagged, and stored. Salvage each unit with its normal operating auxiliary equipment. Transport salvaged equipment and fixtures, including motors and machines, to a designated storage area as directed by the Contracting Officer. Do not remove equipment until approved. Do not offer low-efficiency equipment for reuse; provide to recycling service for disassembly and recycling of parts.

3.1.11.1 Preparation for Storage

Remove water, dirt, dust, and foreign matter from units; drain tanks, piping and fixtures; if previously used to store flammable, explosive, or other dangerous liquids, steam clean interiors. Seal openings with caps, plates, or plugs. Secure motors attached by flexible connections to the unit. Change lubricating systems with the proper oil or grease.

3.1.11.2 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

3.1.11.3 Ducts

Classify removed duct work as scrap metal.

3.1.11.4 Fixtures, Motors and Machines

Remove and salvage fixtures, motors and machines associated with plumbing, heating, air conditioning, refrigeration, and other mechanical system installations. Salvage, box and store auxiliary units and accessories with the main motor and machines. Tag salvaged items for identification, storage, and protection from damage. Classify non-porcelain broken, damaged, or otherwise unserviceable units and not caused to be broken, damaged, or otherwise unserviceable as debris to be disposed of by the Contractor. Salvage and crush porcelain plumbing fixtures unsuitable for reuse.

3.1.12 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.12.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.12.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.12.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.12.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.2 DISPOSITION OF MATERIAL

3.2.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.2.2 Reuse of Materials and Equipment

Remove and store materials and equipment listed in the Demolition and Deconstruction Plan to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

3.2.3 Salvaged Materials and Equipment

Remove materials and equipment that are listed in the Demolition and Deconstruction Plan to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site.

a. Salvage items and material to the maximum extent possible.

b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.

c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers. Deliver the following items reserved as property of the Government to the areas designated in the Demolition and Deconstruction Plan.

d. Remove the items designated in the Demolition and Deconstruction Plan as property of the using service prior to commencement of work under this contract.

e. Remove historical items in a manner to prevent damage. Deliver the following historical items to the Government for disposition: Corner stones, contents of corner stones, and document boxes wherever located on the site.

f. Remove and capture all Class I ODS refrigerants in accordance with the Clean Air Act Amendment of 1990.

3.2.4 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Remove recovered ODS from Government property and dispose of in accordance with 40 CFR 82. Dispose products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

3.2.4.1 Special Instructions

No more than one type of ODS is permitted in each container. Apply a warning/hazardous label to the containers in accordance with Department of Transportation regulations. Provide a tag with the following information on all cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS:

a. Activity name and unit identification code

b. Activity point of contact and phone number

c. Type of ODS and pounds of ODS contained

d. Date of shipment

e. National stock number (for information, call (804) 279-4525).

3.2.4.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.2.5 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.2.6 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible materials in a sanitary fill located off the site.

3.3 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.4 DISPOSAL OF REMOVED MATERIALS

3.4.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations as contractually specified in the Waste Management Plan.

3.4.2 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.4.3 Removal from Government Property

Transport waste materials removed from demolished and deconstructed structures, except waste soil, from Government property for legal disposal. Dispose of waste soil as directed.

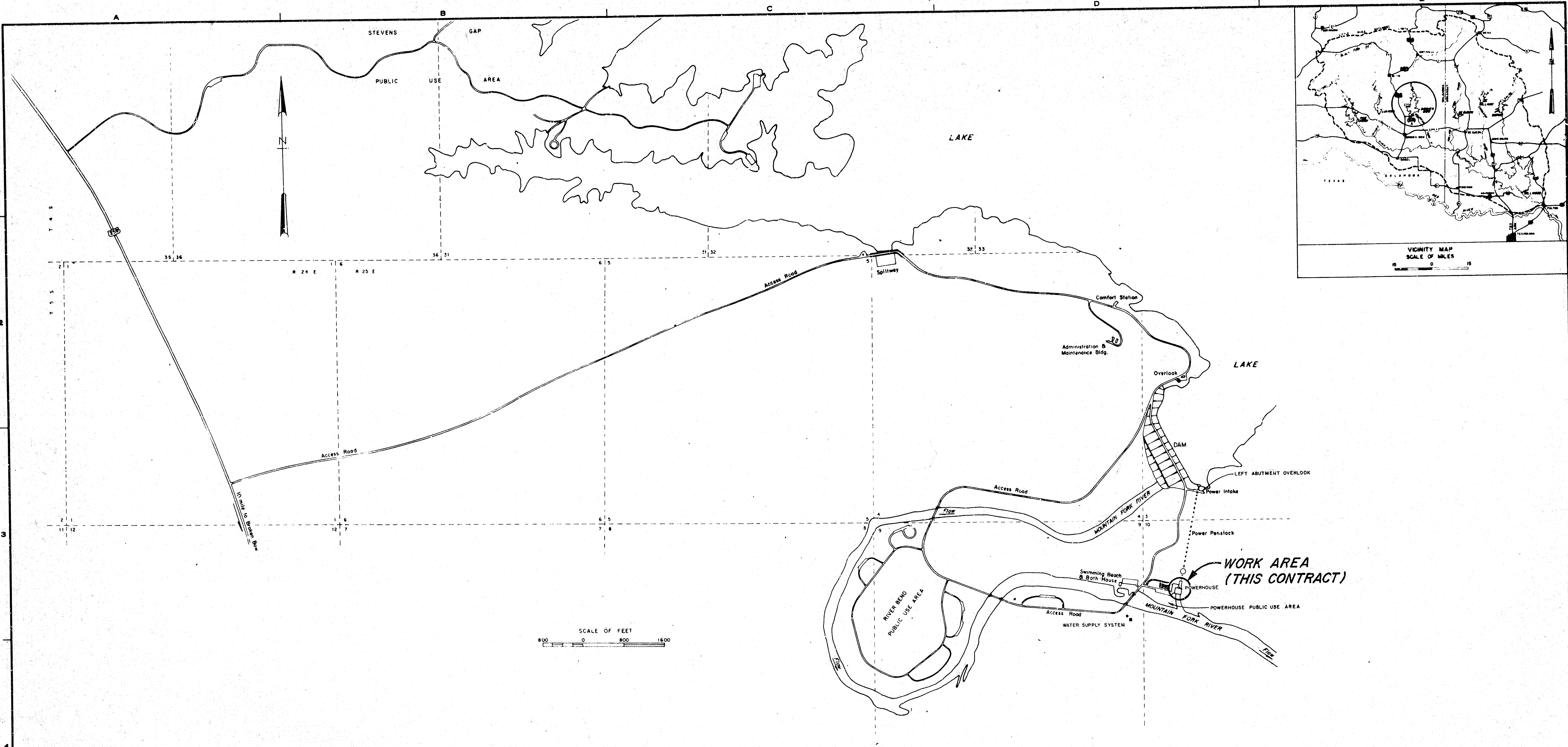
3.5 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

APPENDIX A1 -

HISTORICAL DRAWINGS



INDEX TO DRAWINGS

DWG. NO.	TITLE
CC720-C34-1/1	PROJECT LOCATION & DRAWING INDEX

INFORMATION DRAWINGS

DWG. No.	TITLE
172-C9-20/28.2	ROOF PLAN AND DETAILS
172-C9-20/29.1	ROOFING & FLASHING DETAILS SHEET 1
172-C9-20/30.1	ROOFING & FLASHING DETAILS SHEET 2

RECORD (AS BUILT) DRAWING

KEY	DATE	CHANGE	REVISION (INDICATED BY △)	APPR.

U. S. ARMY ENGINEER DISTRICT, TULSA
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED BY: Don Trotter
DRAWN BY: Reese
CHECKED BY: Dean English
SUBMITTED: Robert J. Tillman
CHIEF GENERAL DESIGN SECTION

RED RIVER WATERSHED — MOUNTAIN FORK RIVER, OKLAHOMA

BROKEN BOW LAKE

REROOFING POWERHOUSE

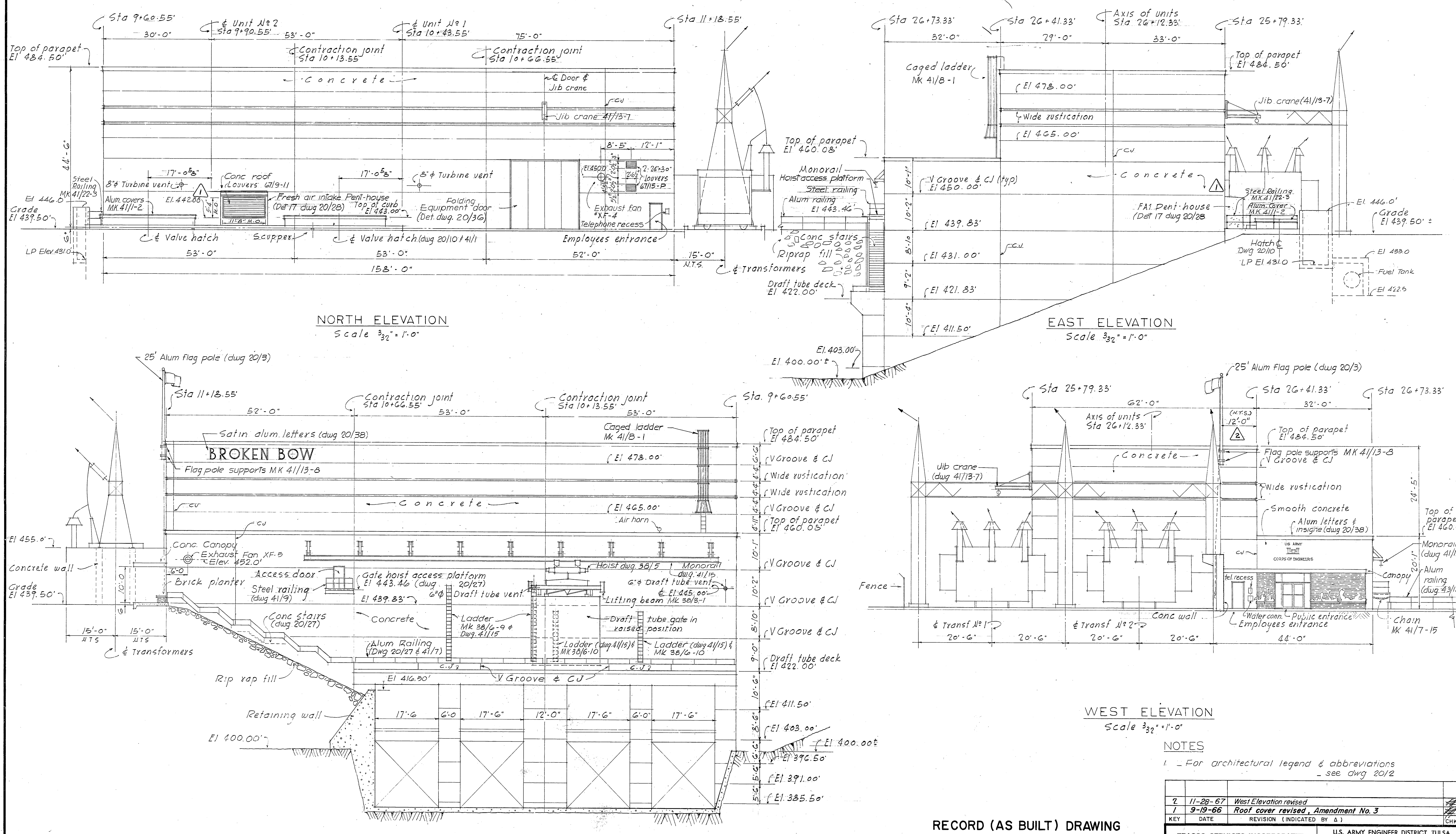
PROJECT LOCATION & DRAWING INDEX

INVITATION NO. DACW56-81-B-0056
SCALE: AS SHOWN
DRAWING NUMBER CC720-C34-1/1
DATE APRIL, 1981

SIGNATURES AFFIXED BELOW INDICATE OFFICIAL RECOMMENDATION AND APPROVAL OF ALL DRAWINGS IN THIS SET AS INDEXED ON THIS SHEET

RECOMMENDED: [signature] CHIEF DESIGN BRANCH
APPROVED: Donald R. Henderson CHIEF ENGINEERING DIVISION

CONTRACT No. DACW56-81-C-0143



NOTES

1. _ For architectural legend & abbreviations _ see dwg 20/2

RECORD (AS BUILT) DRAWING

GRAPHIC SCALE

3/32" = 1'-0" 8' 4' 0' 8' 16' 24'

KEY	DATE	REVISION (INDICATED BY Δ)	CHKD.
2	11-28-67	West Elevation revised	
1	9-19-66	Roof cover revised, Amendment No. 3	

EBASCO SERVICES INCORPORATED
NEW YORK, NEW YORK

U.S. ARMY ENGINEER DISTRICT, TULSA
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED: D.H.
DRAWN: D.H.
CHECKED: RC
SUBMITTED: EBASCO SERVICES, INC.
RECOMMENDED: CHIEF, HYDRO DES BR – SW DIV.
RECOMMENDED: CHIEF, ENGINEERING DIVISION – SW DIV.
APPROVED: CHIEF, ENGRG. DIV. – TULSA DIST., FOR THE DISTRICT ENGINEER

RED RIVER BASIN — MOUNTAIN FORK RIVER, OKLAHOMA

BROKEN BOW DAM
POWERHOUSE ARCHITECTURAL
EXTERIOR ELEVATIONS

JULY 1966

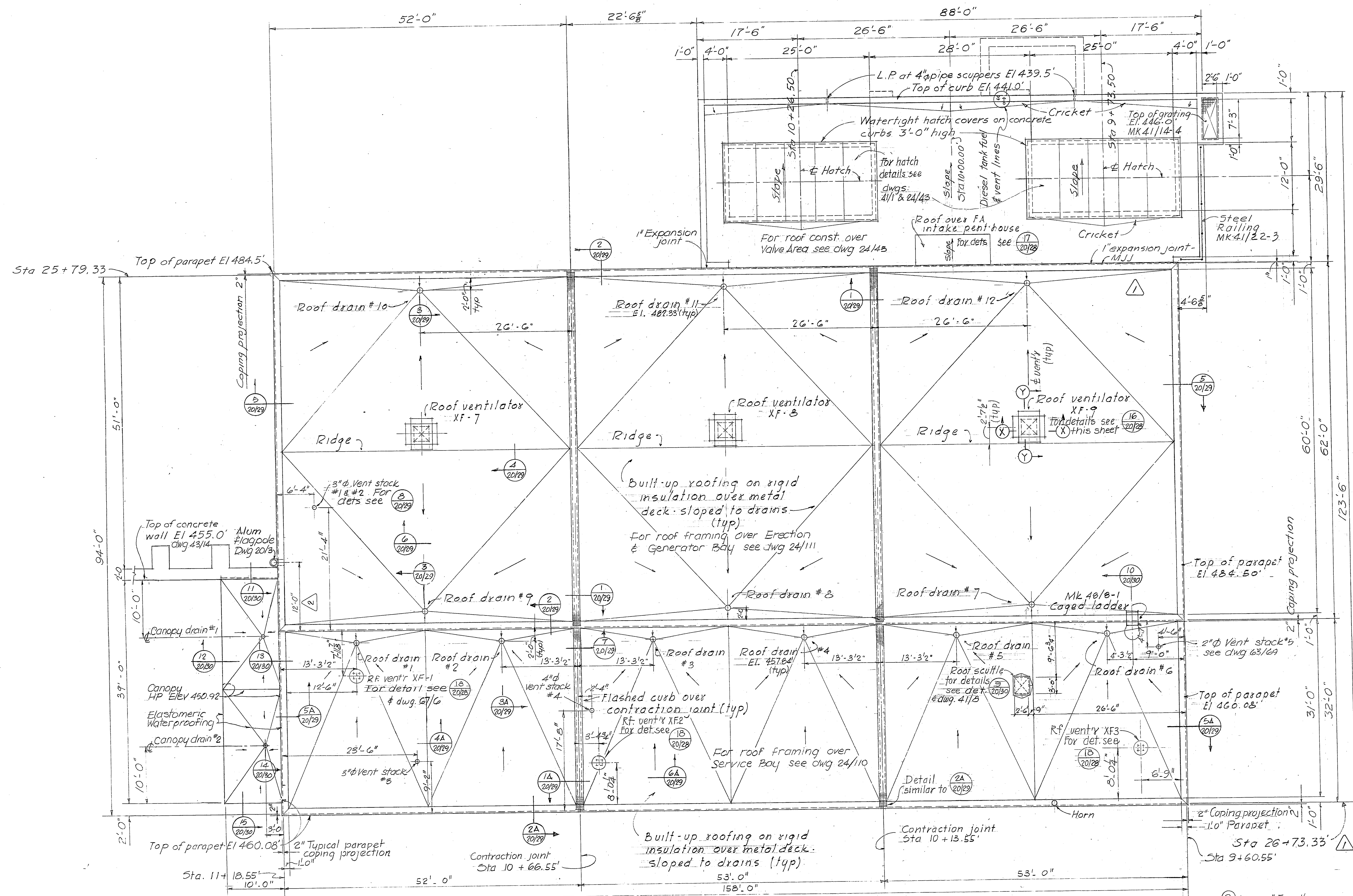
SCALE: 3/32" = 1'-0"

DRAWING NUMBER 172-C9-20/1.2

SHEET OF

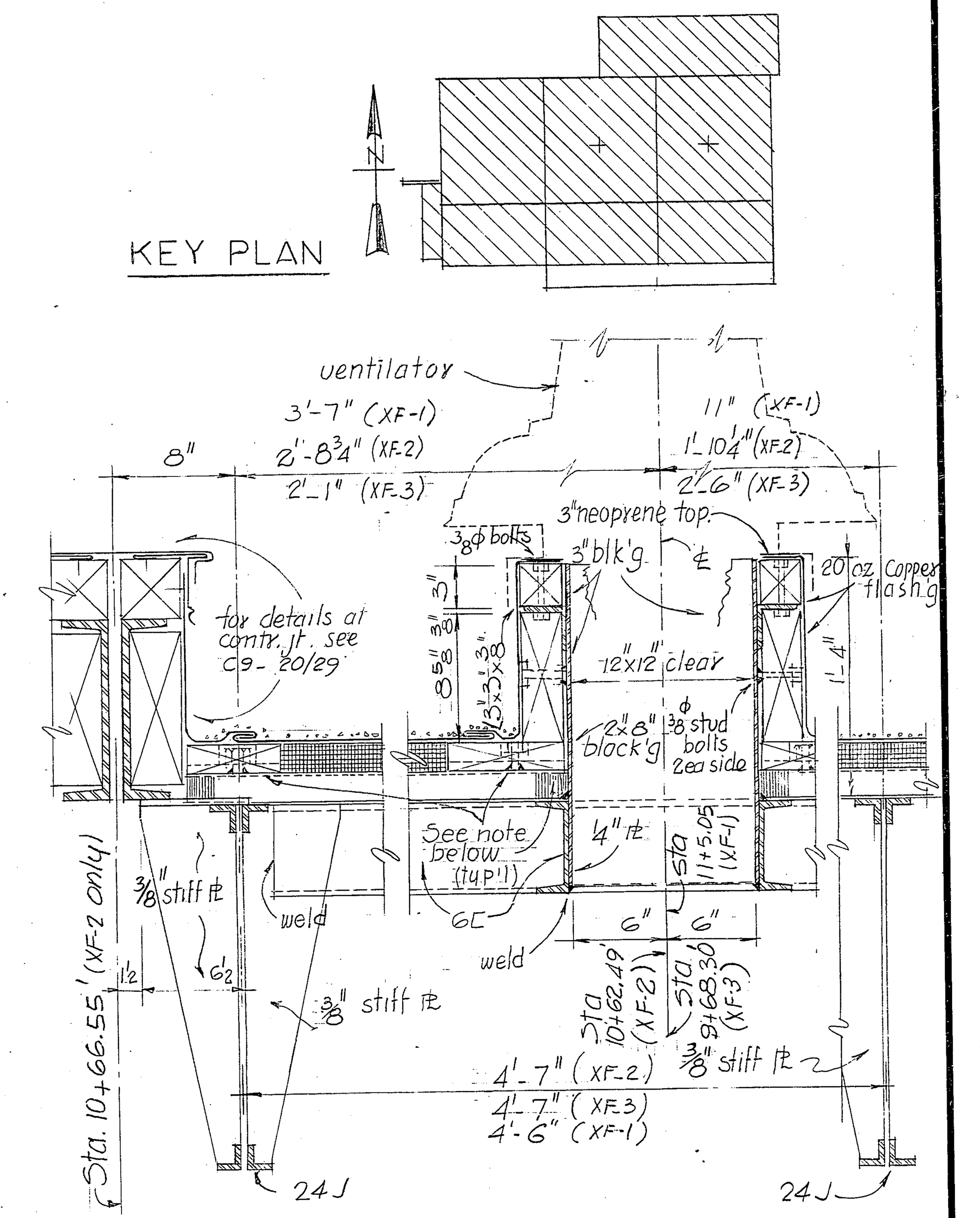
INVITATION NO. DACW56-67-B-0014

CONTRACT NO. 67-C-0069



ROOF PLAN

Scale 1/8"=1'-0"



DETAIL 18

ROOF VENTILATOR DET.

Scale 1 1/2" = 1'-0"

See det. dwg 24/110

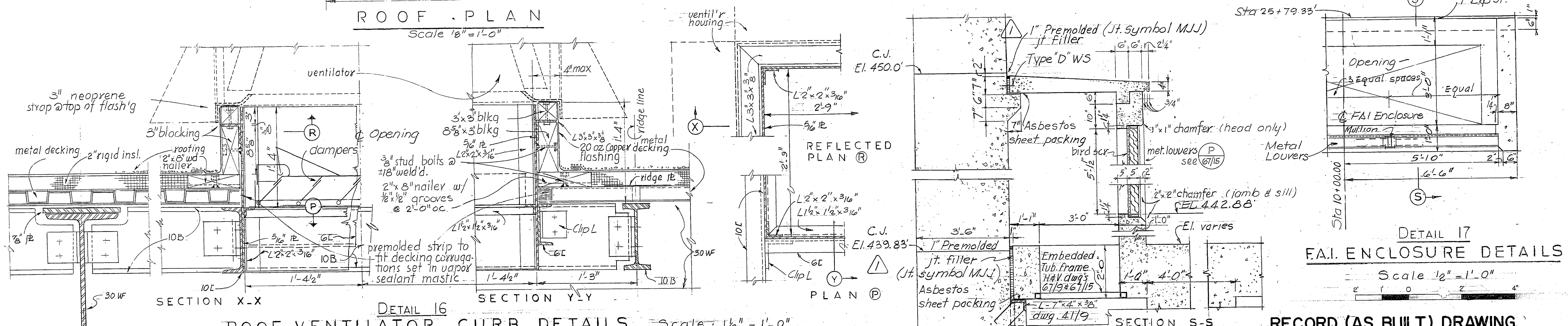
NOTES

1. For legend & abbreviations - see dwg. 20/2
2. For Erect. & Gen. Bay roof framg. see dwg 24/111
3. For Service Bay roof framing - see dwg. 24/110
4. For Valve Area roof construction - see dwg. 24/43
5. For Entrance Canopy const det. - see dwg. 24/26
6. For roofing details - see dwgs 24/29 & 24/30
7. For roof drainage details - see dwg 65/61
8. For roof vent, F.A.I. etc. details - see dwg 67/6
9. Wood nailer to have 1/2"x1/2" grooves 2'-0" o.c.; to be bolted with 3/8" stud bolts welded to deck.
10. Peripheral premolded filler strip to fit decking corrugations & to be set in vapor sealant mastic.
11. All wood blocking to be pressure treated (see specs)
12. For details of joint shapes & symbols see Dwg. 20/42
13. Payment
    Building construction item 22

GRAPHIC SCALES

1/8" = 1'-0"

1 1/2" = 1'-0"



DETAIL 16

ROOF VENTILATOR CURB DETAILS Scale 1 1/2" = 1'-0"

For ventilators XF-7, 8 & 9 see sect. B-B & R-R dwg. 24/111

DETAIL 17

F.A.I. ENCLOSURE DETAILS

Scale 1/2" = 1'-0"

RECORD (AS BUILT) DRAWING

2	11-28-67	Flagpole on "Roof Plan" relocated
1	3-16-67	Added joint shape symbols & note 12

EBASCO SERVICES INCORPORATED
NEW YORK, NEW YORK

U.S. ARMY ENGINEER DISTRICT, TULSA
CORPS OF ENGINEERS
TULSA, OKLAHOMA

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RED RIVER BASIN — MOUNTAIN FORK RIVER, OKLAHOMA

BROKEN BOW DAM
POWERHOUSE ARCHITECTURAL
ROOF PLAN AND DETAILS

JULY 1966

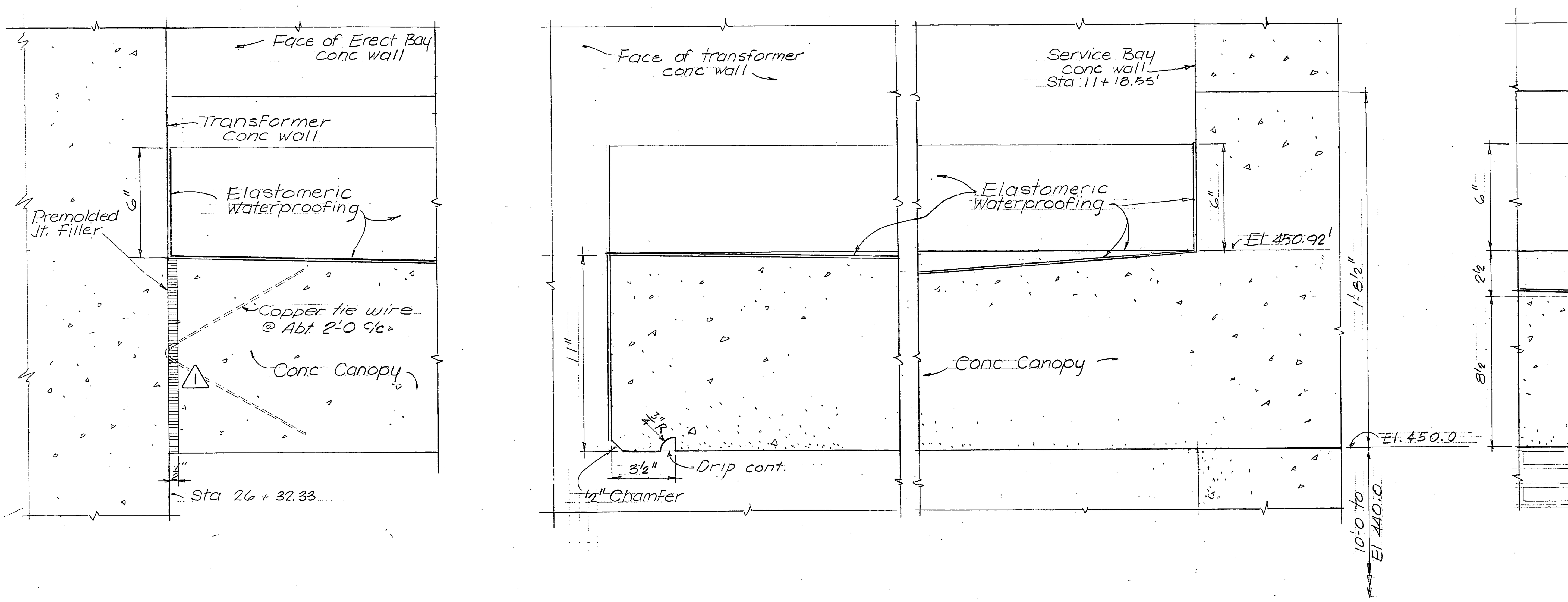
SCALE: As Noted

DRAWING NUMBER 172-C9-20/28.2

INVITATION NO. DACW56-67-B-0014

CONTRACT NO. 67-C-0069

CONTRACT NO. 67-0069



DETAIL 11
Ref. Dwg. 20/28

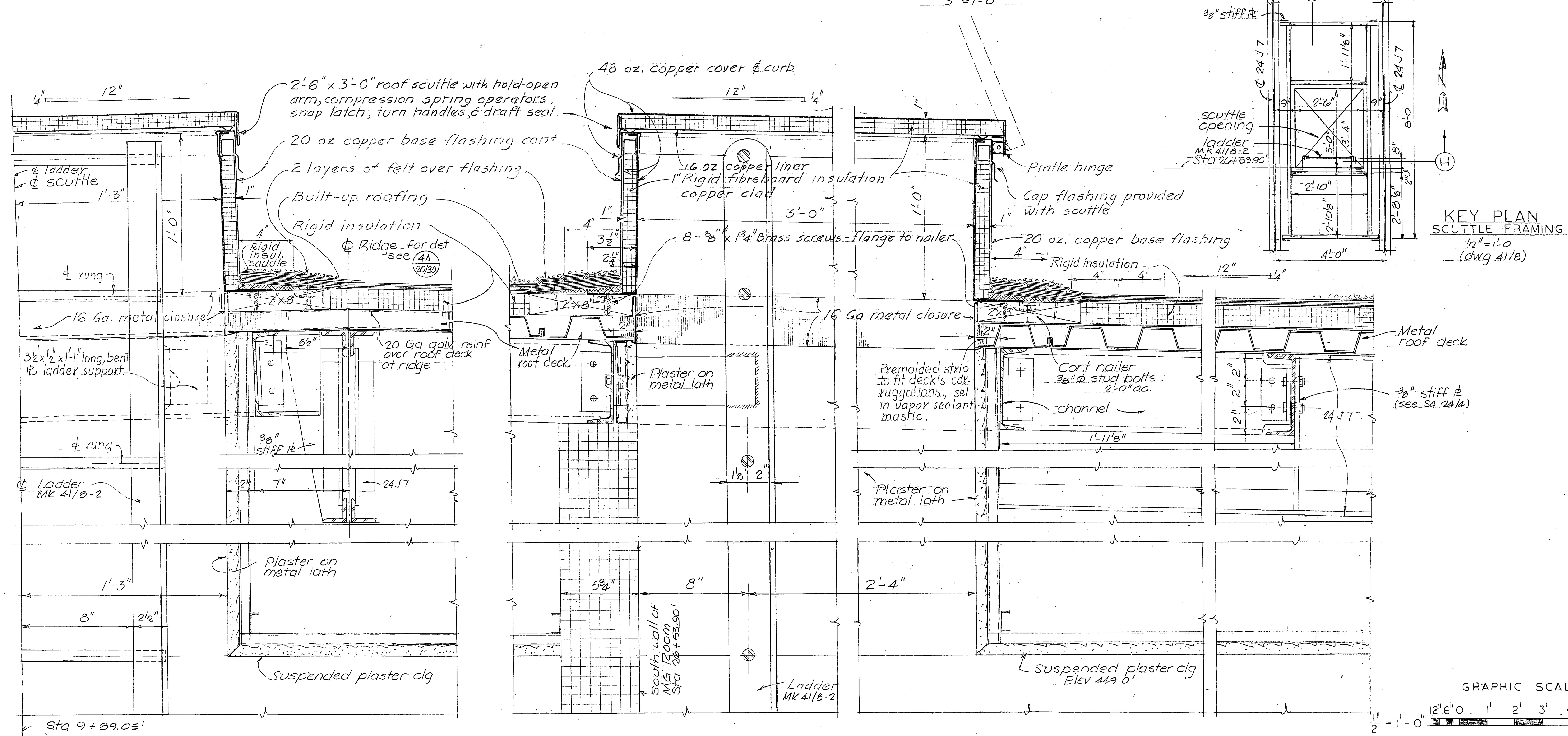
DETAIL 12
Ref. Dwg. 20/28

DETAIL 13
Ref. Dwg. 20/28

DETAIL 14
Ref. Dwg. 20/28

DETAIL 15
Ref. Dwg. 20/28

CANOPY DETAILS
3" = 1'-0"

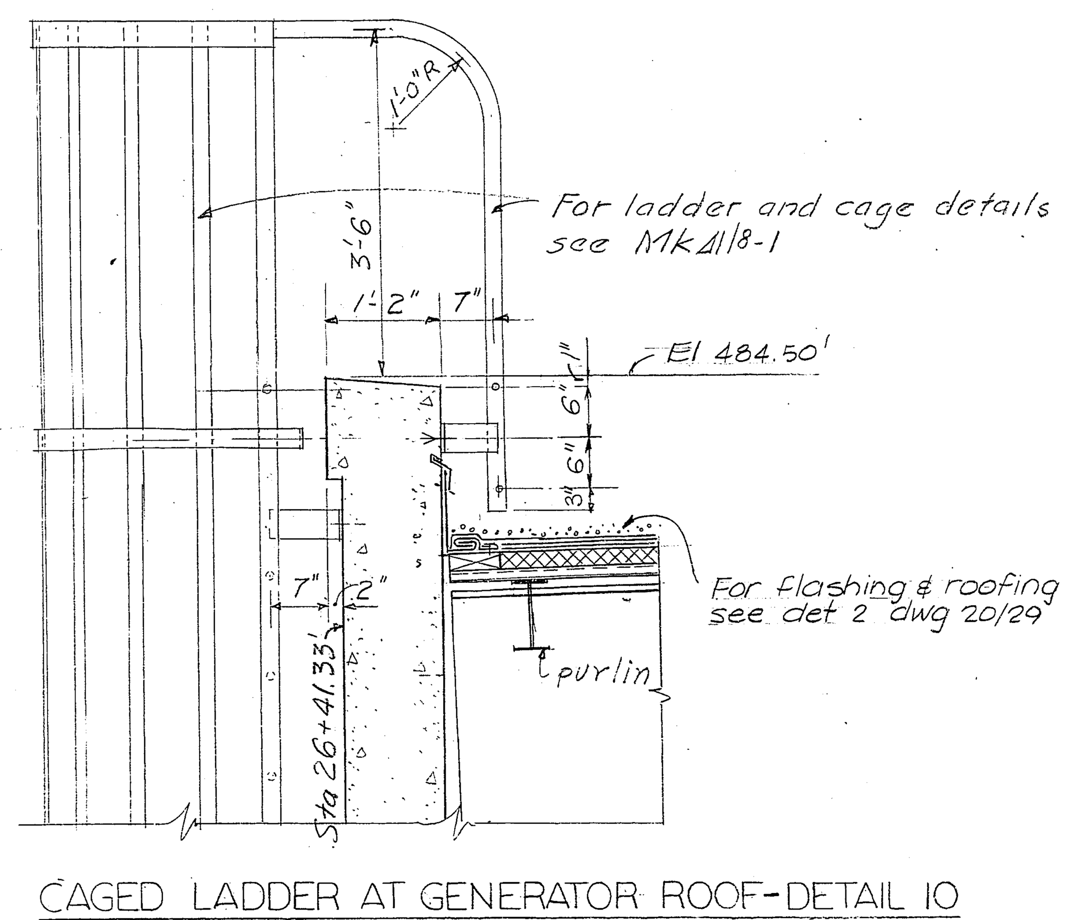


SECTION H-H

SECTION J-J

SCUTTLE DETAIL-9
3" = 1'-0"
See dwg. 41/8

KEY PLAN
SCUTTLE FRAMING
1/2" = 1'-0
(dwg 41/8)



CAGED LADDER AT GENERATOR ROOF-DETAIL 10
3/4" = 1'-0"

RECORD (AS BUILT) DRAWING

NOTES:

1. For legend & abbreviations – see dwg 20/2
2. For roof plan – see dwg 20/28
3. Payment Building construction item (22)

GRAPHIC SCALE

1/2" = 1'-0" 12" 6" 0 1' 2' 3' 4' 5'
3/4" = 1'-0" 12" 6" 0 1' 2' 3'
3" = 1'-0" 0 3" 6" 9" 12"

KEY	DATE	REVISION (INDICATED BY △)	CHKD.
1	1-31-67	Revised pre-molded Jt. filler, & deleted key.	

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U.S. ARMY ENGINEER DISTRICT, TULSA
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RED RIVER BASIN — MOUNTAIN FORK RIVER, OKLAHOMA

BROKEN BOW DAM
POWERHOUSE ARCHITECTURAL
ROOFING & FLASHING DETAILS
SHEET 2

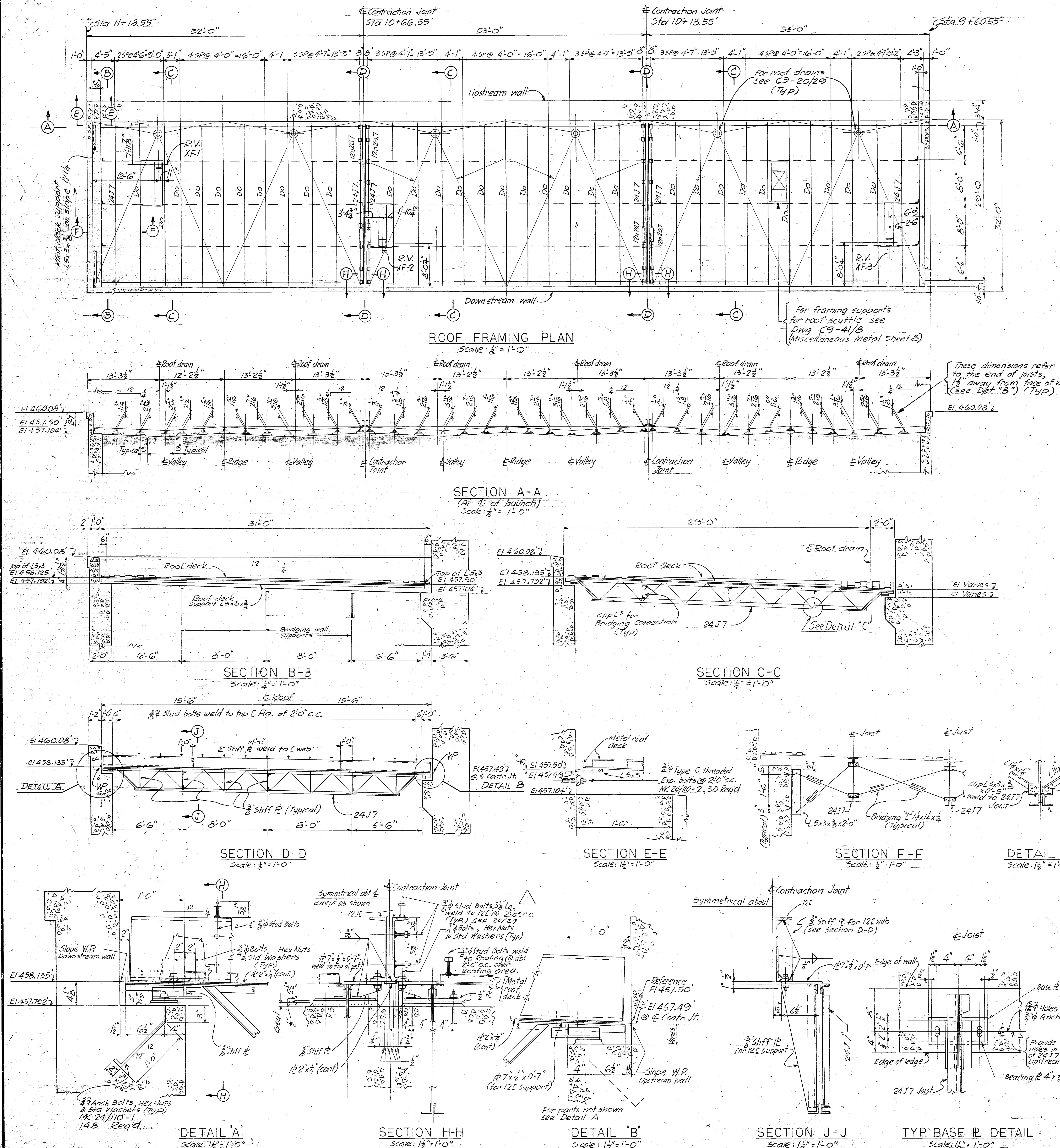
JULY 1966

SCALE: AS NOTED

DRAWING NUMBER 172-C9-20/30.1

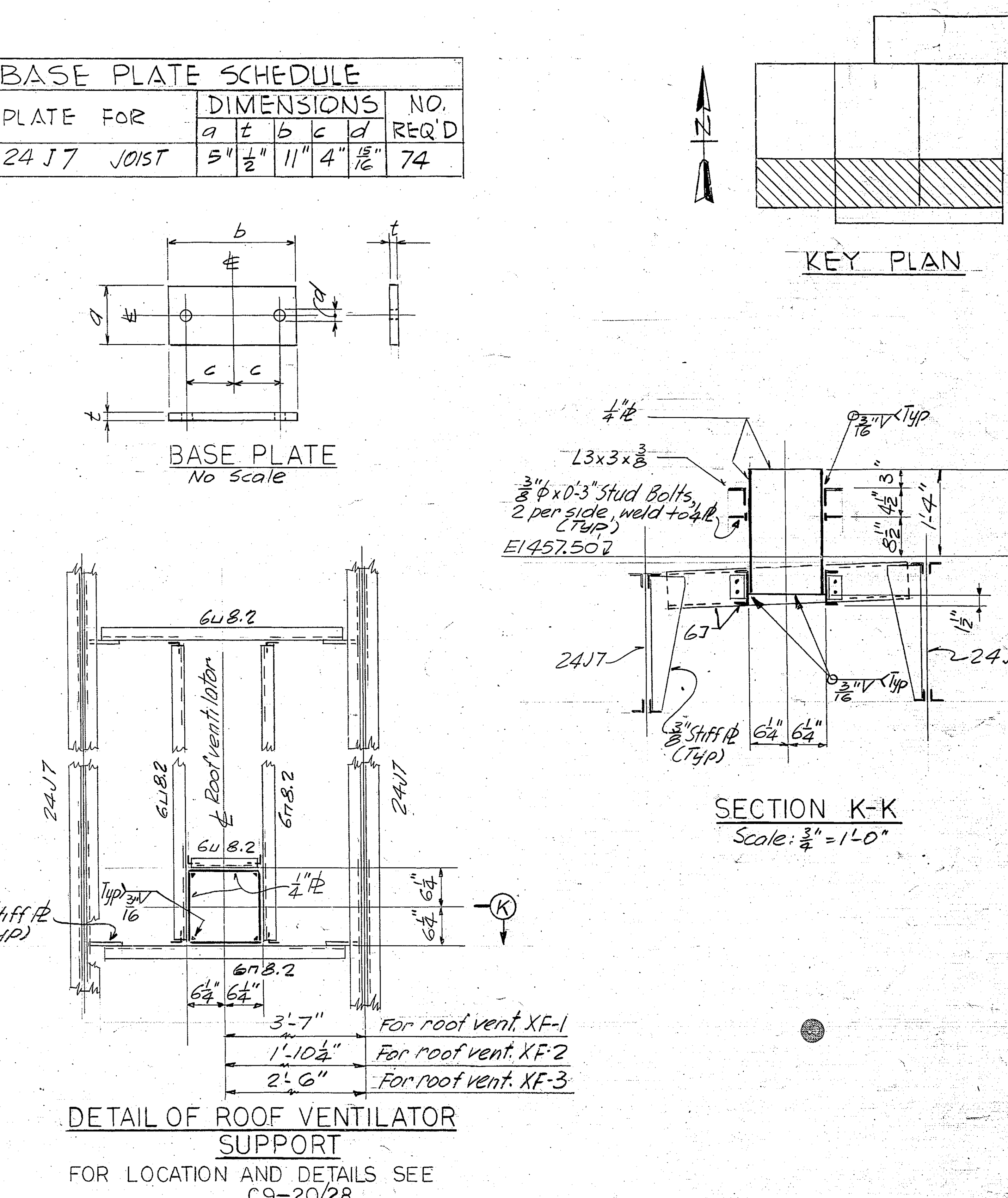
INVITATION NO. DACW56-67-B-0014

CONTRACT NO. 67-C-0069



BASE PLATE SCHEDULE

BASE PLATE FOR	DIMENSIONS a	t	b	c	d	NO. REQ'D
24 J 7 JOIST	5"	1/2"	11"	4"	13/16"	74



NOTES:

1. 13/16"φ Holes for 3/4"φ Bolts, except as noted.
2. Field connections shall be bolted with A 307 bolts.
3. Reference elevation (W.P.) for roof ridge L.P. is at the ℄ of contraction joint at the face of upstream wall.
4. Bearing plates shall be grouted in place after dead load is applied.
5. Asterisks (✱) indicate dimensions that will be furnished after purchase of joists and insulation.
6. ~~Payment, roof framing item No. 22~~
7. ~~Payment, building construction, item No. 22~~

GRAPHIC SCALES

SCALE: 1/8 INCH = 1 FOOT

SCALE: 1/4 INCH = 1 FOOT

SCALE: 1 1/2 INCHES = 1 FOOT

RECORD (AS BUILT) DRAWING

KEY	DATE	REVISION (INDICATED BY △)	CHKD.
1	9-19-66	Section H-H and notes revised, Amendment No. 3	

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NEW YORK, NEW YORK

U.S. ARMY ENGINEER DISTRICT, TULSA
CORPS OF ENGINEERS
TULSA, OKLAHOMA

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APPROVED: CHIEF, ENGRG. DIV.—TULSA DIST., FOR THE DISTRICT ENGINEER

RED RIVER BASIN — MOUNTAIN FORK RIVER, OKLAHOMA

BROKEN BOW DAM
POWERHOUSE STRUCTURAL

SERVICE BAY ROOF FRAMING

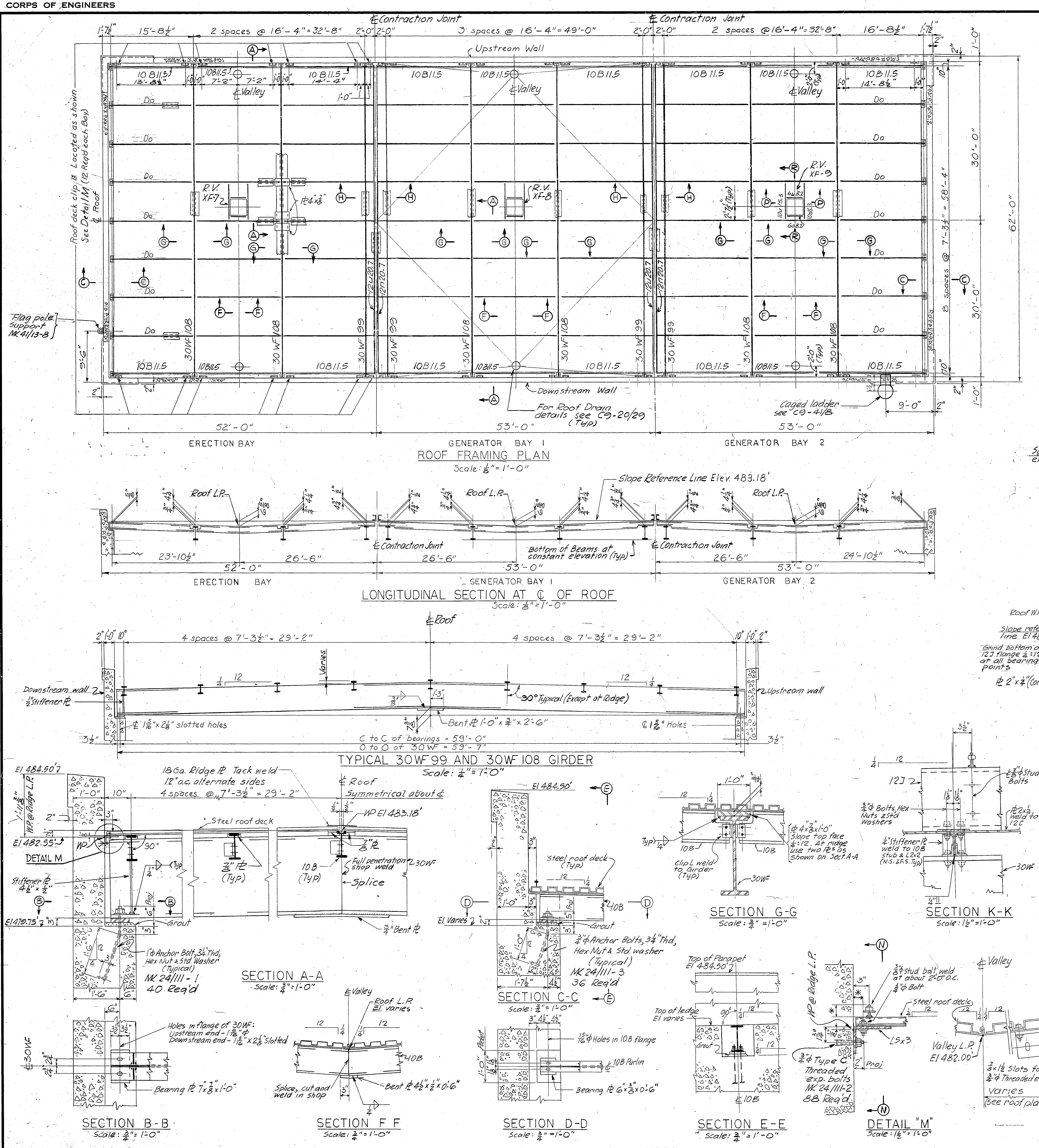
JULY 1966

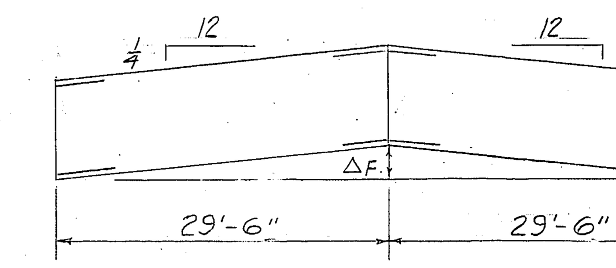
SCALE: As shown

DRAWING NUMBER 172-C9-24/110.1

INVITATION NO. DACW56-67-B-0014

CONTRACT NO. 67-C-0069

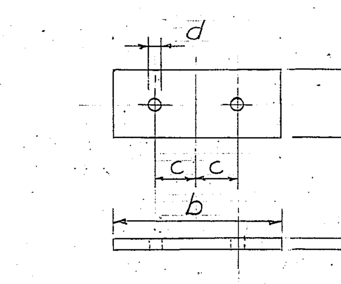




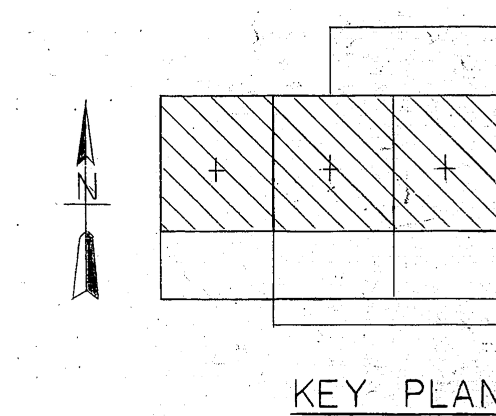
FABRICATED CAMBER		Δ_F 7 3/8"
DEFLECTED UNDER D.L.	30WF99	1/2"
	30WF108	7/8"
DEFLECTED UNDER D.L.+L.L.	30WF99	1 3/8"
	30WF108	2 3/8"
PERMANENT CAMBER	30WF99	Δ_F - $\Delta_{D.L.}$ 6 7/8"
	30WF108	6 1/2"

CAMBER DIAGRAM - 30WF99 & 108
No Scale

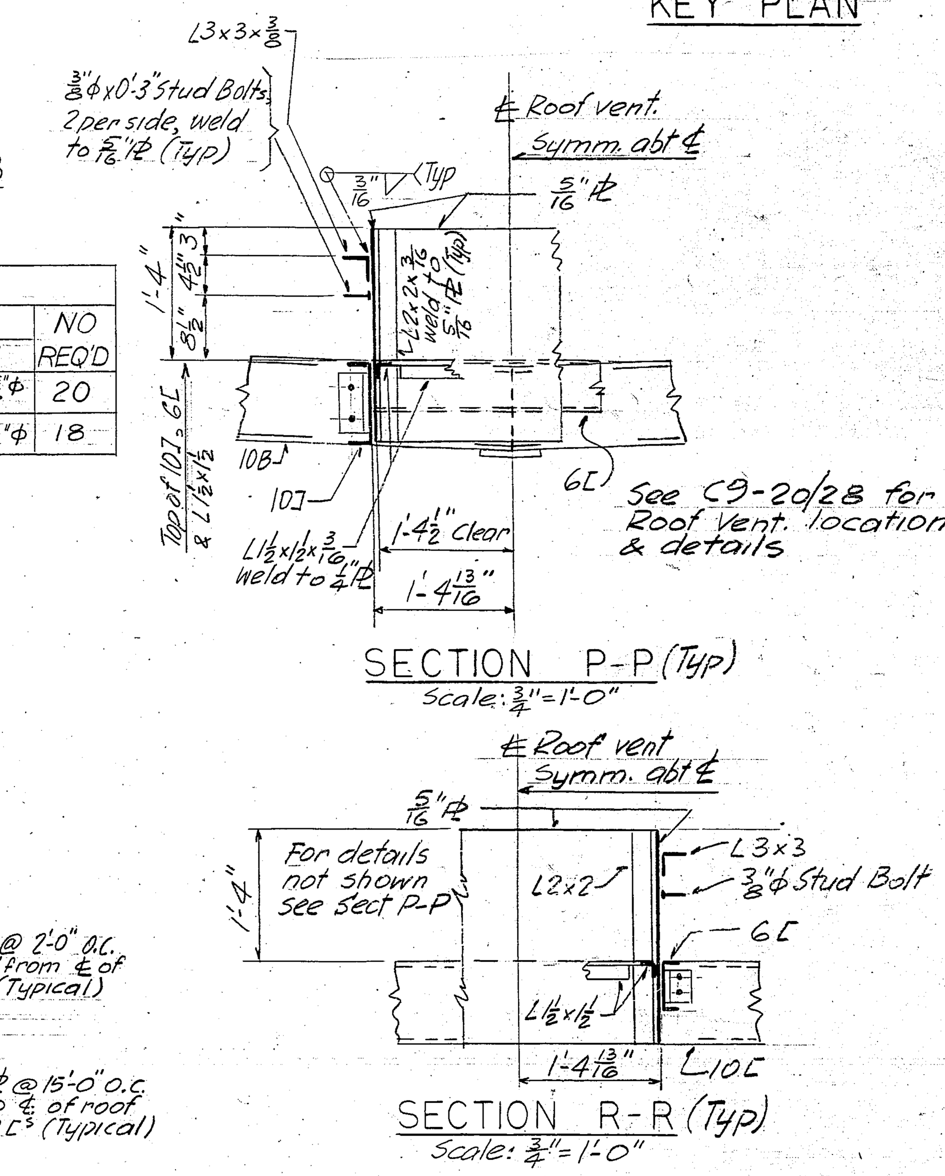
BEARING PLATE SCHEDULE						
BEARING ℘ FOR	DIMENSIONS					NO REQ'D
	a	t	b	c	d	
30WF99 & 108	7"	7/8"	1'-0"	2 3/4"	1 3/16"φ	20
10B11.5 Purlins	6"	3/8"	6"	1 3/8"	11/16"φ	18

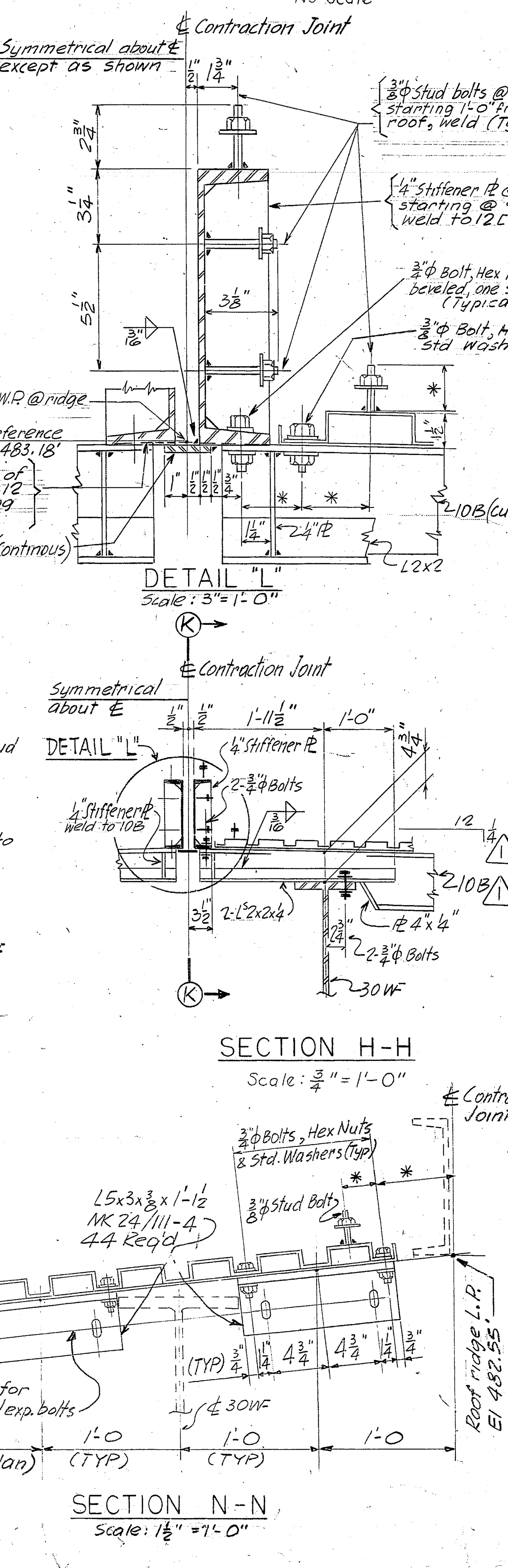


BEARING PLATE
No Scale



KEY PLAN





GRAPHIC SCALES
SCALE: 1/8 INCH = 1 FOOT
SCALE: 1/4 INCH = 1 FOOT
SCALE: 3/4 INCH = 1 FOOT
SCALE: 1 1/2 INCH = 1 FOOT
SCALE: 3 INCH = 1 FOOT

NOTES:

1. 13/16"φ Holes for 3/4"φ Bolts, except as noted.
2. Field connections shall be bolted.
3. Reference elevation (W.P.) for roof ridge L.P. is at the ¢ of contraction joint at the face of up/downstream wall.
4. Bearing plates shall be grouted in place after dead load is applied.
5. Asterisks (*) indicate dimensions that will be furnished after purchase of insulation.
6. Payment, roof framing item No. 22A.
7. Payment, building construction, item No. 22.

RECORD (AS BUILT) DRAWING

KEY	DATE	REVISION (INDICATED BY Δ)	CHK'D
1	11-7-66	Amend. No. 3 (9-19-66)	

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U.S. ARMY ENGINEER DISTRICT, TULSA, CORPS OF ENGINEERS, TULSA, OKLAHOMA
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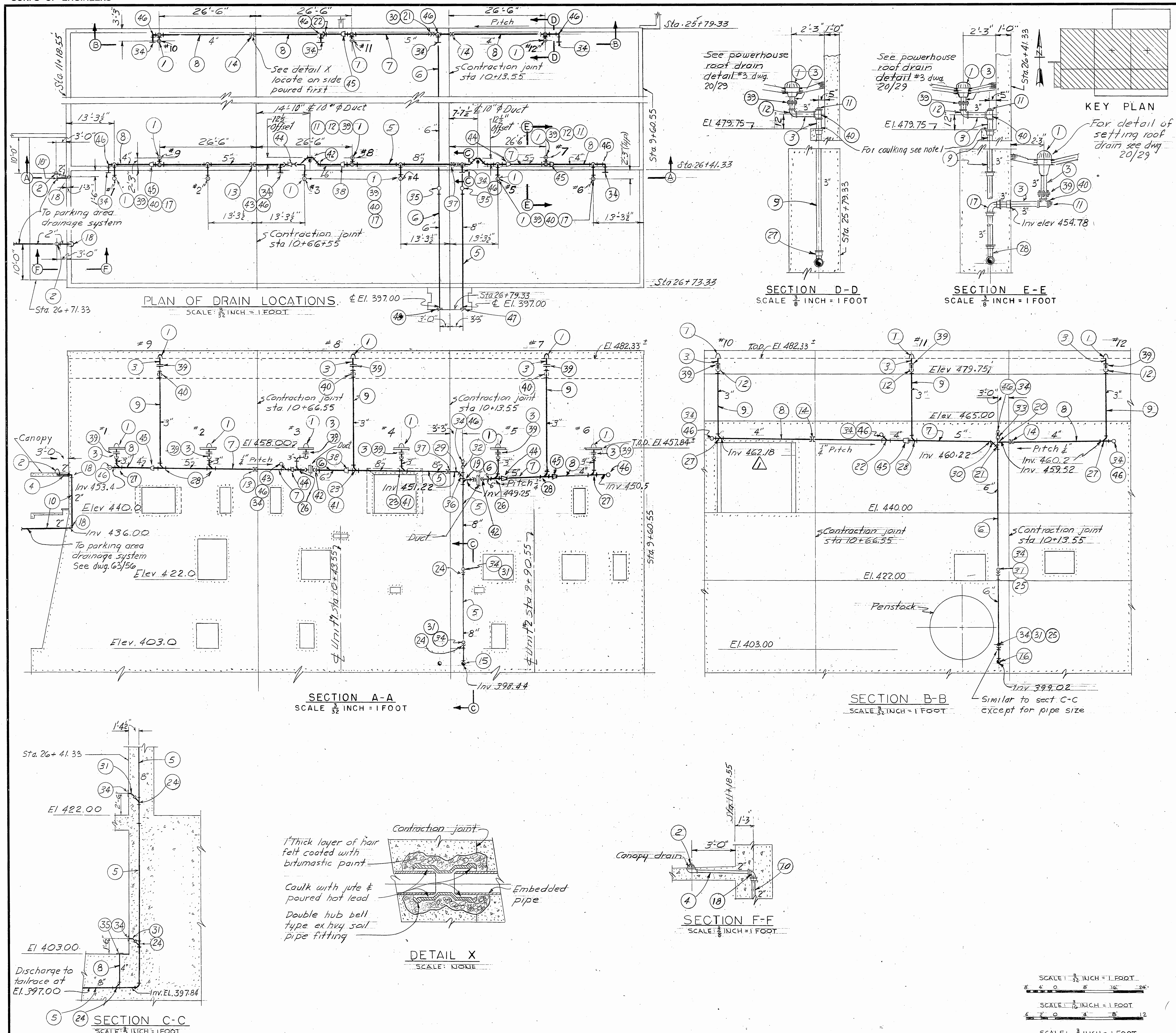
APPROVED: CHIEF, ENGRG DIV-TULSA DIST., FOR THE DISTRICT ENGINEER
INVITATION NO. DACW 56-67-B-0014

RED RIVER BASIN — MOUNTAIN FORK RIVER, OKLAHOMA
BROKEN BOW DAM
POWERHOUSE STRUCTURAL
GENERATOR BAY & ERECTION BAY
ROOF FRAMING
JULY 1966
SCALE: As shown
DRAWING NUMBER 172-C9-24/111.1

CONTRACT NO. 67-C-0069



LIST OF MATERIAL

PART NO	MATL	QUAN	DESCRIPTION	BID ITEM
63/61-1	C I	12	3" Roof Drain Thd Outlet	43
2	"	2	2" Canopy Drain Cornice 90° Thd. Outlet Drain	43
3	Steel	85 L F	3" Galv Steel Pipe Schedule 40	43
4	"	10 L F	2" " " " " "	43
5	C I	20 LSH 6 LDH	8" Soil Pipe Bell & Spigot Extra Heavy	43
6	"	32 LSH 5 LDH	6" " " " " " "	43
7	"	10 LSH 4 LDH	5" " " " " " "	43
8	"	17 LSH 6 LDH	4 " " " " " "	43
9	"	24 SH 10 DH	3" " " " " " "	43
10	"	10 SH 2 DH	2" " " " " " "	43
11	"	12	3" Std Square Head Pipe Plug Screwed	43
12	"	6	3"-90° Drainage Elbow Galv Screwed	43
13	"	1	5" Double Hub Extra Heavy SPF	43
14	"	2	4" " " " " "	43
15	"	1	8"-(90°) Sanitary Sweep Extra Heavy SPF	43
16	"	1	6" " " " " " "	43
17	"	6	3" " " " " " "	43
18	"	4	2" " " " " " "	43
19	"	1	8"x6" Single Sanitary T-branch Ex Hvy SPF	43
20	"	1	6"x4" " " " " "	43
21	"	1	6"x5" Single Y-branch Extra Heavy SPF	43
22	"	1	4" " " " " "	43
23	"	2	8"x3" Single Y-branch Ex. Hvy. S.P.F.	43
24	"	3	8"x4" Single Y-branch ExHvy S.P.F	43
25	"	20	6"x4" " " " "	43
26	"	2	6"x3" Single Comb Y & 1/8 Bend Ex Hvy SPF	43
27	"	4	4x3" " " " " " "	43
28	"	2	5x3" " " " " " "	43
29	"	1	8"-1/8 Bend Extra Heavy SPF	43
30	"	1	5" " " " " "	43
31	"	6	4" " " " " "	43
32	"	1	8"x4" Reducer Extra Heavy SPF	43
33	"	1	6"x4" " " " "	43
34	"	12	4" Wall Cleanout With Round Access Cover	43
35	"	2	4" Floor Level CO with Anchorage Lugs	43
36	"	1	8" Single Y-Branch Extra Heavy SPF	43
37	"	1	8" Double Hub Extra Heavy SPF	43
38	"	1	8"x6" Reducer Extra Heavy SPF	43
39	C.I.	13	3" Scrwd Flgd. Union	43
40	C.I.	12	3"-90° Galv. Scrd Drainage Single T.Y. Branch	43
41	C.I.	2	3"-1/8 Bend Extra Hvy S.P.F.	43
42	C.I.	8	6"-1/8 Bend Extra Hvy. S.P.F	43
43	C.I.	1	5"x4" Single Y Branch Ex.Hvy. S.P.F.	43
44	C.I.	2	6"x5" Reducer Ex.Hvy S.P.F	43
45	C.I.	3	5"x4" " " " " "	43
46	C.I.	8	4" Sanitary Long Sweep Ex.Hvy S.P.F	43
47	C.I.	1	8" Backwater Valve	43
48	C.I.	1	6" " " " " " " " " "	43
49	Lead	1120 Lbs	Caulking Lead for Soil Pipe & Fittings	43
50	Oakum	120 Lbs	" " Oakum " " " " " "	43

REFERENCE DRAWINGS:

172-C9-63/6 - Drainage & Unwatering Sys. Schematic Diagram
172-C9-20/28 - Roof Plan & Details
172-C9-20/29 - Roof & Flashing Details SH #1

NOTES:

1. For method of caulking steel pipe in C.I. bell see detail "X" dwg 63/53

RECORD (AS BUILT) DRAWING

1	10-28-70	A.B.	Revised as constructed.	[illegible]	
KEY	DATE	CHANGE	REVISION (INDICATED BY Δ)	APPR.	

EBASCO SERVICES INCORPORATED
NEW YORK, NEW YORK

U.S. ARMY ENGINEER DISTRICT, TULSA
CORPS OF ENGINEERS
TULSA, OKLAHOMA

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RECOMMENDED: [signature]
CHIEF, ENGINEERING DIVISION — SW DIV.
APPROVED: [signature]
CHIEF, ENGRG. DIV.-TULSA DIST. FOR THE DISTRICT ENGINEER

RED RIVER BASIN — MOUNTAIN FORK RIVER, OKLAHOMA

BROKEN BOW DAM
POWERHOUSE MECHANICAL

DRAINAGE - ROOF DRAIN PIPING

JULY 1966

SCALE: AS NOTED

DRAWING NUMBER 172 - C9 - 63/61.1

INVITATION NO. DACW56-67-B-0014

CONTRACT NO. 67-C-0069

APPENDIX A2 -

2001 AS-BUILT DRAWINGS

US Army Corps
of Engineers
Tulsa District

Broken Bow Power House
Repair Built-up Roof

Red River Basin
Mountain Fork River, Oklahoma

RECORD DRAWING "AS-BUILT"

IFB NO. DACA56-98-B-0007
CONTRACT NO. DACA56-98-C-0025
DRAWING CODE: CC720-C39-0/1

0/1

LIST OF ABBREVIATIONS

ABV	ABOVE
ACI	AMERICAN CONCRETE INSTITUTE
ADD'L	ADDITIONAL
ADD.	ADDITION OR ADDITIVE
AFF	ABOVE FINISH FLOOR
AISC	AMERICAN INSTITUTE OF STEEL CONSTRUCTION
ALT	ALTERNATE
ALUM	ALUMINUM
ANCH	ANCHOR
APA	AMERICAN PLYWOOD ASSOCIATION
ARCH'L	ARCHITECTURAL
ARCH.	ARCHITECTURAL OR ARCHITECT
ASM	SMACNA ARCHITECTURAL SHEET METAL MANUAL, 1979 EDITION
ASTM	AMERICAN SOCIETY FOR TESTING OF MATERIALS
A.B.	ANCHOR BOLT
BD	BOARD
BLDG	BUILDING
BLK'G	BLOCKING
BM	BEAM
BOTT	BOTTOM
BRNG	BEARING
BRN'G	BEARING
BUR	BUILT-UP ROOF(ING)
CAB'T	CABINET
CCT	CIRCUIT
CER	CERAMIC
CFM	CUBIC FEET PER MINUTE
CI	CAST IRON
CJ	CONTROL JOINT
CLG	CEILING
CLO	CLOSET
CLR	CLEAR
CL	CENTERLINE
℄	CENTERLINE
CMU	CONCRETE MASONRY UNIT
COL	COLUMN
COMPR	COMPRESSIBLE
COMP	COMPOSITION
CONC	CONCRETE
CONF	CONFER OR CONFERENCE
CONNX	CONNECTION
CONTR	CONTRACTION OR CONTRACTOR
CONT	CONTINUE OR CONTINUOUS
CRC	COLD ROLLED CHANNEL
CR	COLD ROLLED
CT	CERAMIC TILE
CW	COLD WATER
C	CENTERLINE
C. TO C.	CENTER TO CENTER
C.I.	CAST IRON
C.J.	CONTROL JOINT
C.M.U.	CONCRETE MASONRY UNIT
C.R.	COLD ROLLED
C.T.	CERAMIC TILE
DC	DIRECT CURRENT
DEMO	DEMOLITION
DEMOL	DEMOLITION
DET	DETAIL
DHW	DOMESTIC HOT WATER
DIAG	DIAGONAL
DIAG'S	DIAGONALS
DN	DOWN
DR	DOOR
DTL	DETAIL
DWD	DRINKING WATER DISPENSER
DWGS	DRAWINGS
DWG	DRAWING
DWL	DOWEL OR DRYWALL
D.C.	DIRECT CURRENT
D.W.D	DRINKING WATER DISPENSER
EA	EACH
EF	EACH FACE OR EXHAUST FAN
EIFS	EXTERIOR INSULATION AND FINISH SYSTEM
ELEC	ELECTRICAL
ELEV	ELEVATION
EL	ELEVATION
EMER	EMERGENCY
EQUIP	EQUIPMENT
EQ	EQUAL
EW	EACH WAY
EW-EF	EACH WAY - EACH FACE
EXIST	EXISTING
EXP	EXPANSION
EXTEN	EXTENSION
EXT	EXTERIOR
E	EAST
FEC	FIRE EXTINGUISHER CABINET
FFE	FINISH FLOOR EXISTING
FIN FLR	FINISH FLOOR
FIN	FINISH
FLR	FLOOR
FRM'G	FRAMING
FTG	FOOTING
FT	FOOT (FEET)
F.E.C	FIRE EXTINGUISHER CABINET
GALV	GALVANIZED
GA	GAUGE (GAGE)
GB	GRAB BAR
GR BM	GRADE BEAM
GSU	GLAZED STRUCTURAL UNITS
GWB	GYPSUM WALL BOARD
GYP BD	GYPSUM BOARD
GYP	GYPSUM
HC	HANDICAPPED
HDWE	HARDWARE
HGHT	HEIGHT
HM	HOLLOW METAL
HND	HAND
HORIZ	HORIZONTAL
HOR	HORIZONTAL
HP	HORSE POWER
HVAC	HEATING VENTILATION AND AIR CONDITIONING
H	HEIGHT
H.C.	HANDICAPPED
H.M.	HOLLOW METAL
INFO	INFORMATION
INSUL	INSULATION
INTERM	INTERMEDIATE
INTER	INTERMEDIATE
INT	INTERIOR
IN.	INCH
JNT	JOINT
JST	JOIST
JT	JOINT
LAV	LAVATORY
LGHT	LIGHT
LH	LEFT HAND
LLH	LONG LEG HORIZONTAL
LLV	LONG LEG VERTICAL
LOC	LOCATE OR LOCATION
LTS	LIGHTS
LT	LIGHT
L	LENGTH OR ANGLE DESIGNATION
MANUF	MANUFACTURER
MAT'L	MATERIAL
MAX	MAXIMUM
MBR	MEMBER
MECH	MECHANICAL
MFGR	MANUFACTURER
MFG	MANUFACTURED OR MANUFACTURER
MG	MIRROR GLASS
MIN	MINIMUM
MJA	SEALANT JOINT SHAPE DETAIL
MPH	MILES PER HOUR
MTL	MATERIAL OR METAL
NIC	NOT IN CONTRACT
NO.	NUMBER
NOM	NOMINAL
NTS	NOT TO SCALE
N	NORTH
N.I.C.	NOT IN CONTRACT
OA	OUTSIDE AIR
OC	ON CENTER
OPP	OPPOSITE
OSL	OUTSIDE LEG
OZ	OUNCE(S)
O.A.	OUTSIDE AIR
O.C.	ON CENTER
PLF	POUNDS PER LINEAL FOOT
PL	PLATE OR PROPERTY LINE
⅊	PLATE OR PROPERTY LINE
PROV	PROVIDE
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PTD	PAPER TOWEL DISPENSER OR PAINTED
PT	POINT
QTY	QUANTITY
R	RADIUS
RAD	RADIUS
RCP	REFLECTED CEILING PLAN
REF	REFER
REINFORC	REINFORCING OR REINFORCE
REINF	REINFORCING OR REINFORCE
REM	REMARK OR REMOVE
REQ'D	REQUIRED
REV	REVERSE OR REVISED OR REVISION
RE	REFER
RH	RIGHT HAND
RM	ROOM
RO	ROUGH OPENING
RPM	REVOLUTIONS PER MINUTE
RTU	ROOF TOP UNIT
RU	ROOFTOP UNIT
R.O.	ROUGH OPENING
SCHED	SCHEDULE
SCH	SCHEDULE
SCR	SHOWER CURTAIN ROD
SCW	SOLID CORE WOOD
SC	SCALE OR SHOWER CURTAIN
SD	SOAP DISPENSER
SECT	SECTION
SF	SQUARE FOOT (FEET)
SHT	SHEET
SIM	SIMILAR
SJ	SAWED JOINT
SMACNA	SHEET METAL AND AIR CONDITIONING CONTRACTORS NATIONAL ASSOCIATION
SMLD	SHELF, METAL, LIGHT DUTY
SPC	SPACE
SPEC	SPECIFICATION
STD	STANDARD
STL	STEEL
STOR	STORAGE
STO	STORAGE
SUPT	SUPPORT
S	SOUTH
S.C.W.	SOLID CORE WOOD
TEMP	TEMPORARY OR TEMPERED OR TEMPERATURE
THRSHLD	THRESHOLD
THRU	THROUGH
THR'D	THREADED
TJB	SEALANT JOINT SHAPE DETAIL
TJD	SEALANT JOINT SHAPE DETAIL
TJE	SEALANT JOINT SHAPE DETAIL
TP	TOWEL PIN
TTD	TOILET TISSUE DISPENSER
TYP	TYPICAL
T	THICKNESS
T.C.	TOP OF CONCRETE
T.O.S.	TOP OF STEEL
T.O.W.	TOP OF WALL
UL	UNDERWRITERS LABORATORY
U.N.O.	UNLESS NOTED OTHERWISE
U/G	UNDERGROUND
VERT	VERTICAL
VEST	VESTIBULE
VIF	VERIFY IN FIELD
VTR	VENT THROUGH ROOF
V.C.T.	VINYL COMPOSITION TILE
V.I.F.	VERIFY IN FIELD
WC	WATER CLOSET
WDW	WINDOW
WD	WOOD
WWF	WELDED WIRE FABRIC
W	WIDTH OR WEST
W.C.	WATER CLOSET
W.P.	WORKING POINT
W/O	WITHOUT
W/	WITH
XJNT	EXPANSION JOINT
X	BY (AS IN 2 X 4)
×	BY (AS IN 2×4)
@	AT
#	POUND OR NUMBER
%	PERCENT
"	INCHES
½	ROUND OR DIAMETER
±	PLUS OR MINUS (APPROXIMATE)
&	AND
'	FEET
+/-	PLUS OR MINUS (APPROXIMATE)

SYMBOLS LEGEND

10 / G1 — REFERENCE TO DETAIL DRAWINGS: TOP NUMBER REFERS TO DETAIL NUMBER. BOTTOM NUMBER REFERS TO THE "SHEET REFERENCE NUMBER" WHERE DETAIL IS SHOWN.

A / G1 G2 — SECTION LETTER; SECTION TITLE; SCALE IN FEET 8' 4' 0 8' 16'; DRAWING SCALE; SHEET NUMBER WHERE SECTION IS DRAWN; SHEET NUMBER WHERE SECTION IS CUT

[5] DRAWING NOTE REFERENCE. THE REFERENCED NOTE IS FOUND ON THE SAME SHEET AS THE REFERENCE KEY. THESE NOTES ARE UNIQUE TO THE SHEET ON WHICH THEY OCCUR AND DO NOT NECESSARILY APPLY TO ANY OTHER SHEET

X — COLUMN GRID IDENTIFIER. NUMBER MATCHES ESTABLISHED COLUMN NUMBERING SYSTEM OF THE EXISTING BUILDING, IF APPLICABLE.

A / G1 G2 — SECTION REFERENCE KEY. TOP CHARACTER IS THE SECTION LETTER. THE BOTTOM RIGHT NUMBER IS THE SHEET NUMBER ON WHICH THE SECTION IS DRAWN. THE BOTTOM LEFT NUMBER IS THE SHEET NUMBER ON WHICH THE SECTION IS CUT. THE ARROW SHOWS THE DIRECTION OF VIEW.

CONNECT TO EXISTING

MATERIALS LEGEND

THE FOLLOWING LEGEND OF MATERIALS APPLIES TO ARCHITECTURAL SHEETS.

- MATERIAL TO BE REMOVED
- NEW CONCRETE MASONRY UNIT
- NEW CONCRETE (SECTION)
- FRAMING LUMBER
- FINISH WOOD (SOLID)
- PLYWOOD
- EARTH
- STEEL - LARGE SCALE
- STEEL - SMALL SCALE
- EXISTING CONSTRUCTION TO REMAIN
- MATERIAL HIDDEN

INDEX OF DRAWINGS

GENERAL NOTES

A. IN ACCOMPLISHING THE REQUIRED WORK, THE TOPS OF WALLS AND INTERIOR OF THE BUILDING WILL BE EXPOSED. PROTECTION SHALL BE ACCOMPLISHED IN ACCORDANCE WITH SPECIFICATION SECTION: "DEMOLITION".

B. COORDINATE ALL DEMOLITION ACTIVITIES WITH THE CORPS OF ENGINEERS PROJECT MANAGER AND THE CONTRACTING OFFICER.

C. ALL DIMENSIONS AND CONDITIONS SHOWN RELATIVE TO THE EXISTING CONSTRUCTION ARE APPROXIMATIONS BASED ON AS-BUILT DRAWINGS AND SITE INVESTIGATIONS. CONTRACTOR SHALL VERIFY INFORMATION SHOWN AND NOTIFY CONTRACTING OFFICER OF ANY DEVIATIONS PRIOR TO BEGINNING WORK.

RECORD DRAWING "AS-BUILT"

SYM	DESCRIPTION	DATE	APPR
1	REVISED AS CONSTRUCTED	6 AUG 01	

$ THINK VALUE ENGINEERING - IT SAVES MONEY $

TULSA CADD

U.S. ARMY ENGINEER DISTRICT
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED BY: J MOORE
DRAWN BY: B ENGLISH
CHECKED BY: J BELL
SUBMITTED BY: J BELL
CHIEF, INST SUP SEC

MOUNTAIN FORK RIVER, OKLAHOMA FY : 98

BROKEN BOW DAM
REPAIR POWER HOUSE BUILT-UP ROOF

GENERAL INFORMATION
INDEX OF DRAWINGS

SCALE: AS SHOWN
DATE: JANUARY 1998
DWG CODE: CC890-C72-0/2
SHEET REFERENCE NUMBER: 0/G1
INVITATION NO DACA56-98-B-0007
CONTRACT NO DACA56-98-C-0025
SHEET B82 OF B85

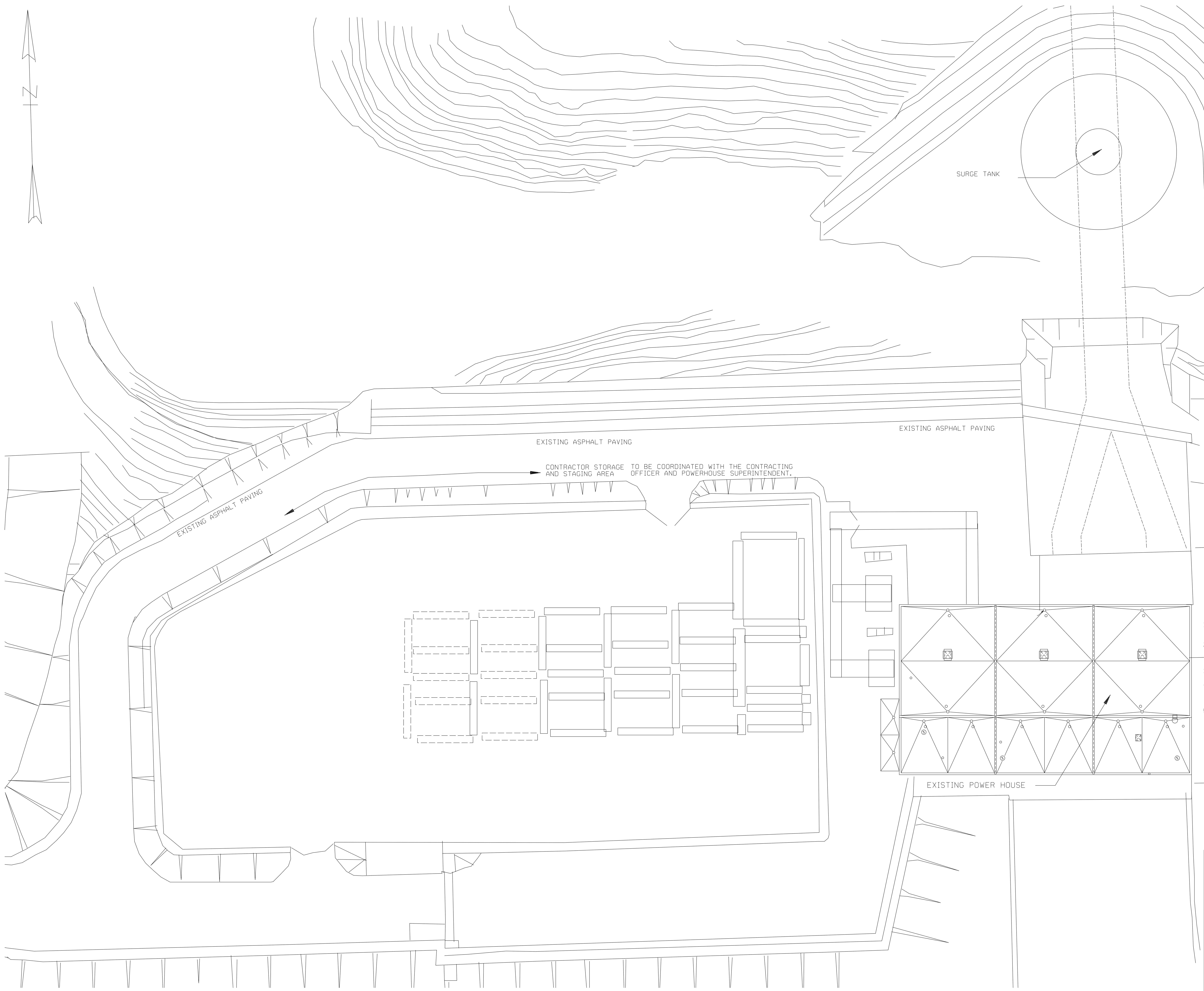
SIGNATURES AFFIXED BELOW INDICATE OFFICIAL RECOMMENDATION AND APPROVAL OF ALL DRAWINGS IN THIS SET INDEXED ON THIS SHEET

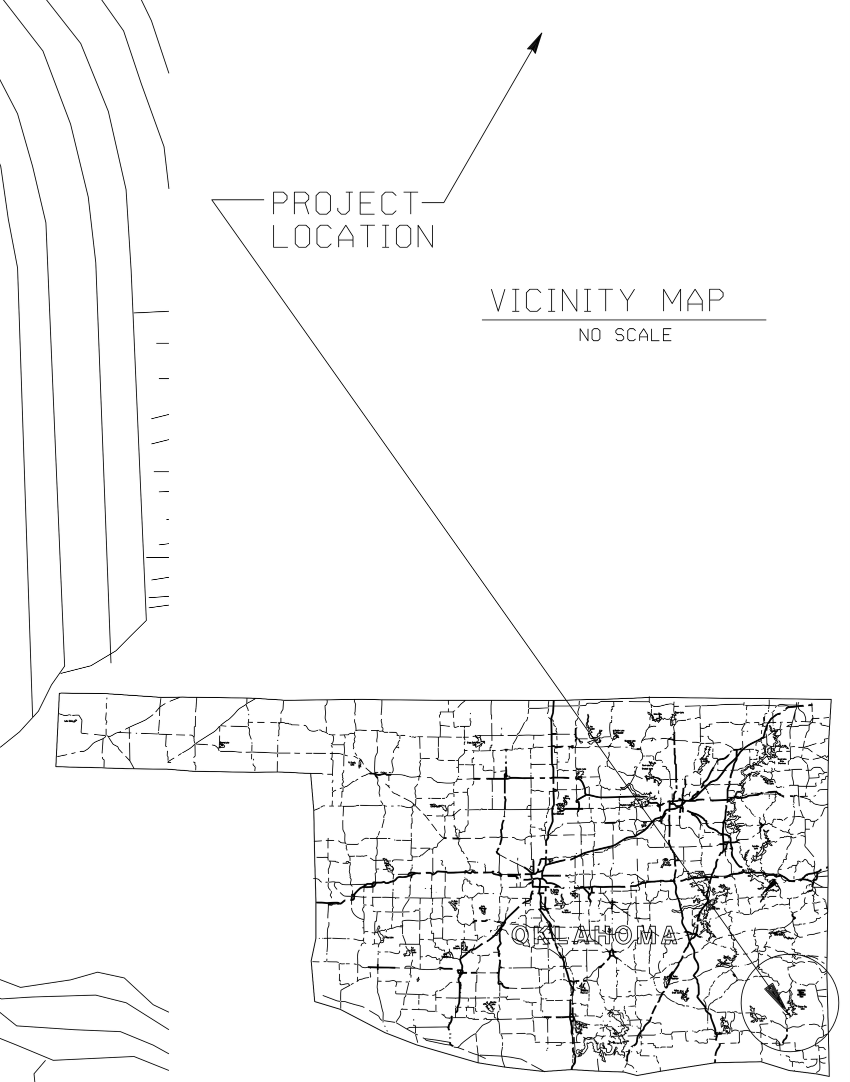
REVIEWED BY:

TITLE: CH. DESIGN BRANCH

APPROVED BY:

TITLE: CH. ENG & CONST DIV. DATE:





GENERAL NOTES

1. GENERAL NOTES SHOWN ON SHEET G1 APPLY TO ALL WORK UNDER THIS CONTRACT.
2. THESE DRAWINGS ARE BASED ON AS-BUILT DRAWINGS. CONDITIONS ACTUALLY FOUND MAY VARY SLIGHTLY FROM THE INFORMATION SHOWN.
3. PROTECT ALL FENCING FROM DAMAGE.
4. COORDINATE ALL PARKING, STORAGE AREAS AND PROCEDURES WITH THE CONTRACTING OFFICER OR POWERHOUSE SUPERINTENDANT.
5. CONTRACTOR SHALL RETURN STAGING AREA TO EXISTING PRIOR TO DEMOBILIZATION.

1/C1 BROKEN BOW POWER HOUSE SITE PLAN

0 10' 20' 40' 60'

NOTE: THIS DRAWING WAS SKETCHED FROM A SCANNED DRAWING AND SCALE, THOUGH CLOSE, IS ONLY APPROXIMATE.

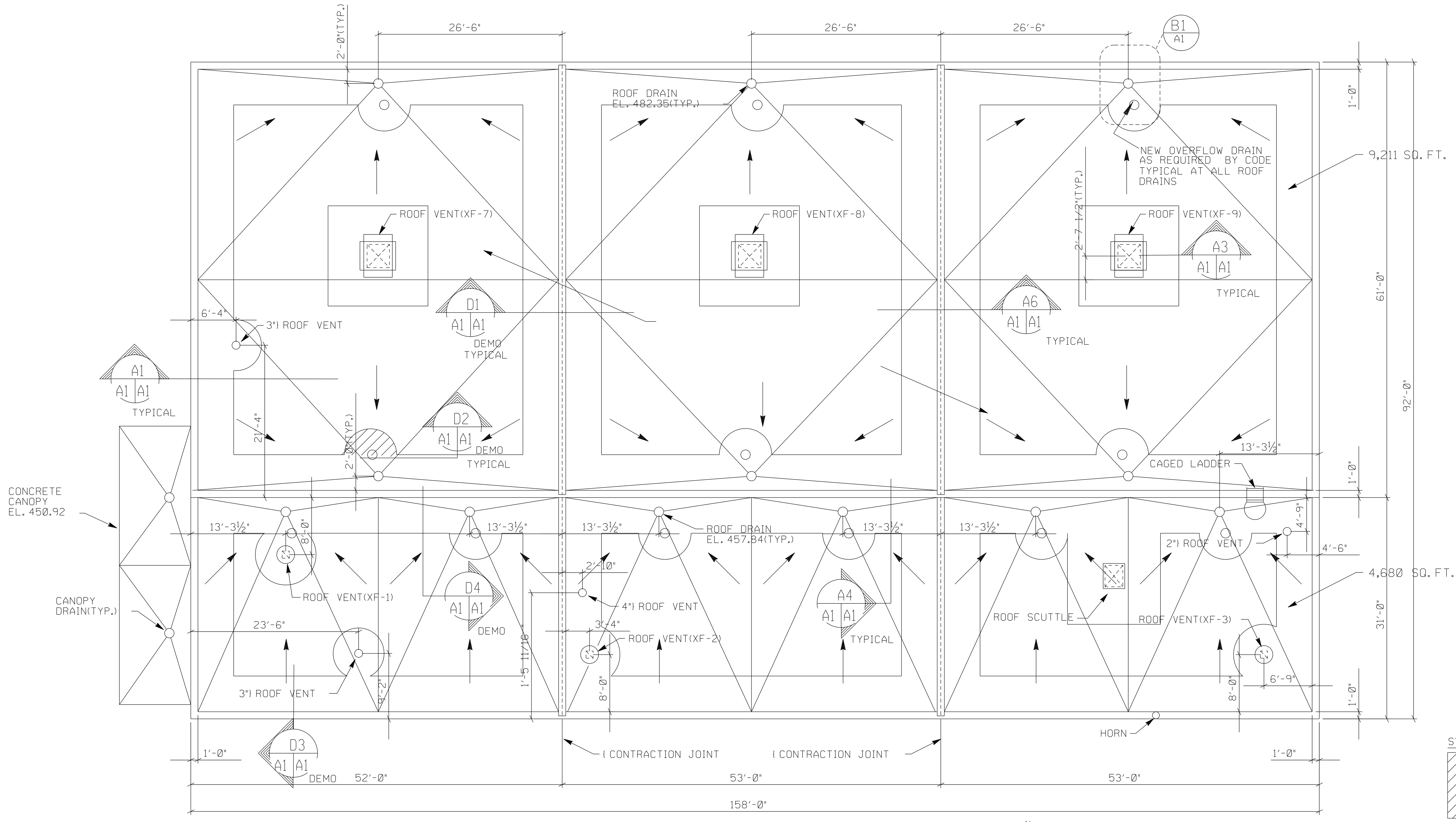
RECORD DRAWING "AS-BUILT"

SYM	DESCRIPTION	DATE	APPR
1	REVISED AS CONSTRUCTED	6 AUG 01	

$ THINK VALUE ENGINEERING - IT SAVES MONEY $

U. S. ARMY ENGINEER DISTRICT
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED BY: J MOORE	MOUNTAIN FORK RIVER, OKLAHOMA FY : 98		
DRAWN BY: B ENGLISH	BROKEN BOW DAM REPAIR POWER HOUSE BUILT-UP ROOF		
CHECKED BY: J BELL	PROJECT LOCATION & SITE MAP		
SUBMITTED BY: J BELL	SCALE: AS SHOWN	SHEET REFERENCE NUMBER	INVITATION NO DACA56-98-B-0007
CHIEF, INST SUP SEC	DATE: JANUARY 1998		CONTRACT NO DACA56-98-C- 0025
	DWG CODE: CC890-C72-1/1	1/C1	SHEET BB3 OF BB5



CONTRACTOR STAGING AREA

CONTRACTOR STAGING AREA WILL BE LOCATED ON PAVEMENT. THE ACTUAL LOCATION AROUND THE FACILITY WILL BE DETERMINED AT THE PRE-CONSTRUCTION CONFERENCE BY NEGOTIATIONS BETWEEN THE CONTRACTOR, THE CONTRACTING OFFICER AND THE PROJECT OFFICE MANAGER.

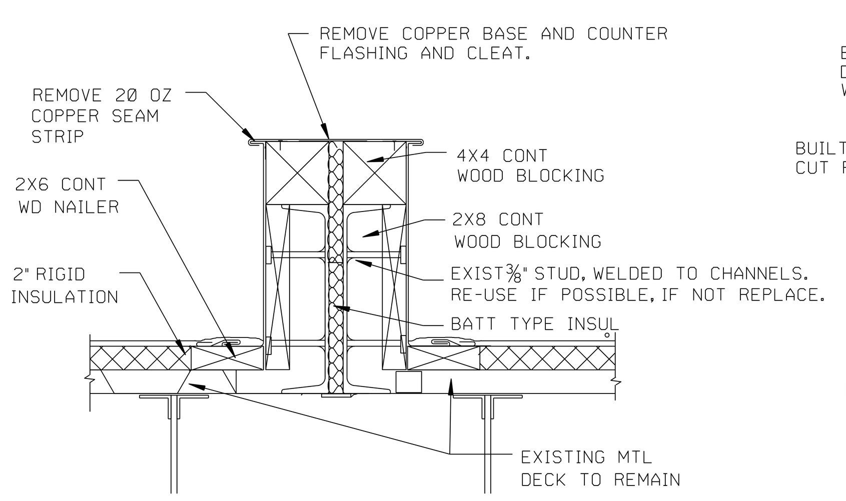
SYMBOL NOTE:

DID NOT REMOVE EXSTG. ROOF & INSUL'N AT SHADED AREAS.

ROOF RECOVER OVER EXISTING SMOOTH SURFACE BUILT-UP ROOF

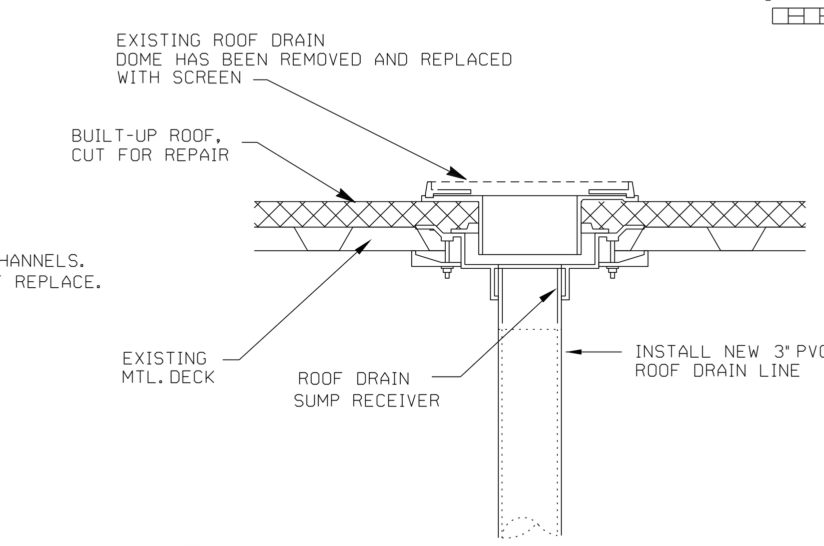
C6 / A1 BROKEN BOW POWERHOUSE ROOF DEMOLITION PLAN

SCALE IN FEET 8' 4' 0 8' 16'



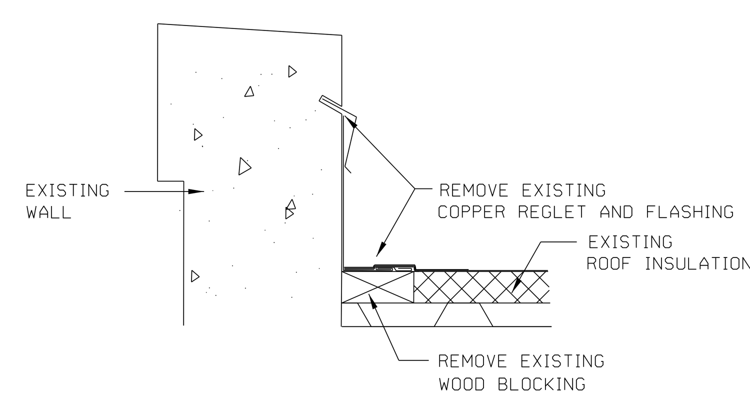
D1 / A1 EXISTING EXPANSION JOINT

NO SCALE



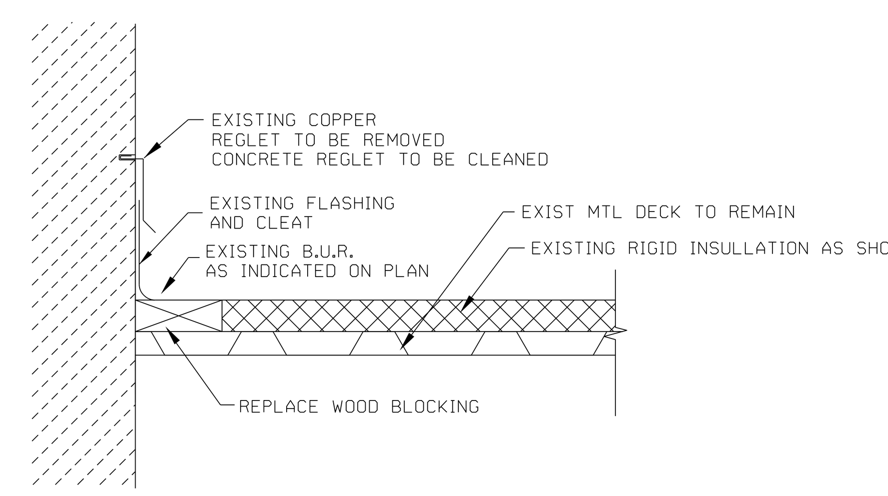
D2 / A1 EXISTING ROOF DRAIN DETAIL

NO SCALE



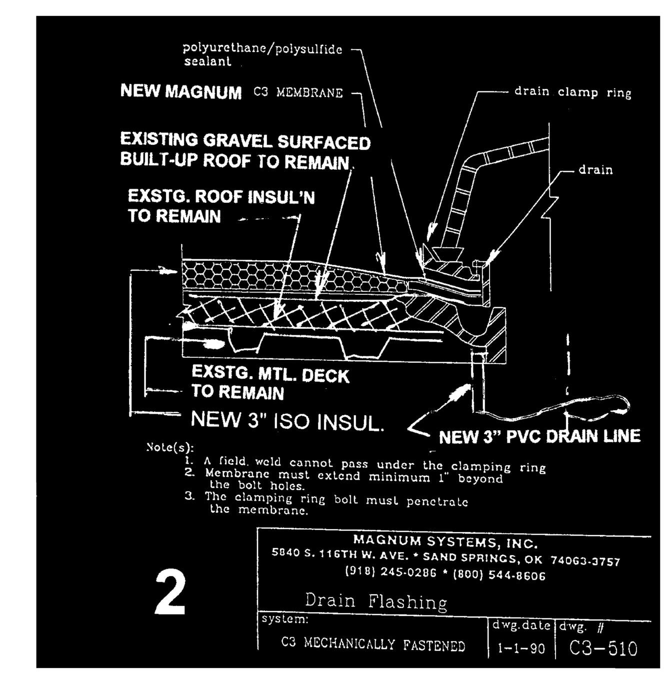
D3 / A1 EXISTING PARAPET DETAIL

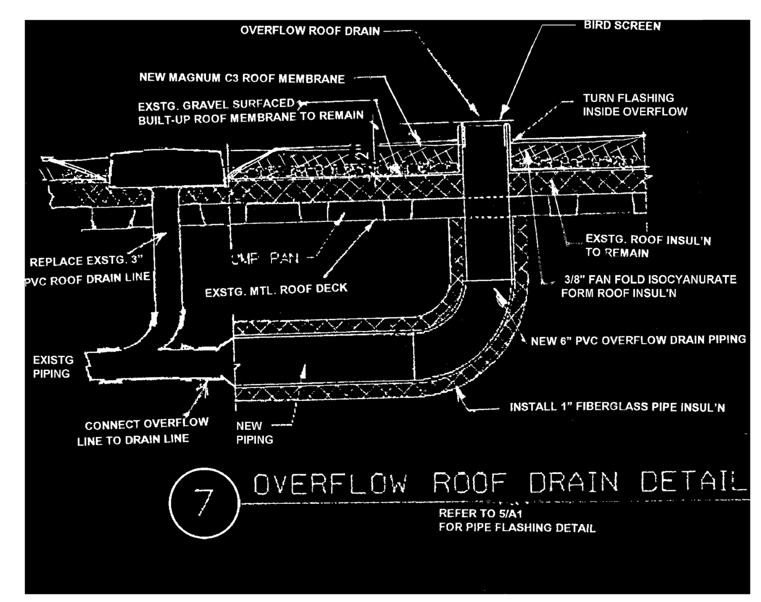
NO SCALE



D4 / A1 EXISTING FLASHING @ WALL

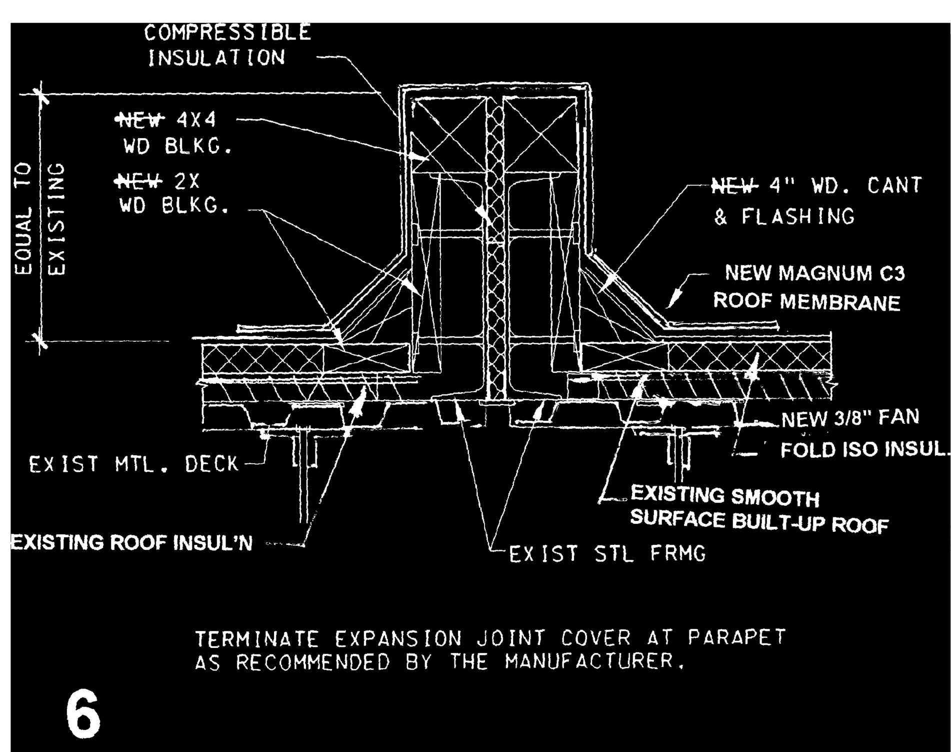
NO SCALE





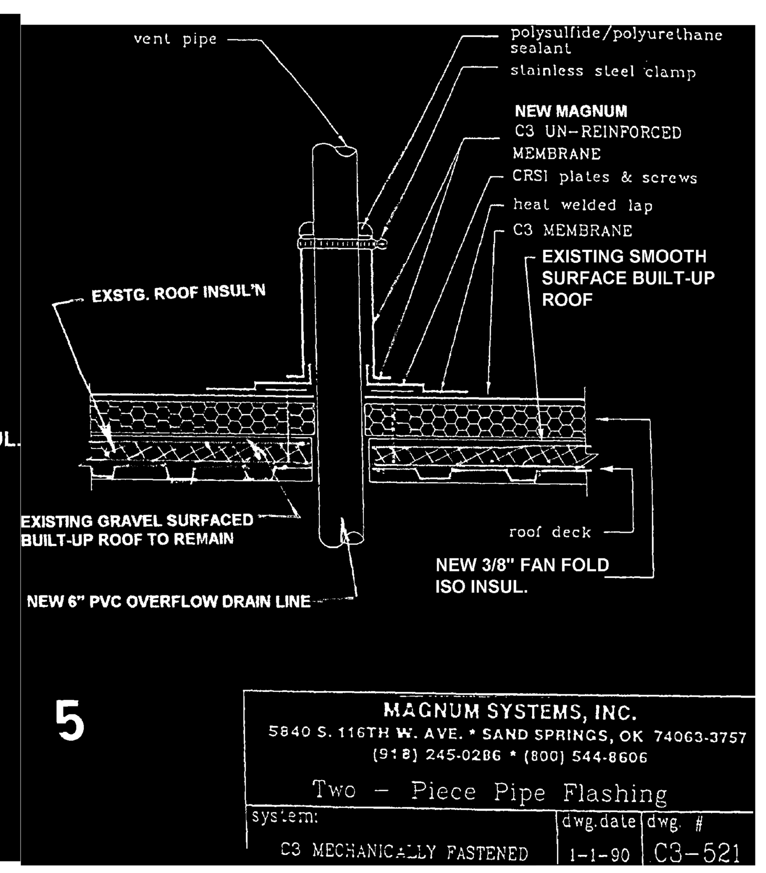
RECORD DRAWING "AS-BUILT"

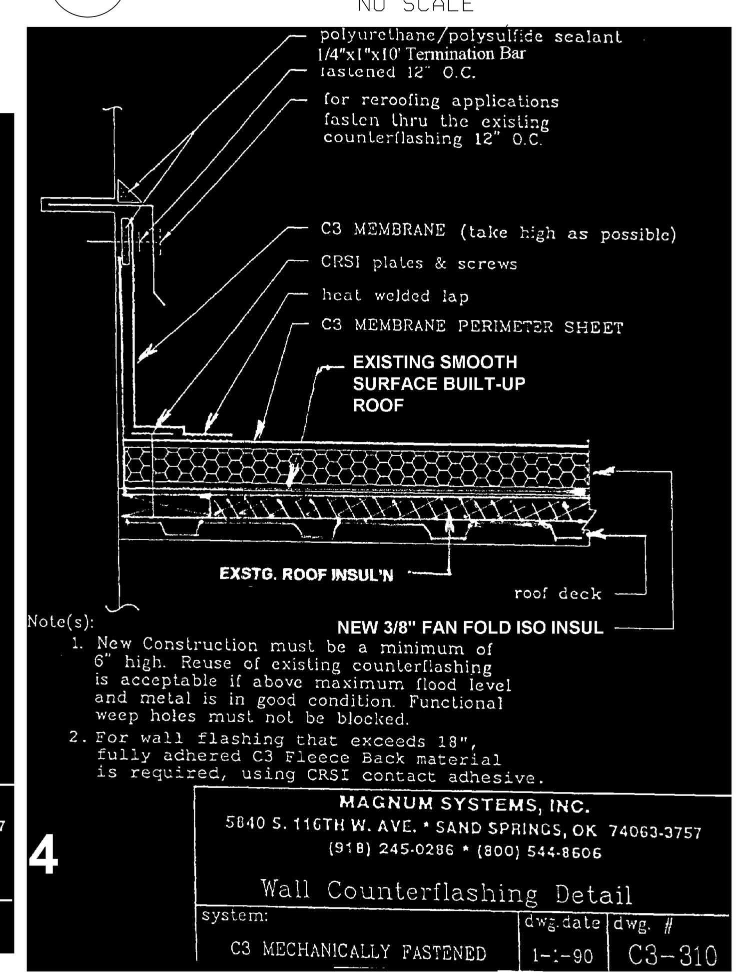
NOTES:
1. ALL SQUARE FOOTAGE CALCULATED FROM INSIDE PARAPET WALL.
2. TOTAL SQUARE FOOTAGE = 13,891.
3. INSULATION SHALL BE R-19 VALUE.

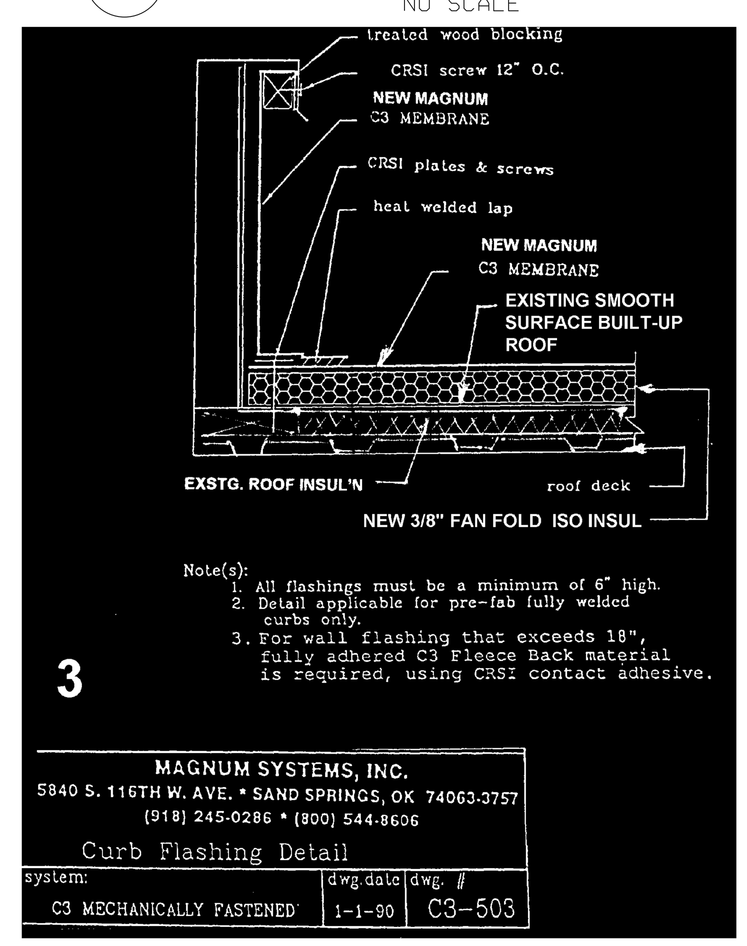


A6 / A1 NEW EXPANSION JOINT

NO SCALE







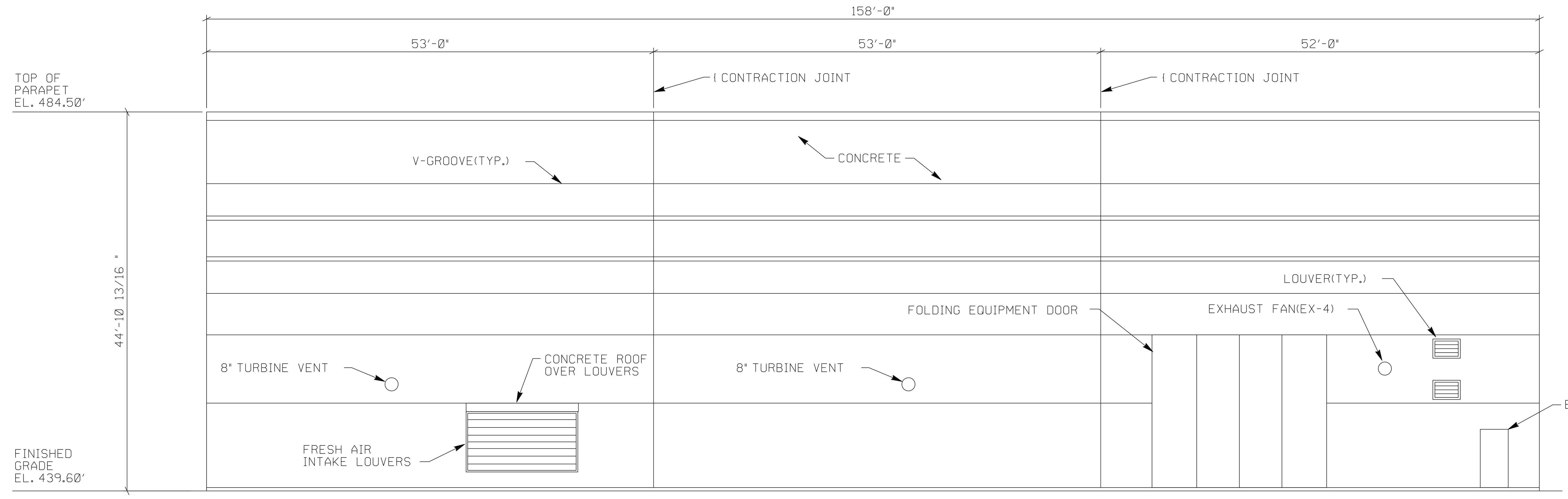
	REVISED AS CONSTRUCTED	15 AUG 01	
SYM	DESCRIPTION	DATE	APPR

$ THINK VALUE ENGINEERING - IT SAVES MONEY $

TULSA CADD

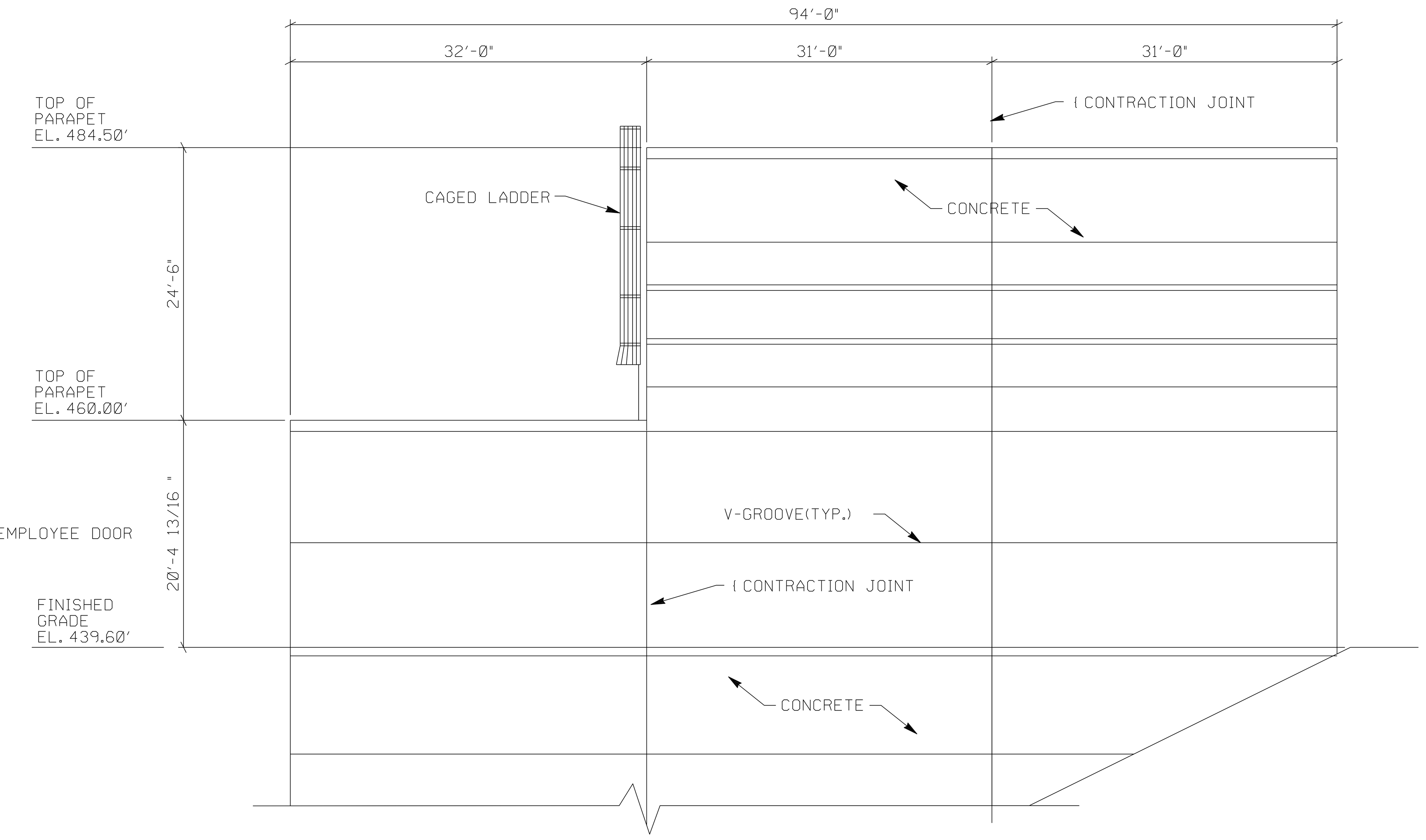
U. S. ARMY ENGINEER DISTRICT
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED BY: J MOORE	MOUNTAIN FORK RIVER, OKLAHOMA		FY : 98
DRAWN BY: B ENGLISH	BROKEN BOW DAM REPAIR POWER HOUSE BUILT-UP ROOF		
CHECKED BY: J BELL	ROOF CONSTRUCTION/DEMOLITION PLAN AND DETAILS		
SUBMITTED BY: J BELL	SCALE: AS SHOWN	SHEET REFERENCE NUMBER	INVITATION NO DACA56-98-B-0007
CHIEF, INST SUP SEC	DATE: JANUARY 1998	20/A1	CONTRACT NO DACA56-98C-0025
	DWG CODE: CC720-C39-20/1		SHEET 84 OF 85



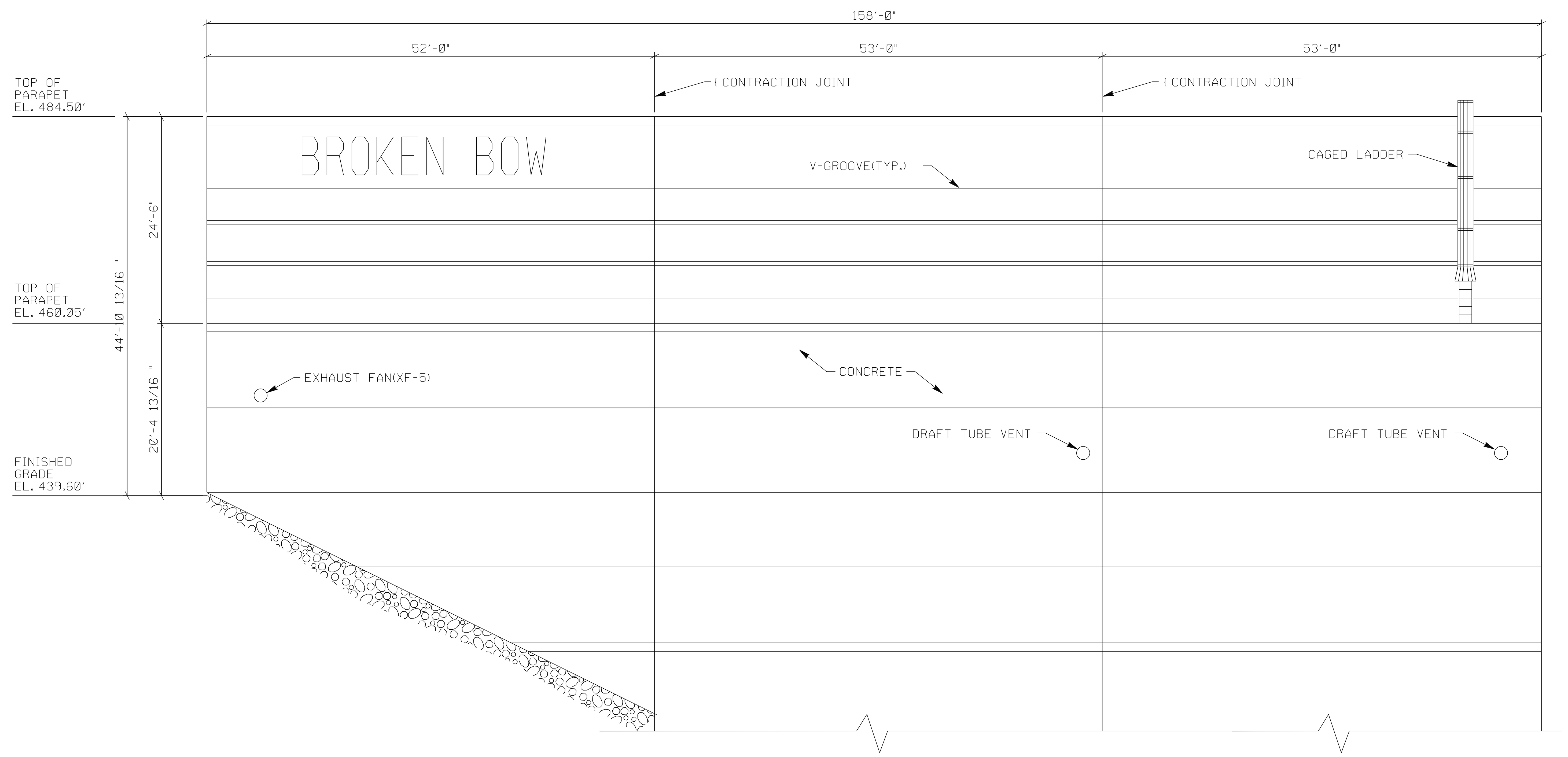
1 A2 BROKEN BOW POWERHOUSE NORTH ELEVATION(FACING DOWNSTREAM)

SCALE IN FEET
8' 4' 0 8' 16'



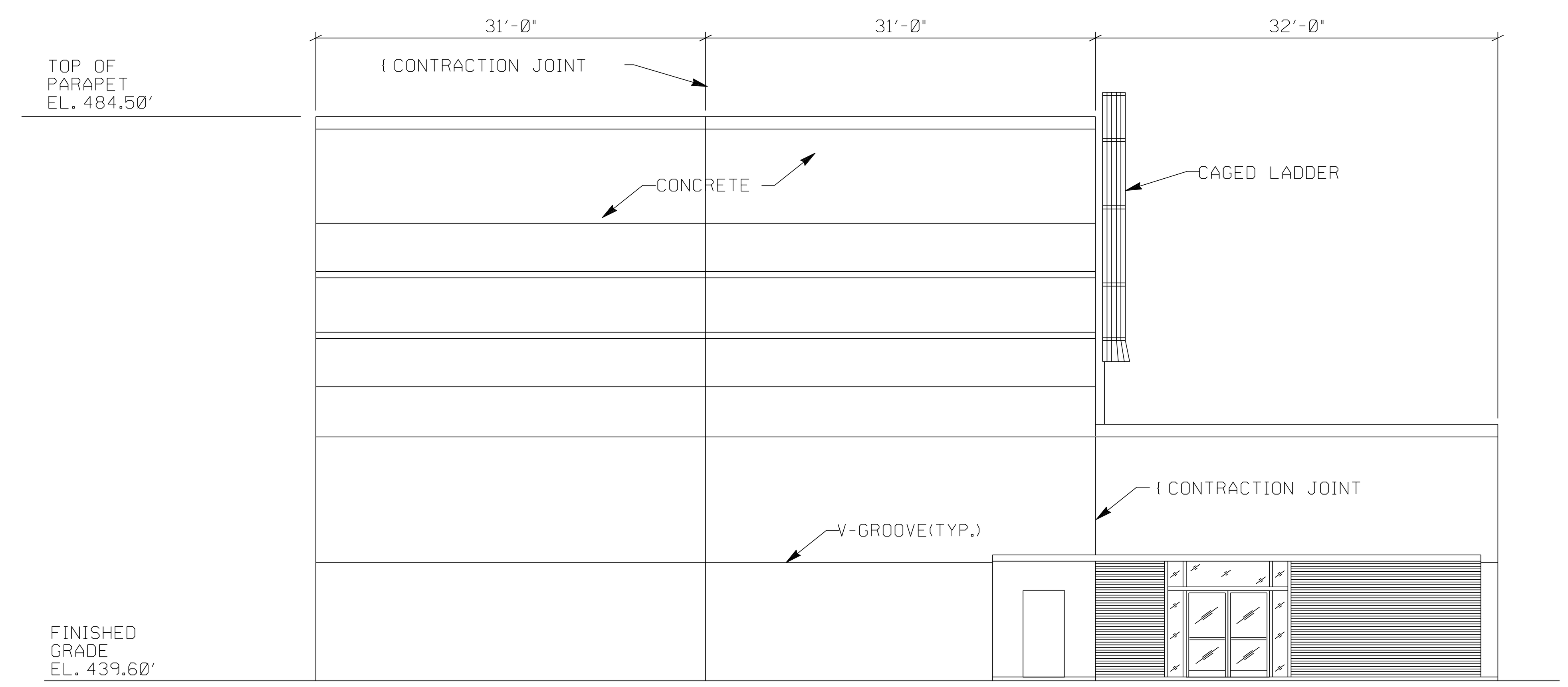
2 A2 BROKEN BOW POWERHOUSE EAST ELEVATION

SCALE IN FEET
8' 4' 0 8' 16'



3 A2 BROKEN BOW POWERHOUSE SOUTH ELEVATION(FACING UPSTREAM)

SCALE IN FEET
8' 4' 0 8' 16'



4 A2 BROKEN BOW POWERHOUSE WEST ELEVATION

SCALE IN FEET
8' 4' 0 8' 16'

RECORD DRAWING "AS-BUILT"

SYM	DESCRIPTION	DATE	APPR
1	REVISED AS CONSTRUCTED	6 AUG 01	

$ THINK VALUE ENGINEERING - IT SAVES MONEY $

TULSA CADD

U.S. ARMY ENGINEER DISTRICT
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED BY: J MOORE
DRAWN BY: B ENGLISH
CHECKED BY: J BELL
SUBMITTED BY: J BELL
CHIEF, INST SUP SEC

MOUNTAIN FORK RIVER, OKLAHOMA FY : 98

BROKEN BOW DAM
REPAIR POWER HOUSE BUILT-UP ROOF

POWER HOUSE ELEVATIONS

SCALE: AS SHOWN
DATE: JANUARY 1998
DWG CODE: CC720-C39-20/2
SHEET REFERENCE NUMBER 20/A2
INVITATION NO DACA56-98-B-0007
CONTRACT NO DACA56-98-C-0025
SHEET B85 OF B85

APPENDIX A3 -

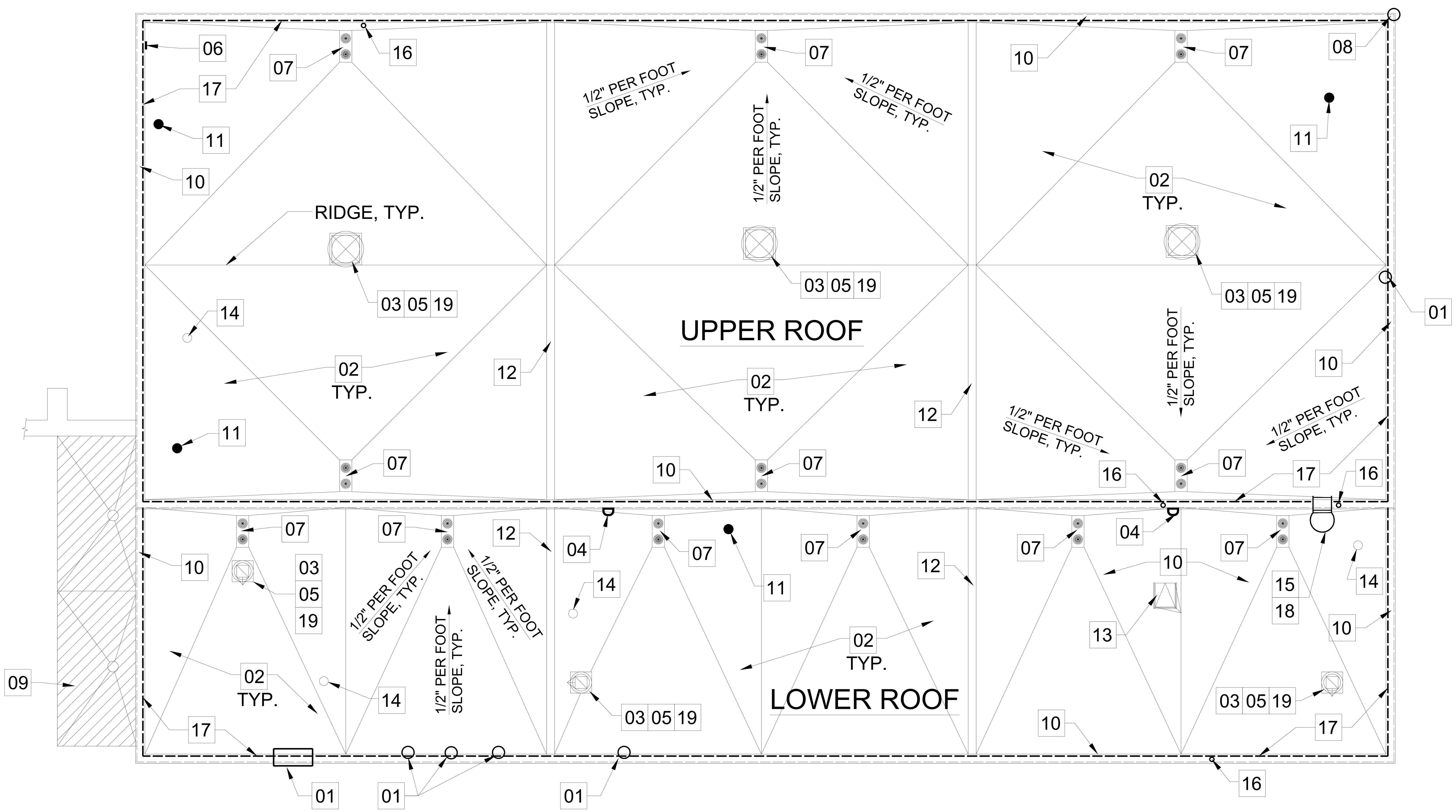
CONCEPT PLAN

GENERAL NOTES

1. CONCEPT ROOF PLAN IS BASED ON AS-BUILT DRAWINGS AND SITE VISIT BY THE GOVERNMENT. CONTRACTOR IS STRONGLY ENCOURAGED TO VISIT THE SITE AND FIELD VERIFY ALL EXISTING CONDITIONS PRIOR TO SUBMITTING BID. NOTIFY GOVERNMENT IMMEDIATELY OF ANY DISCREPANCIES OR INCONSISTENCIES.
2. REFER TO SPECIFICATION SECTION 01 11 00 SUMMARY OF WORK FOR ADDITIONAL INFORMATION AND FOR DESCRIPTIONS OF OPTIONS 1, 2 AND 3.
3. REFER TO 07/21/25 SITE VISIT PHOTOS AND PHOTO KEY PLAN IN APPENDIX A4 FOR ADDITIONAL INFORMATION.
4. COMPONENT AND CLADDING DESIGN WIND LOADS SHALL BE CALCULATED AND SUBMITTED INACCORDANCE WITH THE WHOLE BUILDING DESIGN GUIDE INCLUDING ASCE 7 AND INTERNATIONAL BUILDING CODE (IBC) VERSIONS AS REFERENCED IN UFC 3-301-01.

CONCEPT ROOF PLAN KEYNOTES

01	EXISTING REPAIR ON TOP OF CONCRETE PARAPET. CONFIRM STABILITY AND MAKE ADDITIONAL REPAIR AS NECESSARY PRIOR TO INSTALLING NEW METAL COPING.
02	REMOVE EXISTING ROOF ASSEMBLY DOWN TO STEEL ROOF DECK AND REPLACE WITH NEW TPO FULLY ADHERED MEMBRANE, COVER BOARD, AND TAPERED INSULATION ASSEMBLY WITH ASSOCIATED FLASHING, TRIM, AND ROOF PENETRATION FLASHING. INSTALL TAPERED INSULATION WITH CRICKETS AS REQUIRED TO MEET MINIMUM 1/2" PER FOOT SLOPE OVER EXISTING STRUCTURE SLOPED AT 1/4" PER FOOT.
03	REMOVE AND STORE EXISTING ROOF VENTILATORS ONSITE AS PER LOCATION DETERMINED BY OPERATIONS STAFF. REINSTALL VENTILATORS TO WORKING CONDITION. SEE OPTION 3 TO REPLACE VENTILATORS.
04	EXISTING TIE-OFF ANCHOR MOUNTED ON WALL TO REMAIN.
05	REMOVE AND REPLACE EXISTING EQUIPMENT CURBS AS PART OF NEW ROOFING WORK.
06	PLUG WITH CUT / EXPOSED WIRE INSIDE. CONFIRM FUNCTION AND CONSIDER REMOVAL OF WIRE AND PATCHING OF WALL TO ACCOMMODATE NEW TPO MEMBRANE ON PARAPET.
07	NEW PRIMARY ROOF DRAIN AND OVERFLOW ROOF DRAIN ASSEMBLY TO REPLACE EXISTING DRAINS. PATCH EXISTING OVERFLOW DRAIN HOLES IN METAL ROOF DECK. REFER TO SPEC. SECTION 01 11 00 SUMMARY OF WORK FOR PLUMBING SCOPE OF WORK DESCRIPTION.
08	REPAIR EXISTING DAMAGE TO CONCRETE PARAPET PRIOR TO INSTALLATION OF NEW METAL COPING.
09	ENTRY CANOPY ROOF IS N.I.C. - NOT IN CONTRACT (SHOWN WITH DIAGONAL HATCH PATTERN).
10	WRAP NEW TPO MEMBRANE OVER NEW 1/4" COVER BOARD OVER EXISTING PARAPET WALLS AND CAP WITH NEW PREFINISHED GALVALUME METAL COPING.
11	RECOMMENDED ROOF CORE LOCATIONS. REFER TO SPEC. SECTION 01 11 00 SUMMARY OF WORK.
12	EXISTING EXPANSION JOINT CURB. ENSURE NEW TPO MEMBRANE OVER THE CURB ALLOWS PROPER EPANSION AND CONTRACTION PER MANUFACTURER'S STANDARD DETAILS.
13	REMOVE EXISTING ROOF HATCH AND CURB AND REPLACE WITH NEW STEEL, PAINTED ROOF HATCH AND ACCOMPANYING SAFETY RAILING, NEW STEEL CURB, AND NEW INTERIOR LADDER MOUNTED SAFETY POST. REMOVE EXISTING EXTERIOR MOUNTED LADDER EXTENSION. RELOCATE EXISTING EQUIPMENT ATTACHED TO LADDER EXTENSION TO ANOTHER LOCATION IN COORDINATION WITH POWERHOUSE STAFF. REROUTE EXISTING CONDUIT AND WIRING AWAY FROM ROOF HATCH CURB AND INSTALL PROPER FLASHING AT ANY NEW ROOF PENETRATIONS.
14	EXISTING ROOF PENETRATION TO REMAIN WITH NEW TPO FLASHING.
15	EXISTING ROOF LADDER TO REMAIN. CLEAN LADDER AND CAGE AND PAINT WITH INDUSTRIAL, RUST-INHIBITIVE COATING. REPAIR CONCRETE VOIDS WHERE ATTACHMENT BOLTS ARE PULLING OUT FROM THE WALL AND FULLY RE-SECURE LADDER TO WALL. SEE OPTION 2 TO REPLACE LADDER.
16	EXISTING PARAPET MOUNTED ITEM TO BE REMOVED AND REATTACHED AFTER ROOFING IS COMPLETED.
17	OPTION 1: ADD NEW GUARDRAILS. REFER TO SPEC. SECTION 01 11 00 SUMMARY OF WORK FOR OPTION 1 TO ADD NEW CODE COMPLIANT GUARDRAILS AT UPPER AND LOWER ROOF PARAPET WALLS
18	OPTION 2: REPLACE EXISTING ROOF LADDER. REFER TO SPEC. SECTION 01 11 00 SUMMARY OF WORK TO REPLACE EXISTING LADDER WITH NEW STEEL ROOF LADDER WITH OSHA COMPLIANT LADDER SAFETY SYSTEM.
19	OPTION 3: REPLACE EXISTING ROOF VENTILATORS. REFER TO SPEC. SECTION 01 11 00 SUMMARY OF WORK TO REPLACE EXISTING ROOF VENTILATORS WITH IN-KIND VENTILATORS.



A1 CONCEPT ROOF PLAN
1/8" = 1'-0"

0 4' 8' 16' 24'
SCALE IN FEET

TRUE NORTH

BCOES SUBMITTAL

MARK | DESCRIPTION | DATE

ISSUE DATE: 27 JANUARY 2026
SOLICITATION NO.: W912BV26RA015
CONTRACT NO.:
DESIGNED BY:
DRAWN BY:
CHECKED BY:
SUBMITTED BY:
SIZE:

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

ENGINEERING & CONSTRUCTION DIVISION
ENGINEERING BRANCH

BROKEN BOW POWERHOUSE
BROKEN BOW, OKLAHOMA
ROOF REPLACEMENT

CONCEPT ROOF PLAN

SHEET ID
G-101

C:\USERS\M5ECDMLH\ONEDRIVE - US ARMY CORPS OF ENGINEERS\DESKTOP\MISC\01_CIVIL WORKS\11_BROKEN BOW\REROOF PROJECT\100 PERC-BCOES\CAD\CONCEPT PLAN_011526.DWG
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APPENDIX A4 -

ENCLOSURE 1: BROKEN BOW MFR PHOTOS

BROKEN BOW DAM: POWERHOUSE ROOF 07/21/25 OBSERVATION PHOTOS

Image 1. Damaged concrete and attempted repairs on roof parapets.

Image 2. Damaged concrete and attempted repairs on roof parapets.

Image 3. Concrete damage at NE corner parapet.

Image 4. Missing bolts at roof ladder upper level.

Image 5. Additional roof/parapet appurtenances.

Image 6. Additional roof/parapet appurtenances.

Image 7. View of west contraction joint curb on lower roof, looking south toward parapet wall.

Image 8. Typical view of roof drain and overflow drains along lower roof north wall,

Image 9. General view of lower roof looking west from center of roof.

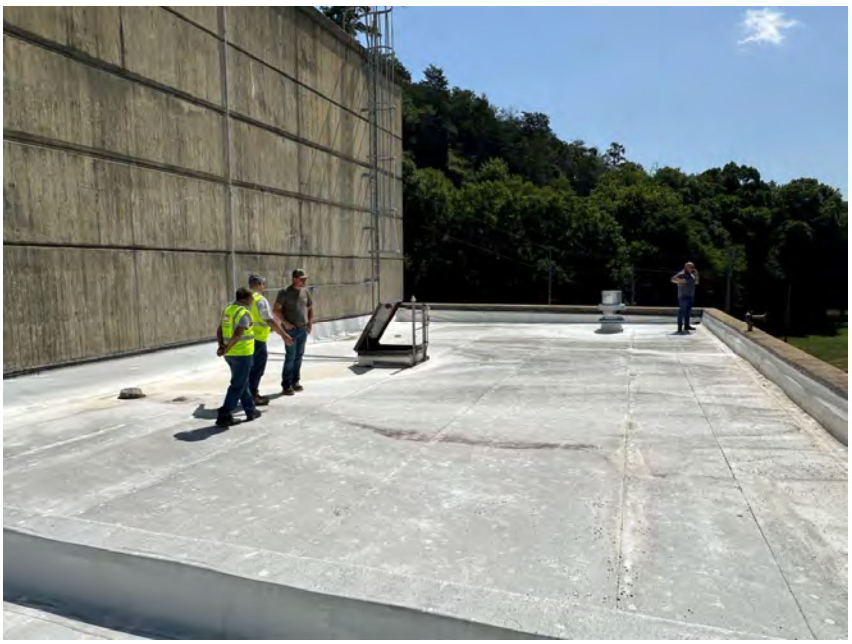


Image 10. General view of lower roof looking east from center of roof.

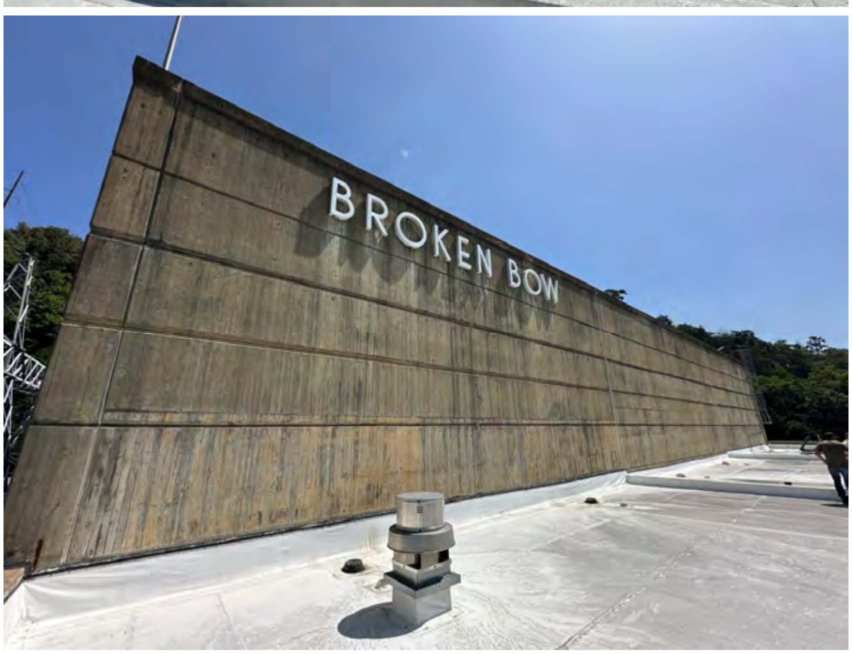


Image 11. View of south face of wall on upper roof, looking north from lower roof.

Image 12. View of roof penetration piping on center section of lower roof.

Image 13. Typical view of roof ventilation adjacent to roof drains on west section of lower roof.

Image 14. View of north end of contraction joint curb abutting south wall of upper roof.

Image 15. View of Roof Hatch.

Image 16. View of Roof Ladder from Lower Roof to Upper Roof.

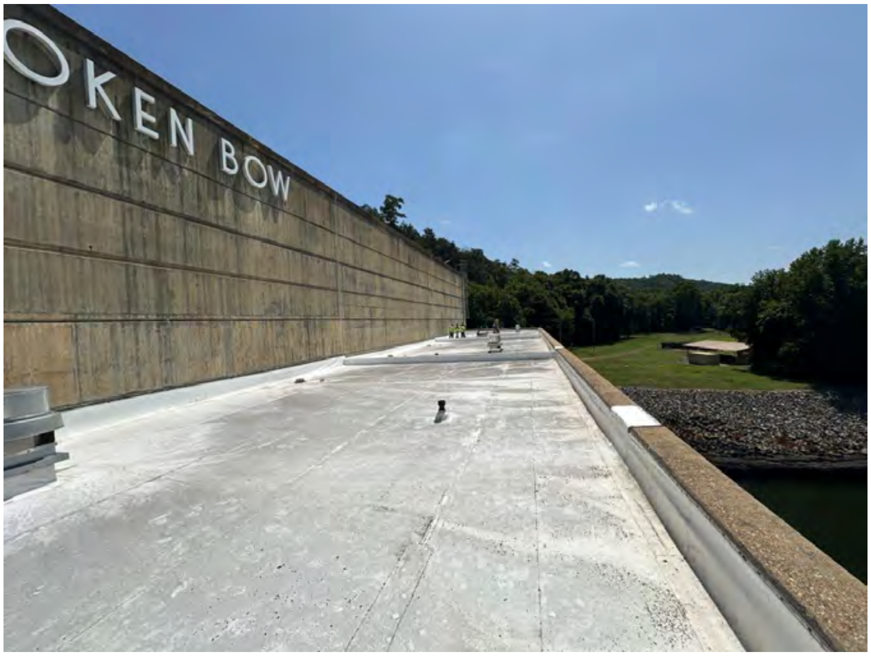


Image 17. View from southwest corner of lower roof.

Image 18. View from southeast corner of lower roof.

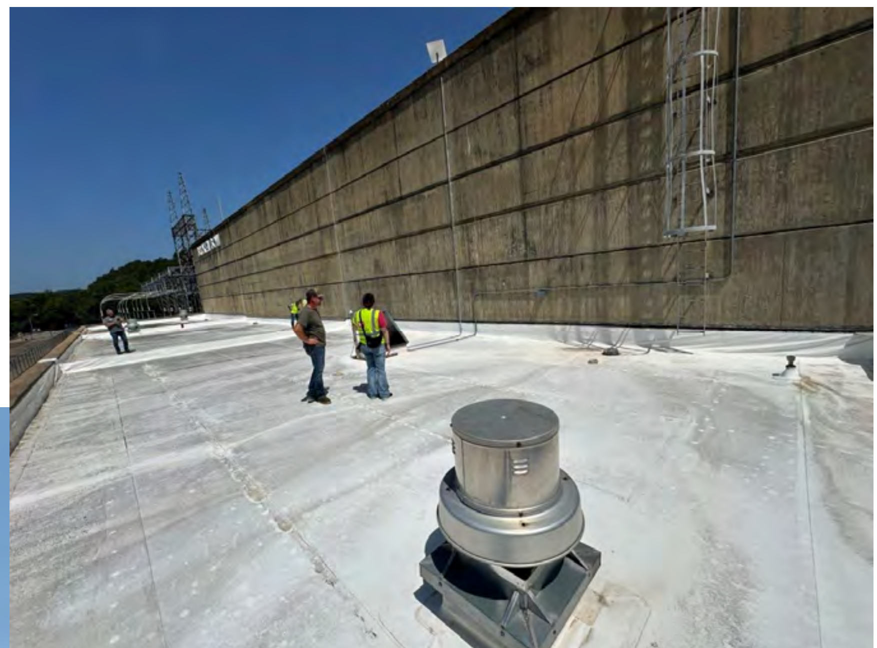


Image 19. View of conduit leading from roof hatch to south wall of upper roof.

Image 20. View down into roof hatch.

Image 21. View of sharp sheet metal at NE corner of lower roof.

Image 22. View of gap in bolt attachment at roof ladder bracket on left side of ladder.

Image 23. View of bolt attachment at roof ladder bracket on right side of ladder.

BROKEN BOW DAM: POWERHOUSE ROOF 07/21/25 OBSERVATION PHOTOS

Image 24. View of SE corner of upper roof at roof ladder access.

Image 25. View of upper roof looking west from the SE corner of the roof.

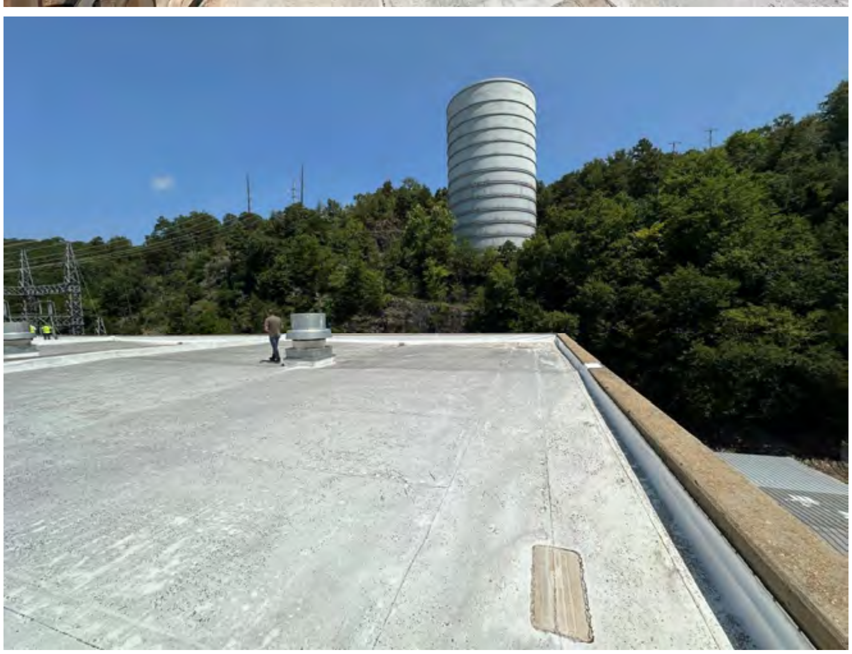


Image 26. View of NE corner of upper roof from SE corner of roof.

Image 27. View of upper roof looking south from the NE corner of the roof.

Image 28. View of upper roof looking SW from the NE corner of the roof.

Image 29. View of degraded roof membrane at NE corner of upper roof.

Image 30. View of roof ventilator in east section of upper roof.

Image 31. View of roof drains on south wall in east section of upper roof, along with antenna on parapet.

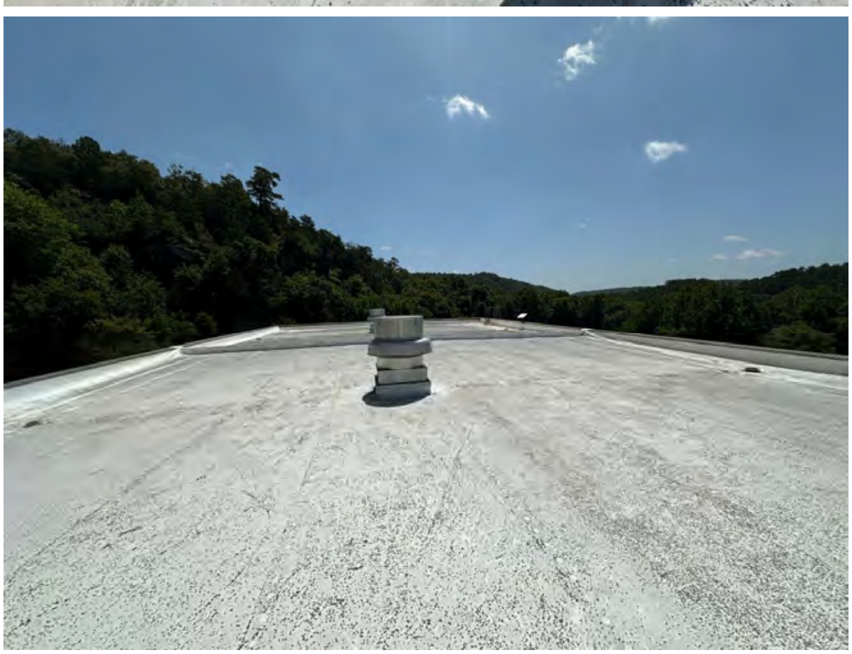
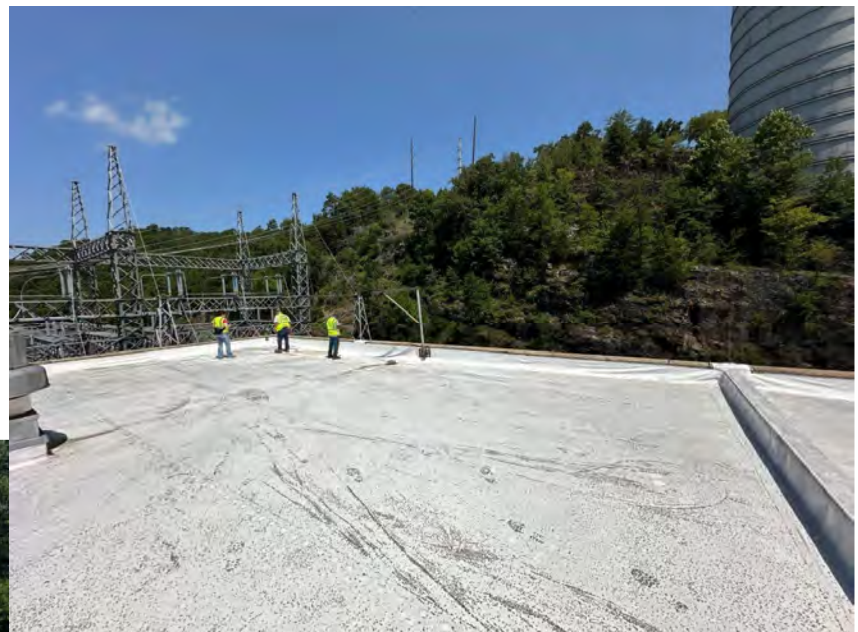


Image 32. View of roof ventilator in center section of upper roof, looking east, along with center section roof drains north and south of the ventilator.

Image 33. View of roof ventilator in west section of upper roof, looking west, along with west section roof drains north and south of the ventilator.

Image 34. View of NW corner of upper roof, looking north.



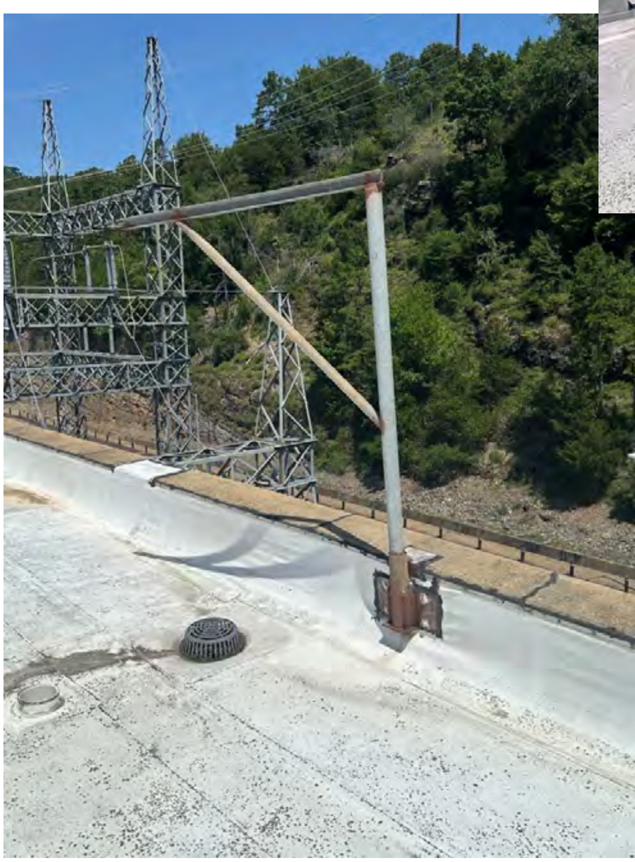


Image 35. View of parapet mounted roof appurtenance at NW corner of upper roof, adjacent to roof drain.

Image 36. View of degraded roof membrane at NW corner of upper roof, looking north.

Image 37. View of degraded roof membrane at NW corner of upper roof.

Image 38. View of degraded roof membrane at SW corner of upper roof.

Image 39. View of north powerhouse wall, east end.

Image 40. View of north powerhouse wall, west end.

Image 41. View of east powerhouse wall showing step up from lower roof to upper roof.

Image 42. View of rip-rap adjacent to east end of powerhouse.

Image 43. View of powerhouse entrance showing drain pipe that was added to divert existing roof drainage pipes from sump to building exterior.

Image 44. View of existing roof drain with very small membrane opening.

Image 45. View showing unprotected opening for overflow roof drainage.

Image 46. Existing D-Ring (1 of 2) utilized for fall protection.

Image 47. View of access into attic space below lower roof.

Image 48. View of new roof drainage piping under lower roof added by powerhouse staff.

Image 49. View of attic space under lower roof showing walkway platform added by operations staff for installation and maintenance of new roof drainage piping.

Image 50. Closeup of roof drainage maintenance platform in attic space.

Image 51. Closeup of roof drain penetrations and new piping in attic space below the lower roof.

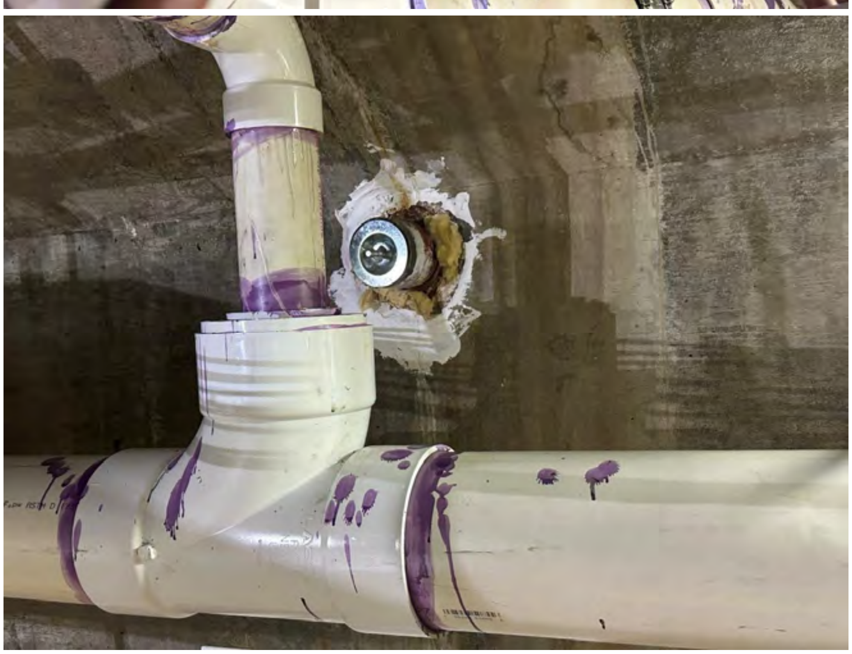


Image 52. Closeup of new roof drain piping in the attic space below the lower roof.

APPENDIX A4 -

ENCLOSURE 2 -

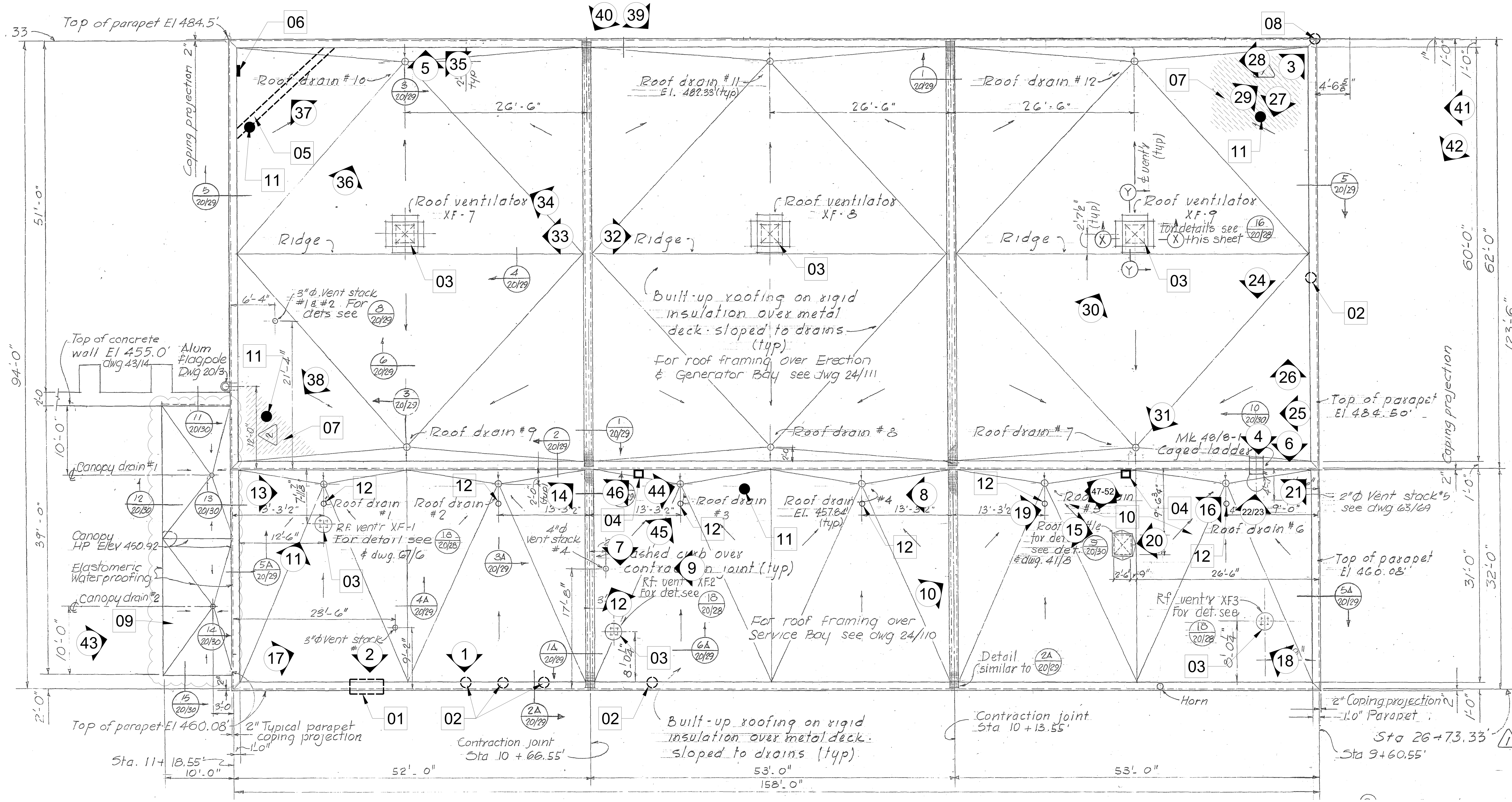
BROKEN BOW MFG PHOTO KEY

LEGEND

XX (photo tag)	PHOTO REFERENCE KEY. REFER MFR ENCLOSURE 1 FOR PHOTOS
XX (box)	ROOF PLAN KEYNOTE

ROOF PLAN KEYNOTES

01	DAMAGE TO CONCRETE PARAPET - COVERED WITH ROOF MEMBRANE.
02	LOCATIONS OF EDGE BREAKOUT ON TOP OF CONCRETE PARAPET WALL, LIKELY DUE TO BOLTING OF THE TERMINATION BAR FOR THE EXISTING ROOF MEMBRANE. REPAIRS HAVE BEEN ATTEMPTED PREVIOUSLY.
03	EXISTING ROOF VENTILATORS REPLACED IN 2010. CONSIDER REPLACEMENT SINCE THEY ARE FIFTEEN YEARS OLD AND WILL NEED TO BE TEMPORARILY REMOVED/REPLACED FOR ROOFING WORK, REGARDLESS.
04	EXISTING TIE-OFF ANCHOR MOUNTED ON WALL.
05	AREA OF ROOF WHERE COVERBOARD HAS EXPERIENCED DETERIORATION, LEAVING A DEPRESSED AREA IN ROOF.
06	PLUG WITH CUT / EXPOSED WIRE INSIDE. CONFIRM FUNCTION AND CONSIDER REMOVAL.
07	AREA OF DEGRADED ROOF MEMBRANE.
08	DAMAGE TO CONCRETE PARAPET - UNREPAIRED.
09	ENTRY CANOPY ROOF IS EXCLUDED FROM THE SCOPE OF WORK (SHOWN CLOUDED).
10	ATTIC SPACE: NO CORROSION OBSERVED, HOWEVER THERE WAS SOME TYPE OF "EFFLORESCENCE" OBSERVED, POSSIBLY FROM MECHANICAL MOISTURE BUILD-UP. REFER PHOTOS 44 - 49.
11	RECOMMENDED ROOF CORE LOCATIONS.
12	LOCATIONS OF OPEN DRAINS NEEDING A NEW DRAIN ASSEMBLY AND COVER.



A1 AS-BUILT ROOF PLAN AND PHOTO KEY

1/8" = 1'-0"

SCALE IN FEET: 0 4' 8' 16' 24'

TRUE NORTH

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

ENGINEERING & CONSTRUCTION DIVISION
ENGINEERING BRANCH

ISSUE DATE: 8 AUGUST 2025

BROKEN BOW POWERHOUSE
BROKEN BOW, OKLAHOMA
ROOF SITE VISIT

EXISTING ROOF PLAN WITH PHOTO TAGS AND SITE OBSERVATION NOTES

SHEET ID

G-101

APPENDIX A5 -

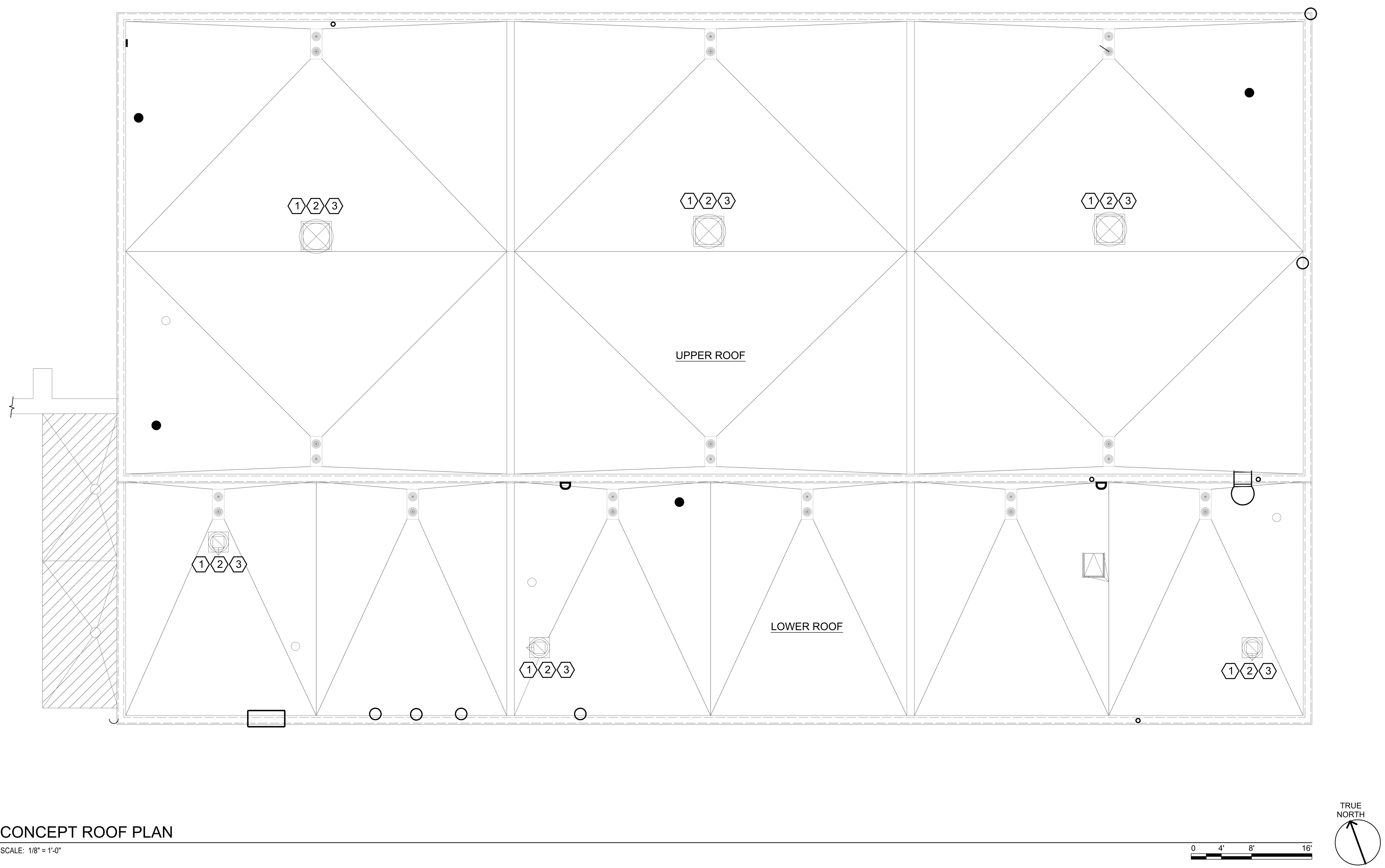
MECHANICAL CONCEPTUAL DESIGN DRAWINGS

GENERAL NOTES

1. CONCEPT ROOF PLAN IS BASED ON AS-BUILT DRAWINGS AND SITE VISIT BY THE GOVERNMENT. CONTRACTOR MUST VISIT SITE AND FIELD VERIFY ALL EXISTING CONDITIONS PRIOR TO SUBMITTING BID. NOTIFY GOVERNMENT IMMEDIATELY OF ANY DISCREPANCIES OR INCONSISTENCIES.
2. REFER TO SPECIFICATION SECTION 01 11 00 SUMMARY OF WORK FOR ADDITIONAL INFORMATION AND FOR DESCRIPTIONS OF OPTIONS 1, 2 AND 3.
3. REFER TO 07/21/25 SITE VISIT PHOTOS AND PHOTO KEY PLAN FOR ADDITIONAL INFORMATION.

CONCEPT ROOF PLAN KEYNOTES

1	VERIFY FAN IS OPERATING PROPERLY. PROVIDE AN INSPECTION REPORT FAN CONDITION. INCLUDE BEARING, MOTOR AND OVERALL LIFE EXPECTANCY.
2	REMOVE EXISTING FAN FOR REINSTALLATION AT A LATER DATE IF BASE BID IS AWARDED. STORE PROPERLY IN CLEAN DRY LOCATION AT THE POWERHOUSE.
3	PROVIDE NEW ROOF CURB IN SAME LOCATION OF APPROXIMATE HEIGHT TO BE 12" ABOVE FINISH ROOF INSULATION.



A2 CONCEPT ROOF PLAN

SCALE: 1/8" = 1'-0"

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. ORNELAS
DRAWN BY:
CHECKED BY:
SUBMITTED BY:
SIZE: ANSI D
ISSUE DATE:
SOLICITATION NO.:
CONTRACT NO.:

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

ENGINEERING & CONSTRUCTION DIVISION
ENGINEERING BRANCH

BROKEN BOW POWERHOUSE
BROKEN BOW, OKLAHOMA
ROOF REPLACEMENT

CONCEPT ROOF PLAN

SHEET ID

M-101

100% READY TO ADVERTISE - NOT FOR CONSTRUCTION

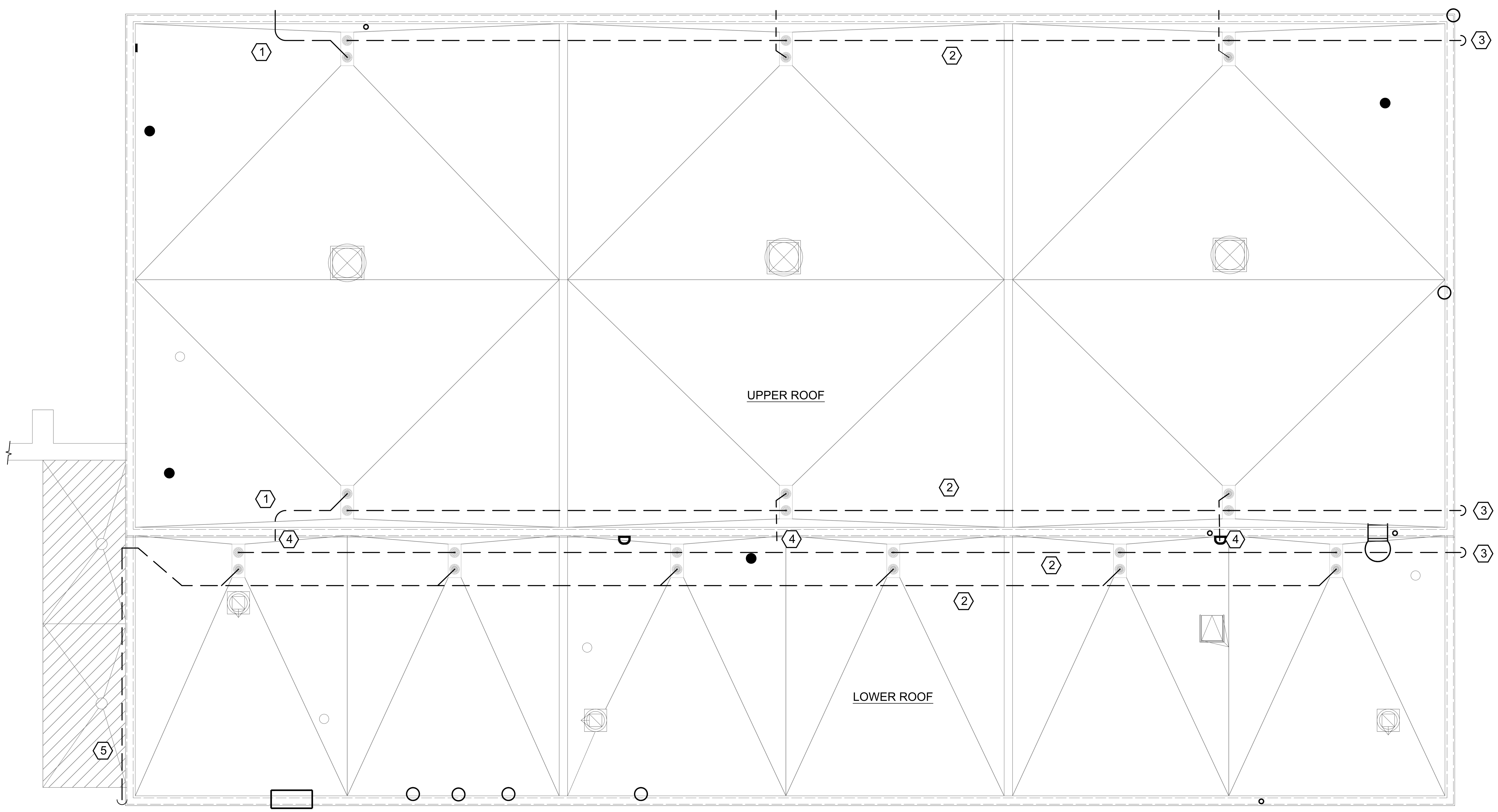
W:\ECD-C\HYDROPOWER\03_PROJECTS\BROKEN_BOW\565819_BB_PHROOFREPL\03_PDT_WORKING_FILES\08_MECHANICAL\CONCEPT PLAN_MECHANICAL.DWG
4:12:54 PM 2/26/2026
M5ECEBGC

GENERAL NOTES

1. CONCEPT ROOF PLAN IS BASED ON AS-BUILT DRAWINGS AND SITE VISIT BY THE GOVERNMENT. CONTRACTOR MUST VISIT SITE AND FIELD VERIFY ALL EXISTING CONDITIONS PRIOR TO SUBMITTING BID. NOTIFY GOVERNMENT IMMEDIATELY OF ANY DISCREPANCIES OR INCONSISTENCIES.
2. REFER TO SPECIFICATION SECTION 01 11 00 SUMMARY OF WORK FOR ADDITIONAL INFORMATION AND FOR DESCRIPTIONS OF OPTIONS 1, 2 AND 3.
3. REFER TO 07/21/25 SITE VISIT PHOTOS AND PHOTO KEY PLAN FOR ADDITIONAL INFORMATION.

CONCEPT ROOF DRAIN PLAN KEYNOTES

1	OFFSET TO AVOID BROKEN BOW SIGN AND MATCH ON OPPOSITE SIDE.
2	SLOPE PIPE TO MEET CODE REQUIREMENTS.
3	DRAINAGE PIPING TO DROP DOWN TO SPLASH PAD.
4	OVERFLOW DRAIN TO DROP TO LOWER ROOF.
5	ROUTE OVERFLOW PIPING ABOVE ENTRY COVER. MATCH ROUTING OF EXISTING PIPING.



A2 CONCEPT ROOF DRAIN PLAN

SCALE: 1/8" = 1'-0"

0 4' 8' 16'

TRUE NORTH

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. ORNELAS
DRAWN BY:
CHECKED BY:
SUBMITTED BY:
SIZE: ANSI D
ISSUE DATE:
SOLICITATION NO.:
CONTRACT NO.:

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TULSA DISTRICT
TULSA, OKLAHOMA

ENGINEERING & CONSTRUCTION DIVISION
ENGINEERING BRANCH

BROKEN BOW POWERHOUSE
BROKEN BOW, OKLAHOMA
ROOF REPLACEMENT

CONCEPT ROOF DRAIN PLAN

SHEET ID
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W:\ECD-C\HYDROPOWER\03_PROJECTS\BROKEN_BOW\565819_BB_PHROOFREPL\03_PDT_WORKING_FILES\08_MECHANICAL\CONCEPT PLAN_MECHANICAL.DWG
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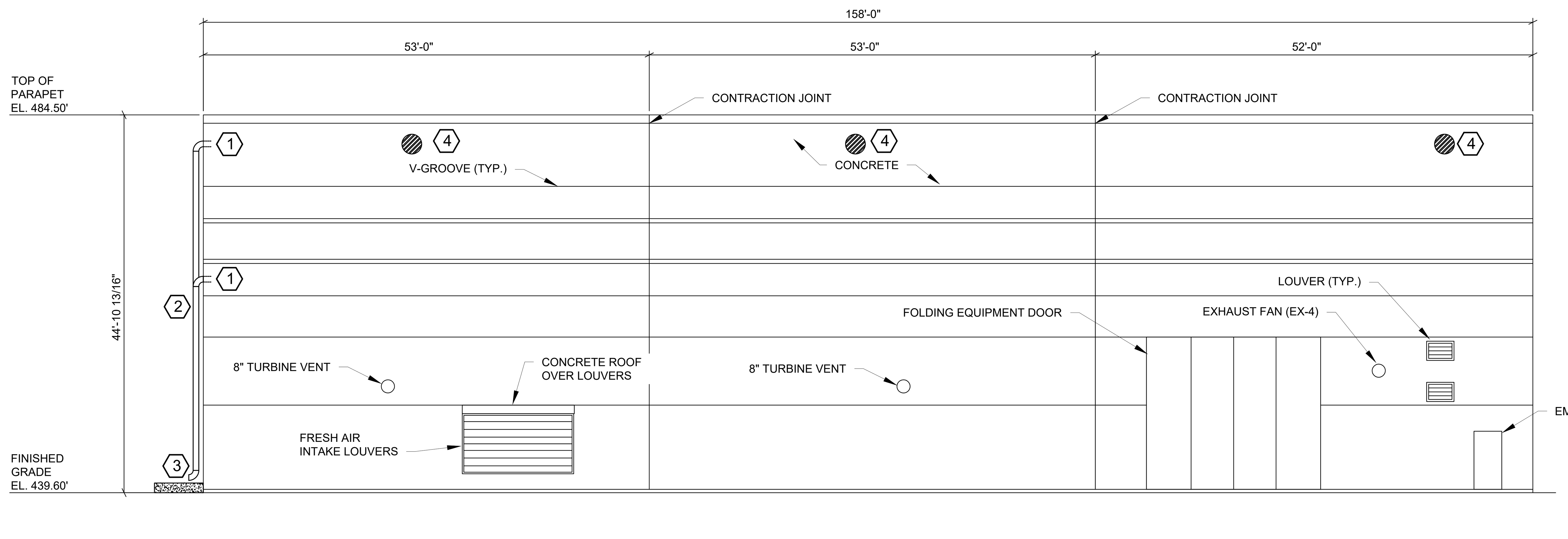
M5ECEBGC

GENERAL NOTES

1. CONCEPT ROOF ELEVATIONS ARE BASED ON AS-BUILT DRAWINGS AND SITE VISIT BY THE GOVERNMENT. CONTRACTOR MUST VISIT SITE AND FIELD VERIFY ALL EXISTING CONDITIONS PRIOR TO SUBMITTING BID. NOTIFY GOVERNMENT IMMEDIATELY OF ANY DISCREPANCIES OR INCONSISTENCIES.
2. REFER TO SPECIFICATION SECTION 01 11 00 SUMMARY OF WORK FOR ADDITIONAL INFORMATION AND FOR DESCRIPTIONS OF OPTIONS 1, 2 AND 3.
3. REFER TO 07/21/25 SITE VISIT PHOTOS AND PHOTO KEY PLAN FOR ADDITIONAL INFORMATION.

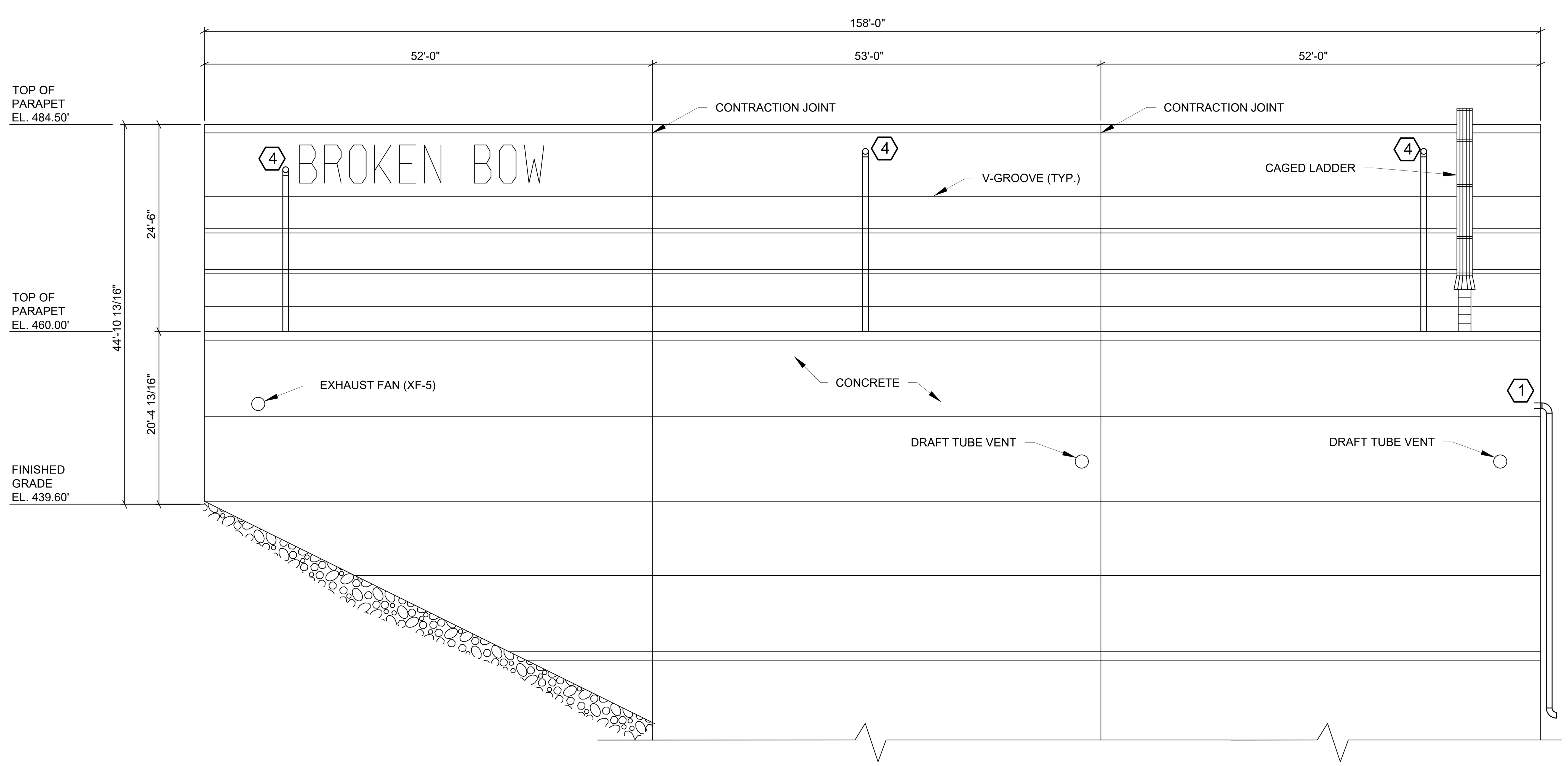
ELEVATIONS KEYNOTES

1	LOWER ROOF AREA FOR DRAINAGE PIPING PENETRATION GENERAL LOCATION. THE LOCATION OF THE PIPING IS A RECOMMENDATION.
2	ROUTE DRAINAGE PIPING DOWN.
3	PROVIDE CONCRETE SPLASH PAD.
4	GENERAL LOCATION FOR UPPER ROOM DRAINAGE PENETRATIONS.



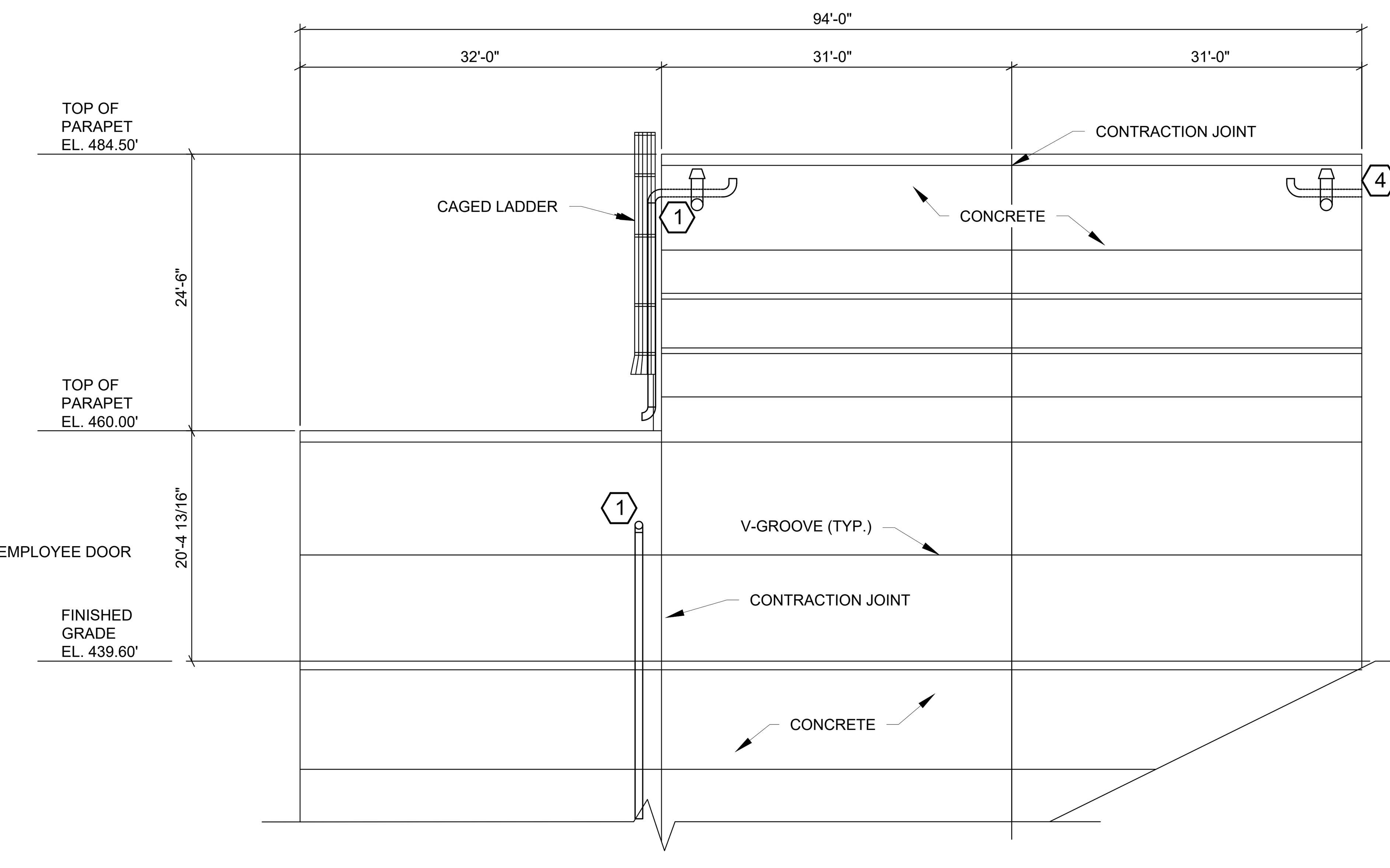
H2 BROKEN BOW POWERHOUSE NORTH ELEVATION (FACING DOWNSTREAM)

SCALE: NTS



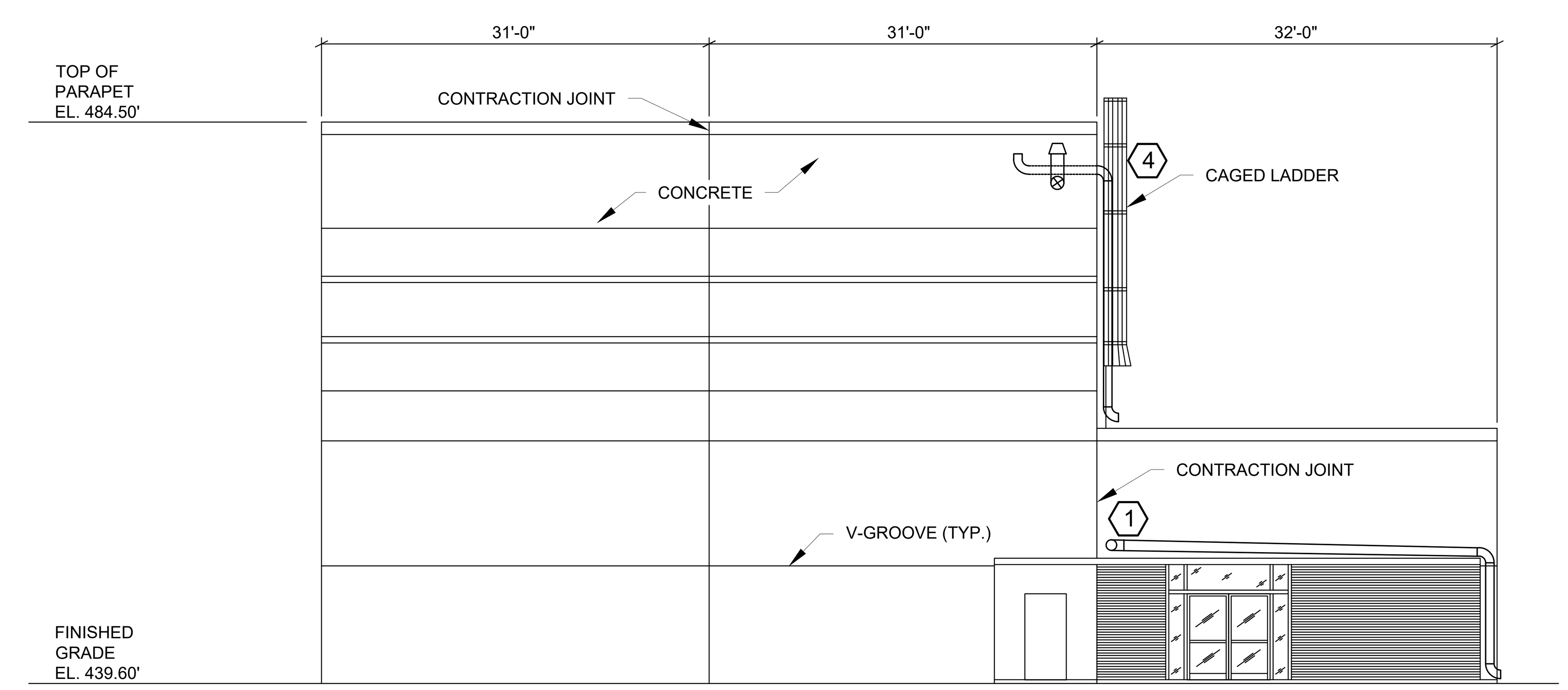
A2 BROKEN BOW POWERHOUSE SOUTH ELEVATION (FACING UPSTREAM)

SCALE: NTS



G14 BROKEN BOW POWERHOUSE WEST ELEVATION

SCALE: NTS



C14 BROKEN BOW POWERHOUSE WEST ELEVATION

SCALE: NTS

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

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CHECKED BY:
SUBMITTED BY:
SIZE: ANSI D
ISSUE DATE:
SOLICITATION NO.:
CONTRACT NO.:

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

ENGINEERING & CONSTRUCTION DIVISION
ENGINEERING BRANCH

BROKEN BOW POWERHOUSE
BROKEN BOW, OKLAHOMA
ROOF REPLACEMENT

CONCEPT ROOF ELEVATIONS

SHEET ID

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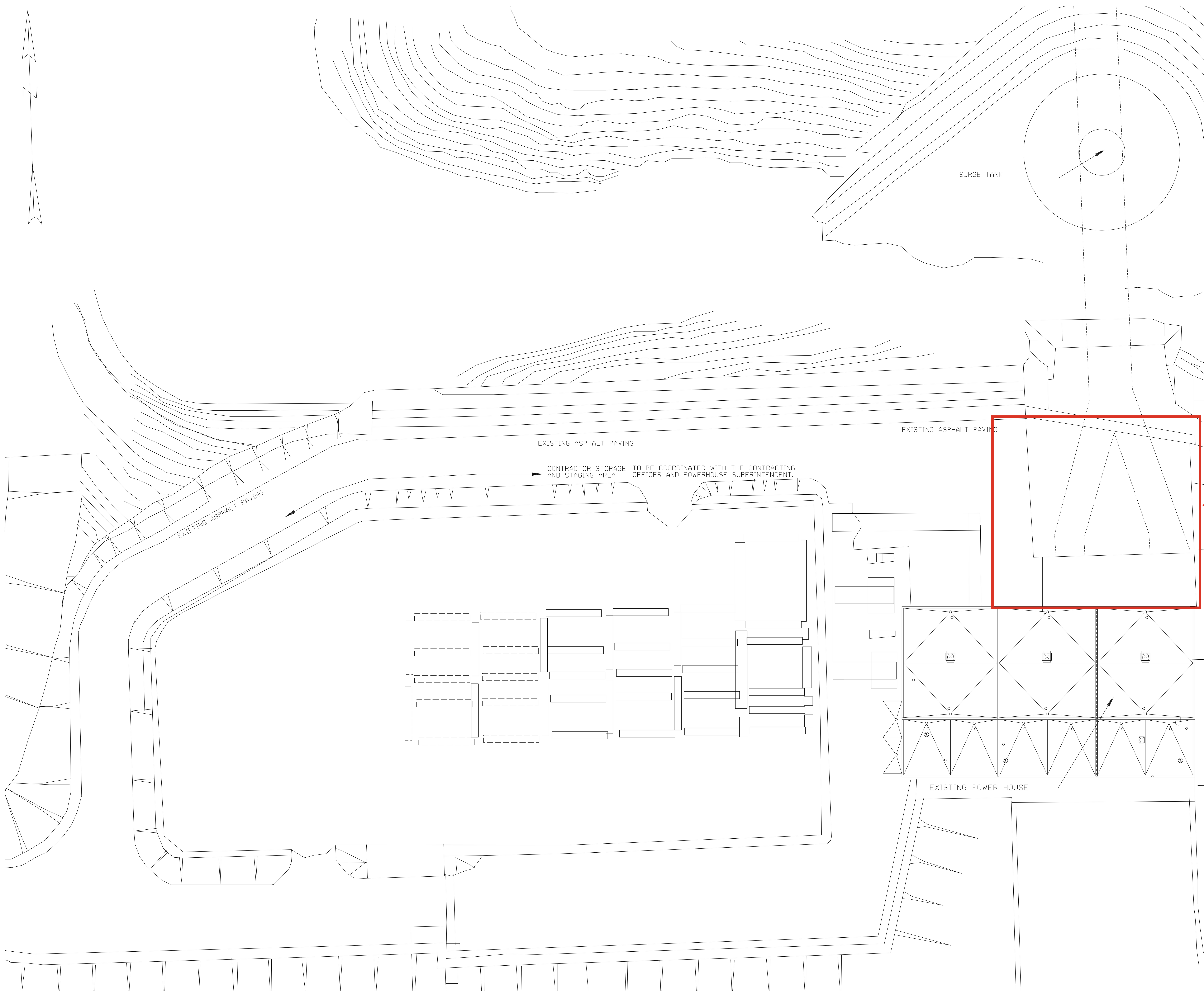
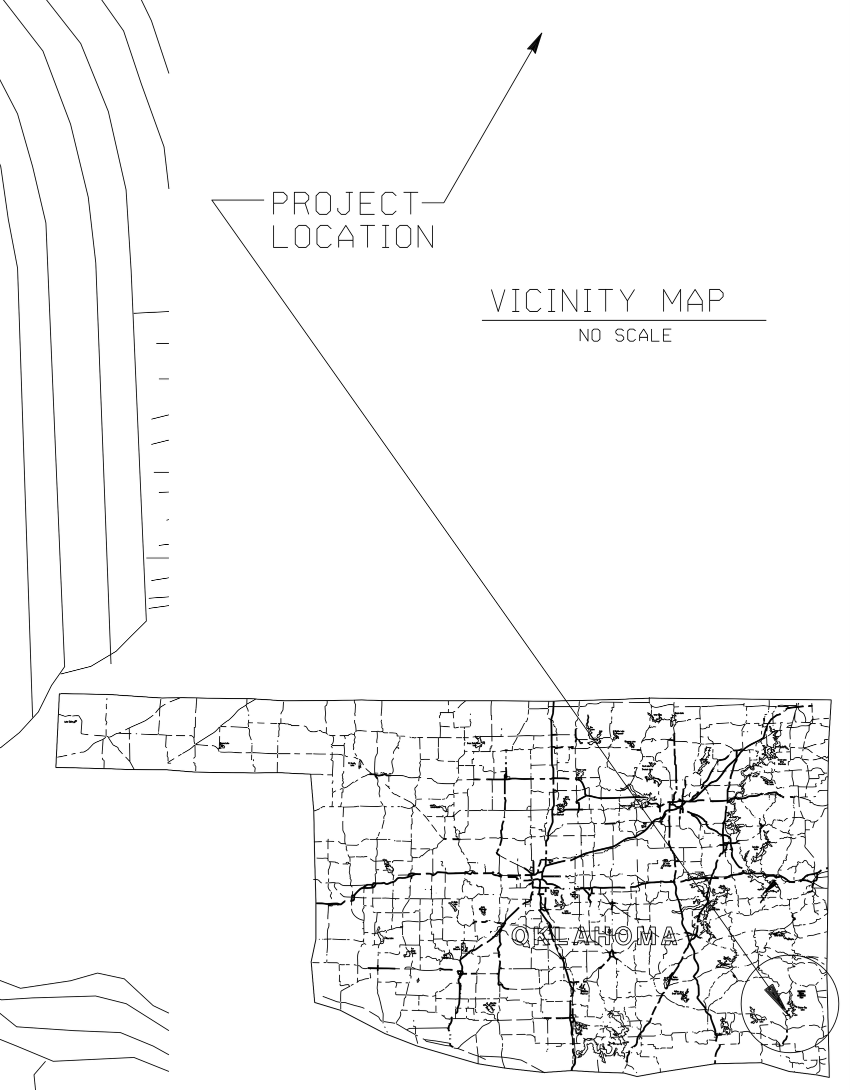
APPENDIX A6 -

SITE PLAN DIAGRAM





1 / C1 BROKEN BOW POWER HOUSE SITE PLAN

0 10' 20' 40' 60'

NOTE: THIS DRAWING WAS SKETCHED FROM A SCANNED DRAWING AND SCALE, THOUGH CLOSE, IS ONLY APPROXIMATE.

GENERAL NOTES

1. THIS DRAWING IS FOR DIAGRAMMATIC PURPOSES ONLY.
2. THESE DRAWINGS ARE BASED ON AS-BUILT DRAWINGS. CONDITIONS ACTUALLY FOUND MAY VARY SLIGHTLY FROM THE INFORMATION SHOWN.
3. PROTECT ALL FENCING FROM DAMAGE.
4. COORDINATE ALL PARKING, STORAGE AREAS AND PROCEDURES WITH THE CONTRACTING OFFICER OR POWERHOUSE SUPERINTENDANT.
5. CONTRACTOR SHALL RETURN STAGING AREA TO EXISTING PRIOR TO DEMOBILIZATION.

RECORD DRAWING "AS-BUILT"

SYM	DESCRIPTION	DATE	APPR
△	REVISED AS CONSTRUCTED	6 AUG 01	

$ THINK VALUE ENGINEERING - IT SAVES MONEY $

U.S. ARMY ENGINEER DISTRICT
CORPS OF ENGINEERS
TULSA, OKLAHOMA

DESIGNED BY: J MOORE	MOUNTAIN FORK RIVER, OKLAHOMA FY : 98		
DRAWN BY: B ENGLISH	BROKEN BOW DAM REPAIR POWER HOUSE BUILT-UP ROOF		
CHECKED BY: J BELL	PROJECT LOCATION & SITE MAP		
SUBMITTED BY: J BELL	SCALE: AS SHOWN	SHEET REFERENCE NUMBER	INVITATION NO DACA56-98-B-0007
CHIEF, INST SUP SEC	DATE: JANUARY 1998	1/C1	CONTRACT NO DACA56-98-C-0025
	DWG CODE: CC890-C72-1/1		SHEET 883 OF 885